

ARKA DATA
Data Driven Drilling Advisory

31 May 2024

The Economic Way to Drill Geothermal Well in Indonesia

By: Yudi Hartono

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Objectives

- Cost Evaluation in Drilling Geothermal Well
- Sharing knowledge for cost saving initiative to provide significant well cost reduction.
- Manage database all good drilling method and practice to achieve drilling cost effective



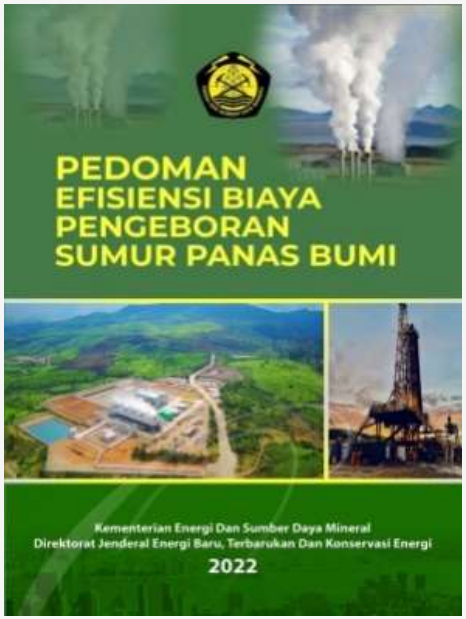






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Book History



Aug 2017 — A non IPM Contract for Exploration Drilling in PT. SMGP, IIGCE Aug 2017

2018-2019 — ITB International Geothermal Workshop (IIGW), presented in pre-workshop session, topic: Drilling Procurement Strategy

23 Dec 2020 — Collaboration between JDS and Enerka to conduct online seminar with the title “The Economic Way To Drill Geothermal Well”.

July 2021 — JDS and EBTKE agreement to translate this seminar to become a book.

9 Dec 2022 — The book of “Pedoman Efisiensi Biaya Pemboran sumur panas bumi” was launched by EBTKE and ESDM.

May 2023 — Book Revision: Re-structure the book content refer to the drilling campaign flow process.

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Content

1 Introduction

2 Cost Efficiency in Geothermal Drilling

3 General Preparations before Drilling

4 Well Design Optimization

5 Efficiency in Drilling Operations

6 Procurement Strategies for Goods and Services

7 Conclusion

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Introduction

1. Installed Capacity of Geothermal Energy and Government Targets
2. Typical Design of Geothermal Wells
3. Drilling Cost Statistics for Geothermal Wells in Indonesia

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Geothermal Installed Capacity & Government Target

- The electrical energy demand is increasing from year to year. In 2023 the average electricity consumption per person in Indonesia reached 1,285kWh/capita or **9.55% increased** from 1,173 kWh/capita in 2022. And in 2024 electricity consumption is targeted to reach 1,408 kWh/capita.

- Ministry of Energy and Mineral Resources (ESDM), it is recorded that the utilized geothermal resources have reached **2.417,7 MW, consisting of 13 Geothermal Power Plan**ts in 11 Geothermal Working Areas.

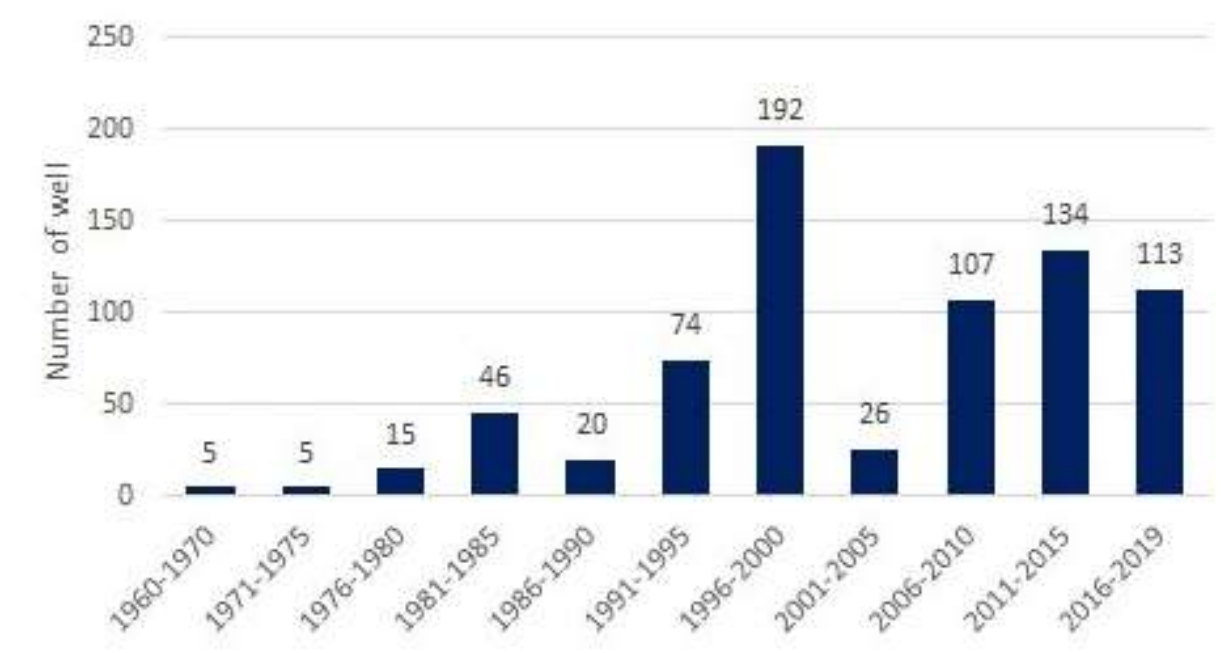
- Indonesia plan to install another **3.3 GW of geothermal energy capacity by 2030**, as stated in state-owned electricity company PLN's latest 10-year Electricity Supply Business Plan (RUPTL). This will require a lot of drilling activities to meet this government objective in boosting electrical energy supply particularly from geothermal sector.



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Geothermal Drilling Activity



2024
2.4 GW

6 Years
± 3.3 GW

2030
5.7 GW

Then (Assumption):

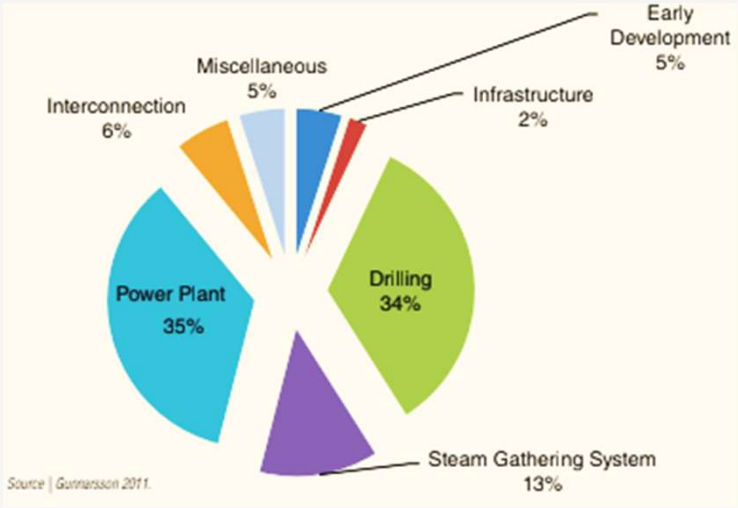
- Every year = 550 MW
- 5-10 MW per well: 55 - 110 Geothermal Production Well per year.
- 60% Success ratio + need to drill injection well, □ **91 - 183 Geothermal Well per year**
- USD 5 mill / well □ **0.46 – 0.92 billion USD drilling business** per year.

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Drilling is Critical Part, Why?

- Drilling cost from exploration to development phase: USD 3210 – 3947 per meter
- Drilling Cost = **30-35%** of Geothermal Project Cost.
- Successfully Drilling Campaign may lead Development of the Geothermal Project.
- Drilling will exist from beginning geothermal project until life of the field.

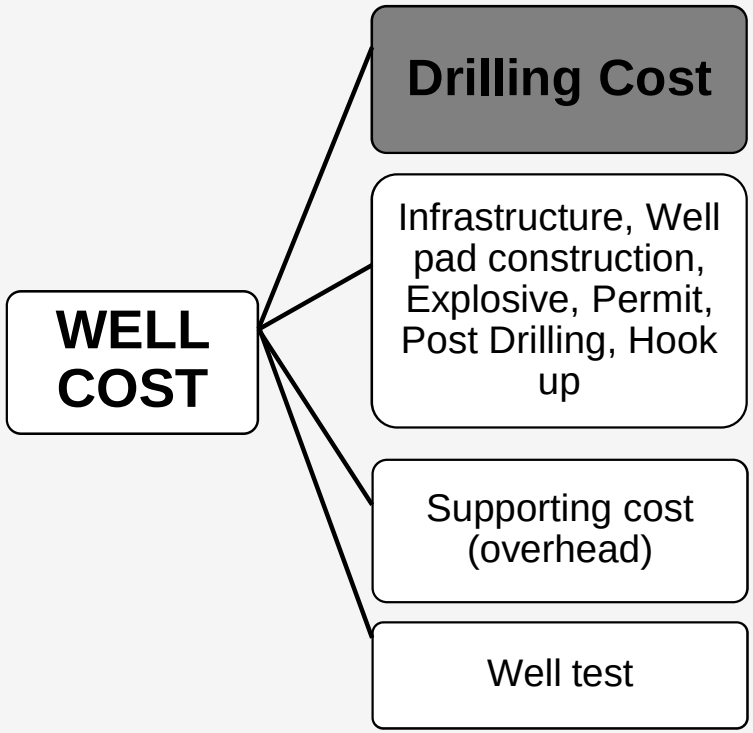


Reducing Drilling Cost -> Significant improve Economic of Field Development

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WELL COST AND DRILLING COST TERMINOLOGY
Cost Structure of Wells and Drilling in Geothermal



A	**TANGIBLE COST**
1	CASING/LINER
2	CASING ACCESSORIES
3	SURFACE WELL EQUIPMENT
4	PAHAT/ BIT
B	**INTANGIBLE COST**
1	KOORDINATOR IPM
2	RIG
3	TOP DRIVE
4	DIRECTIONAL DRILLIING
5	CEMENTING & MATERIAL
6	MONITORING H2S
7	WATER PUMP
8	GENERAL SERVICES
9	INSPEKSI TUBULAR GOOD
10	WELL HEAD/BOTTOM FLANGE WELD
11	CASING INSTALLATION
12	WELL SITE GEOLOGY
13	IPAL - WASTE MANAGEMENT
14	AERATED DRILLING
15	FORMATION EVALUATION CORING
16	FORMATION EVALUATION LOGGING
17	FORMATION EVALUATION MUD LOGGING UNIT
18	RIG INSPECTION
19	FISHING TOOLS, MILLING & WHIPSTOCK
20	AUGING
21	REAL TIME OPERATION
22	SUPERVISION
23	PERMITS & FEES
24	FUEL/ BBM
C	**MOVING (MOB, DEMOB, MOVING, SKIDDING)**

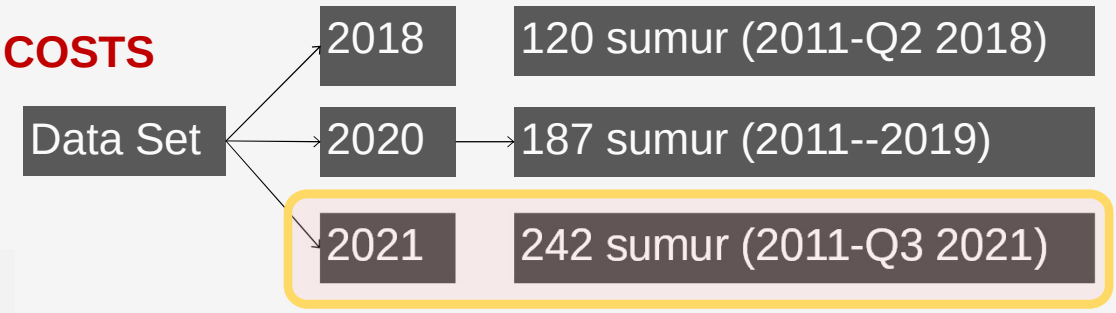
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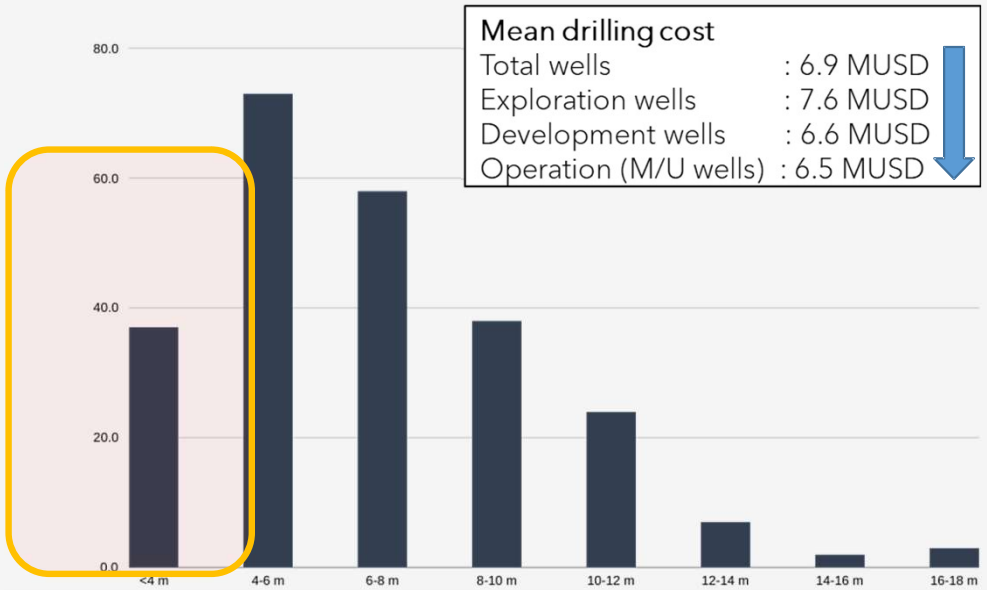
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STATUS OF GEOTHERMAL DRILLING COSTS (2018-2021)
Update on Geothermal Drilling Cost Evaluation

BIAYA PENGEBORAN

- Biaya pengeboran merupakan salah satu performance indikator (KPI) dalam kegiatan pengeboran a.l. *drilling unit cost* (USD/m), *well completion* (m/day), *well output* (MW).
- Parameter yang dipertimbangkan berupa biaya pengeboran (dinormalisasi), kedalaman, tipe kontrak, jenis perusahaan, tahapan kegiatan.
- *Range* biaya pengeboran panas bumi di Indonesia sebesar 1,3 juta USD (kedalaman 498 m) hingga 18 juta USD (kedalaman 2900 m).
- Rata-rata biaya pengeboran 6,9 juta USD/sumur pada akhir 2021 dengan rincian:
  - Sumur eksplorasi : 7,6 juta USD
  - Sumur pengembangan : 6,6 juta USD
  - Sumur operasi (M/U) : 6,5 juta USD





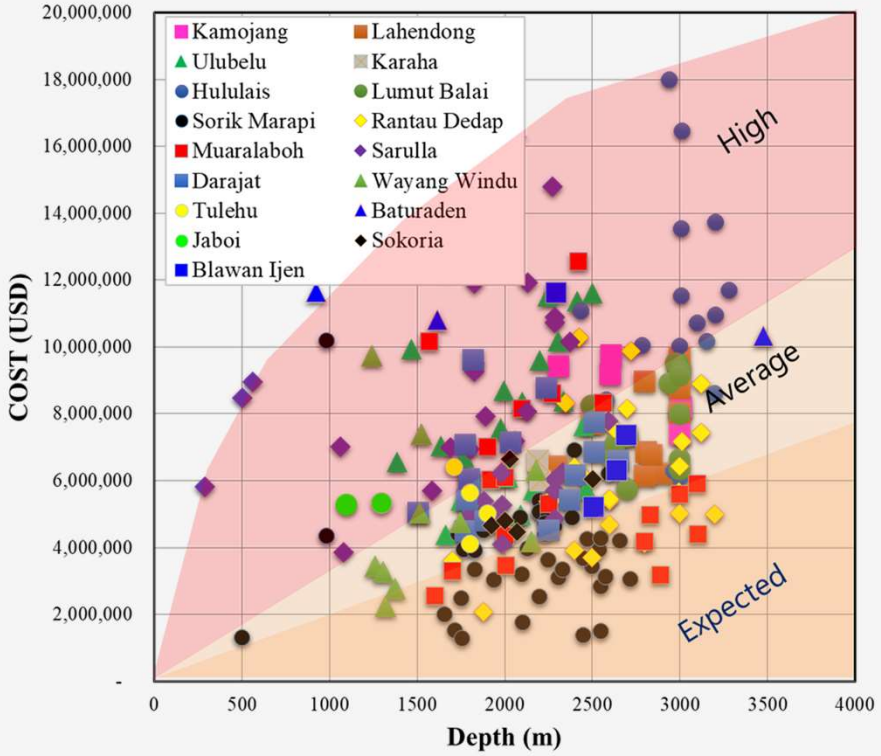
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WELL DEPTH DISTRIBUTION WITH COSTS (CORRECTED US PPI)

Kedalaman dan Biaya Pengeboran 2011-2021

- Terdapat 242 sumur yang dilengkapi dengan realisasi biaya
- Perbandingan dilakukan dengan mempertimbangkan koreksi indeks US PPI



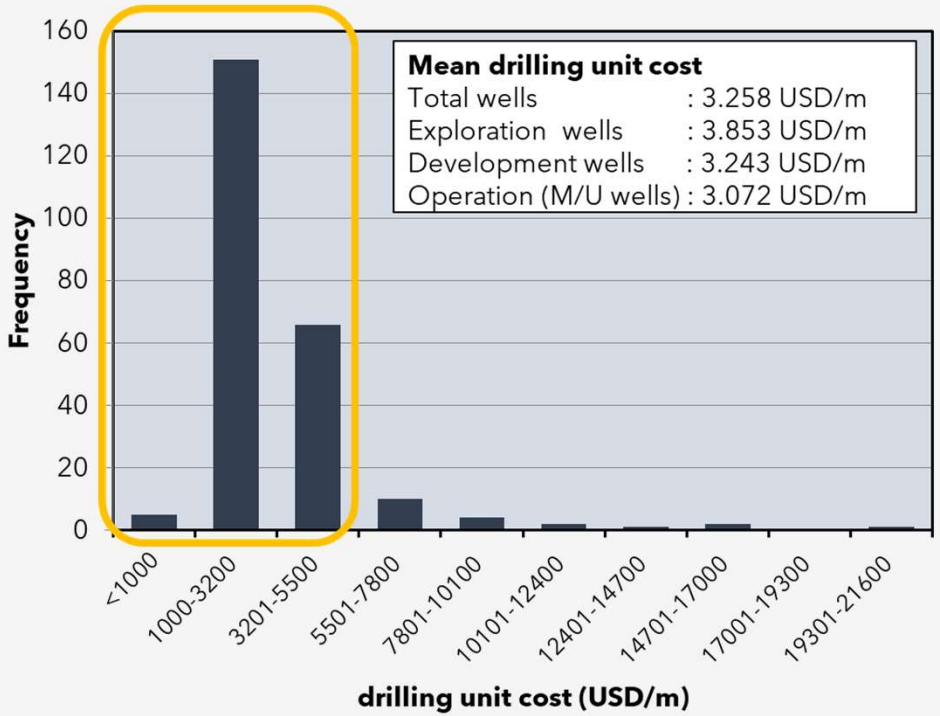
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STATUS OF GEOTHERMAL DRILLING COSTS (2021 UPDATE)
Update on Geothermal Drilling Cost Evaluation (Unit Cost)

BIAYA PENGEBORAN (unit cost USD/meter)

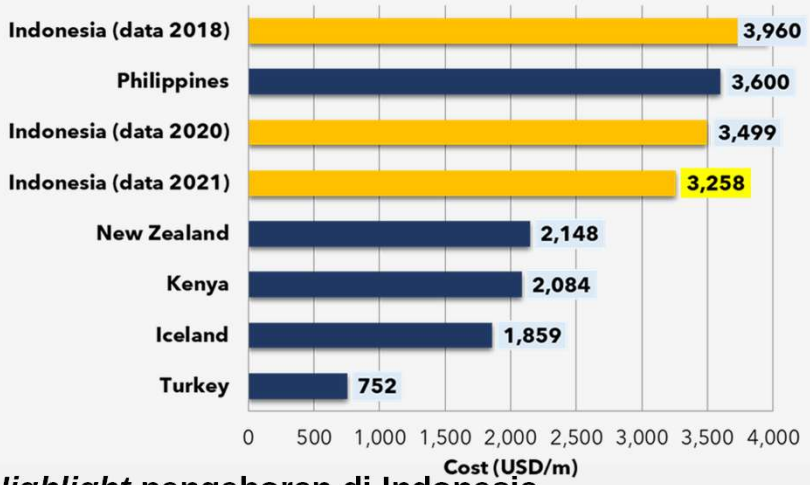
- Rata-rata unit cost 3,258 USD/m
  - Sumur eksplorasi : 3.853 USD/m
  - Sumur pengembangan : 3.243 USD/m
  - Sumur operasi (M/U) : 3.072 USD/m
  - Biaya pengeboran tahap eksplorasi hingga ke operasi (make/up) menunjukkan trend menurun. Mengindikasikan adanya *learning curve*.
  - Pengeboran eksplorasi relatif paling mahal mengingat Badan Usaha dalam tahap memperoleh informasi awal lapangannya.
  - Pada tahap pengembangan dan operasi, pada umumnya biaya pengeboran turun, karena Badan Usaha telah memahami karakeristik lapangan, melakukan efisiensi terhadap biaya *services* yang tidak diperlukan.



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STATUS OF GEOTHERMAL DRILLING COSTS (2018-2021)
Country Comparison



***Highlight* pengeboran di Indonesia**

- Biaya mobilisasi dari yard rig hingga ke lokasi yang *remote* (jarak, pengadaan alat angkat tambahan).
- *Geological setting* pegunungan.
- Pada umumnya menggunakan perjanjian kontraktual konvensional *(daily rate*).

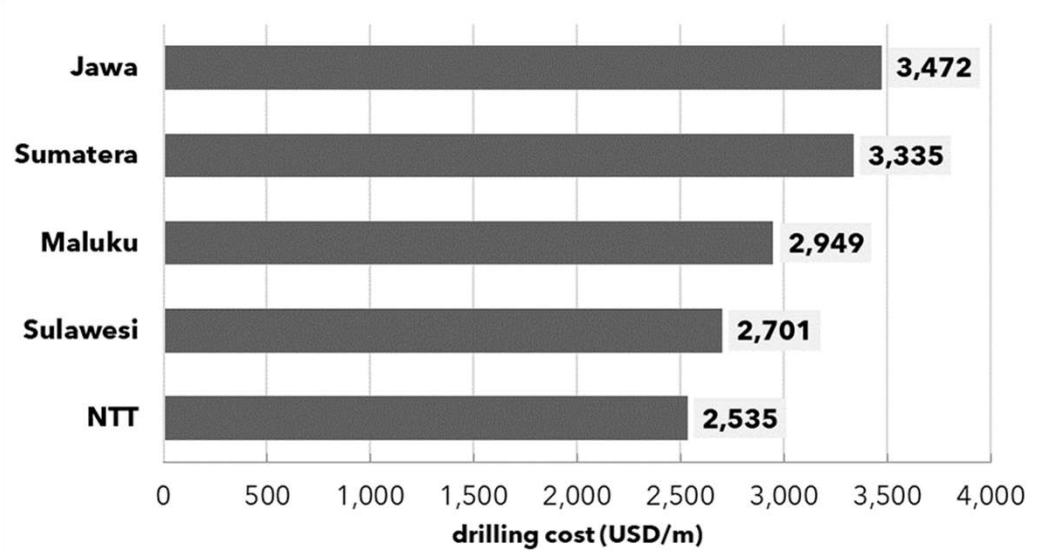
Perbandingan antar negara

- Pengeboran panas bumi di Filipina paling mahal, mengingat sistem panas bumi yang dalam dengan topografi pegunungan.
- Pengeboran di Kenya dan Islandia relatif lebih murah. Biaya pengeboran di Kenya yang dilakukan pengembang yang memiliki rig sekitar 3,5 juta USD/sumur dan dengan sewa rig sekitar 6,5 juta USD/ sumur (Ngugi, 2013) .
- Biaya pengeboran di Islandia berkisar antara 2,5-4,8 juta USD/sumur. Hal ini dikarenakan pengembang panas bumi mempergunakan *hybrid drilling contract*, yaitu gabungan *daily rate, meterage & lump sum* dengan biaya 1000-2000 USD/m (Palsson, 2017).
- Pengeboran di Turki lebih murah: memiliki sistem panas bumi pada *geological setting* yang relatif datar yang didominasi oleh batuan carbonat. Pemboran menggunakan **Rig 1080 HP untuk TD 3200 m-MD, harga sewa hanya USD 11500 per hari**

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GEOTHERMAL DRILLING COST STATUS (2021)
Regional Comparison



- 10 sumur pengeboran yang dievaluasi di Sulawesi didominasi oleh sumur *make-up* dengan menggunakan skema semi IPM.
- 5 sumur yang dievaluasi di NTT dilaksanakan hingga kedalaman 2100 m, menggunakan skema *bundled*.

Perbandingan antar pulau/wilayah

- Perbedaan kedalaman di Jawa dan Sumatera secara umum tidak signifikan yaitu dengan rata-rata kedalaman di Jawa sebesar 2150 sedangkan di sumatera sebesar 2250 m.
- Kedalaman reservoir di Jawa berkisar 650 – 2300 m sedangkan di Sumater berkisar 1000 - 2900 m.
- Tingginya biaya di Jawa diperkirakan karena jumlah sumur di Jawa yang dievaluasi jumlahnya sangat sedikit m (41 sumur) dibandingkan jumlah sumur di Sumatera (182 sumur).
- 4 sumur panas bumi di Maluku dibor dengan menggunakan skema IPM pada kedalaman maksimum 1900 m.
- Sebanyak 52% dari sumur yang dibor selama kurun 2020-2021 di Sumatera dilakukan tanpa menyewa kontraktor rig pengeboran namun dengan rig yang dimiliki oleh Badan Usaha.

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TREND OF GEOTHERMAL DRILLING COSTS (2011-2021)
Comparison of Fluctuations in Geothermal Drilling Costs



Tren tahunan biaya pengeboran

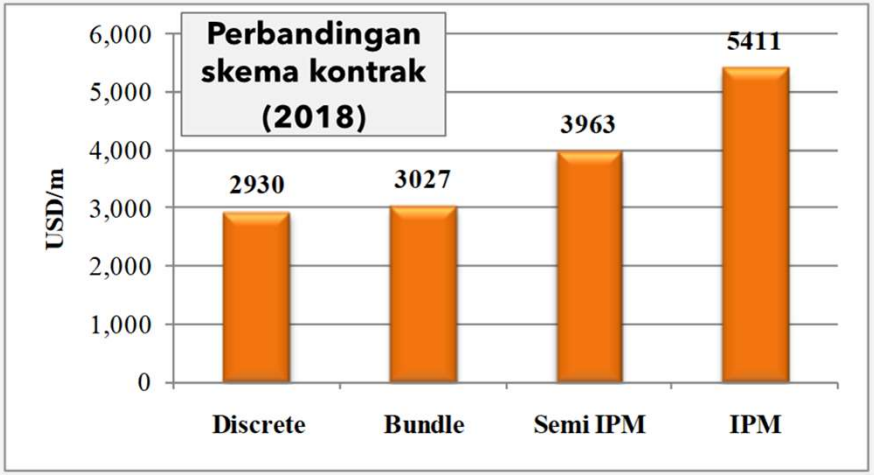
Evaluasi unit cost menunjukkan *trend* :

- Komponen biaya rig pengeboran mengalami penurunan akibat trend penurunan komoditas harga minyak dan berkurangnya frekuensi proyek pengeboran (migas & geothermal). Trend harga minyak turun dari 90-110 USD/bbl (2011-2012) menjadi 50-70 USD/bbl (2017-2019).
- Selain itu, seiring semakin banyaknya tenaga kerja di bidang panas bumi, membuat *developer* dapat merekrut tenaga kerja pada perusahaan masing-masing, dan beralih dari skema kontrak pengeboran Full IPM menjadi Semi-IPM, *bundle* dan *discrete* yang relatif lebih murah serta penggunaan rig sendiri.
- Dengan semakin banyaknya tenaga kerja dan rig pengeboran di bidang panas bumi, proyek pengeboran panas bumi akan semakin terdorong untuk lebih *cost-effective.*
- **Kementerian ESDM mendorong biaya pengeboran panas bumi semakin murah (<2000 USD/m)**

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COMPARISON OF GEOTHERMAL DRILLING CONTRACTS (2018 STUDY)
Commercial/Contractual Schemes for Implementing Geothermal Drilling



Evaluasi skema kontraktual

- Pengeboran tahun 2012-2016 mayoritas menggunakan skema semi IPM dan full IPM pada kedalaman rata-rata 2180 m.
- Pengeboran tahun 2017-2019 mayoritas menggunakan skema *discrete* dan *bundled* pada kedalaman rata-rata 2290 m.
- Kontrak *discrete* dan *bundled* relatif lebih murah pada kisaran kedalaman yang hampir sama dibandingkan kontrak Semi IPM dan IPM.

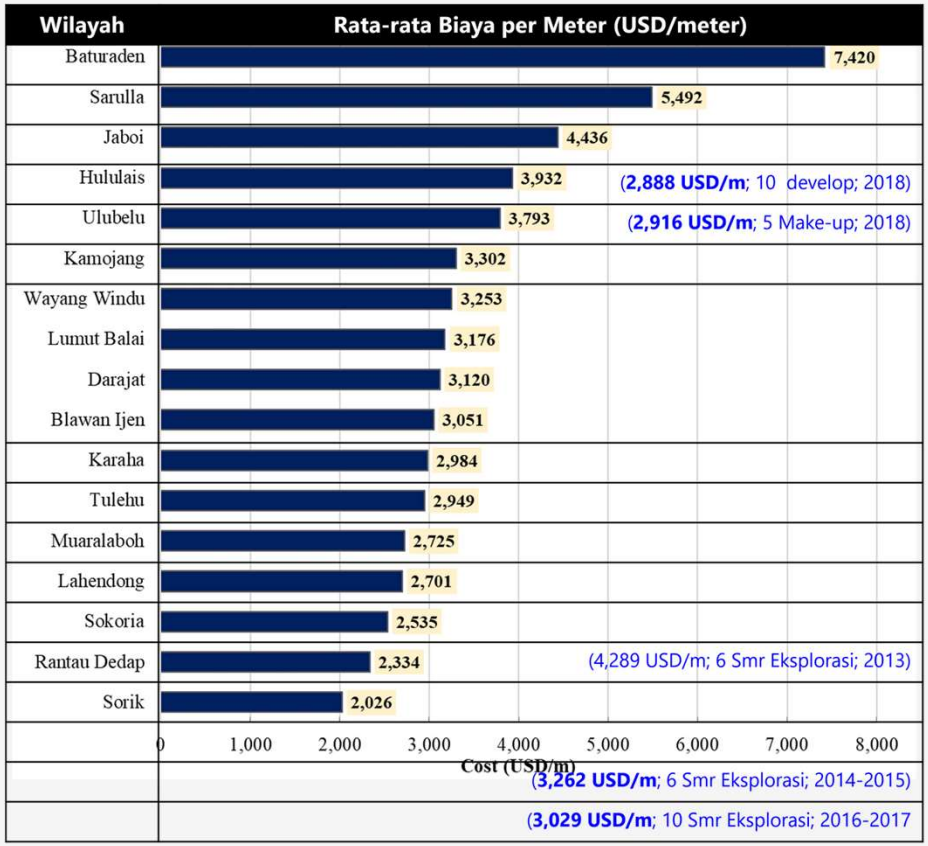
Tren tahunan biaya pengeboran

- Baturaden (2018), Sarulla (2014-2016) dan Jaboi dibor hingga kedalaman rata-rata 1700-1800 m dengan menggunakan **skema kontrak full IPM**.
- Rantau Dedap, Sokoria, Lahendong, Muaralaboh, dan Sorik Marapi dibor hingga kedalaman rata-rata 2100 – 2300 m mayoritas dengan menggunakan ***discrete* dan *bundled* kontrak**. Sorik Marapi tahun 2021 dibor dengan menggunakan rig sendiri, sehingga menurunkan biaya pengeboran.
- Faktor-faktor operasi seperti gangguan pelaksanaan pengeboran yang spesifik dan berbeda pada masing-masing lokasi (formasi batuan yang dijumpai, *loss circulation*, *shallow gas*, jepitan, dll).

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Cost Comparison of Geothermal Drilling Across Fields in Indonesia



Jumlah Sumur	Tahun	Keterangan
3	2018	IPM, Eksplorasi, Jawa
33	2014-2017	IPM, Pengembangan, Sumatera
2	2017-2018	IPM, Eksplorasi, P. Sabang
22	2015-2018	Semi-IPM, Pengembangan, Sumatera
22	2014-2016	Semi-IPM, Make-up, Sumatera
5	2015	Semi-IPM, Make-up, Jawa
10	2018	Semi IPM, Make-up, Jawa
13	2014-2018	Semi-IPM, Pengembangan, Sumatera
16	2011-2021	Discrete, Make-up, Jawa
4	2020-2021	Full-IPM & Bundling, Eksplorasi, Jawa
3	2014 & 2020	Semi-IPM & Discrete, Pengembangan, Jawa
4	2017-2018	IPM, Eksplorasi (appraisal), Maluku
13	2017-2018	Discrete, Pengembangan, Jawa
10	2015-2016	Semi-IPM, Make-up, Sulawesi
5	2017-2018	Bundled, Pengembangan, NTT
26	2018-2021	Discrete, Pengembangan, Sumatera
44	2016-2019, 2020-2021	Bundled, Semi IPM& IPM, Pengembangan, Sumatera. 2021 dengan rig sendiri

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FACTORS AFFECTING GEOTHERMAL DRILLING - Operation & Material Selection

FAKTOR YANG MEMPENGARUHI OPERASIONAL PENGEBORAN

- **Formasi Batuan:** *Boulders, meta sediment, reactive clays, surface loss circulation, sheeting joints*
- **Kegagalan Peralatan:** *Twist off drill pipe*
- **Hilang lumpur**
- ***Stuck pipe, lost in hole, fishing & sidetracking***
- ***Well control*** **:** *Steam kick, shallow gases (CO2 & H2S)*
- **Pengetesan yang tidak direncanakan**
- **Isu konstruksi well pad**

Mitigasi kendala operasional/ *drilling hazard* melalui peningkatan kualitas data geosains **(peranan Eksplorasi Panas Bumi oleh Pemerintah)**

Bongkah metasedimen (Lapangan PGE)

Reactive Clay (PGE)



Umum terjadi pada Lapagan Hululais (PnGE)

Seleksi lokasi wellpad di Sarulla dekat dengan fumarol (*crater*) kurang stabil

Stuck pipe sering terjadi (di Sarulla *pengeboran* 20 dari 33 sumur mengalami *stuck pipe*)

Sarulla (pad WJR, pad NIL 1N dan NIL 3)

Pertimbangan seleksi kualitas material sumur panas bumi:

- Temperatur dan tekanan sumur → seleksi manufaktur casing (tipe ERW/SAW/ Seamless) maupun material casing (*carbon steel/ corrosion resistance alloy*)
- Koneksi casing → *buttress thread/ premium connection.*
- Properti kimia fluida → korosi akibat pH fluida

Case sumur LHD-23:

- pH 2,8
- Temperatur 178 °C
- WHP 37-80 bar
- Desain casing bukan untuk sumur asam

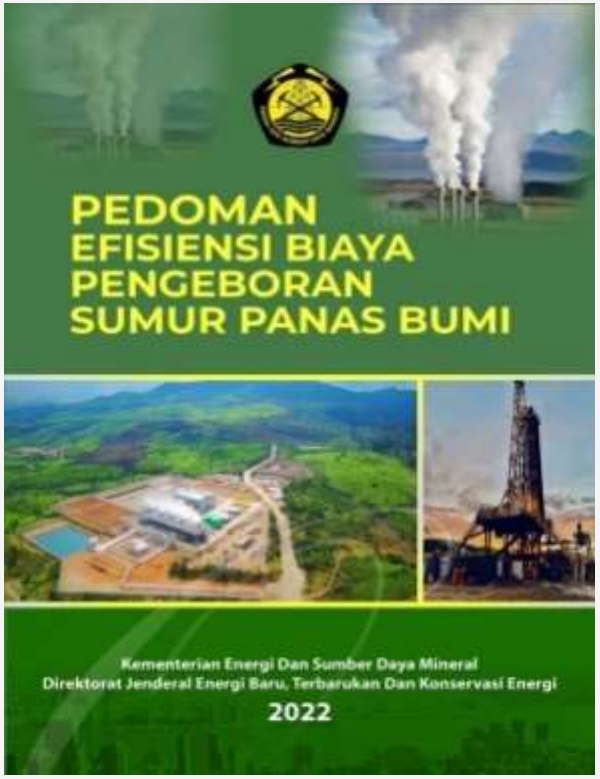


Penipisan casing dan kebocoran side valve

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Lesson Learned & Improvement

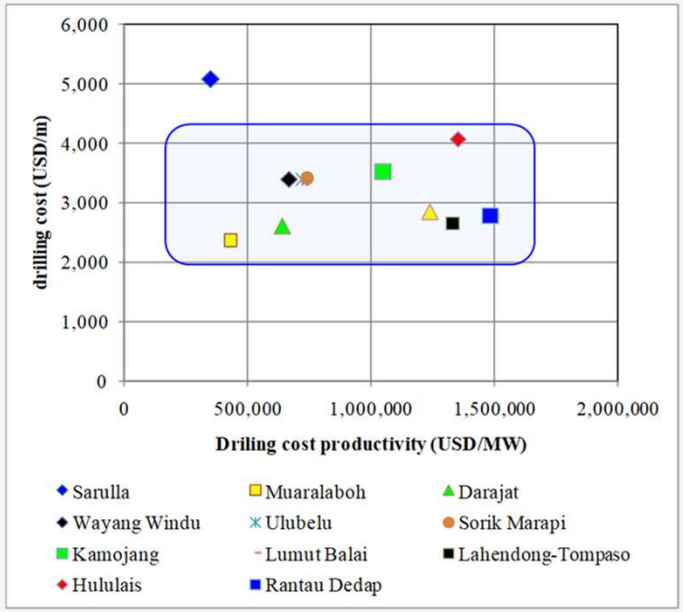


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Drilling Cost for Various Well Capaties

Lapangan	Kedalaman sumur (m)	Biaya (Juta USD)	Kapasitas sumur (MW)	Avg cost juta USD/MW	Cost (USD /m)	Jumlah sumur dievaluasi
Kamojang	2308-3009	8,1 – 9,7	4 - 16,8	1,05	3516	4
Lahendong-Tompaso	2300-3003	6,1 – 8,8	2 – 14,8	1,33	2646	5
Ulubelu	1979-3000	5,7 – 9,7	5 – 30	0,72	3388	8
Lumut Balai	2495-2632	6,6 – 8,2	4,3 – 7,5	1,24	2851	4
Hululais	1668-3280	8,4 – 17,9	4,8 – 15,5	1,35	4067	5
Sorik Marapi	2089-2541	3,9 – 9,9	5 - 15	0,74	3409	6
Rantau Dedap	2248-3123	3,6 – 10,2	2 - 20	1,48	2782	10
Muaralaboh	1600-3103	2,5 – 10,1	5 - 40	**0,43**	2362	11
Sarulla	1799-2128	4,9 – 16,2	14 – 62*	**0,35**	5068	10
Darajat	1775-2652	4,5 – 6,8	4 - 37	0,64	2610	8
Wayang Windu	1244-1747	4,7 – 9,7	6,7 - 18	0,67	4691	4



Keterangan:

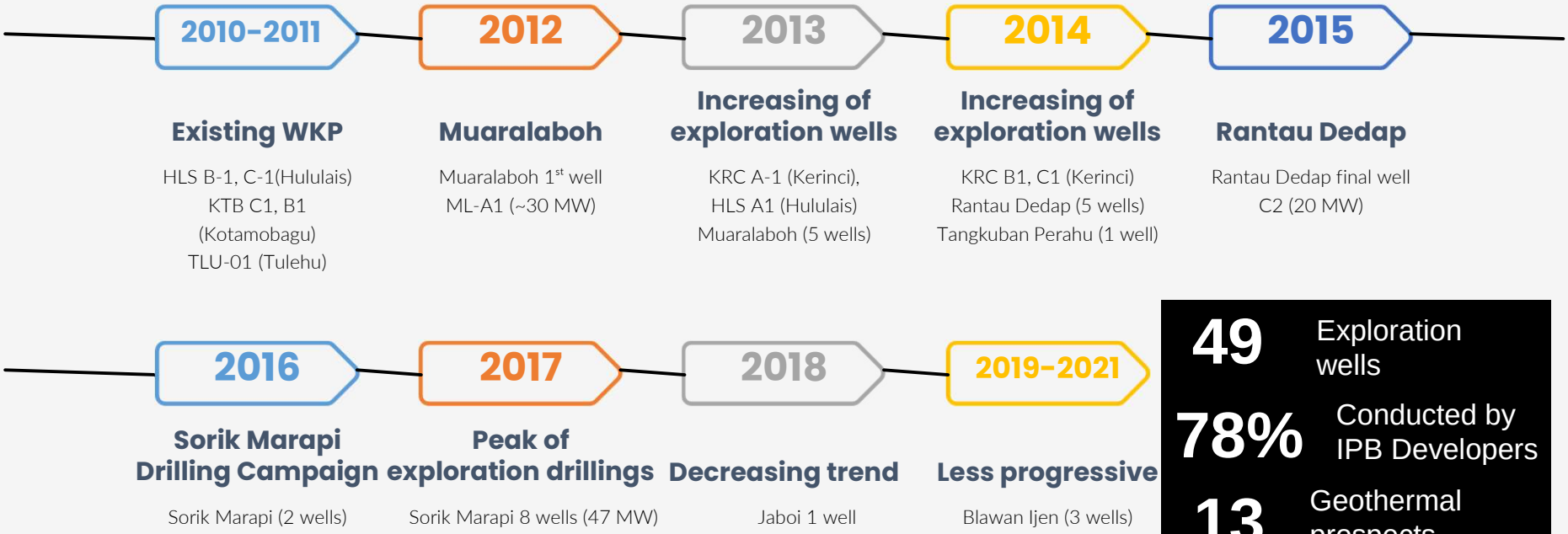
1. Sumur yang dipergunakan merupakan sumur yang memiliki output, terkoneksi ke PLTP, memiliki biaya pengeboran.
2. * Sumur di Sarulla dihitung dengan mempertimbangkan brine untuk bottoming & binary.

Sumur produktif di Indonesia menunjukkan biaya pengeboran/ MW berkisar 400.000 – 1,5 juta USD/sumur

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Exploration Well Drilling Timeline



49 Exploration wells

78% Conducted by IPB Developers

13 Geothermal prospects

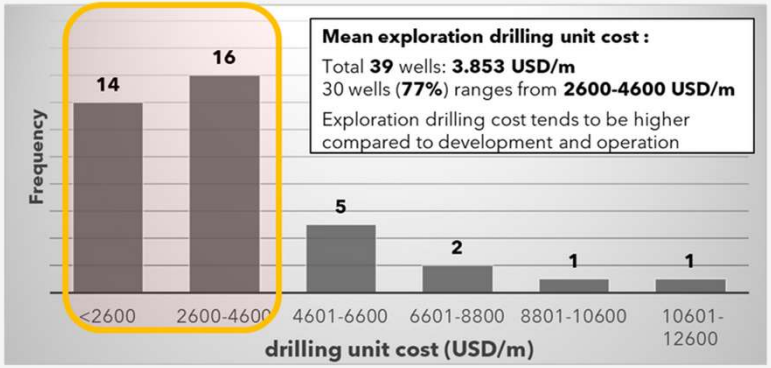
186 MW well output tested from steam flow

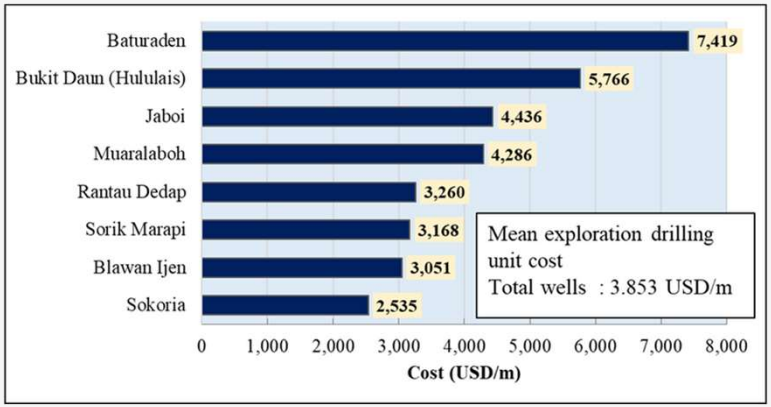
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Exploration Well Drilling Costs

7,6 MUSD	Exploration well drilling cost from 39 wells available (2011-2021)
1,2-12 MUSD	Exploration well drilling cost range
	Average exploration drilling unit cost (2011-2021)
500-3477 m	Exploration well depths (successful & unsuccessful)
Drilling Contracts	Full IPM is preferable, however discrete & bundled are the cheapest
Wellpads	1-5 well pads during exploration with 2-10 wells/prospect

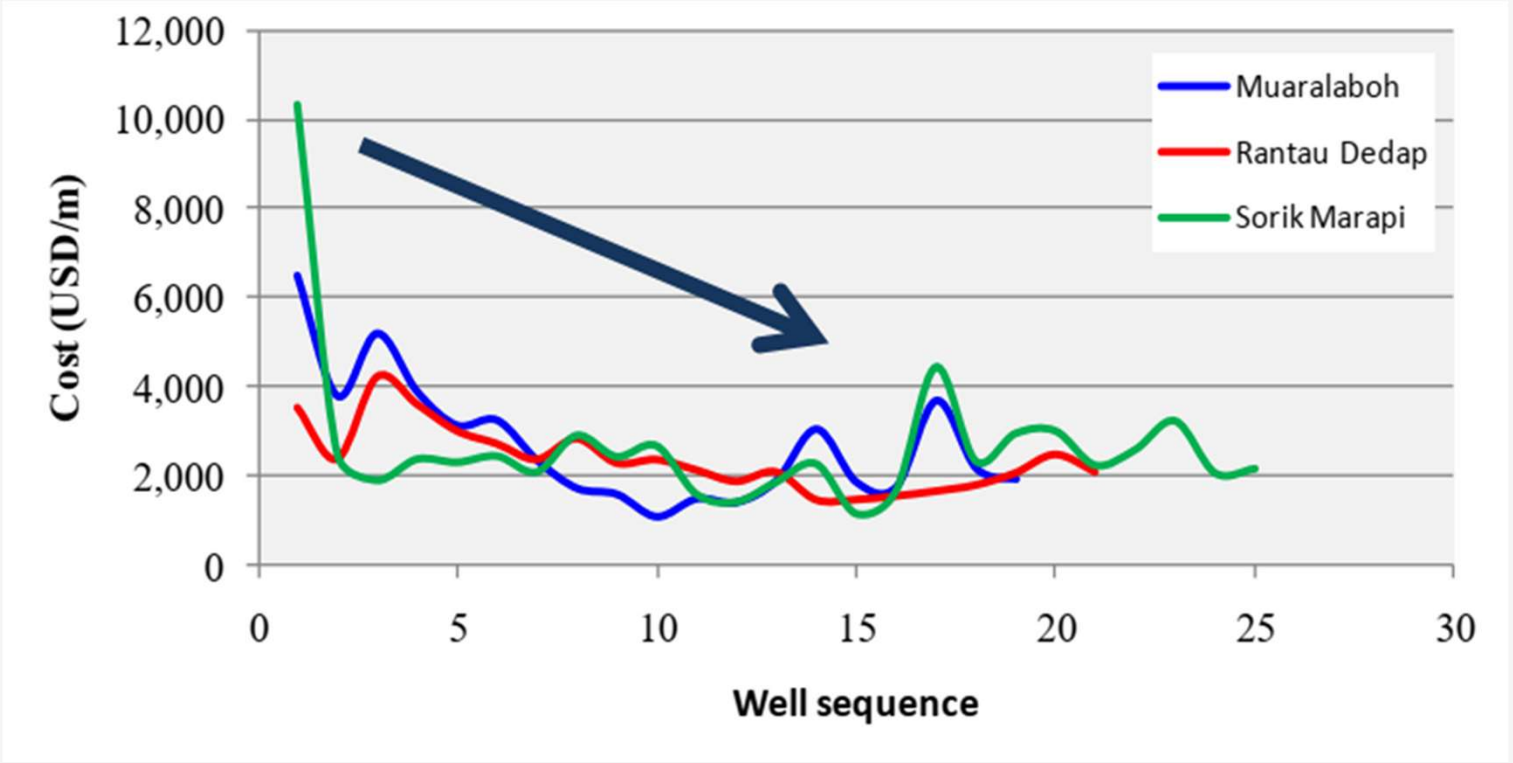




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Learning Curve



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Best Geothermal Drilling Performance by 2024

Well Name Operator	Year Drilled	Rig HP	TD, m-MD	Meter / Day	USD / Meter	Performance Category
SMT-H2 Batu Raden	2018		3,477		2,971	The deepest well
HLS E3 Hululais	2017		3280		3,564	The deepest well
Star Energy						Fasters Meter / day
Central Java	2022	1500	3,009	93.48	1,146	Fasters Meter / day and Cheapest USD / Meter
South Sumatra		1000	2,200	93.66	1,176	Fasters Meter / day
South Sumatra		1700	2,715	93.41	1,148	Cheapest USD / Meter

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How to achieve drilling cost below USD 1000 per meter

Perbandingan Biaya Sumur: Rig 1500 HP dengan Rig 1000 HP, 3 joint

Sumur	Nama Rig	Rig HP	Tahun	TD	CPM	Total Cost	Drilling Days	Cost saving	Nomalize w/ Rig 1000 HP		
									Well Cost	CPM	
				m-MD	USD	USD	Days	USD/Day	USD	USD	%
SLR-J-7D	PRA-09	1500	2022	3,009	1,146	3,448,314	65	20,990	2,083,964	693	60%
SLR-J-7E	PRA-09	1500	2022	3,004	1,397	4,196,588	55	20,990	3,042,138	1,013	72%
SLR-G-29B	PRA-09	1500	2022	3,000	1,330	3,990,000	51	20,990	2,919,510	973	73%
SLR-G-29C	PRA-09	1500	2023	2,905	1,321	3,837,505	40	20,990	2,997,905	1,032	78%
								Average Cost Per Meter		927	

Description	1500 HP	1000 HP 3 Joint	Saving USD/day
Daily Rate	38,000	26,000	12,000
Fuel Consumption	19,065	10,075	8,990
	Total Saving/ Day		20,990

Perbandingan Biaya Sumur: Rig 1500 HP dengan Rig 1000 HP, 2 joint

Sumur	Nama Rig	Rig HP	Tahun	TD	CPM	Total Cost	Drilling Days	Cost saving	Nomalize w/ Rig 1000 HP		
									Well Cost	CPM	
				m-MD	USD	USD	Days	USD/Day	USD	USD	%
SLR-J-7D	PRA-09	1500	2022	3,009	1,146	3,448,314	65	26,990	1,800,066	598	52%
SLR-J-7E	PRA-09	1500	2022	3,004	1,397	4,196,588	55	26,990	2,864,741	954	68%
SLR-G-29B	PRA-09	1500	2022	3,000	1,330	3,990,000	51	26,990	2,769,981	923	69%
SLR-G-29C	PRA-09	1500	2023	2,905	1,321	3,837,505	40	26,990	2,949,780	1,015	77%
								Average Cost Per Meter		871	

Description	1500 HP	1000 HP 2 Joint	Saving USD/day
Daily Rate	38,000	20,000	18,000
Fuel Consumption	19,065	10,075	8,990
	Total Saving/ Day		26,990

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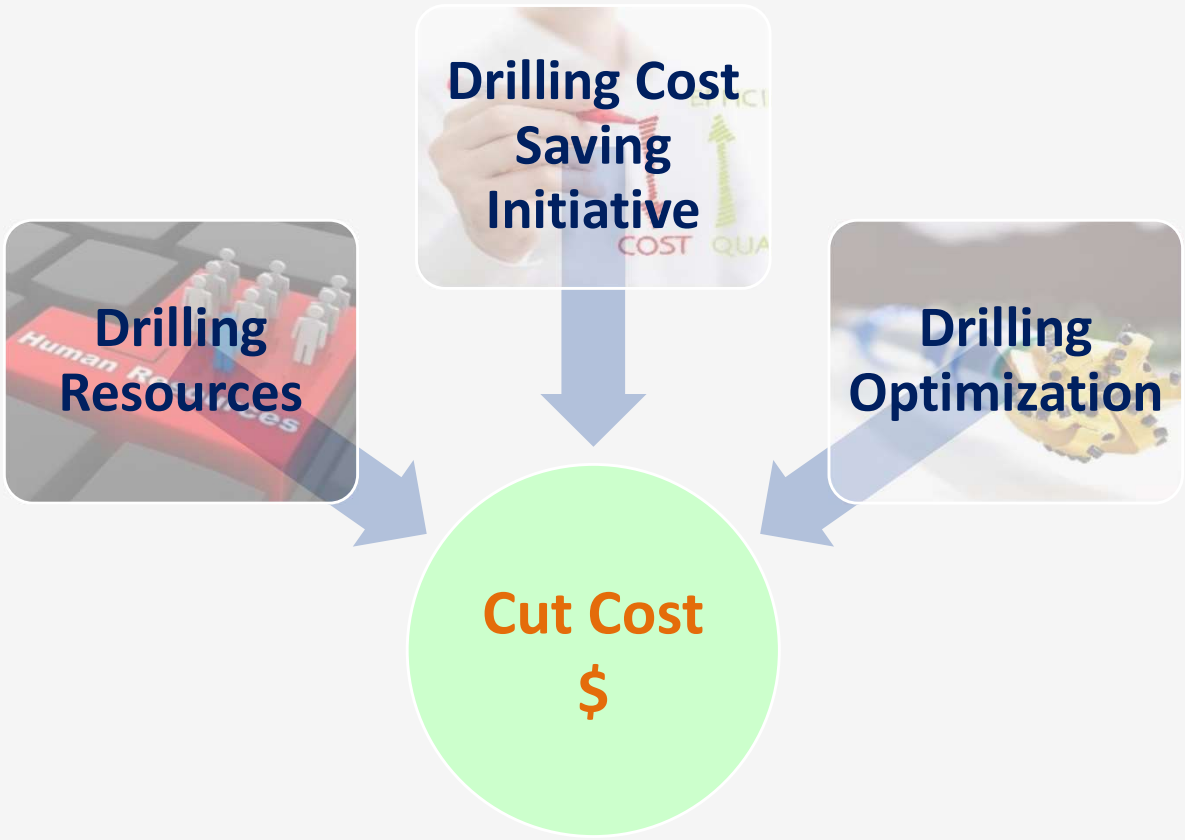
Cost Efficiency in Geothermal Drilling

1. Components of Geothermal Well Drilling Costs
2. Cost-Saving in Drilling: When to Start?

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Cost Efficiency in Geothermal Drilling

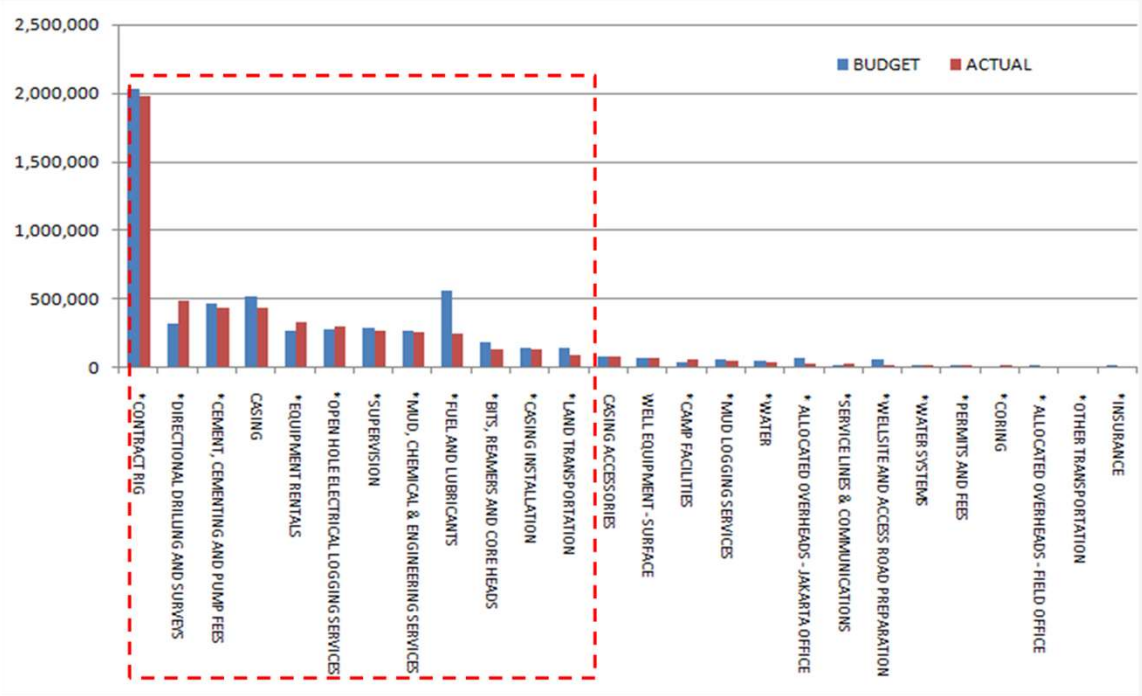
Drilling Geothermal Well below USD 3 Million

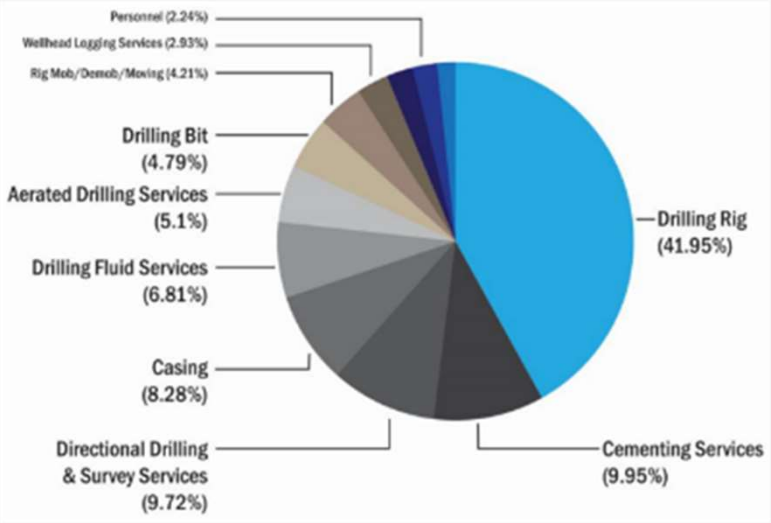


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Cost Efficiency in Geothermal Drilling

Major Well Cost Component





More attention to the major well cost component

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Cost Efficiency in Geothermal Drilling

Major Well Cost Component


Mud and Cementing


Drilling Bit



Fuel Consumption


Stuck Pipe


Heavy Equipment


Drilling Rig


Casing & Installation

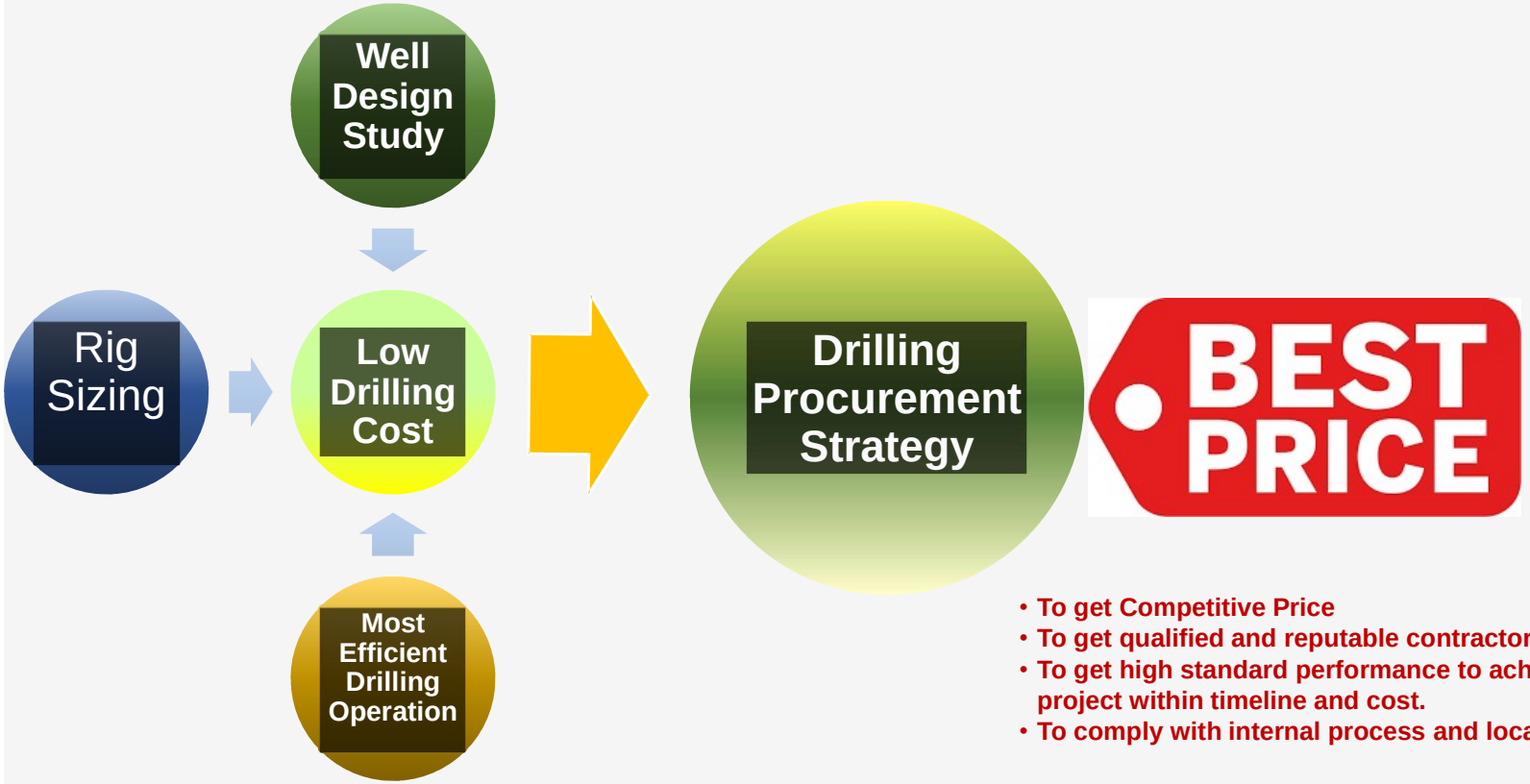

Directional Drilling


Aerated Drilling

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Cost Efficiency in Geothermal Drilling

Drilling Cost Reduction Method



- To get Competitive Price
- To get qualified and reputable contractors: HSE, Technical and Finantial
- To get high standard performance to achieve the project objective: Deliver project within timeline and cost.
- To comply with internal process and local regulation.

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When We could cut WELL COST ?

Well Design & Planning

- Surface location
- Well TD
- Casing Depth
- Wellhead & BOP
- Drilling Program
- Drilling Risk assessment
- BHA failure study
- Bit selection study
- Contingency Plan
- HSE bridging document
- ERP
- Drilling Cost Guide
- DDR format
- Pre Spud meeting

Procurement Process

- Drilling Service and Procurement List
- Market survey
- Remuneration Table
- Vendor List
- SOW
- Tender priority rank
- General Services
- Identify problem on procurement process
- Local Hire
- Local Content Regulation
- Master List Facility

Drilling Operation

- Project schedule
- Instruction to deliver
- Personnel mobilization
- Pre-install conductor
- Pre-weld wellhead
- Rig Mobilization
- Rig flat time
- ROP enhancement
- Drilling Parameters optimization
- Simultaneous Operation
- Do offline job

Well Cost





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Cost Efficiency in Geothermal Drilling

Drilling Cost Saving Initiative

Pre Drilling



1. Optimize well TD refer to Feed Zone
2. Procurement strategy for Drilling Material and Services
3. Proper drilling rig selection
4. Proper Casing Specification: Connection, sources
5. Cement Optimization
6. BHA Optimization for Hard Drilling
7. New Technology Drilling Bit
8. Wellhead connection type
9. Well Pad location and design
10. Optimum Water Supply design
11. Stuck pipe recovery technic
12. Mob/Demob strategy
13. Purchase Rig Option

Drilling Execution

1. Rig inspection reduce NPT
2. Do offline job
3. Faster Pipe handling
4. Stuck pipe prevention
5. Optimize on hire and off hire date
6. Efficient on Drilling material usage.
7. Tight HSE policy
8. Prevent from damaging environment

Post Drilling

1. Capture lesson learn
2. Tight invoice checking
3. Minimize warehouse space after drilling

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General Preparations before Drilling

1. Buidling a Drilling Team
2. Drilling Rig Selection and Specification
3. Drilling Location Preparation
4. Water Supply Preparation for Drilling

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General Preparations before Drilling

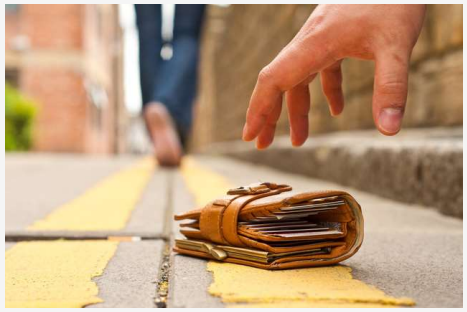
1 Create a Drilling Team

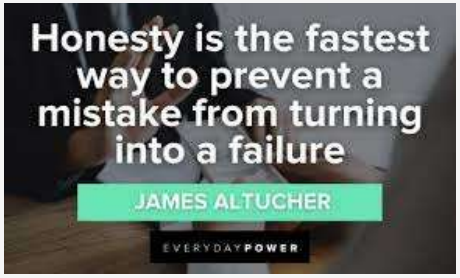
- Honesty
- Ideal Geothermal Drilling Organization Chart
- Hiring Keys for Drilling Personnel
- Drilling Team Qualifications

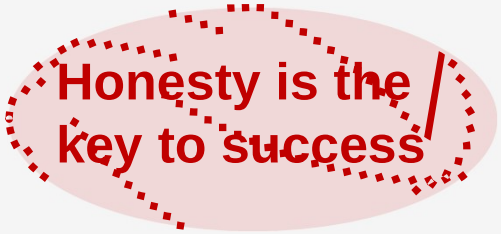
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General Preparation Before Drilling

Create a Drilling Team - HONESTY













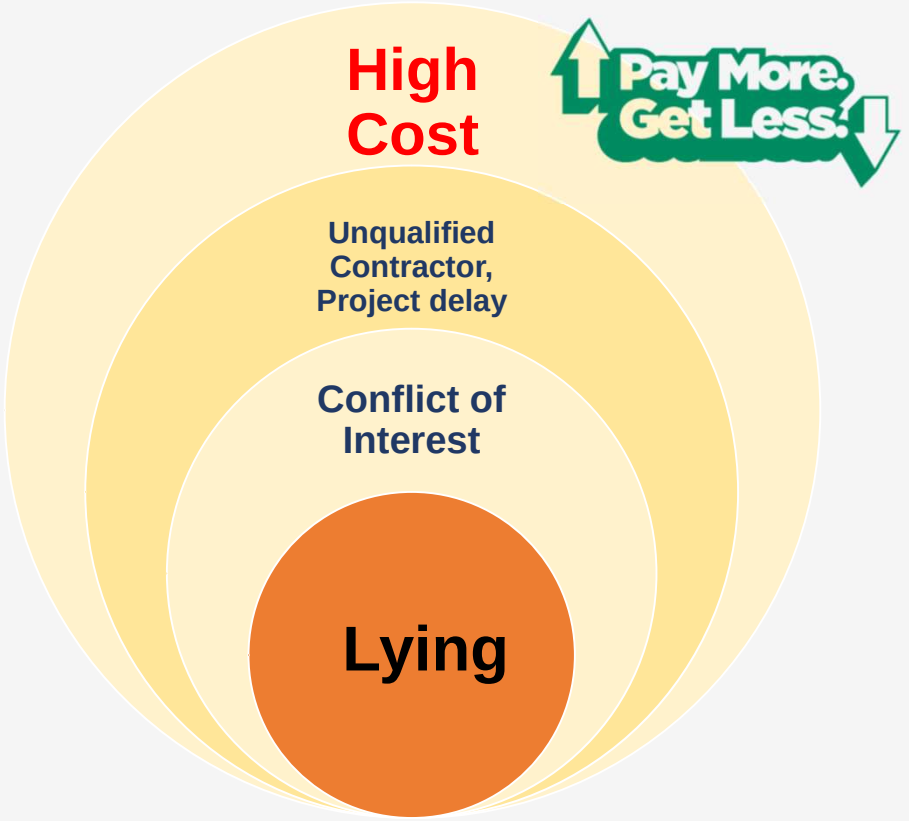


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General Preparation Before Drilling

Honesty and Professionalism

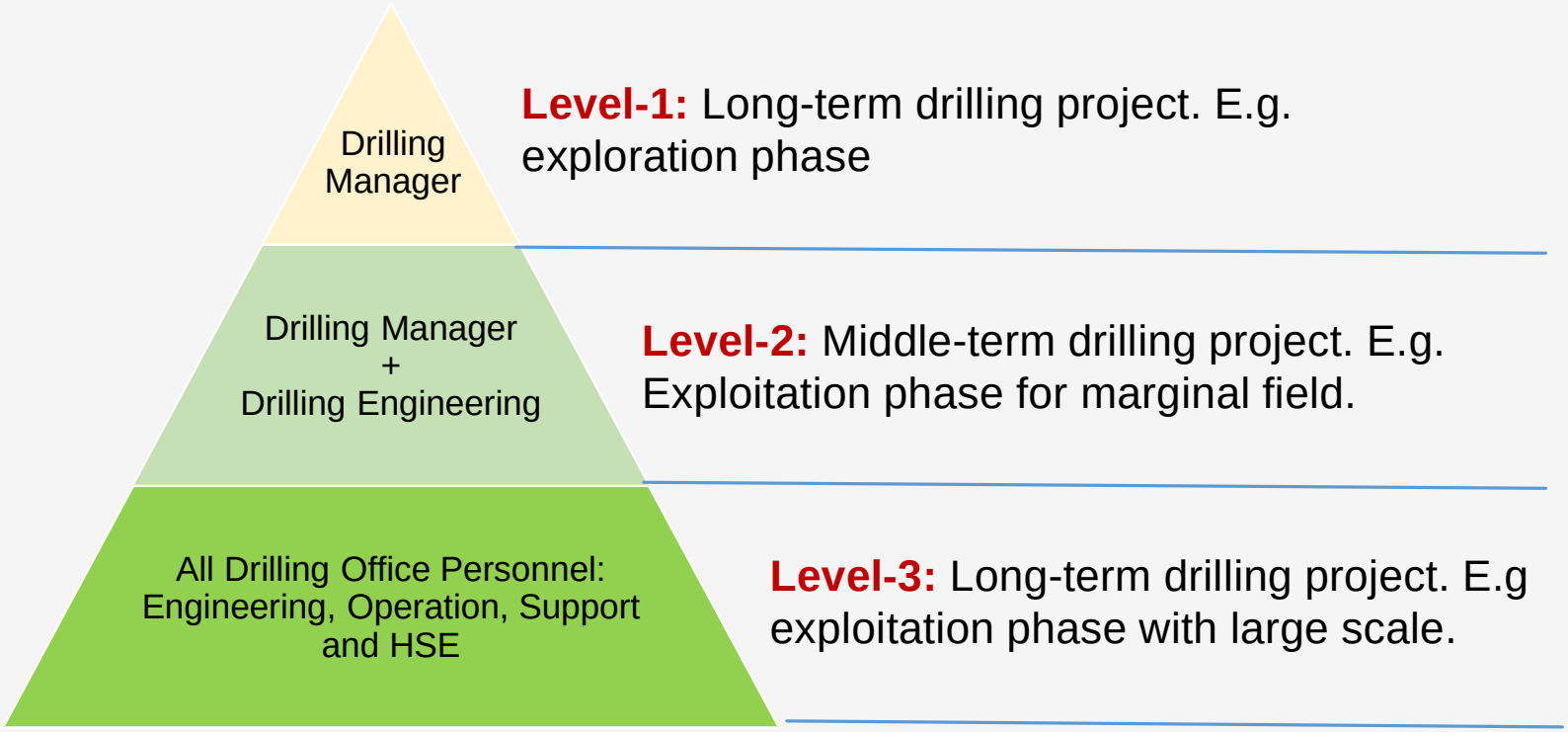
- Selecting a drilling team
- Designing the wells
- Arrange the AFE / Budget
- Define the drilling procurement strategy
- Conduct the tender process
- Execute drilling operation
- Review dan checking invoices
- Post drilling review
- Etc.



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General Preparation Before Drilling

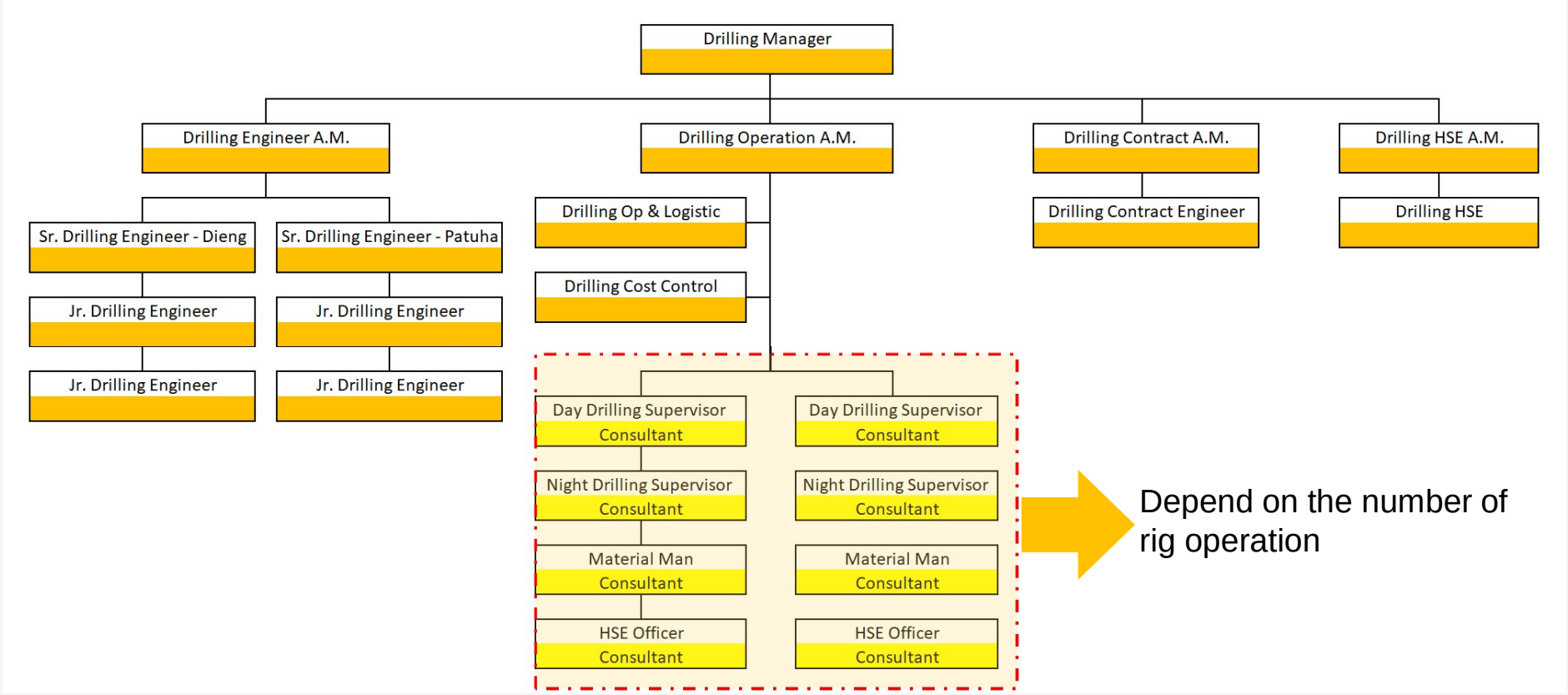
Hiring Keys for Drilling Office Personnel



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General Preparation Before Drilling

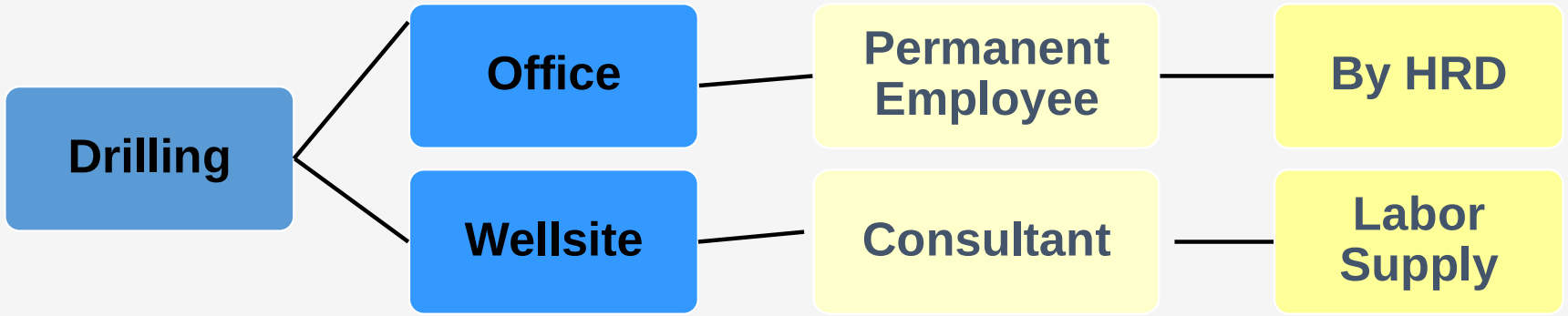
Ideal Geothermal Drilling Organization Chart



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General Preparation Before Drilling

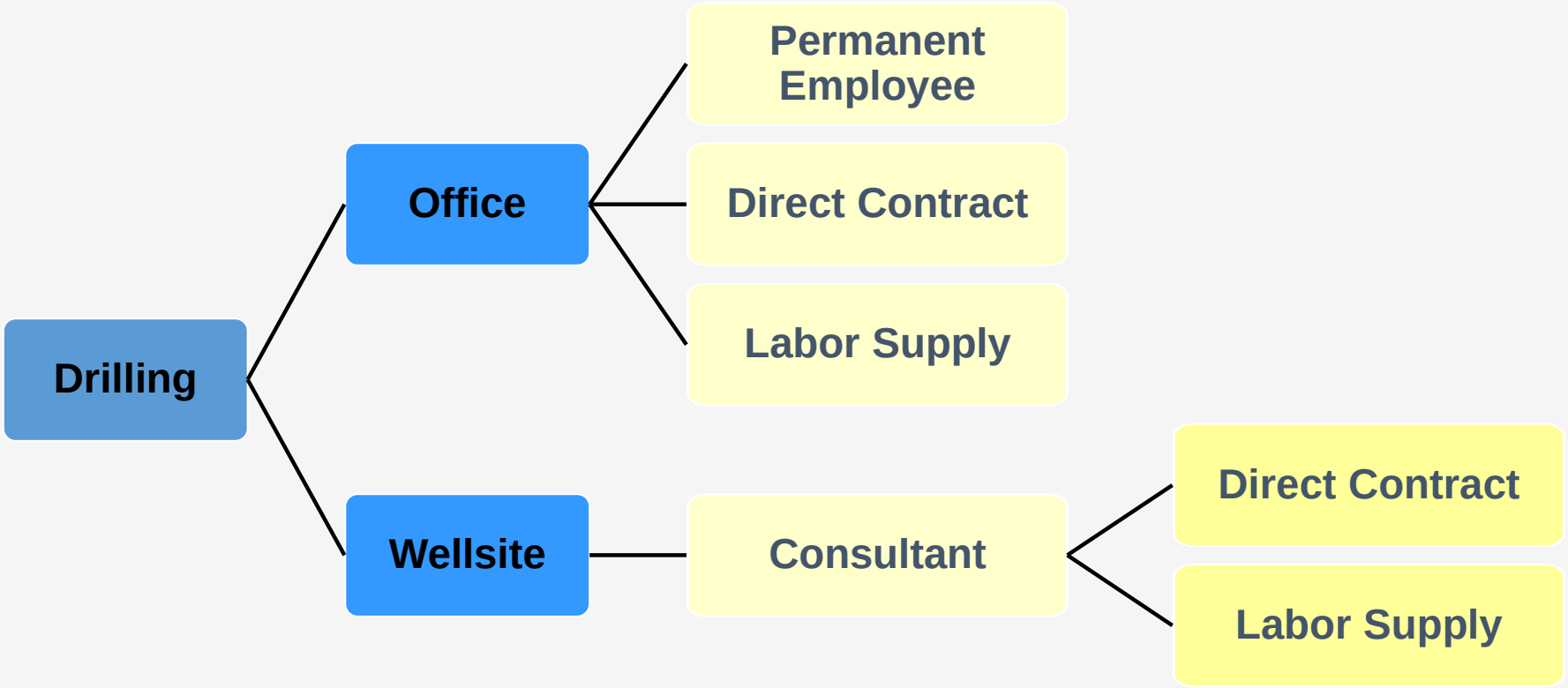
Drilling Personnel and Status



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General Preparation Before Drilling

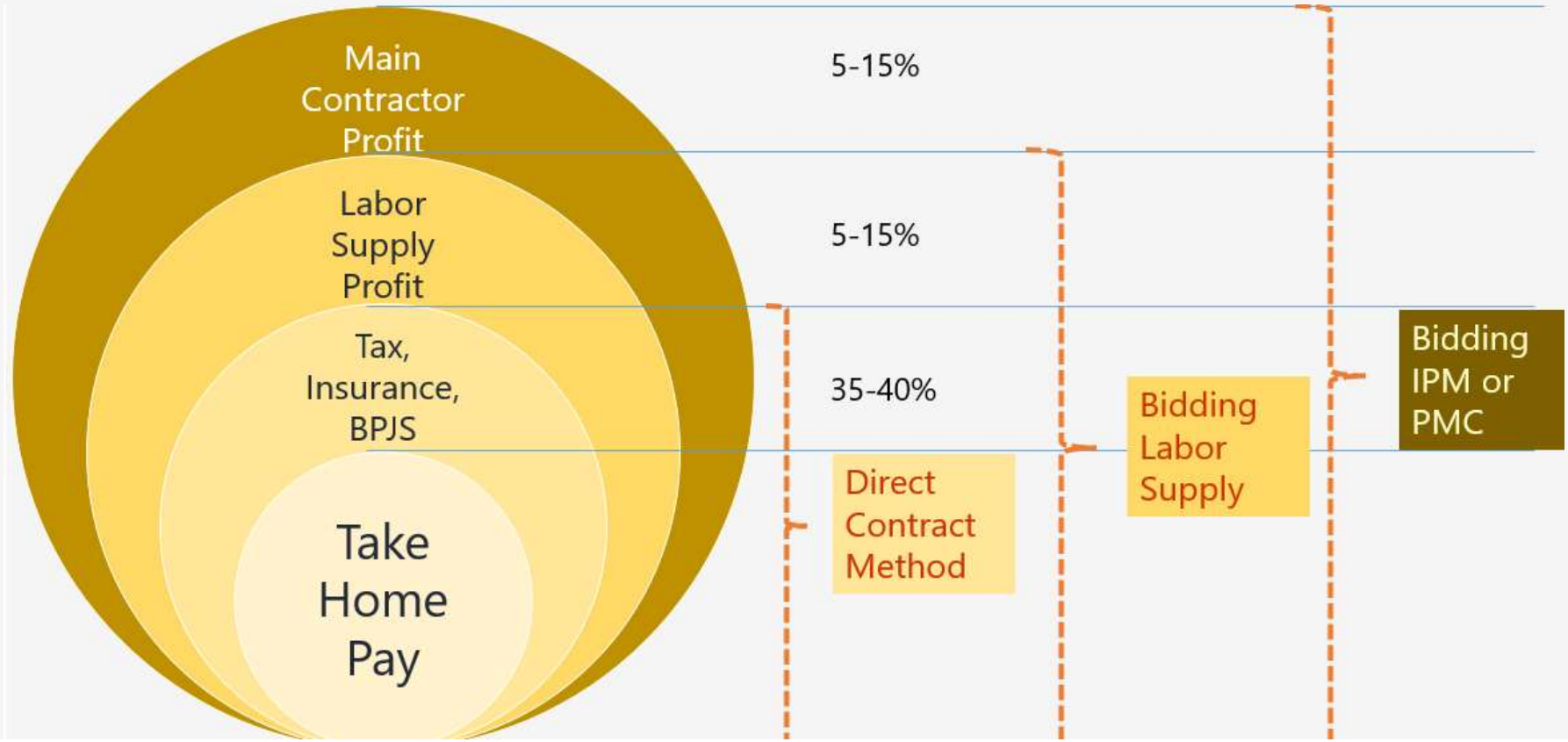
Best Drilling Personnel and Organization Chart



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General Preparation Before Drilling

Geothermal Drilling Personnel (Hiring Method)



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General Preparation Before Drilling

Cost Comparison vs Hiring Process Method

No	Drilling Personnel Position	Daily Rate USD	Days per Mo	Monthly Salary	Direct Contract Tax + 35%	Labor Supply Provit = 15%	PMC/IPM Provit = 15%
1	Day Drilling Supervisor	1,000	16	16,000	5,600	3,240	3,726
2	Day Drilling Supervisor	1,000	16	16,000	5,600	3,240	3,726
3	Night Drilling Supervisor	500	16	8,000	2,800	1,620	1,863
4	Night Drilling Supervisor	500	16	8,000	2,800	1,620	1,863
5	Material Man	200	16	3,200	1,120	648	745
6	Material Man	200	16	3,200	1,120	648	745
7	HSE Officer	200	16	3,200	1,120	648	745
8	HSE Officer	200	16	3,200	1,120	648	745
			Total	60,800	21,280	12,312	14,159
				Commulative	82,080	94,392	108,551
					Cost Saving / month	12,312	26,471
					Cost Saving / Year	**147,744**	**317,650**

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General Preparation Before Drilling

Drilling Team Qualifications

1. At least 10 years experience and executed a lot of drilling wells (engineering, operation, support and HSE) at various places.

1. Have done various drilling optimization studies and cost efficiencies.

1. Experiences in handling various drilling procurement process, have wide networking and strong negotiation skill.

1. Have good relationship with other operator, service company and all level of government authorities.

1. Never involved in any conflict of interest and corruption.

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General Preparations before Drilling

2 Drilling Rig Selection and Specification

- Rig Horsepower
- Rig Equipment
- Heavy Equipment
- Rig moving: Skid and Walking
- Rig Fuel Consumption
- Own Rig

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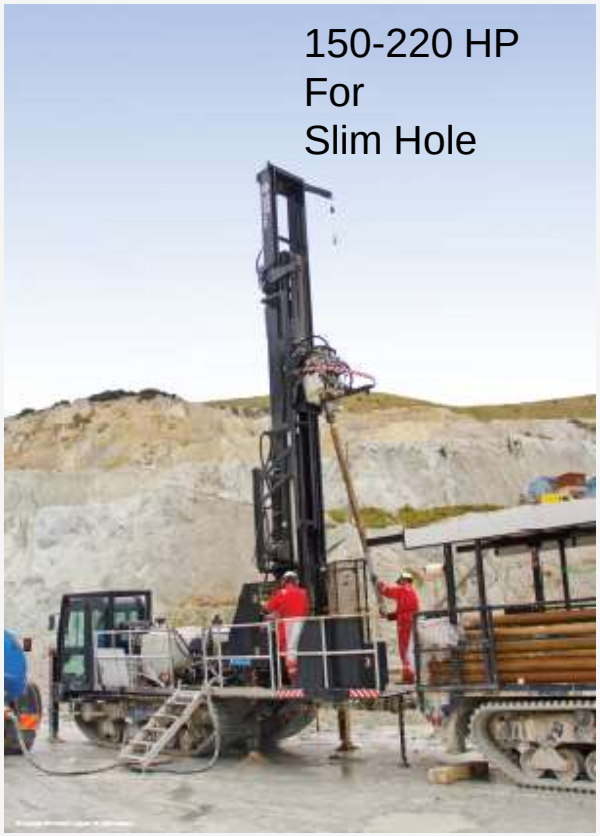
General Preparation Before Drilling

Drilling Rig Selection: Big Hole, Standards Hole and Slim Hole



Rig For Big Hole and Standard Hole:

- 1000 HP
- 1300 HP
- 1500 HP
- 2000 HP



150-220 HP For Slim Hole

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General Preparation Before Drilling

Onshore Drilling Rig Overview – Main Rig Component


Drawwork


Top Drive System




Rotary Table


Drill Pipe


Rig Skid System


Hawk Jaw


Rig Walking System

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General Preparation Before Drilling

Rig Horsepower Selection

Crude Oil Prices vs Daily Rig Rated in Turkey



Rig Horsepower Utilization vs Drilling Total Depth:

750-900 HP (100-120 Ton):
- 1000 – 1500 m Depth

1000 (350 Ton):
- 1500 – 3200 m Depth

1350-1500 HP (350-500 Ton):
- 3200 – 5000 m Depth

2000 HP (500 Ton):
- Deep well, superheated well

Daily Drilling Rig Cost in Turkey (1350 – 2000 HP):

- **$ 11000 - 20000**

Daily Drilling Rig Cost in Indonesia (1000 – 2000 HP):

- **$ 19000 - 30000**

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General Preparation Before Drilling

Rig Horsepower: 1000 HP -> 2000 HP

RIG CAPACITY	1000 HP	1500 HP	2000 HP
Pulling Capacity, Klbs	**600**	**700-800**	**1000**
Drill Pipe Tensile (II), Klbs: • 5”, 19.5#, G-105, S135 □Z140 • 5.5”, 21.9-24.7#, TSS105 □ S135 • 5.875”, 23.4-28.7#, S135 □ V150 • 5.5”, 25.2-34.02#, S135 □ VM165	**436 - 582** 483 – 704 691 – 895 760 – 1062	**436 - 582** **483 – 704** **691** – 895 **760** – 1062	**436 - 582** **483 – 704** **691 – 895** **760 – 1062**
• Maximum Surface Torque • DP Connection: NC50 • Hi Torque DP Connection: XT-50, 55	• 2500 m-MD □ **26000 – 27000 ft-lbs** • **30700 ft-lbs** • 46000 – 52800 ft-lbs		
Rotary Table Size	27.5”	37.5”	37.5”
• TDS-350 Ton, 800 HP • TDS-500 Ton, 1000 HP	• **700 Klbs, 37000 ft-lbs** • 1000 Klbs, 37500 ft-lbs		
No of Loads for RM, RU and RD			
Loads Size and Weight			
Soil Bearing Load			

- **More loads for Rig Moving**
- **Bigger and Heavier**
- **Higher number of soil bearing**



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General Preparation Before Drilling

Rig Horsepower: 1000 HP -> 2000 HP

Rig HP dan Pemilihan Drill Pipe yang sesuai:

a. Rig 1000 HP, daya angkat **600 Klbs:** -> Drill Pipe: 5”, 19.5#, G-105, S135 atau Z140

b. Rig 1500 HP, daya angkat **700-800 Klbs:** -> Drill Pipe: 5.5”, 21.9-24.7#, TSS105 -> S135

c. Rig 2000 HP, daya angkat **1000 Klbs:** **->** Drill Pipe: 5.5”, 25.2-34.02#, S135 -> VM165

Apabila Rig hanya mempunyai 5” DP, 19.5 ppg, G-105 (kapasitas tensile = 436 klbs), effisiensi terhadap daya angkat rig sbb:

- Rig 1000 HP: 73%
- Rig 1500 HP: 55%
- Rig 2000 HP: 44%

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General Preparation Before Drilling

Rig Horsepower: 1000 HP -> 2000 HP

Berdasarkan pengalaman dari bebarapa Lokasi pemboran di Indonesia, berikut kemampuan terdalam pemboran berdasarkan Rig HP:

- 750-900 HP (100-120 Ton): 1000 – 1500 m kedalaman sumur
- 1000 HP (350 Ton): 1500 – 2500 m kedalaman sumur
- 1350-1500 HP (350-500 Ton): 2200 – 3300 m kedalaman sumur.
  - Catatan: sumur dg kedalaman 2700-3300 m-MD, Rig Kontraktor agar mempunyai 5” DP, 24 ppf, Grade G-105. Kombinasi 5” DP, **24 ppf**, Grade G-105 (bagian atas) dan 5” DP, 19.5 ppf Grade G-105 (bagian bawah) untuk menghandle torsi dan beban angkat dipermukaan yang besar.
  - Pemakain 5” DP Grade S-135 TIDAK disarankan, rentan lingkungan acid, bukan sour service.

- 2000 HP (500 Ton): 3500 – 5000 m kedalaman sumur.
  - Untuk sumur-sumur Super Critical (kedalaman sumur dengan BHT lebih dari 375 deg C).

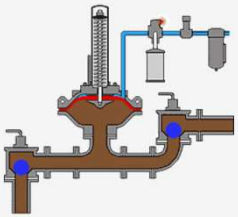
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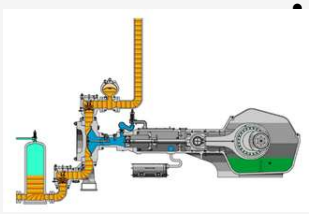
General Preparation Before Drilling

Mud Pump









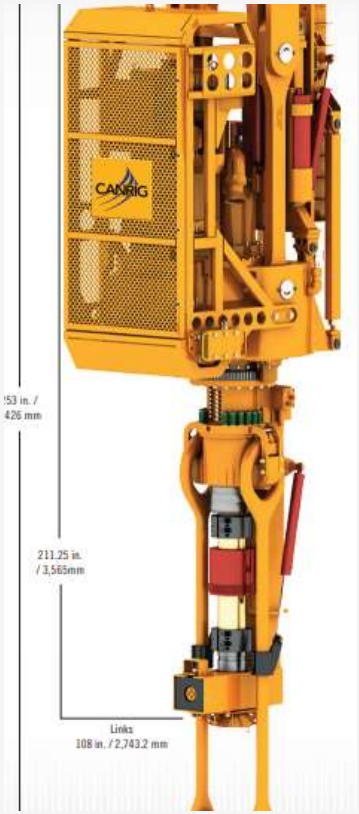
- Memerlukan jumlah pompa lebih banyak untuk mendapatkan laju aliran yang lebih besar dengan tekanan pompa yang relative rendah.
- Rig menyediakan pompa lumpur berjumlah **4 set**: 2 set untuk drilling, 1 dedicate untuk pompa ke annulus dan 1 set untuk back up.
- Pada umumnya rig mempunyai 3 pompa yang sudah teritegrasi dengan sistem tangki lumpur. Untuk 1 pompa tambahan sebaiknya pompa independent yang bisa ditie in kedalam annulus dan untuk back up selama drilling.
- Kami menyarankan memakai pompa lumpur dengan kapasitas tenaga 1300 hp. Pompa:
  - akan menggunakan satu size liner pompa untuk pemboran semua hole section. Apabila kurang dari 1300 hp, kita harus menganti liner pompa yang lebih kecil untuk memberikan tekanan pompa yang mencukupi kebutuhan pemboran. Tentunya hal ini kadang menimbulkan tambahan **rig time dan cost**.

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General Preparation Before Drilling

Pemilihan Top Drive System (TDS)







Three (3) aspects to select TDS to be installed on the Rig:

a. **TDS Type**: Hydraulic top drives vs Electric
b. **Tonnage**: 250-350 Ton vs 500 Ton
c. **Horse-Power**: 800 HP vs 1000 HP

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General Preparation Before Drilling

Pemilihan Top Drive System (TDS)

Untuk menentukan kapasitas TDS, kita perlu mempertimbangkan keperluan kapasitas saat operasi pemboran:

a. Drilling RPM: **40 – 80 (saat dengan rotary BHA)**

b. Maximum Continuous Torque (Maksimun torsi secara terus menerus):

- Torsi maximum saat pemboran dengan kedalaman hingga 2600 m-MD -> **23,000 ft-lbs**
- Torsi maximum saat pemboran dengan kedalaman hingga 3300 m-MD -> **29,000 ft-lbs**

c. Maksimun torsi untuk menyambung pipa pemboran:

- Torsi untuk MU 5” DP 19.5 ppf -> 30,730 – 42,700 ft-lbs
- Torsi untuk MU 5” DP 25.5 ppf -> 38,036 - **49,200 ft-lbs**
- Torsi untuk MU 5” HWDP NC 50 -> 29,680 - 30,700 ft-lbs
- Torsi untuk MU 6-3/4” DC -> 32,300 – 35,500 ft-lbs
- Torsi untuk MU 8-1/2” DC -> 53,300 – 58,600 ft-lbs
- Torsi untuk MU 9-1/2” DC -> 88,600 - 97,500 ft-lbs
- Torsi untuk MU Casing 7” – 20” -> **11,500 – 48,150 ft-lbs**

d. Beban angkat maximum:

- 5”, 19.5#, G-105, S135 -> Z140: 436 – 582 klbs atau 198 – 264 Ton
- Casing 7” – 20” -> 240 – 263 klbs atau **109 – 120 Ton**

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General Preparation Before Drilling

Kajian Top Drive System (TDS): 250 Ton

Description	Unit	Specification					
Manufacturer		Canrig	Canrig	NOV	Canrig	Bentec	Bentec
Type		T250	T250	TDS-10SH	C275	TD-250-C	TD-275-HT
Hoisting and Rotating	Ton	150 or 250	150 or 250	250	275	250	275
Drilling Motor				AC-Induction Motor		AC-Induction Motor	AC-Induction Motor
Motor Rating	hp	380	380	400	**600**	672	672
Working Height	M			4.66		5.64	5.71
Weight	Ton	6.2	6.2	8.1	8.2	10.8	11.3
Gear		12.47:1	10.96:1	13.1:1	9.38:1	Hellical Gear, 15:1	Hellical Gear, 15:1
Max. Continuous Torque	ft-lbs	23,000	21,000	22,288	**30,000**	23,809	24,912
Speed at Max. Cont. Torque	ft-lbs	90	100	85	105	100	100
Max. Speed	Rpm	176	176	182	225	200	200
Max. Make & Break out Torque	ft-lbs	35,000	32,000	42,680	60,000	44,254	51,500
Static Brake Torque	ft-lbs			50,000		51,500	51,500
Mainshaft Quill	In			3"		2-1/2"	2-1/2"
Working Pressure	Psi			7,500		7,500	7,500
Min. Connection Range	In			4		2-1/2"	2-1/2"
Max. Connection Range	In			6-5/8"		7-1/2"	7-3/4"
IBOP Connection				6-5/8"		NC 50	NC 50
Cooling System				Local Blower		Local Blower	Local Blower
Hydraulic Power				Onboard		Onboard	Onboard
Suitable Elevator Links	Ton			250 T		150, 250 & 350	150, 250 & 350
Min. Temperature	Deg C	(45)	(45)	(40)	(45)	(45)	(45)
Max. Temperature	Deg C	55	55	55	55	55	55
Footprint (Width x Depth)	2	6.2 X 1.1	6.2 X 1.1		5.08 X 1.24	1.4 x 1.6	1.4 x 1.88

TDS 250-275 Ton ini tidak bisa dipakai untuk sumur dalam:

- daya angkat memadai namun kemampuan torsi masih belum mencukupi.

TDS 250-275 Ton ini bisa dipertimbangkan apabila pemboran kita dengan kedalaman dangkal kurang dari 1500 m-MD.

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General Preparation Before Drilling

Kajian Top Drive System (TDS): 350 - 400 Ton

Description	Unit	Specification			
Manufacturer		Canrig	Canrig	NOV	Bentec
Type		T350/400	T350/400	IDS-350PE	TD-350-HT
Hoisting and Rotating	Ton	350/401	350/400	350	350
Drilling Motor				AC-Induction Motor	AC-Induction Motor
Motor Rating	hp	570	770	1000	1030
Working Height	M			6.34	6.35
Weight	Ton	11.3	12.3	14.9	15.7
Gear		11.11:1	11.11:1	12.6:1	Hellical Gear, 14:1
Max. Continuous Torque	ft-lbs	30,000	37,500	37,000	46,500
Speed at Max. Cont. Torque	ft-lbs	90	115	145	115
Max. Speed	Rpm	220	220	212	230
Max. Make & Break out Torque	ft-lbs	45,000	58,000	65,000	73,760
Static Brake Torque	ft-lbs			55,000	77,440
Mainshaft Quill	In			3"	3"
Working Pressure	Psi			7,500	7,500
Min. Connection Range	In			4	2-1/2"
Max. Connection Range	In			8-1/2"	8-1/2"
IBOP Connection				6-5/8"	NC 61
Cooling System				Local Blower	Local Blower
Hydraulic Power				Onboard	Onboard
Suitable Elevator Links	Ton			250 T	150, 250 & 350
Min. Temperature	Deg C	(45)	(40)	(40)	(45)
Max. Temperature	Deg C	55	55	55	55
Footprint (Width x Depth)	2	6.02 X 1.23	6.4 X 1.8		1.66 X 1.55

Kajian 350-400 Ton TDS:



- Daya angkat memadai, kemampuan torsi mencukupi apabila TDS dengan kekuatan motor ratingnya lebih besar dari 770 hp.
- merupakan tipe yang paling effisien untuk digunakan dalam pemboran sumur panas bumi dengan kedalaman 2000 – 3300 m-MD

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General Preparation Before Drilling

Kajian Top Drive System (TDS): 500 Ton

Description	Unit	Specification								
Manufacturer		NOV	Canrig	NOV	Bentec	Canrig	Canrig	Canrig	NOV	Canrig
Type		TDS-115A	T500	TDS-115H	TD-500-XT	T500-HT	C500	C500	TDS-11HD	T500-HT
Hoisting and Rotating	Ton	500	500	500	500	500	500	500	500	500
Drilling Motor		AC-Induction Motor		AC-Magnet Motor	AC-Induction Motor				AC-Induction Motor	
Motor Rating	hp	800	940	1100	1140	1270	1150	1150	1200	1270
Working Height	M	5.8		8.9	6.35				6.6	
Weight	Ton	11.5	18.1	12.5	15.7	18.6	13.2	15.5	13.6	18.6
Gear		10.5:1	8:1	10.56:1	Hellical Gear, 14:1	8:1	6.81:1	8.65:1	10.56:1	8:1
Max. Continuous Torque	ft-lbs	37,500	52,000	51,000	52,000	70,000	51,400	65,300	58,800	70,000
Speed at Max. Cont. Torque	ft-lbs	110	100	110	115	100	125	90	110	100
Max. Speed	Rpm	228	220	228	230	220	265	225	228	220
Max. Make & Break out Torque	ft-lbs	75,000	65,000	75,000	73,760	100,000	95,000	95,000	75,000	100,000
Static Brake Torque	ft-lbs	50,000		50,000	77,444				50,000	
Mainshaft Quill	In	3"		3"	3"				3"	
Working Pressure	Psi	7,500		7,500	7,500				7,500	
Min. Connection Range	In	4"		4"	2-1/2"				4"	
Max. Connection Range	In	8-1/2"		8-1/2"	8-1/2"				8-1/2"	
IBOP Connection		6-5/8"		6-5/8"	NC 61				6-5/8"	
Cooling System		Local Blower		Local Blower	Local Blower				Local Blower	
Hydraulic Power		Onboard		Onboard	Onboard				Onboard	
Suitable Elevator Links	Ton	250T, 350T, 500T		250T, 350T, 500T	150, 250 & 350				250T, 350T, 500T	
Min. Temperature	Deg C	(40)	(40)	(40)	(45)	(40)	(45)	(45)	(40)	(40)
Max. Temperature	Deg C	55	55	55	55	55	55	55	55	55
Footprint (Width x Depth)	2		6.52 X 1.32		1.66 X 1.55	6.35 X 1.34	6.8 X 1.44	6.8 X 1.44		6.35 X 1.34

Heavier -> More Rig Power -> More Fuel

Dengan motor rating minimum 800 hp, TDS 500 Ton: daya angkat memadai, kemampuan torsi mencukupi. Namun ada beberapa TDS 500 Ton ini mempunyai berat lebih dari TDS 350 Ton, sehingga diperlukan daya angkat yang lebih untuk menaikan dan menurunkan TDS, tentunya akan memerlukan konsumsi bahan bakar yang lebih.

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General Preparation Before Drilling

Kajian Top Drive System (TDS): > 750 Ton

Specifikasi melebihi kebutuhan -> De-Efisiensi

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General Preparation Before Drilling

Rig Floor Handling Equipment



- **Manual Tong**: mampu menyambung (Make Up) semua size komponen BHA secara manual dengan menggunakan tenaga drawwork, namun memerlukan waktu yang lama.



- **Hawk Jaw**, tipe **JUNIOR**: mampu menyambung (Make Up) komponen BHA hingga ukuran 8-1/2”, secara Hydraulic/Pneumatic sehingga singkat waktu untuk Make Up.
- **Hawk Jaw**, tipe **SENIOR**: mampu menyambung (Make Up) komponen BHA hingga ukuran 9-1/2”, secara Hydraulic/Pneumatic sehingga singkat waktu untuk Make Up.

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General Preparation Before Drilling

Make Up time for Each BHA Component – Use Senior Type of Hawk Jaw

BHA Component	Using Hawk Jaw (Mins)	Using Power Tong (Mins)	Rig Time Saving (mins)	No Joints / Stands	Total Saving	Remark
9-1/2” DC	10	30	20	3	60	
8-1/2” DC	10	30	20	3	60	
6-1/2” DC	5	20	15	1	15	
5” HWDP	5	10	5	15	75	
5” DP	5	10	5	29	145	

No	BHA Component	OD
		in
1	26" Bit	26
2	9-5/8" Mud Motor	9.625
3	9 1/2" Catcher Sub w/ Float Sub	9.5
4	24" IBS Stabilizer	9.5
6	Shock Sub 9 1/2"	9.5
8	9-1/2" NMDC	9.25
9	9-1/2" MWD XEM 800	8.0
10	9-1/2" MWD Gap Sub	8.0
11	9-1/2" NMDC (Optional)	8.0
12	1 x 9-1/2" Pony NMDC (Optional)	8.0
13	X/O (7-5/8" REG Pin x 6-5/8" REG)	8.0
14	7 x 8" DC	8.0
15	8" Hydraulic jars	8.0
16	1 x 8" DC	8.0
17	X/O 6 5/8" RP x 4 1/2" IFB	6.4
18	1 x 6 3/4" DC	6.75
19	15 x 5" HWDP	5.0
20		

No	BHA Component
1	17-1/2" Bit
2	9-5/8" Mud Motor
3	9 1/2" Catcher Sub w/ Float Sub
4	15" IBS Stabilizer
8	9-1/2" NMDC
9	9-1/2" MWD XEM 800
10	9-1/2" MWD Gap Sub
11	9-1/2" NMDC (Optional)
12	1 x 9-1/2" Pony NMDC (Optional)
13	X/O (7-5/8" REG Pin x 6-5/8" REG)
14	7 x 8" DC
15	8" Hydraulic jars
16	2 x 8" DC
17	X/O 6 5/8" RP x 4 1/2" IFB
18	1 x 6 3/4" DC
19	15 x 5" HWDP
20	

No	BHA Component
1	12-1/4" PDC Bit
2	8" Mud Motor
3	8" Catcher Sub w/ Float Sub
4	11-3/4" IBS Stabilizer
8	8" NMDC
9	8" MWD XEM 800
10	8" MWD Gap Sub
11	8" NMDC (Optional)
12	1 x 8" Pony NMDC (Optional)
13	1 x 8" DC
14	X/O 6 5/8" RP x 4 1/2" IFB
15	1 x 6-3/4" DC
16	10 x 5" HWDP
17	6.5" Hyd Jar
18	15 x 5" HWDP
19	
20	

Potential Cost Saving

- Per BHA Run = 355 mins = 6 hours
- 6 BHA run per well -> 36 hours
- Cost saving = 36/24 * USD 70K = **USD 105 K**
- **22 wells drilling campaign = USD 2.31 Million or 0.5 well cost**

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General Preparation Before Drilling

Optimum Rig HP vs Well Depth

Rig 1000 HP	Rig 1500 HP
• **1000 m-MD □ 3300 m-MD** • 5”, 19.5#, G-105 • DP Connection: NC50 • TDS-350 Ton, 800-1000 HP • **4 x 1000 or 4 x 1300 HP Mud Pumps** • **Hawk jaw, senior type □ capable to MU up to 9-1/2” DC** • New Generation Rig • Fast skid system or Walking Rig	• **3300 m-MD □ 4000 m-MD** • Main DP: 5”, 19.5#, G-105 • Additional DP for tension limit: ◦ 5”, **24#**, G-105 ◦ 5.5”, 21.9-24.7#, TSS105 • DP Connection: NC50 or higher • TDS-350 Ton, 800 - 1000 HP • **4 Mud Pumps:** ◦ **4 x 1300 HP Mud Pumps or** ◦ **3 x 1500 HP + 1 x 1300 HP Mud Pumps** • **Hawk jaw, senior type □ capable to MU up to 9-1/2” DC** • New Generation Rig • Fast skid system or Walking Rig

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General Preparation Before Drilling

Type, Total, dan Capacity of Crane

Kebutuhan minimum crane: 2 set 50T, jenis Roughter (Mobile) dan 1 Set 80T, jenis Crawler dimana penempatan Crane sebagai berikut:

- 1 set 50T Roughter Crane dan 1 Set 80T Crawler Crane akan ditempatkan dilokasi well pad pemboran.
- 1 Set 50T Roughter Crane akan ditempatkan di area casing yard atau di sekitar Gudang.

Note:
Untuk Lokasi pemboran Panas Bumi yang ada di Indonesia pada umumnya di Lokasi pengunungan, kami sarankan memakai alat berat yang relative baru, dimana umur alat berat kuarng dari 3 tahun

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General Preparation Before Drilling

Main Heavy Equipment For Rig Moving & Daily Operation



Forklift



Roughter / Rough Terrain / Mobile Crane

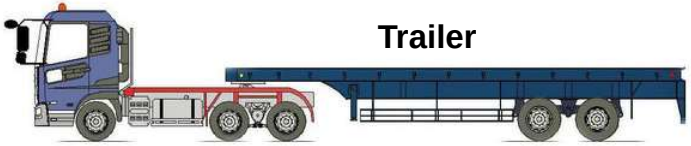
Crawler Crane





Low Bed

Prime Mover



Trailer

Hi Bed

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General Preparation Before Drilling

Main Heavy Equipment For Rig Moving & Daily Operation

Heavy Equipment		Capacity	Qty Rig Moving	Qty Daily Rig Operation	Remark
	Forklift	8-10 Ton	2 Set	2 Set	10 Ton preferable. During mobilization along high slope access road, need to be standing by to help pushing the trailer.
	Mobile Crane	50 Ton	2 Set	2 Set	It is easier to move from one place to other place
	Mobile Crane	25-30 T Optional	1 Set	Not Required	Smaller Tonage Crane for rigging up / setting up equipment
	Crawler Crane	80 Ton	1 Set	1 Set	Some Rig Equipment need 80 T Crane to be lifted → it will more safe and faster rather than 2 x 50 T Crane
	Low Bed	6 X 6	As Required	1Set Prime Mover + 1Set Low Bed Trailer + 1Set Hi Bed Trailer	To transport taller Rig Equipment
	Hi Bed	6 X 6	As Required		

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General Preparation Before Drilling

Optional Heavy Equipment For Rig Moving


Winch Truck


Vacuum Truck



Water Truck


Dump Truck


Back Hoe

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General Preparation Before Drilling

Kajian Kemampuan dan Kecepatan Rig Moving

Kegiatan perpindahan rig (Rig Moving) ini dapat dikategorikan dalam 4 jenis:

a. **Rig Mobilisasi**: perpindahan rig dari lokasi awal ke lokasi sumur pertama pemboran.
b. **Rig Moving didalam satu wellpad**: cara perpindahan rig dalam satu wellpad bisa dengan **proses skidding atau walking (berjalan)** tergantung kemampuan rignya
c. **Rig Moving antar wellpad**: Perpindahan rig ini, rig harus direbahkan dan dibongkar untuk dapat dipindahkan ke Lokasi wellpad berikutnya untuk di rig-up.
d. **Rig DeMobilisasi**: perpindahan rig dari lokasi sumur terakhir pemboran ke lokasi yang diinginkan oleh Kontraktor rig.

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General Preparation Before Drilling

Rig Skid vs Walking System

Rig Skid and Walking System	Days to spud
 • Without Drill string stand up the rig mast	Rig Release and ready to spud the next well: • 5 days (additional rig time to LD drill pipe and Re-MU (2 day)) • **USD 370 K**
 • With Drill string stand up the rig mast	Rig Release and ready to spud the next well: • 3 days • **USD 210 K**
 • With Drill string stand up the rig mast	Rig Release and ready to spud the next well: • 0.5 days • **USD 35 K**

If **WITHOUT** skid system, we have to:

- LD Pipe
- RD
- RM
- RU
- MU Pipe

- **10 Days or USD 700 K Costly**

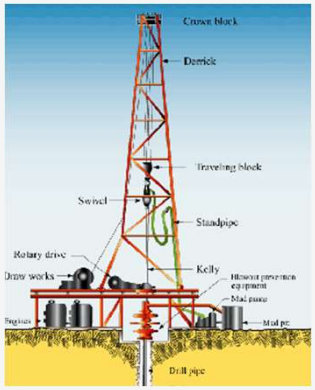
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General Preparation Before Drilling

Type Of Skid Rail









Skid Rail Beam Layout



Skid Rail:

Structural steelwork that provides smooth flat surfaces for the drilling equipment set to move in an X-Y planes to allow access over all well slots

Efficient Rig Skid System:

- Capable skid the rig with Drill String still stand in Derrick.
- Sufficient number of skid rail set, so the rig could skid continuously without waiting move the skid rail set from backside and set in the front end.
- Use Automatic Lock and Unlock system
- Use long cylinder hydraulic system for long stroke to skid faster.
- Have independent hydraulic power system, instead of being combined with Rig Hydraulic system

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General Preparation Before Drilling

RIG RENUMERATION TABLE STRUCTURE

1. MOBILIZATION & DEMOBILIZATION

	Description	Unit	Est Total Unit	Unit price (USD)	Total Price (USD)
		a	b	c	b x c
1	Mobilization	Lump sum	1		
2	Demobilization	Lump sum	1		
3	Moving Inter Well Pad (max. 14 days)	Lump sum	3		
4	Moving On The Same Well Pad (max. 5 days)	Lump sum	6		
				Sub Total 1	-

2. RIG BUNDLING OPERATION RATE

	Description	Unit	Est Total Unit	Unit price (USD)	Total Price (USD)
		a	b	c	b x c
1	Operation Daily Rate (ODR)	Day	650		
2	Standby Rate (75% x ODR)	Day	-		
3	Standby Minimum Crew Rate (25% x ODR)	Day	-		
4	Repair Rate (50% x ODR)	Day	-		
				Sub Total 2	-

3. WASTE MANAGEMENT SERVICE per WELL

1	Drilling Waste Treatment Service (domestic waste & drill cuttings)	Well	10		
				Sub Total 3	-

CONTRACT VALUE ESTIMATE

	Description	Total Price
1.	MOBILIZATION & DEMOBILIZATION	
2.	RIG BUNDLING OPERATION RATE	
3.	WASTE MANAGEMENT SERVICE	
	Estimated Total Rig Bundling Value	

Rig Moving:

- Each Rig type has deferent Rig Moving method and time duration for moving
- To include fuel consumption to the daily rig rate.

General Preparation Before Drilling

RIG FUEL CONSUMPTION

- **1500 HP New Generation Rig: 4000 – 6000 L/day**
- **1500 HP Old Rig: 8000-12000 L/day**
- **To be considered into rig tender commercial evaluation**

1500 HP OR 2000 HP Rig

- Well design require 1500 HP Rig,
- During drilling rig tender, some of Rig Contractor offer us 2000 HP Rig with Rig Price similar with 1500 HP Rig.



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General Preparation Before Drilling

RIG FUEL CONSUMPTION

Fuel Price

- Basic price + Handling
- USD 0.9-1.2 per Liter

Equipment Daily Usage

- Rig: 4-12 KL
- Air Drilling: 5-8KL
- Heavy Equipment: 1-2 KL
- Water Pump: 1 KL
- Base Camp: 0.2 KL
- Vehicles: 0.3 KL

- Total Consumption: 11.5 – 23.5 KL/day

How to Reduce

- **Use New Generation Rig.**
- **< 3 years age of Heavy Equipment**
- **Proper Piping for water supply**

- **Convert Electricity source from GenSet to PLN**

- Digital Measurement
- Fuel Handling SOP
- Regular fuel sample check
- Dedicated Fuel Truck
- Always Lock fuel tank line
- Daily Usage Report

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General Preparation Before Drilling

RIG RENUMERATION TABLE STRUCTURE

C. HSD Fuel Supply For Employer Consumption

Description	Unit price (USD/LITER)	Employer Service Equipment Fuel Consumption per Day (Estimated) (Liter/Day)	Rig and Camp Equipment Fuel Consumption Per Day (Liter/Day)	Handling fee per liter (%)	Total Days (Day)	Total Cost (USD)
	a	b	c	d	e	((a+(axd))x(b+c))xe
High Speed Diesel	1	2000			650	

- Company is suggested to procure fuel itself.
- The above table is added if the company policy require Rig to supply fuel.
- Since every Rig type has deferent Fuel Consumption
  - Option-1: To include fuel consumption to the daily rig rate
  - Option-2: Rig is required to fill maximum fuel consumption per day and company only pay the actual usage of the fuel. Rig will pay any fuel consumption exceeded the sated maximum fuel/day.

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General Preparation Before Drilling

Fuel Cost Component (Pertamina)

LAMPIRAN C-1

Rincian Biaya

NO	QTY/BLN	QTY/THN	QTY/ 5 THN	UOM	DESKRIPSI	PROPINSI	SUPPLY POINT	HARGA BBM (Rp./liter)	100% MOPS	Besaran Diskon (7% MOPS)	Harga setelah Diskon	HANDLING FEE (Lama)	HANDLING FEE (Baru)	TOTAL HARGA (Rp./liter)	TOTAL HARGA (Rp./bulan)	TOTAL HARGA (Rp./tahun)	TOTAL HARGA (Rp./5 tahun)	TERM
1	82.700	992.400	4.962.000	L	AG ULUBELU	Lampung	TBBM Panjang	9.300	6.174	432,18	8.867,82	755	631	9.499	785.552.414	9.426.629.969	47.133.144.846	FRANCO
2	206.451	4.057.412	20.187.060	L	PROYEK LUMUT BALAI	Sumatera Selatan	TBBM Lahat	9.300	6.174	432,18	8.867,82	496	486	9.364	3.150.466.603	37.805.599.234	189.027.996.169	FRANCO
3	226.036	2.712.432	13.562.160	L	PROYEK HULULAIS	Bengkulu	TBBM Pulau Baai	9.300	6.174	432,18	8.867,82	615	605	9.473	2.141.158.342	25.694.380.098	128.471.900.491	FRANCO
4	1.000	12.000	60.000	L	PROYEK BUKIT DAUN	Bengkulu	TBBM Pulau Baai	9.300	6.174	432,18	8.867,82	615	583	9.431	9.430.850	113.169.840	565.849.200	FRANCO
5	1.000	12.000	60.000	L	PPROYEK SUNGAI PEMUH	Jambi	TBBM Jambi	9.300	6.174	432,18	8.867,82	535	535	9.403	9.402.820	112.833.840	564.169.200	FRANCO
6	258.750	3.105.000	15.525.000	L	AG KAMOJANG	Jawa Barat	TBBM Padalarang	9.300	6.174	432,18	8.867,82	305	307	9.175	2.371.984.675	28.487.816.100	142.439.080.500	FRANCO
7	5.000	60.000	300.000	L	PROYEK KARAHA	Jawa Barat	TBBM Padalarang	9.300	6.174	432,18	8.867,82	412	412	9.280	46.399.100	556.789.200	2.783.946.000	FRANCO
8	237.867	2.854.404	14.272.020	L	AG LAHENDONG	Sulawesi Utara	TBBM Bitung	9.400	6.174	432,18	8.867,82	385	385	9.253	2.224.727.235	26.696.726.819	133.483.634.096	FRANCO
9	9.733	116.796	583.980	L	AG SIBAYAK	Sumatera Utara	TBBM Belawan	9.300	6.174	432,18	8.867,82	500	469	9.337	90.875.269	1.090.503.229	5.452.516.144	FRANCO
10	29.700	360.400	1.752.000	L	PROSPEK SEULAWAH AGAM	Banda Aceh	TBBM Krueng Raya	9.300	6.174	432,18	8.867,82	460	271	9.139	266.853.544	3.202.242.528	16.011.212.640	FRANCO
11	29.200	350.400	1.752.000	L	PROSPEK GUNUNG LAWU	Jawa Tengah	TBBM Boyolali	9.300	6.174	432,18	8.867,82	650	281	9.149	267.145.544	3.205.746.528	16.028.732.640	FRANCO
Total (exclude Pajak)	1.216.537	14.603.244	73.016.220	L													891.962.181.939	
Terbilang							Enam Ratus Delapan Puluh Satu Milyar Sembilan Ratus Enam Puluh Dua Juta Seratus Delapan Puluh Satu Ribu Sembilan Ratus Dua Puluh Rupiah											

KETERANGAN:
- Harga BBM belum termasuk PPN, PPh, dan PBBKB sesuai wilayah masing-masing.
- Harga Handling Fee belum termasuk PPN
- Harga Handling Fee berlaku untuk minimum volume delivery 5 KL (Kapasitas Mobil Tanki Terkecil)

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General Preparation Before Drilling

PERTAMINA FUEL Cost Component - Basic Price (issued every 14 days period)

Harga Solar Industri Pertamina

Wilayah	Harga Solar Industri
Wilayah I (Sumatera, Jawa, Bali, dan Madura)	Industri : Rp8.114,53 per liter
	Tambang : Rp8.416,90 per liter
	Shipping : Rp8.380,40 per liter
Wilayah II (Kalimantan)	Industri : Rp8.114,98 per liter
	Tambang : Rp8.559,40 per liter
	Shipping : Rp8.544,65 per liter
Wilayah III (Sulawesi, NTB)	Industri : Rp8.225,69 per liter
	Tambang : Rp8.532,20 per liter
	Shipping : Rp8.495,20 per liter
Wilayah IV (Maluku, NTT, Papua)	Industri : Rp8.392,43 per liter
	Tambang : Rp8.705,15 per liter
	Shipping : Rp8.667,40 per liter

- Basic Price
  1. Do request for price discount to Pertamina -> 5 – 20%
  2. Do price comparison with other fuel supplier i.e Shell, Petronas etc
- VAT
  1. Fix refer to Basic price
- PBBKB
  1. Fix refer to Basic price
- Transport
  1. Fix price

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General Preparation Before Drilling

Chance to Get Pertamina Fuel Price Discount

If company do direct purchase fuel, could request to Pertamina Fuel price Discount. Below are possibility % price discount:

- Up to 20%: Government or BUMN direct purchase and Cash deposit in advance.
- Up to 10%: Private Company direct purchase and Cash deposit in advance.
- Up to 5%: Private Company direct purchase by invoice system.

- Up to 10%: To Pertamina Fuel agent

	Target volume Tahun 2019 (Rata-Rata Perbulan)	Diskon Berlaku 2020
Realisasi target	≥ 40.000 KL	21% MOPS

Target Volume Periode Tahun 2020 – 2023 (Rata-Rata Perbulan)	Diskon Berlaku Periode Tahun 2021 – 2024
40.000 – 45.000 KL	23% MOPS
> 45.000 KL	25% MOPS

Nota Kesepahaman antara
SKK Migas dan PT Pertamina

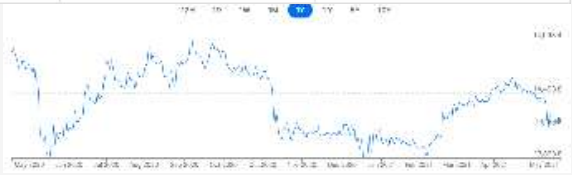
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General Preparation Before Drilling

RIG FUEL Supply – Cost Saving

Description	No Price Discount	5% Price Discount	10% Price Discount	20% Price Discount	23% Price Discount
Basic Price (Industrial), IDR/L	21,100.0	20,045.0	18,990.0	16,880.0	16,247.0
Basic Price $/L	1.314	1.248	1.183	1.051	1.012
Consumption L/Day	12,000	12,000	12,000	12,000	12,000
Drilling Days Per Well	35	35	35	35	35
Cost Saving / Well	None	27,595.44	55,190.88	110,381.76	126,939.03

Note: Discount only for basic price, not apply to VAT, PBBKB and Transport



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General Preparation Before Drilling

Rig Equipment Capability

Rig Equipment Capability	Function	Check List
Overhead BOP Crane	Reduce flat rig time: faster NU & ND BOP	
BOP Test Stump	Reduce flat rig time (Offline job) for BOP pressure test.	
Split Type Rotary Table	Capable run pre-welded wellhead with production casing	
Rig Skidding/ Walking Devices	Easier to rig move between well in same well pad.	
Rig Skid/ Walking with DP stands	Reduce rig time to LD and re-MU DP stands	
Drill String MU equipment Hawk Jaw, senior type	Faster and saver MU drill string up to 9-1/2" DC	
Offline MU DP stand	Reduce flat rig time: Offline MU stand DP	
DP hard banding	Avoid rig time to LD DP and replace.	
DP internal coating	Prevent DP washout, avoid extra BHA trip	
Complete Fishing Tool and Inspected	Rig to provide complete and detail fishing tool, including capable to fish internal parts of downhole equipment	

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Data Driven Drilling Advisory

General Preparation Before Drilling

Rig Equipment Capability

Rig Equipment Capability	Function	Check List
Sufficient Heavy Equipment: At rig site: • Minimum 1 Forklift, 2 Crawler Crane, 1 High bed, 1 Tronton, 1 Winch Truck.	Avoid delay rig time due to waiting on heavy equipment	
Fast Rig Down, Rig Moving and Rigging Up.	Reduce stand by rate during Rig Down, Rig Moving and Rigging Up.	
New Generation Rig	Less Fuel Consumption, better drilling performance	
Optimum Rig HP Utilization	**1000 (350 Ton):** 1500 – 3200 m Depth **1350-1500 HP (350-500 Ton):** 3300 – 4500 m Depth	

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General Preparation Before Drilling

Opsi Pembelian Rig Bekas (Second).

Pertimbangan Opsi Mempunyai Rig Sendiri apabila program pemboran yang banyak dan berkelanjutan:

- Dari market survey pertengahan tahun 2020, harga rig 1000-1500 HP bekas berkisar USD 2.5 juta – 4.5 juta.
- Biaya rig menjadi lebih murah, dimana sewa Rig merupakan komponen terbesar dari biaya pemboran.
- Dapat meniadakan aktivitas mobilisasi dan demobilisasi Rig yang mahal dan mengurangi resiko masalah komunity.
- Fleksibilitas jadwal pemboran, dikarenakan rig sudah tersedia dilokasi.
- Mendorong perusahaan untuk menambah Jumlah sumur pemboran karena dengan rig sendiri mampu menekan biaya sumur.

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General Preparation Before Drilling

Own 1000 – 1500 HP Used Rig
Cases: KS Orka bought DATI 231 Rig

Description	Rig Rental Services	Own Rig
Rig Purchase Price	None	USD 2.5 Million
Rig Upgrade	None, included in Mob cost	USD 500 K
Mobilization	USD 1 Million	USD 1 Million
Demobilization	USD 0.5 Million	USD 100 K
Daily Rate	USD 25-30 K	USD 13 K: • Rig Management: USD 5.5 K • Catering: USD 3.5 K • Maintenance: USD 4.0 K

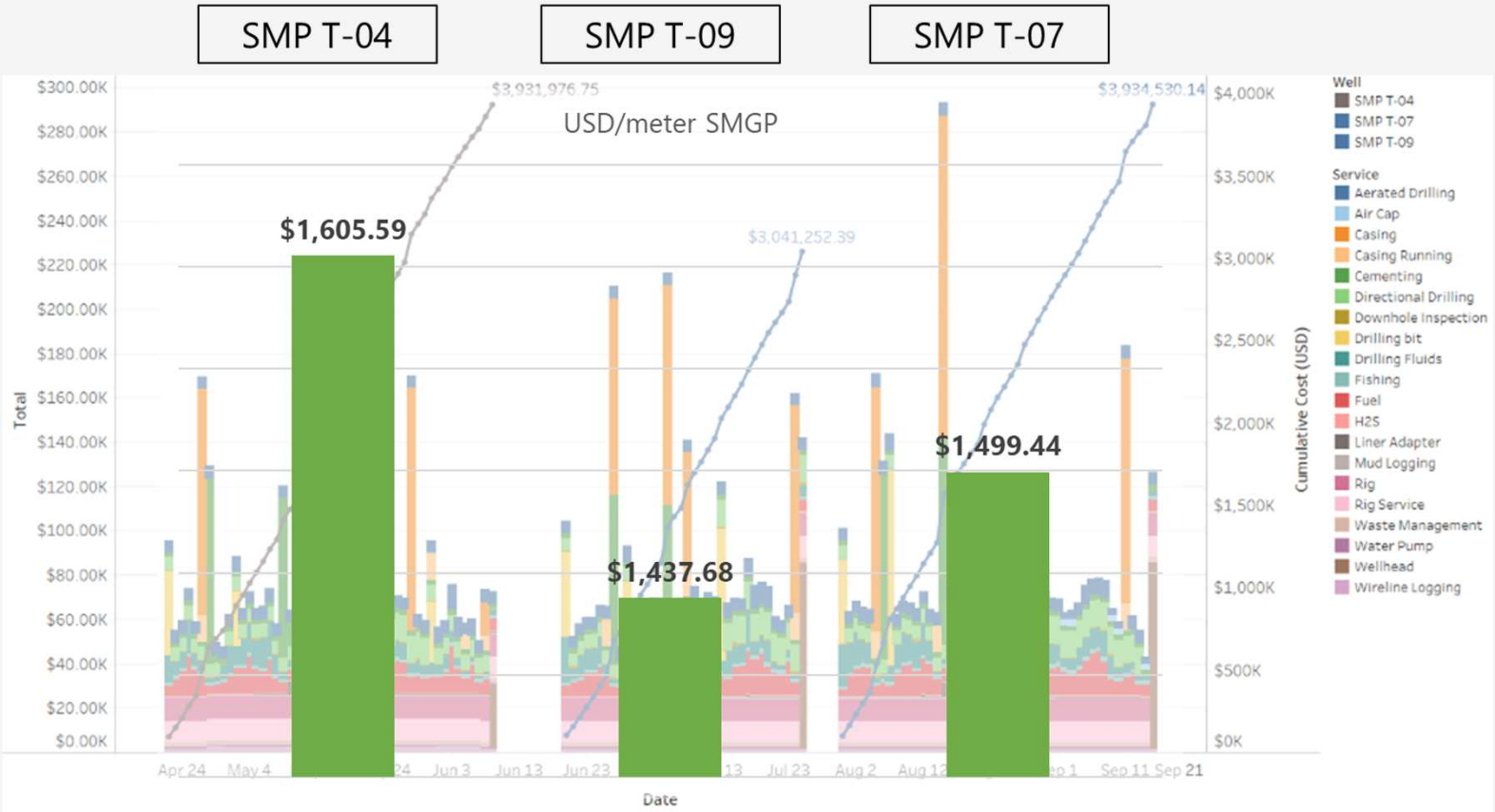
- Total Rig cost balance after **173 rig operating days** = **5 wells**
  - **Drill 6th well onward -> Save Rig Cost USD 15 K daily**
  - **or equivalent with cost reduction of USD 525 K per well**



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General Preparation Before Drilling

SMGP Year 2020 Drilling Statistics – Use Own Rig



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General Preparations before Drilling

3

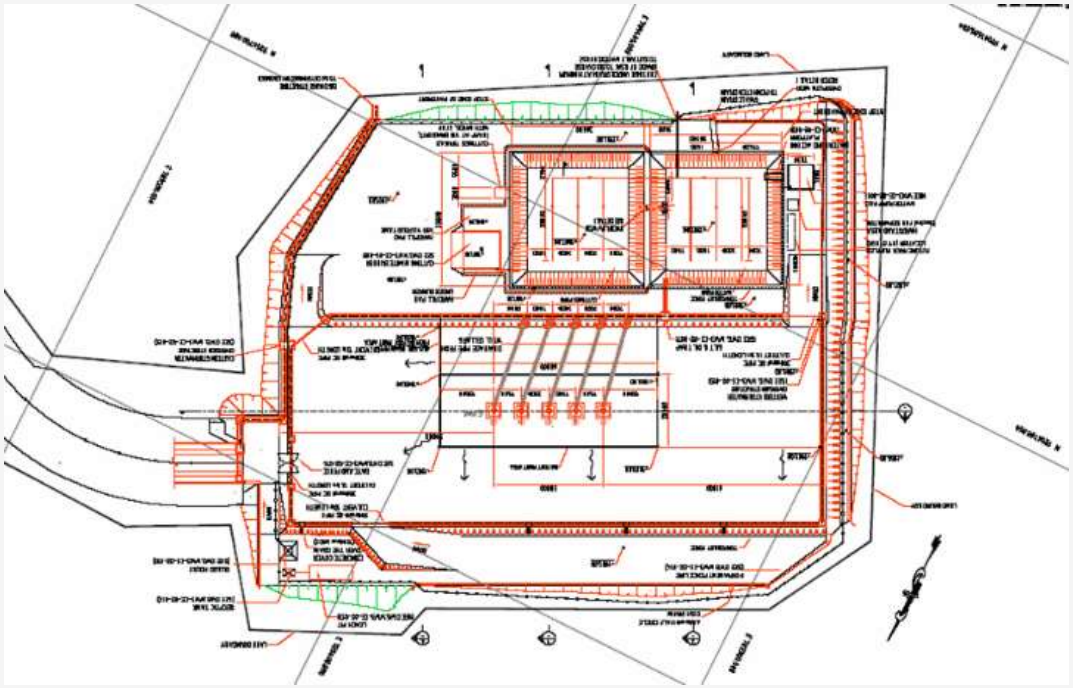
Drilling Location Preparation

- Elevated Wellpad Design
- Mud Pond Size and Design
- Cellar Design and distance between Cellar
- 30” Conductor Design

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Data Driven Drilling Advisory

General Preparation Before Drilling

Effective Wellpad Design



Typical Rig Site Layout

- 5000-8000 m3 water pond
- Lower elevation for cutting and water pond
- Pre design for well test pipe line layout
- Proper drainage and control box
- Direct Under ground drainage from Cellar to cutting Pond



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General Preparation Before Drilling

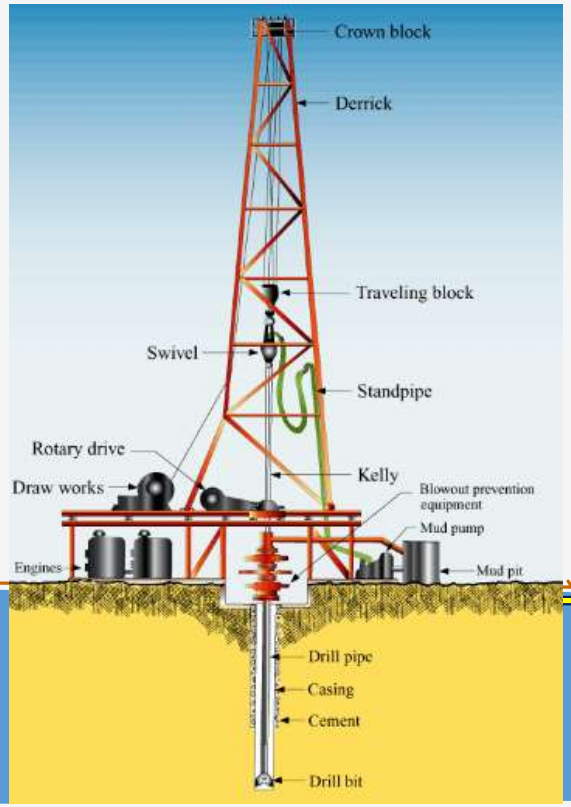
Elevated Well-Pad Design

Place for Mini Camps



Access for Heavy Equipment





- Mini camp area could be higher elevation than rig elevation.
- Distance between Cellar -> Well accessibility after being drilled
- Water flow by gravity to water pond.
- SIMOP: Pipeline layout not interfere with Rig Equipment layout.

Drainage Line from Cellar to Mud Pond

Settling Pond

Water and Mud Pond

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General Preparation Before Drilling

Layout higher elevation for Rig Mini Camps

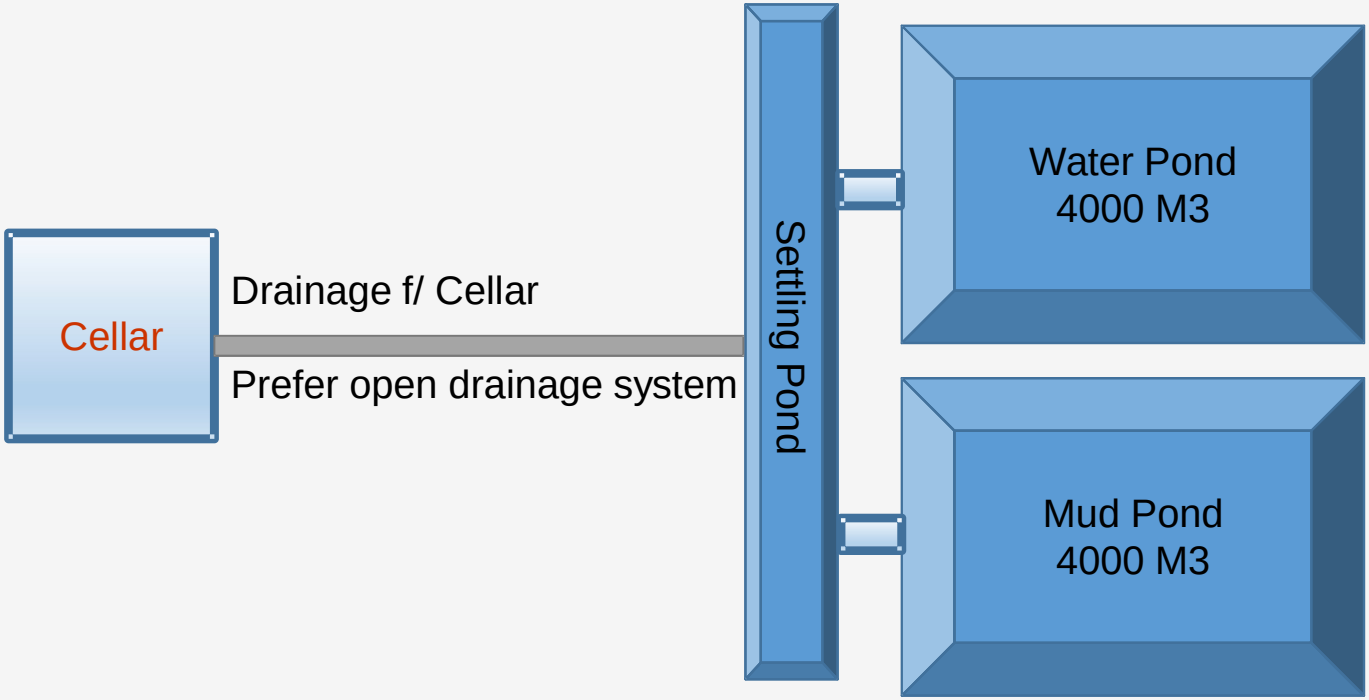




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General Preparation Before Drilling

Mud Pond Size and Design



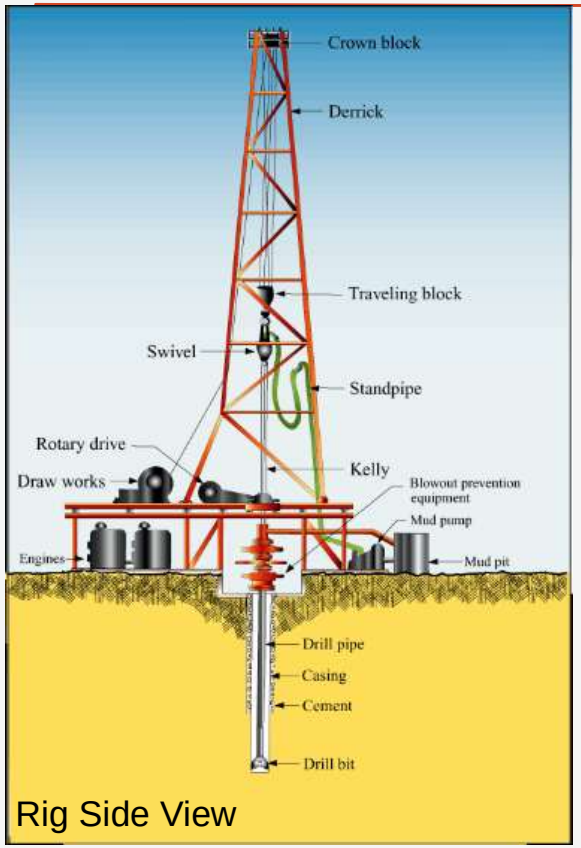
Optional (due to limited area). Could be supply from other closest Wellpad Pond

Mandatory

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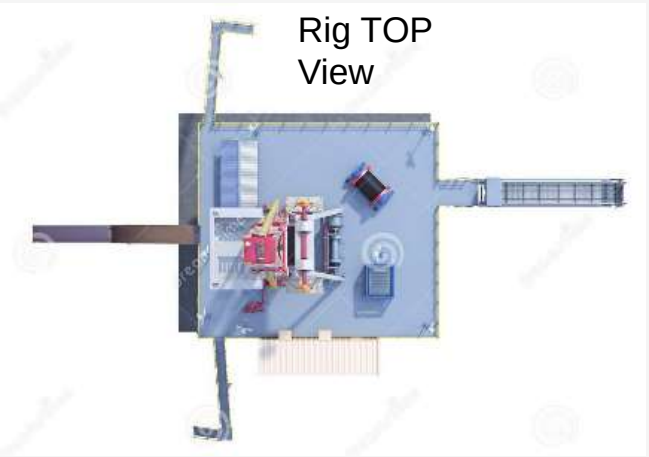
General Preparation Before Drilling

Rig Picture



Rig Side View


Rig Side View


Rig TOP View

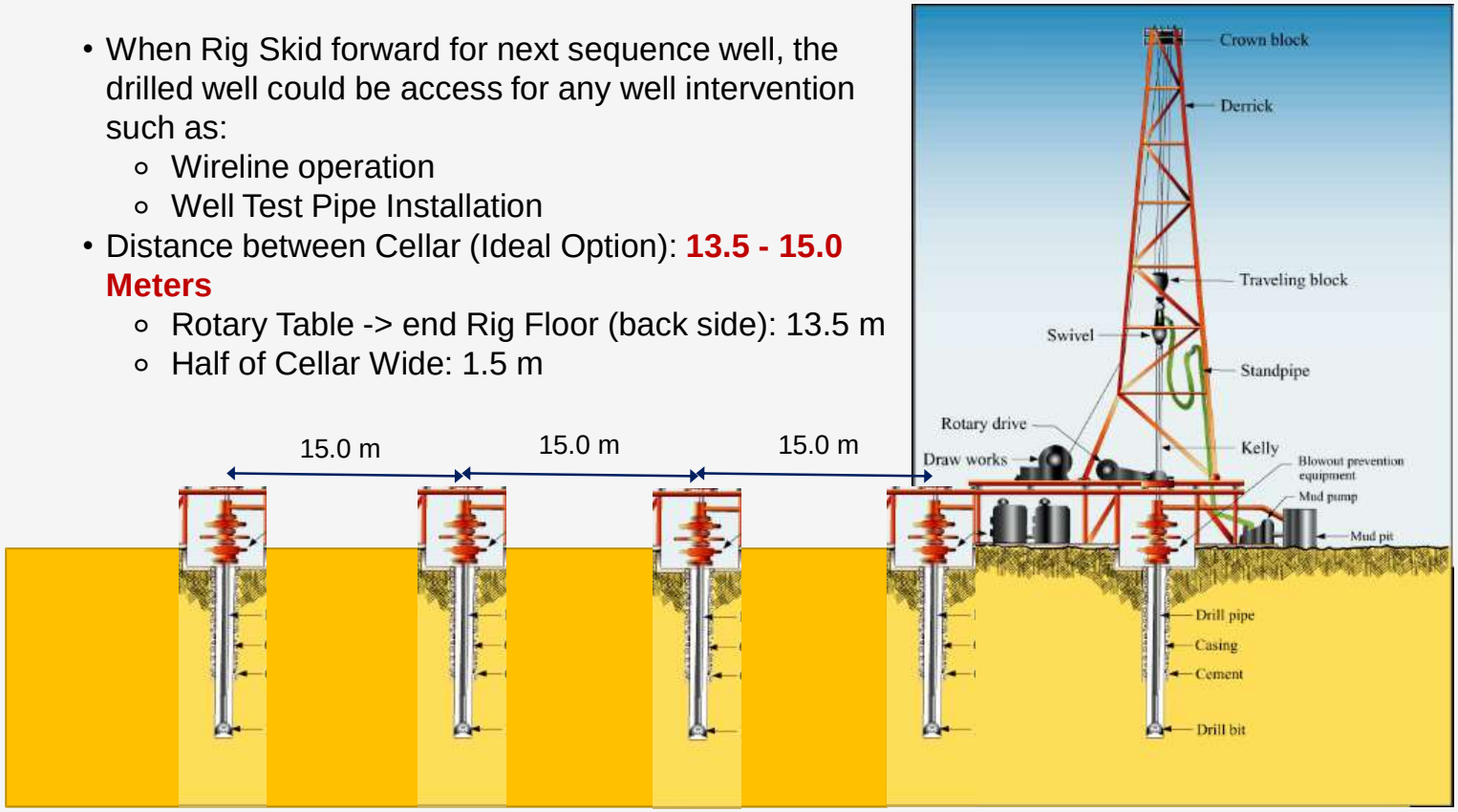
- Rig Floor Size
- Post Drilling Activity

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General Preparation Before Drilling

Rig Picture: Side View for Determining Distance between Cellars

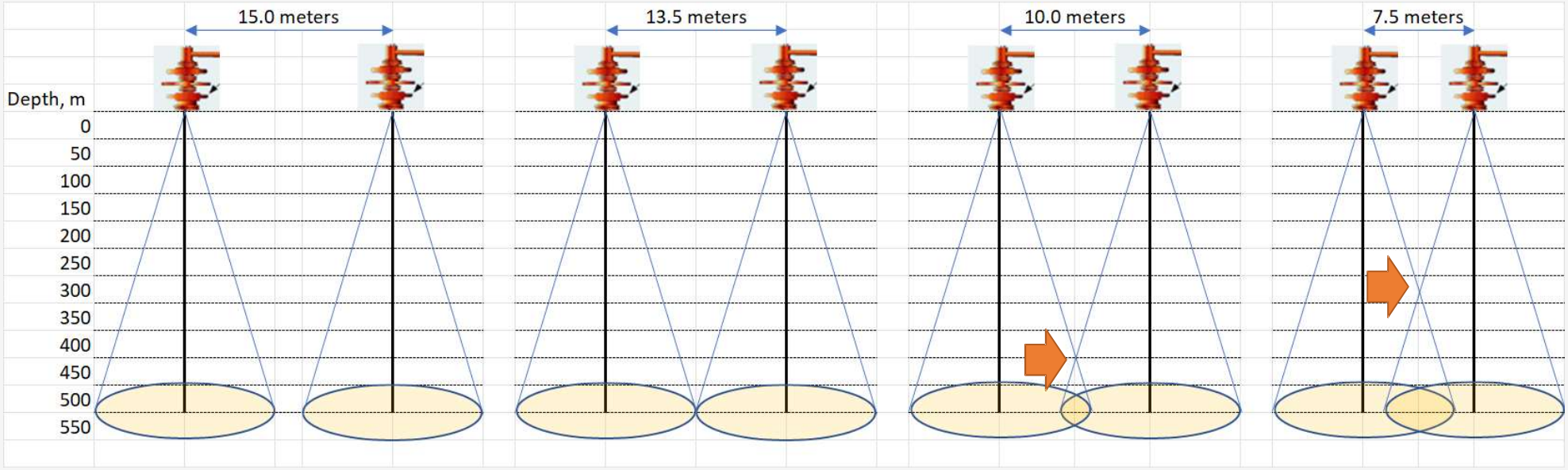
- When Rig Skid forward for next sequence well, the drilled well could be access for any well intervention such as:
  - Wireline operation
  - Well Test Pipe Installation
- Distance between Cellar (Ideal Option): **13.5 - 15.0 Meters**
  - Rotary Table -> end Rig Floor (back side): 13.5 m
  - Half of Cellar Wide: 1.5 m



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General Preparation Before Drilling

Surface Anticollision

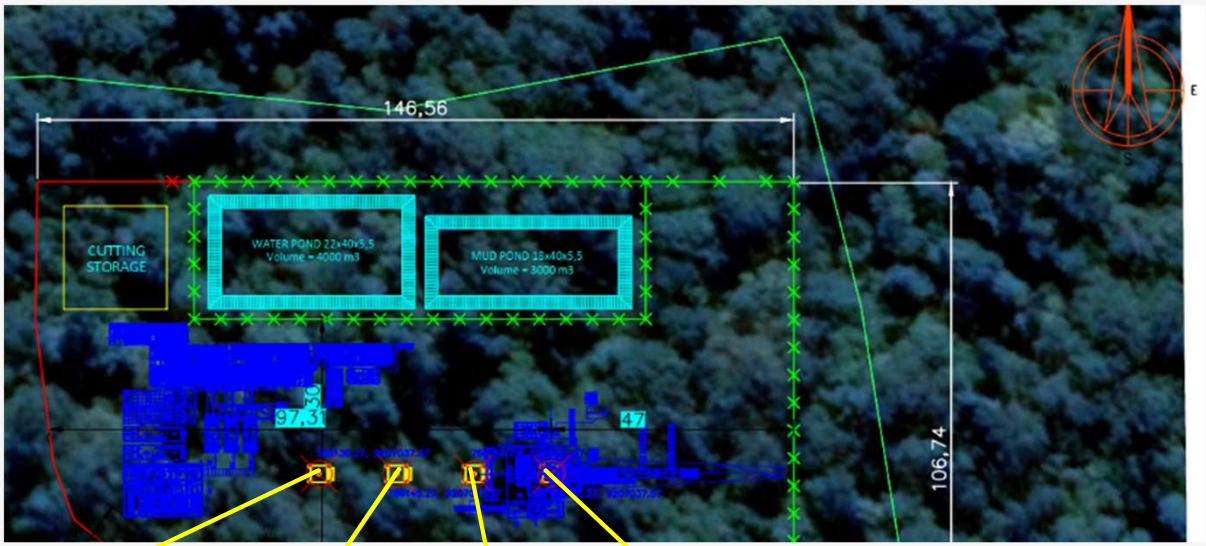


SF > 1 SF = 1 SF < 1 SF < 1

SF = Separation Factor
SF = (Ellipse Radius-1 + Ellipse Radius-2)/ Jarak lubang

General Preparation Before Drilling

PAD BB: Cellar Assignment



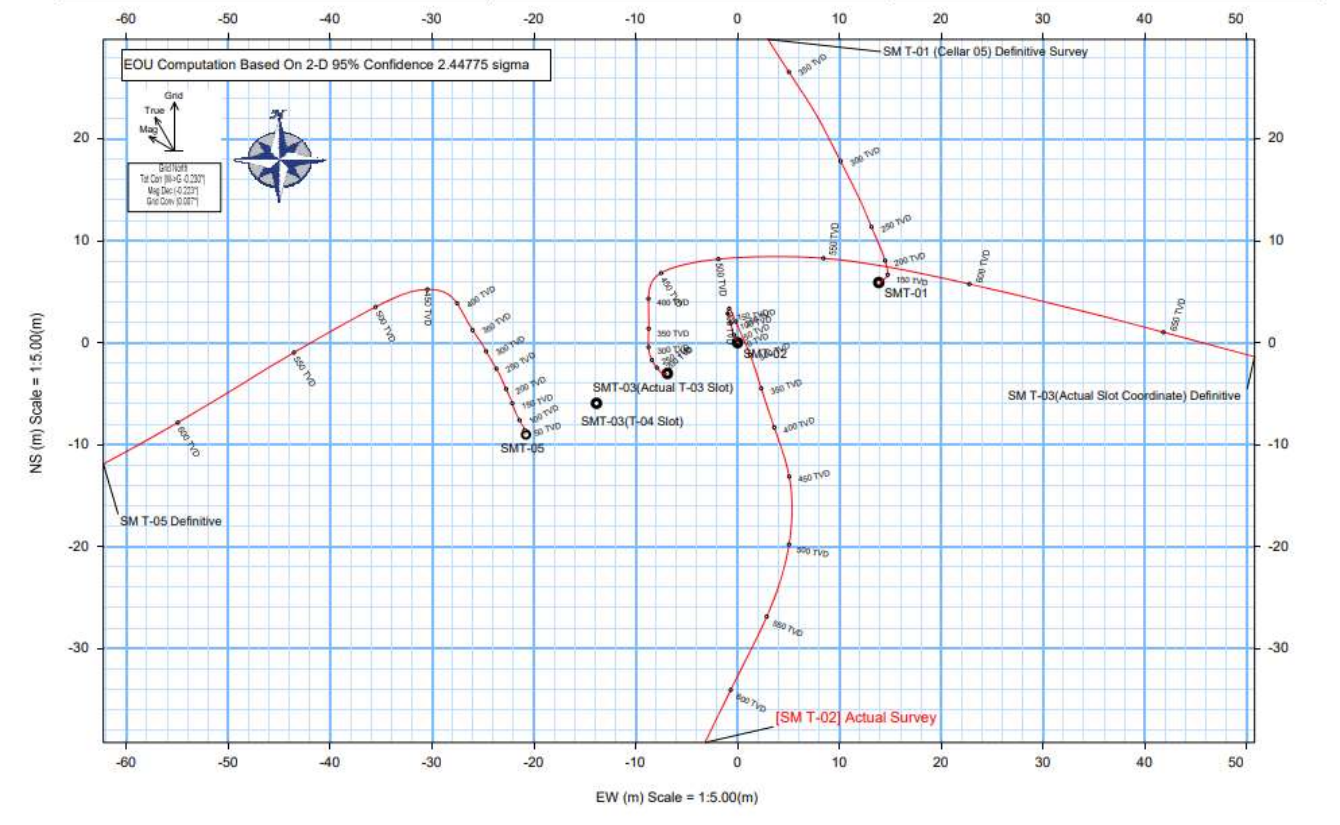
Well	Az
PTH-BB-9	215
PTH-BB-9A	168
PTH-BB-9B	245
PTH-BB-9C	135

- Cellar assignment refer to Well Trajectory
- Avoid any Crossing as possible
- Check well sequence to easy Rig Skidding
- Check Wellhead Height to avoid any obstruction during Rig Skidding

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General Preparation Before Drilling

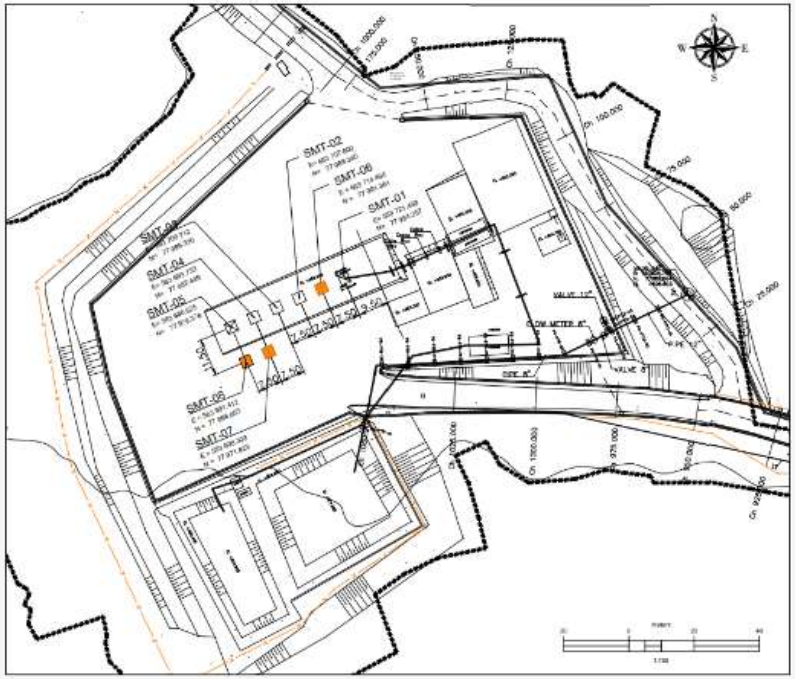
Cellar Assignment, Optimized ?



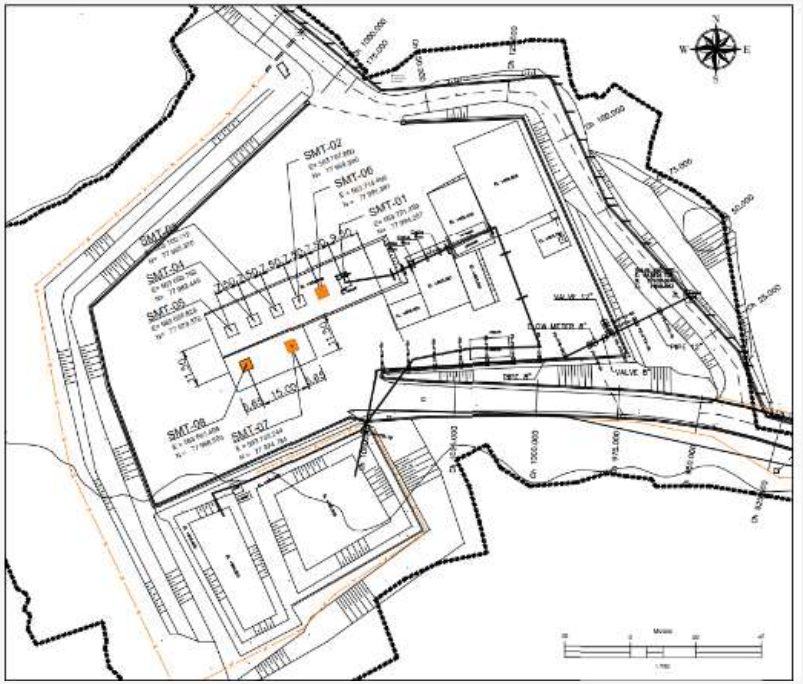
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General Preparation Before Drilling

Additional Cellar to the Existing Pad. Is it safe?



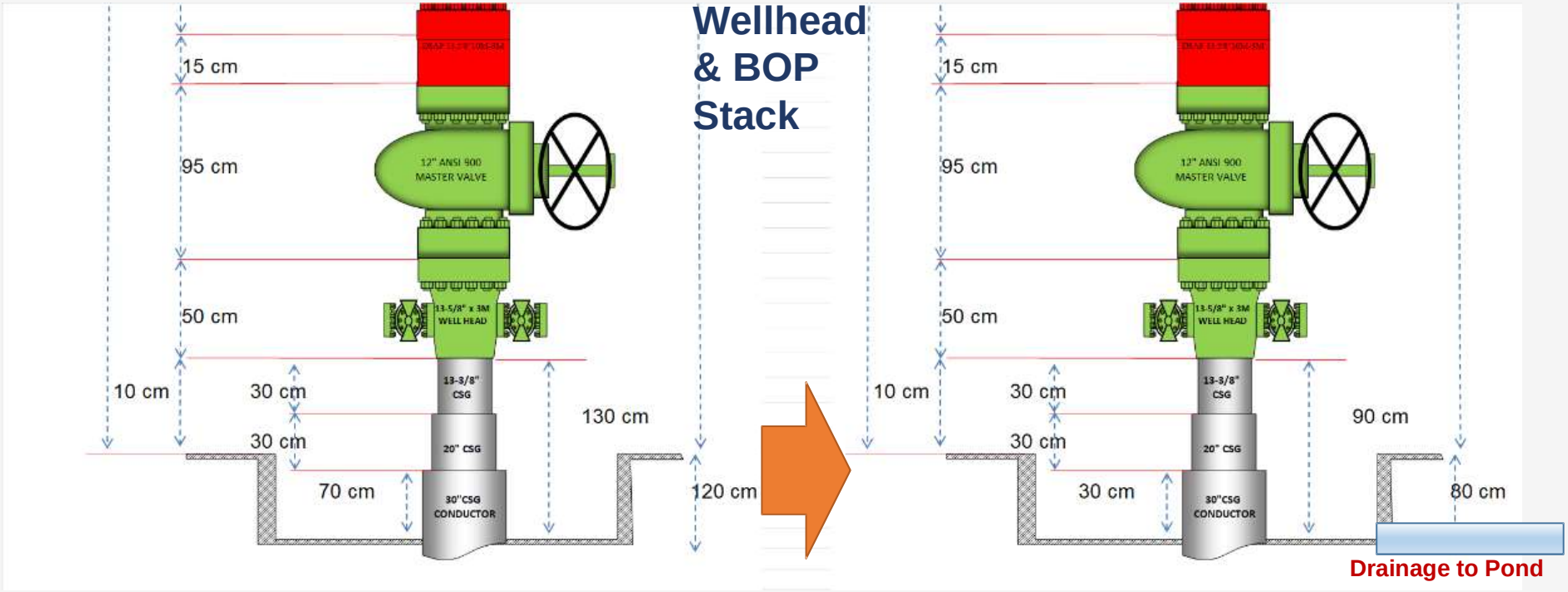
Well Collision Risk - High



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General Preparation Before Drilling

Cellars Design-2: 3m X 3m X 0.8m Depth



3m X 3m X 1.2m depth

3m X 3m X 0.8m depth

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General Preparation Before Drilling

Cellars Design-2: 3m X 3m X 0.8m Depth



Advantage:

- More simple -> Less Construction Cost
- Always dry cellar
- No risk accumulated H2S



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General Preparation Before Drilling

30” Conductor Specification and Installation

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General Preparation Before Drilling

30” CASING CONDUCTOR SPECIFICATION

Manufacture	Grade	Lenght	Weight	Connection
• Spiral • ERW • SAW • Seamless	• X-52 • X-56 • K-55	• R-3	• 310 ppf • 133 ppf • 129 ppf	• Bevel End • Quick Connector

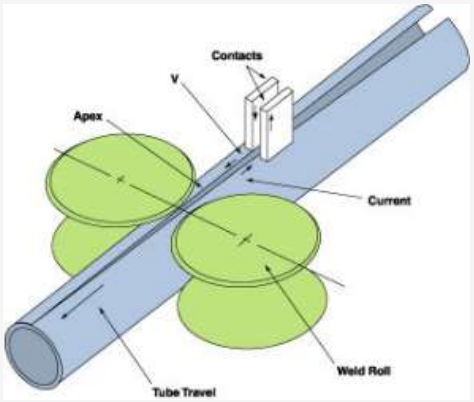
MANUFACTURED BY ROLLING METAL PLATE:

- ERW: Electric Resistance Welded
- SAW: Submerge Arc Welding
- LSAW: Longitudinal Submerge Arc Welding

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General Preparation Before Drilling

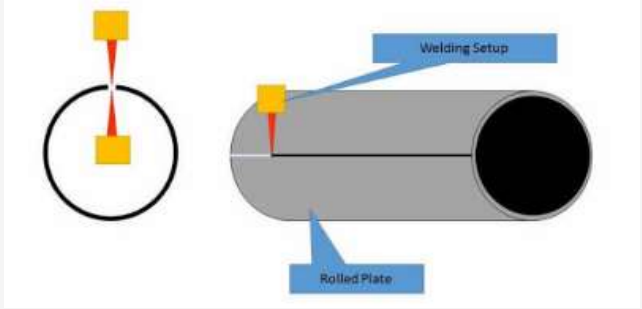
30” CASING CONDUCTOR MANUFACTURE PROCESS



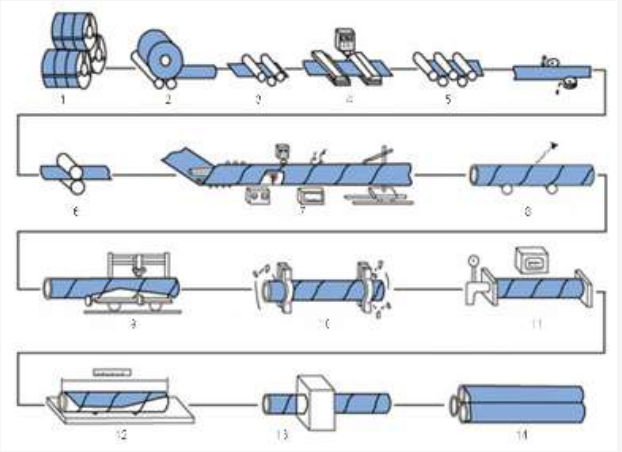
ERW: Electric Resistance Welded



SAW: Submerge Arc Welding



LSAW: Longitudinal Submerge Arc Welding
DSAW: LSAW with double seam weld in inside and outside



Spiral SAW

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General Preparation Before Drilling

CASING MANUFACTURE PROCESS

ERW: Electric Resistance Welded

- Manufactured by rolling metal and then welding it longitudinally across its length

SAW: Submerge Arc Welding

LSAW: Longitudinal Submerge Arc Welding,

- Single seam weld on the outer surface

DSAW: LSAW with double seam weld in inside and outside

SPIRAL SAW



SEAMLESS:

- Manufactured by extruding the metal to the desired length. 20-30% higher price than ERW casing

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General Preparation Before Drilling

Ketebalan Pipa 30” Konduktor

Pemilihan ketebalan pipa 30” Konduktor harus disesuaikan dengan metoda pemasangan pipa konduktor tersebut:

- **PILING**, sebaiknya memakai pipa dengan ketebalan 1” untuk mencegah kolap saat pemasangan (hammer piling).
- **BOR dan CEMENT**, sebaiknya memakai pipa dengan ketebalan 0.75” atau 0.5”. Dengan cara ini tidak ada resiko konduktor kolap saat pemasangan.

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General Preparation Before Drilling

30” Cost Comparison

30” Casing	Welded	USD/Jt
1” Wall Thickness	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Stronger pipe, for piling installation



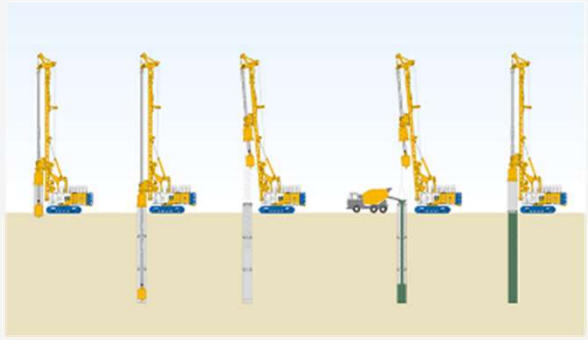
30” Casing	Welded	USD/Jt
0.75” Wall Thickness	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Thin pipe, for Auger installation



30” Casing	Welded	USD/Jt
0.5” Wall Thickness	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Thin pipe, for Auger installation



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General Preparation Before Drilling

30” Conductor Setting Depth

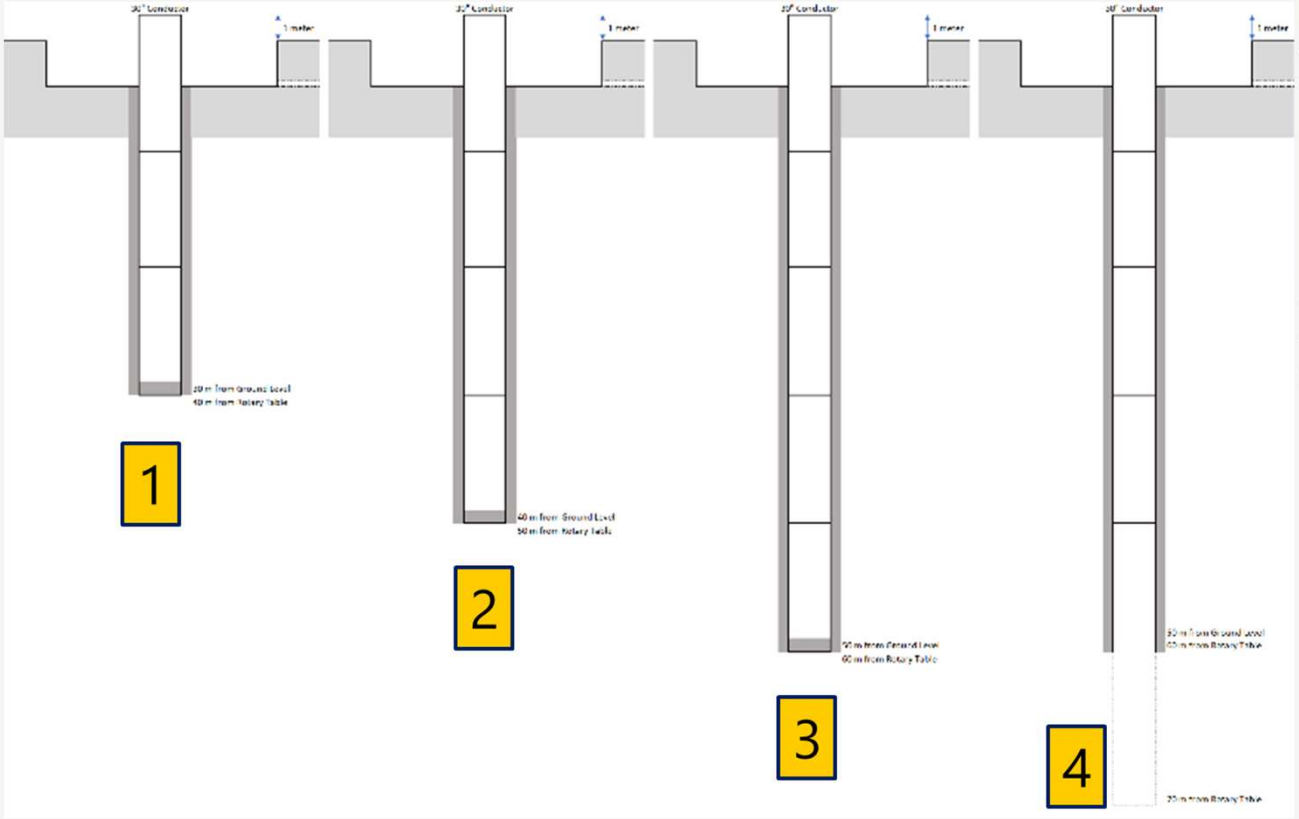
Kedalaman Pipa Konduktor sebaiknya mempertimbangkan hal-hal berikut ini:

- Kecukupan ruang kedalaman untuk merangkai BHA lubang awal, misalnya directional BHA untuk lubang permukaan 26”.
- Resiko hilang lumpur dimana lumpur yang hilang bisa leakage (bocor) ke permukaan dan bisa menimbulkan pencemaran lingkungan.
- Bisa menutupi/mengkover zona losses permukaan.
- Kemampuan alat pemasang konduktor.

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General Preparation Before Drilling

Conductor Depth Illustration - 26” BHA Length



26" Directional BHA:

9-1/2" MM + 9-1/2" EM MWD Tool

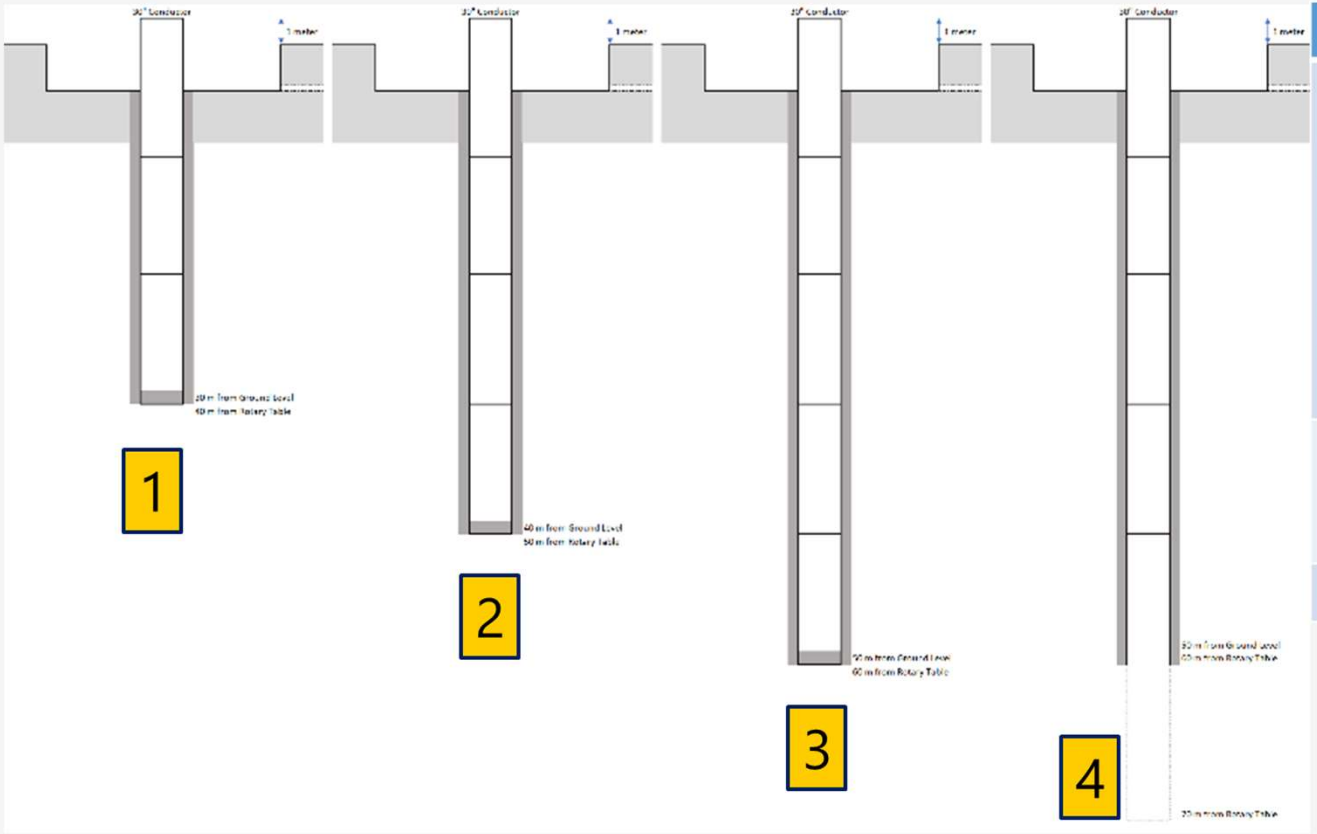
No.	BHA Component	OD	ID	Connection			LENGTH	Cumm
		in	in	Up	Down	Conn.	m	Length
1	26" Bit	26	TFA	7⅝ Reg		P	0.5	0.5
2	9-5/8" Mud Motor	9⅝	4.75	7⅝ Reg	7⅝ Reg	B/B	10.7	11.2
3	9 1/2" Catcher Sub w/ Float Sub	9½	3	7⅝ Reg	7⅝ Reg	B/P	1	12.2
4	24" IBS Stabilizer	9½	3	7⅝ Reg	7⅝ Reg	B/P	1.5	13.7
5	Shock Sub 9 1/2"	9½	3	7⅝ Reg	7⅝ Reg	B/P	2	15.7
6	9-1/2" NMDC	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	24.8
7	9-1/2" MWD XEM 800	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	33.9
8	9-1/2" MWD Gap Sub	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	43
9	9-1/2" NMDC (Optional)	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	52.1
10	9-1/2" Pony NMDC (Optional)	9½	3	7⅝ Reg	7⅝ Reg	B/P	3.0	55.1
11	X/O (7 5/8" REG Pin x 6 5/8" REG)	9½	3	6⅝ Reg	7⅝ Reg	B/P	3.0	58.1
12	7 x 8" DC	8	2.812	6⅝ Reg	6⅝ Reg	B/P	63.7	121.8
13	8" Hydraulic Jars	8	2.812	6⅝ Reg	6⅝ Reg	B/P	10.7	132.5
14	8" DC	8	2.812	6⅝ Reg	6⅝ Reg	B/P	9.1	141.6
15	X/O 6 5/8" RP x 4 1/2" IFB	8	2.812	4½ IF	6⅝ Reg	B/P	3.0	144.6
16	6 3/4" DC	6¾	2.813	4½ IF	4½ IF	B/P	9.1	153.7
17	15 x 5" HWDP	5	3	4½ IF	4½ IF	B/P	136.5	290.2
18								

- 26” Directional BHA length = > 58 m
- Option-3 and 4, we could run 26” Directional BHA without any drill out rotary BHA
- Rig time saving = 20 hours

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General Preparation Before Drilling

Conductor Depth Illustration - 26” BHA Length



Option-1 & 2	Option-3 & 4
MU Rotary BHA RIH Drill to 70 m POOH MU DD BHA RIH Drill to TD POOH	20 hours MU DD BHA RIH Drill to TD POOH
Additional cost for Auger and Casing =	Rig time saving = 20 hours USD 58 K
Potential Cost Saving:	

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General Preparation Before Drilling

Advantages to set Deeper 30” Conductor Depth

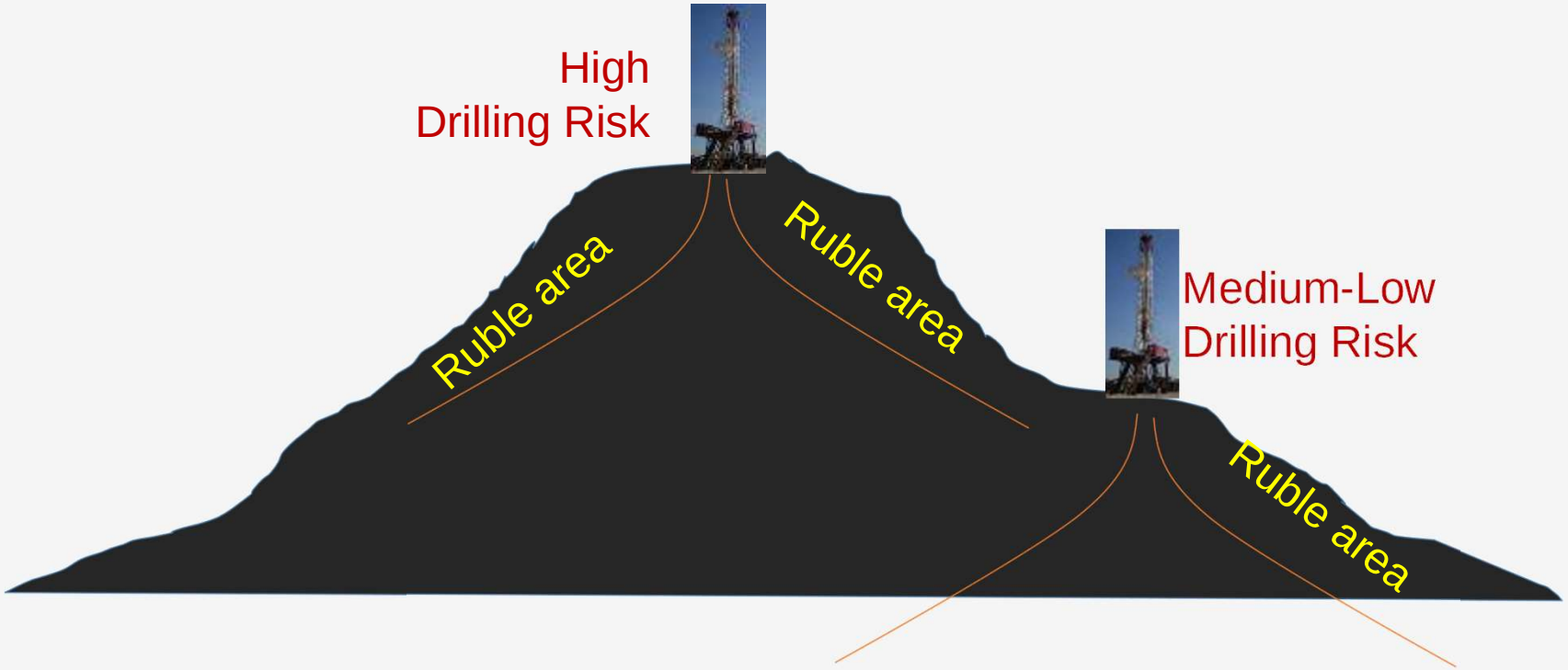
- Rig time saving since it could run directional BHA instead of initiated with Rotary BHA to provide sufficient space for Directional BHA
- If any surface losses, has been cured during Augering operation which much less operating daily rate.

Note: However, it need to confirm what the maximum depth that Auger could drill.

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General Preparation Before Drilling

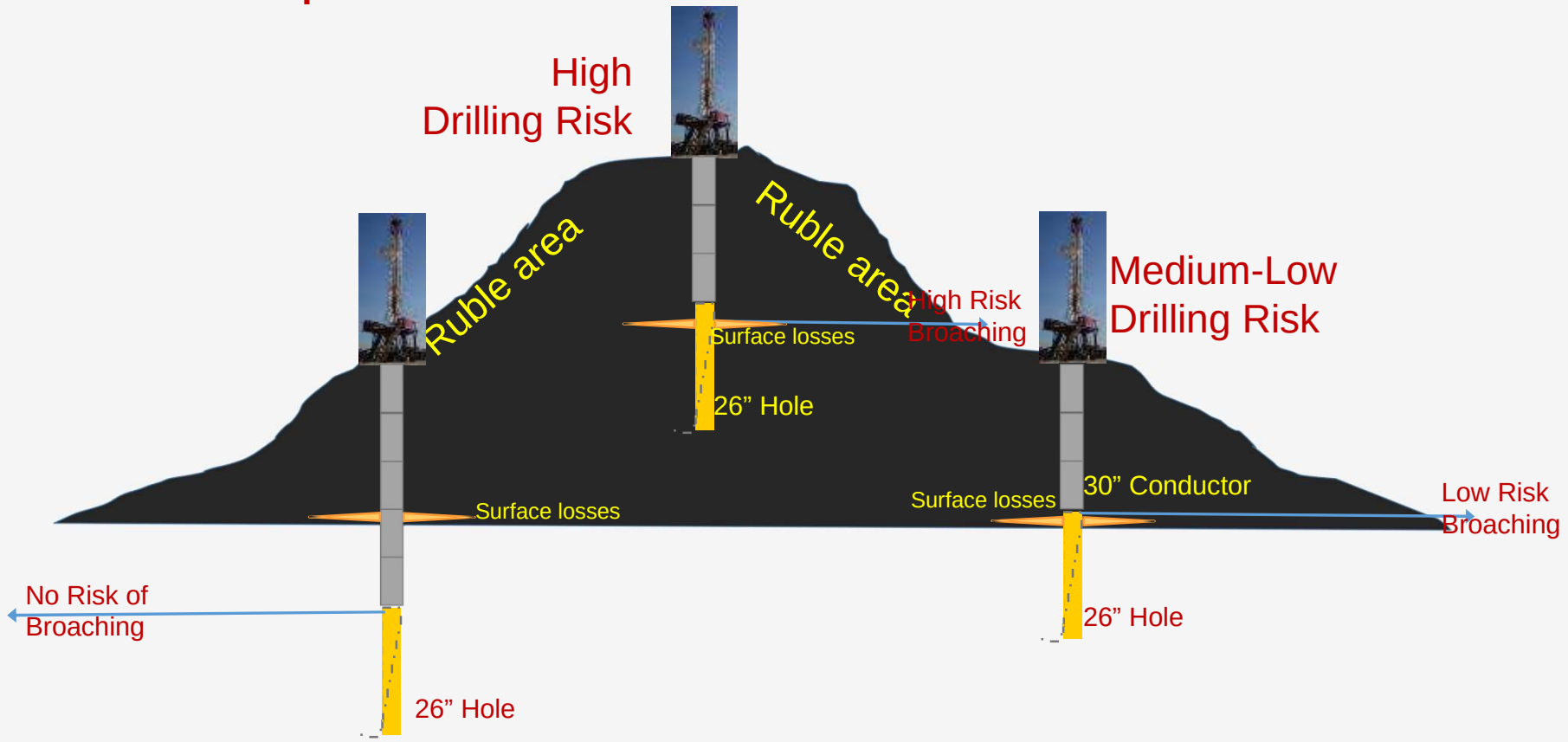
Wellpad Surface Location



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General Preparation Before Drilling

30” Conductor Depth

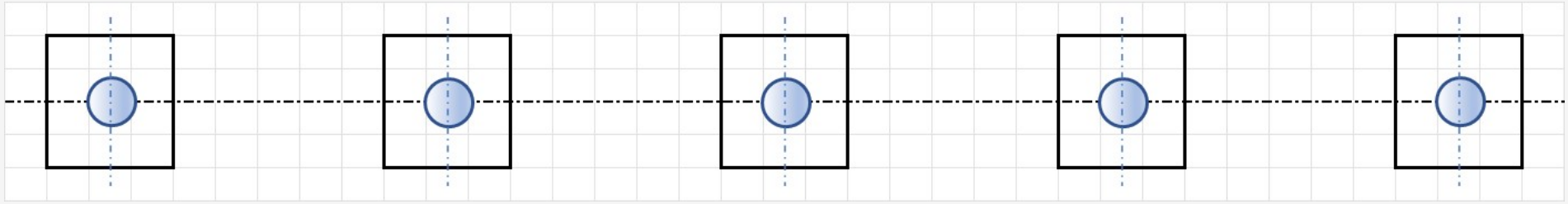


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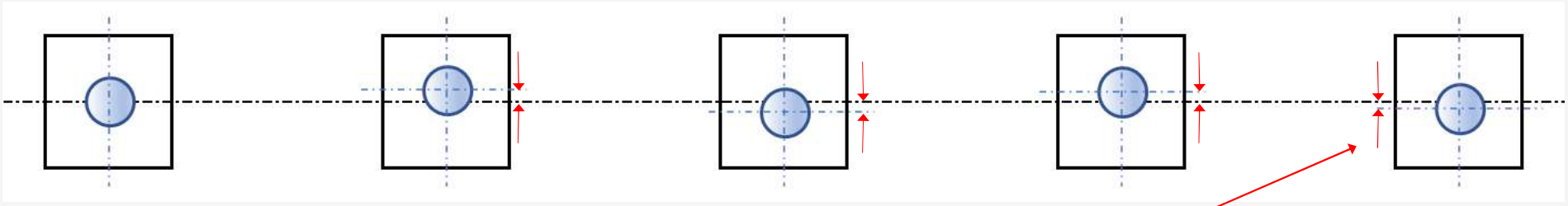
General Preparation Before Drilling

Cellar Off Center and Rig Alignment

Case-1: All Conductor Center Align



Case-2: Alignment Conductor Off-Center



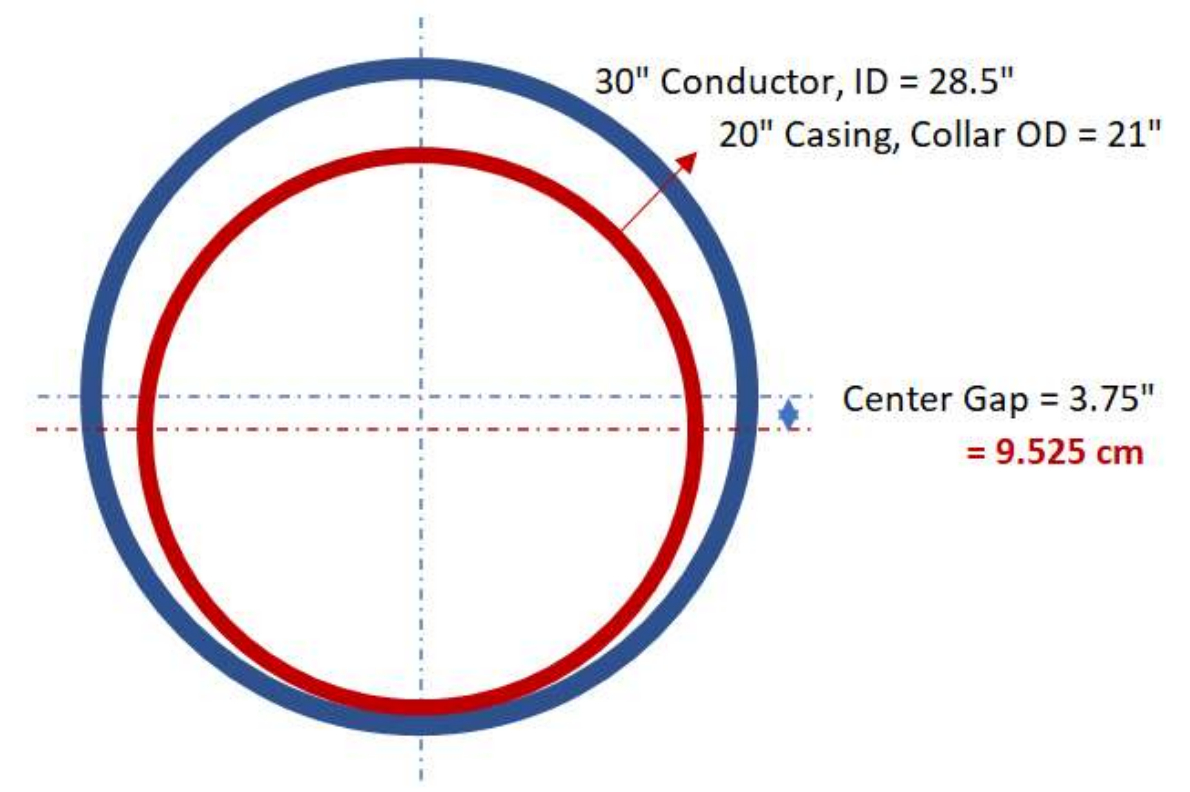
Case-2:

- **Do alignment line to have Minimum Off Center of Conductor Center**
- **Set the center of skidding rail to have Tolerable Off Center deviation up to 10 cm**

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General Preparation Before Drilling

Maximum Conductor - Rig Off Center



- Refer to 30” Conductor ID and 20” Collar OD, maximum off center between Casing Conductor Center and Rig Center < **10 Cm**
- When running 20” casing, it will push the 20” Casing Center to alignment with Rig Center.

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General Preparation Before Drilling

Cellar Off Center and Rig Alignment

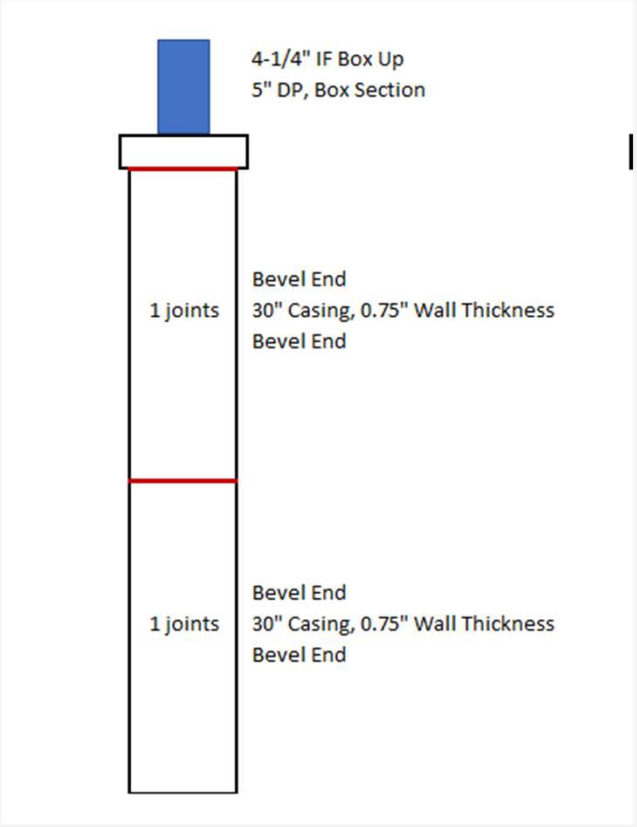
Off center pipa konduktor ini bisa dimitigasi saat pekerjaan pemasangan pipa konduktor:

- Jangan gunakan titik semua cellar dari sistem koordinat yang telah diberikan oleh baik team drilling atau dari team Sub-Surface.
- Titik koordinat hanya untuk menentukan posisi 2 cellar paling ujung.
- Kemudian untuk menentukan posisi beberapa cellar yang ditengah, dengan menarik garis lurus dari titik tangah dari 2 cellar paling ujung.
- Kemudian titik-titik cellar bagian tengah ditentukan dari jarak antar cellar.
- Setelah semua konduktor terpasang, lakukan GPS ulang untuk memberikan definitive koordinat masing-masing cellar/sumur. Definitive Koordinat ini yang akan dipakai oleh team drilling dan Sub-Surface dalam perencanaan dan drilling program.

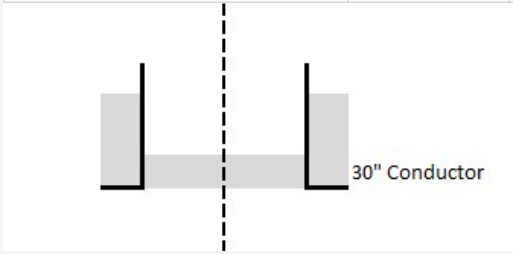
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General Preparation Before Drilling

Pre-Welded 2 Joints of 30” Conductors



30" CONDUCTOR INTALLATION - COST COMPARISON					
Using Auger System	Rp 7.175.000.000	14300	USD Rate	$ 501.748	Cost for 3 wells
				$ 167.249	Cost for 1 well
IF USING RIG					
Cost Description	Unit Price	Qty	Unit	Total Cost/ Well	
Rig Operation Cost	$ 80.000	1	Day	$ 80.000	
36" Hole Opener Service	$ 15.000	1	LS/well	$ 15.000	
36" Hole Opener Rental	$ 150	40	Day	$ 6.000	
36" Hole Opener Cone Replacement	$ 20.000	0,33	LS	$ 6.667	
36" Hole Opener Mob/Demob	$ 8.000	0,33	LS	$ 2.667	
Cementing	$ 250	60	bbls	$ 15.000	
				$ 125.333	
			Price escalation, 10%	$ 18.800	
			TOTAL COST TO INSTALL 30" CONDUCTOR BY RIG	**$ 144.133**	
				$ -23.116	Cost saving per well
				$ -69.348	Cost saving for 3 wells



- Do not use Rig to install 30” Conductor if hard surface formation

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General Preparations before Drilling

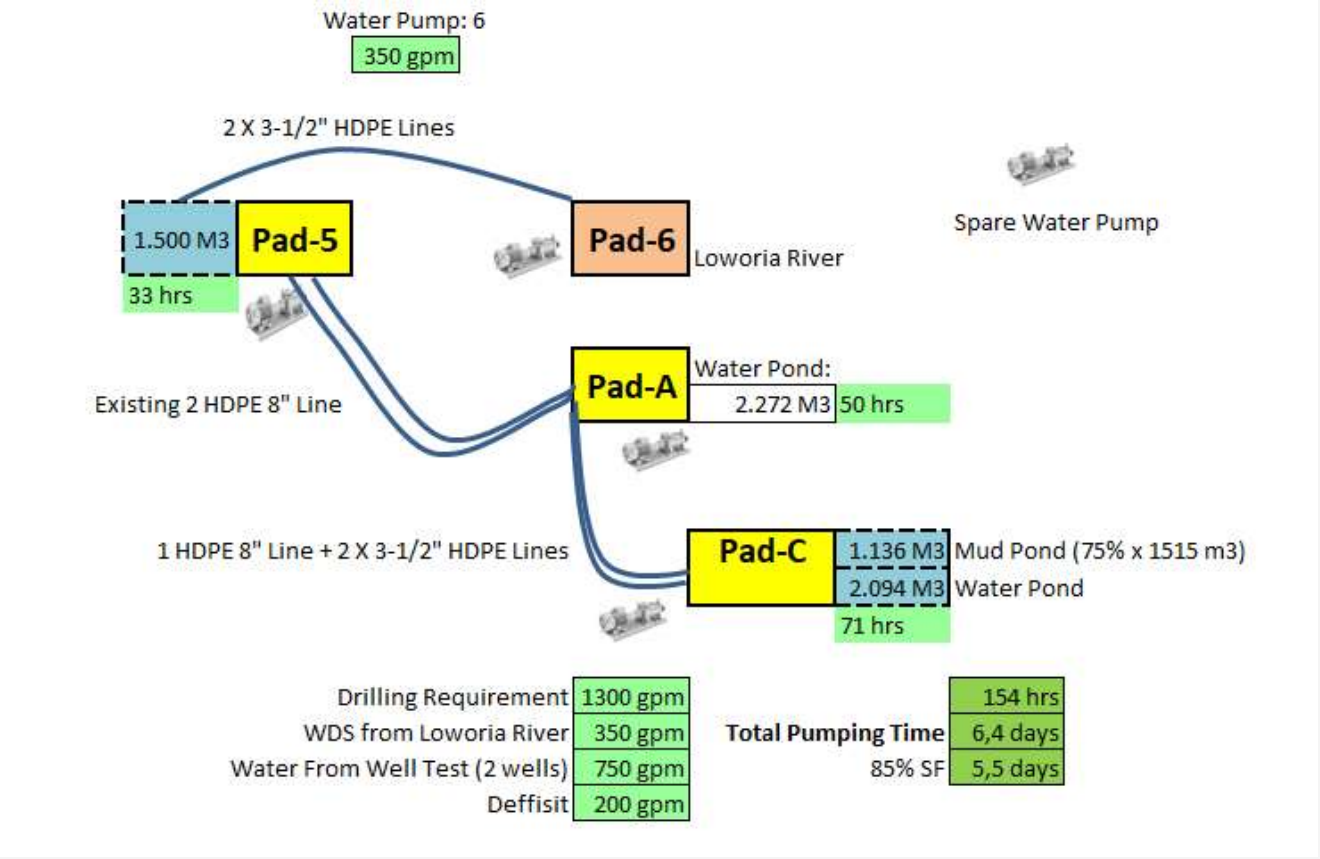
4 Water Supply Preparation for Drilling

- Water Pipe Line
- Steel Pipe and HDPE Water line

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Data Driven Drilling Advisory

General Preparation Before Drilling

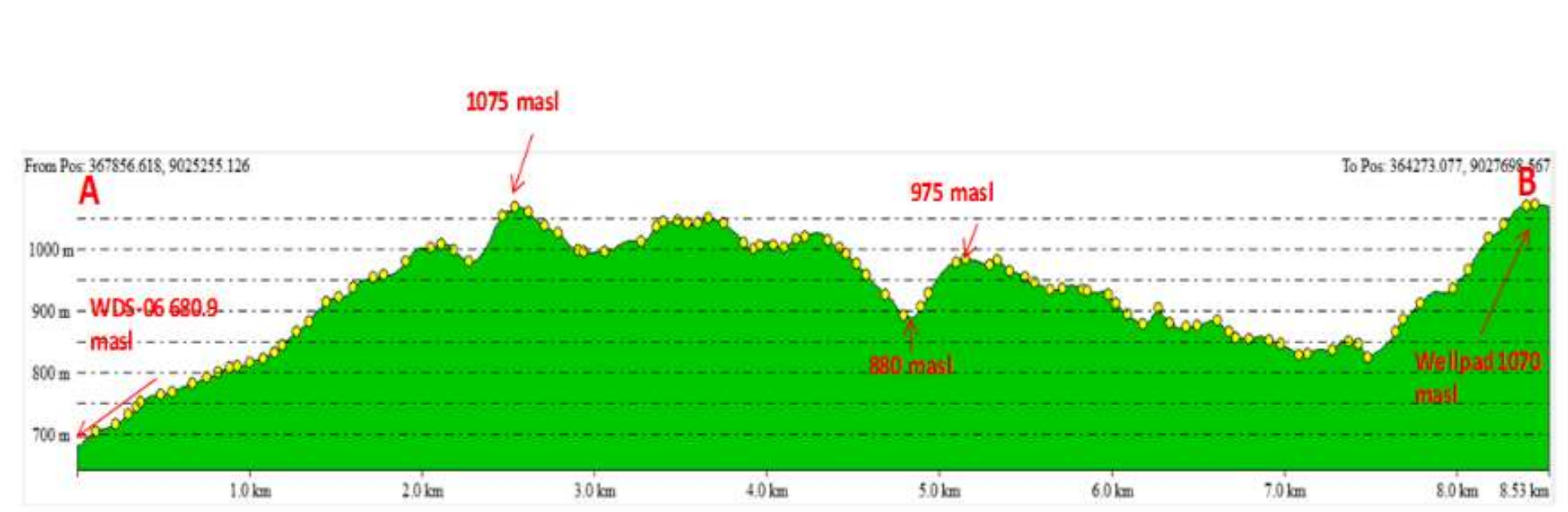
WATER SUPPLY – 1300 GPM



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General Preparation Before Drilling

Water Supply

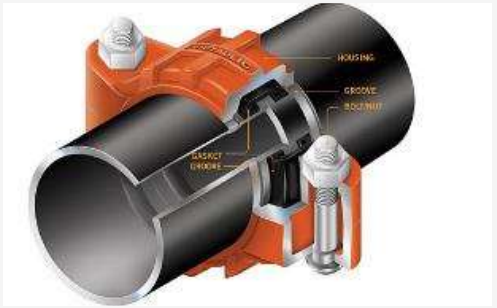


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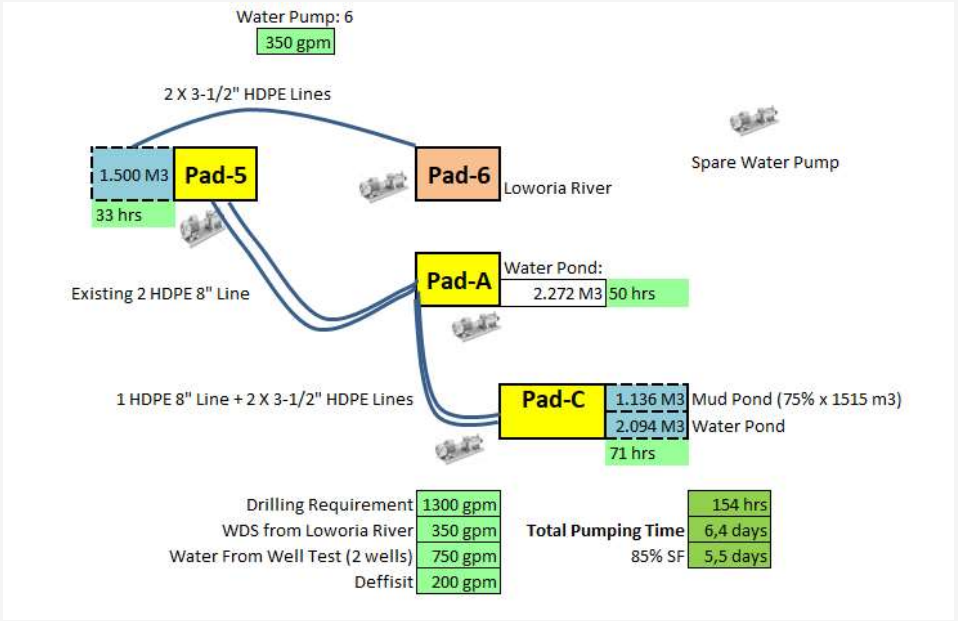
General Preparation Before Drilling

WATER SUPPLY – 1300 GPM

- Water Sources:
  - Water pH?
  - Water Flow rate.
  - Distance and elevation
  - Community issue
- Pipe Line Sizes:
  - 2 lines of 8”ODx6”ID HDPE pipe OR
  - 2 line of steel pipe
- Water Pond Storage
  - 8000 – 10000 m3
  - If no water source -> bigger water pons -> up 100.000 m3







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General Preparation Before Drilling

WATER SUPPLY – 1400 GPM

WATER PIPE LINE: Use Galvanize or Steel Pipe Line instead of HDPE:

- Less pressure drop due to friction
- Higher Burst Pressure capacity -> allow to utilize maximum water pump **FLOW RATE** and **PRESSURE capacity**.
  - **Water Pump: 600 psi**
  - **Steel Pipe: ±1000 psi**
  - **HDPE Pipe: 300 psi.**
- Less water pump requirement -> Less Personnel requirement -> Less Community issue
- Easy to repair if any pipe leak






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Well Design Optimization

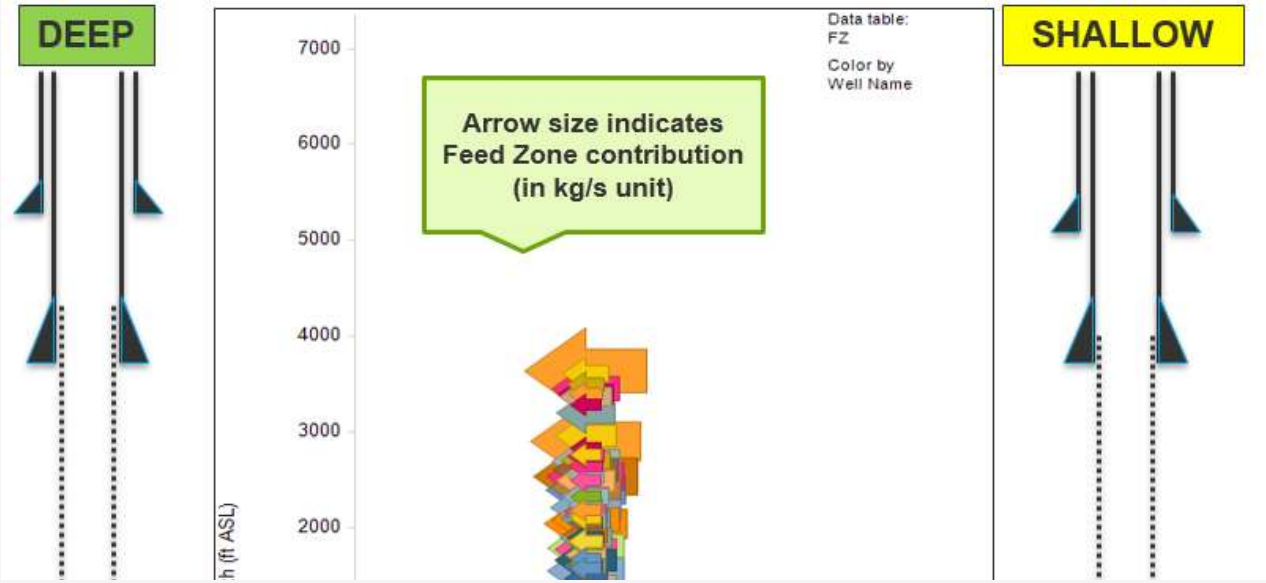
1. Optimization of Total Depth (TD) for Wells Based on Feed Zone Data
2. Selection of Design and Size for Geothermal Wells
3. Casing Specifications and Procurement Strategy
4. Efficiency in Wellhead Installation
5. Selection of Latest Drilling Bit Technology
6. Optimum Design of Bottom Hole Assembly (BHA)

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Well Design Optimization

1 Optimization of Total Depth (TD) for Wells Based on Feed Zone Data

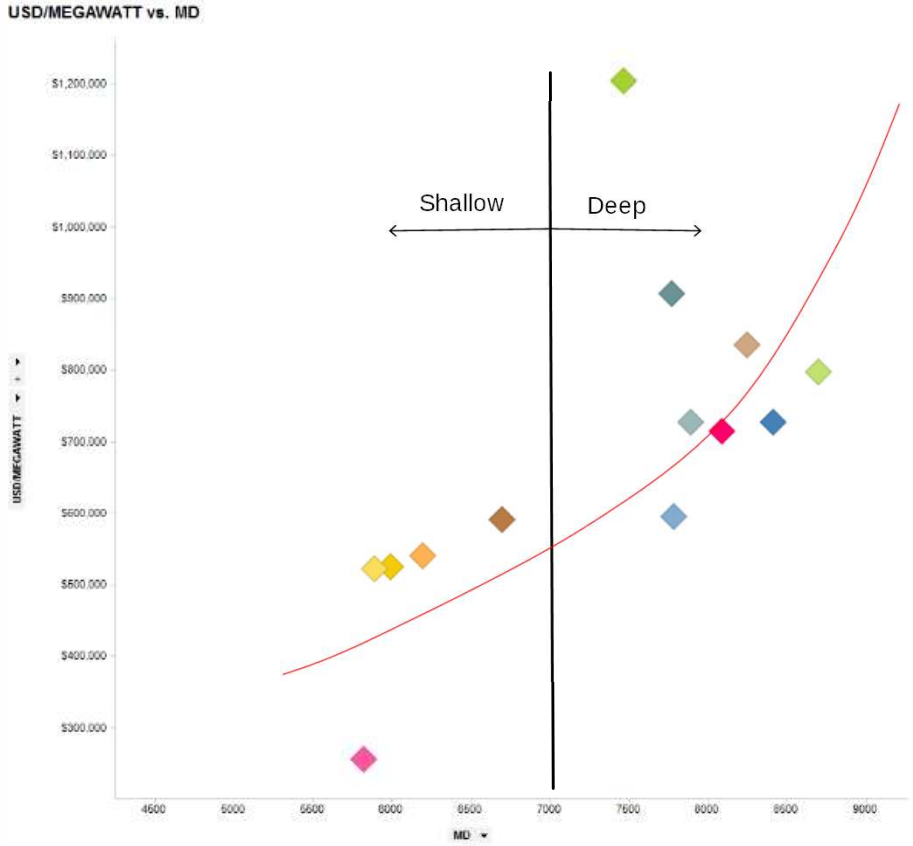
- DEEPER Total Depth -> COSTLY
- Optimize well TD refer to Feed Zone study



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Well Design Optimization

Well Cost Per Megawatt

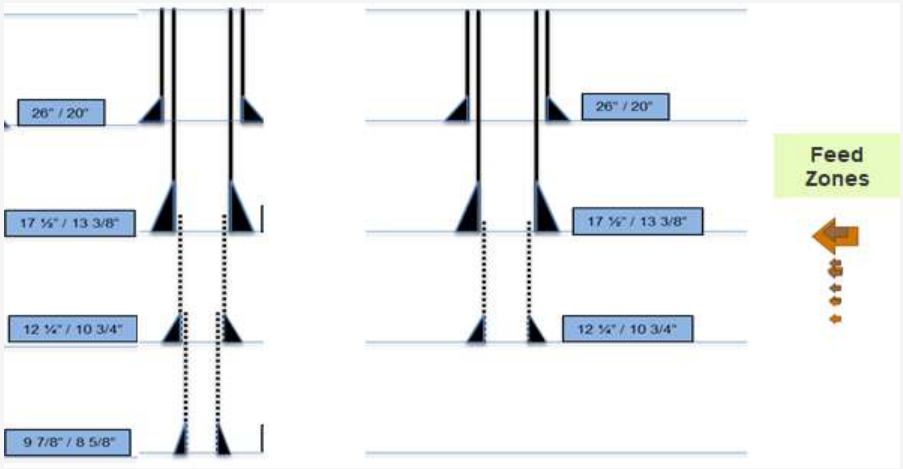


Hence, Cost/MW increased with depth (more drilling cost with almost no increase in production gain).

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Pad XX Case Study (ABC 28 & ABC 30)

Well Cost Per Megawatt





Drilling shallow well at Pad XX will reduce well cost by $700k per well (13% saving) and reduce the Cost/MW to $578 k/MW **(13% reduction).**

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Well Design Optimization

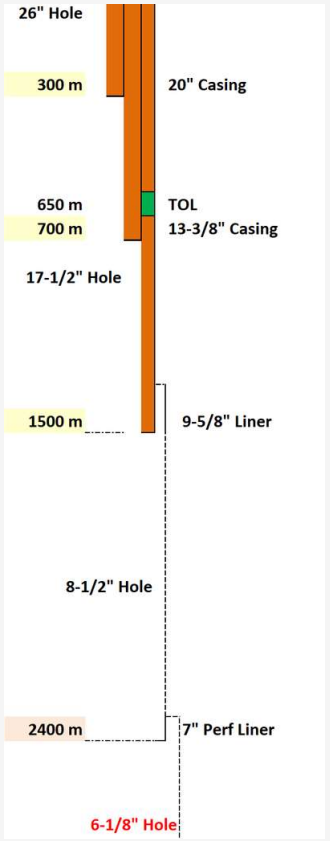
2 Selection of Design and Size for Geothermal Wells

- Big Bore Hole
- Standard Hole

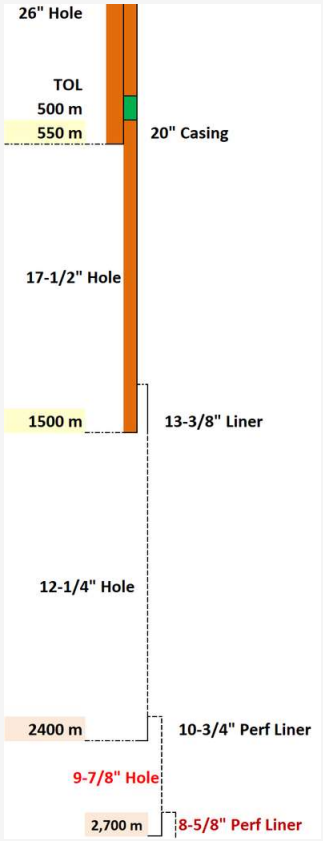
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Well Design Optimization

Well Design: STANDARD Hole OR BIG BORE Hole



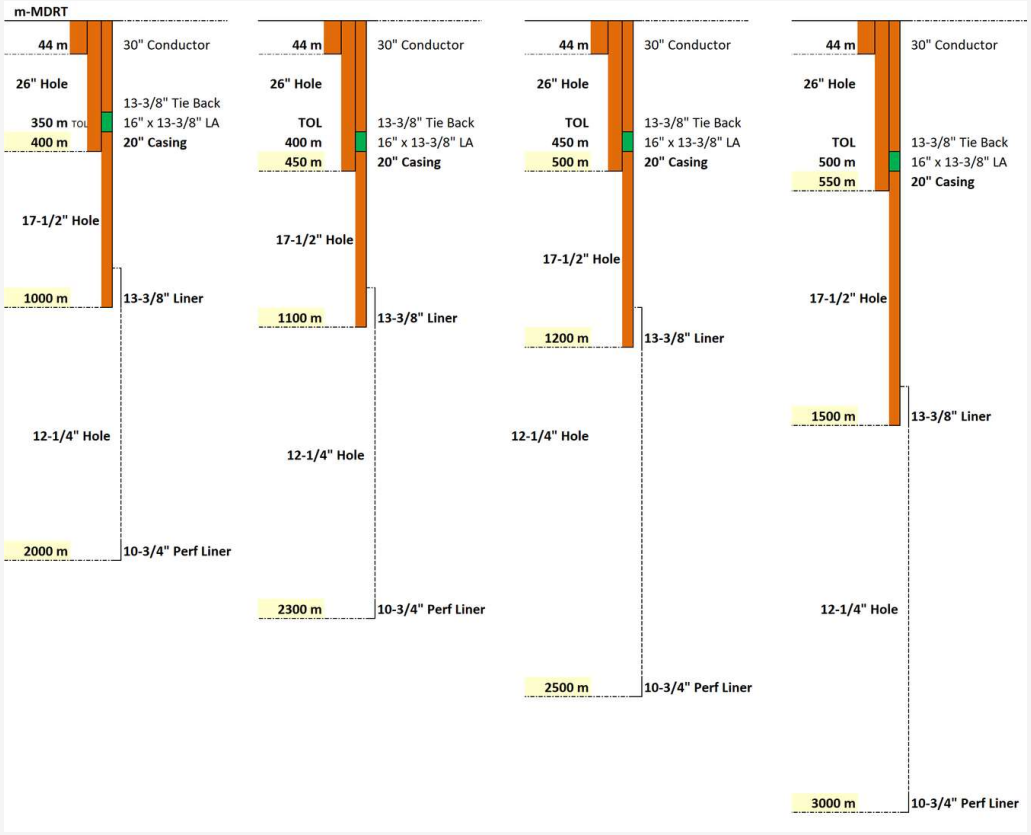
Standard Hole	Comparison Description	Big Bore Hole
ABOVE TOR (Top Of Reservoir)		
4	Number of casing size & Wellhead	3
4	BOP and Wellhead installation, Cementing Job, Dumping Cement, WOC	3
4	Minimum BHA Size and Trip	3
4	No of Bit Size	3
BELOW TOR (Top Of Reservoir)		
1 up to 2	Number of casing size	1 up to 3
8-1/2 & 6-1/8	Hole Size, in	12-1/4", 9-7/8", 7-7/8"
5" and 3-1/2"	Drill Pipe Size	5" only
6-1/2" & 4-3/4"	MM and MWD	8" and 6-1/2"
Less	Production Flow	Bigger



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Well Design Optimization

Ideal Well Design: Big Bore Casing Schematic, Dual String

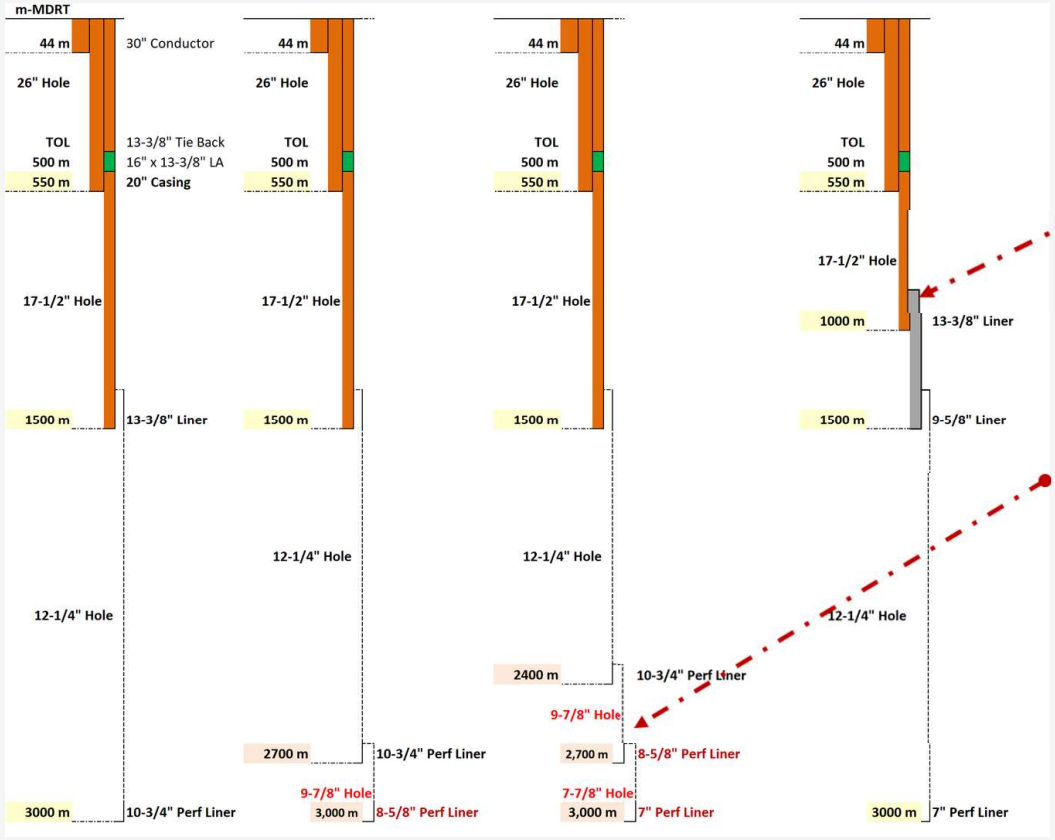


- 20” casing depth (400 – 550 m-MD) depend on 13-3/8” production casing depth.
- 13-3/8” production setting depth will be advised by Sub-surface team.
- 13-3/8” shoe depth as deep as possible to reduce long 12-1/4” hole section.

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Well Design Optimization

Well Design: INTERMEDIATE – PRODUCTION Hole Section with Contingency Plan



INTERMEDIATE Hole: If 13-3/8” Liner is un-able set at top of reservoir -> Contingency Plan:

- Drill 12-1/4” to top of reservoir and set 9-5/8” Liner and cemented.

PRODUCTION Hole: If any problem during drilling, and need set perforated liner earlier -> Contingency Plan:

- Drill 12-1/4” hole as deep as possible and set 10-3/4” perforated casing.
- If not reach TD -> Drill 9-7/8” hole as deep as possible and set 8-5/8” perforated casing.
- If not reach TD -> Drill 7-7/8” hole as deep as possible and set 7” perforated casing.

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Well Design Optimization

3 Casing Specifications and Procurement Strategy

- Seamless or Non-Seamless (Spiral, ERW, SAW)
- Casing Connection Type
- Comparison between Long string vs Tie Back Design
- Casing Liner Adapter Design
- Casing Procurement Strategy

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Casing Specification

Manufacture	Grade	Lenght	Weight	Connection Type
Non Seamless: • Spiral • ERW • SAW Seamless	• X-52 • X-56 • K-55 • L-80 • N-80 • C-90 • C-95 • P110 • Cr13	• R-2 • R-3	• 310 ppf • 133 ppf • 129 ppf • 72 ppf • 68 ppf • 53.5 ppf • 47 ppf • etc	• Bevel End • BTC, STC, LTC • SCC (Special Clearance Coupling) • Semi Premium • Quick Connector • Premium

MANUFACTURED BY ROLLING METAL PLATE **(30” & 20”):**

- ERW: Electric Resistance Welded
- SAW: Submerge Arc Welding
- LSAW: Longitudinal Submerge Arc Welding

SEAMLESS **(13-3/8”, 10-3/4”, 8-5/8” and 7”)**:

- Manufactured by extruding the metal to the desired length.
- 20-30% higher price than ERW casing



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SEAMLESS VS ERW CASING

Seamless Casing	Rolled Metal Plate
Advantage: • Good Casing Integrity • Gas Well • Suitable for high pressure	Advantage: • Cheaper • Shorter Delivery Time
Dis-Advantage: • Expensive • Longer Delivery Time	Dis-Advantage: • Fair Casing Integrity • Not suitable for high pressure • Poor CO2 and H2S resistance

- 13-3/8” Production Casing
- 10-3/4”, 8-5/8”, 7” Perforated Liner

- **30” Conductor**
- **20” Surface Casing**



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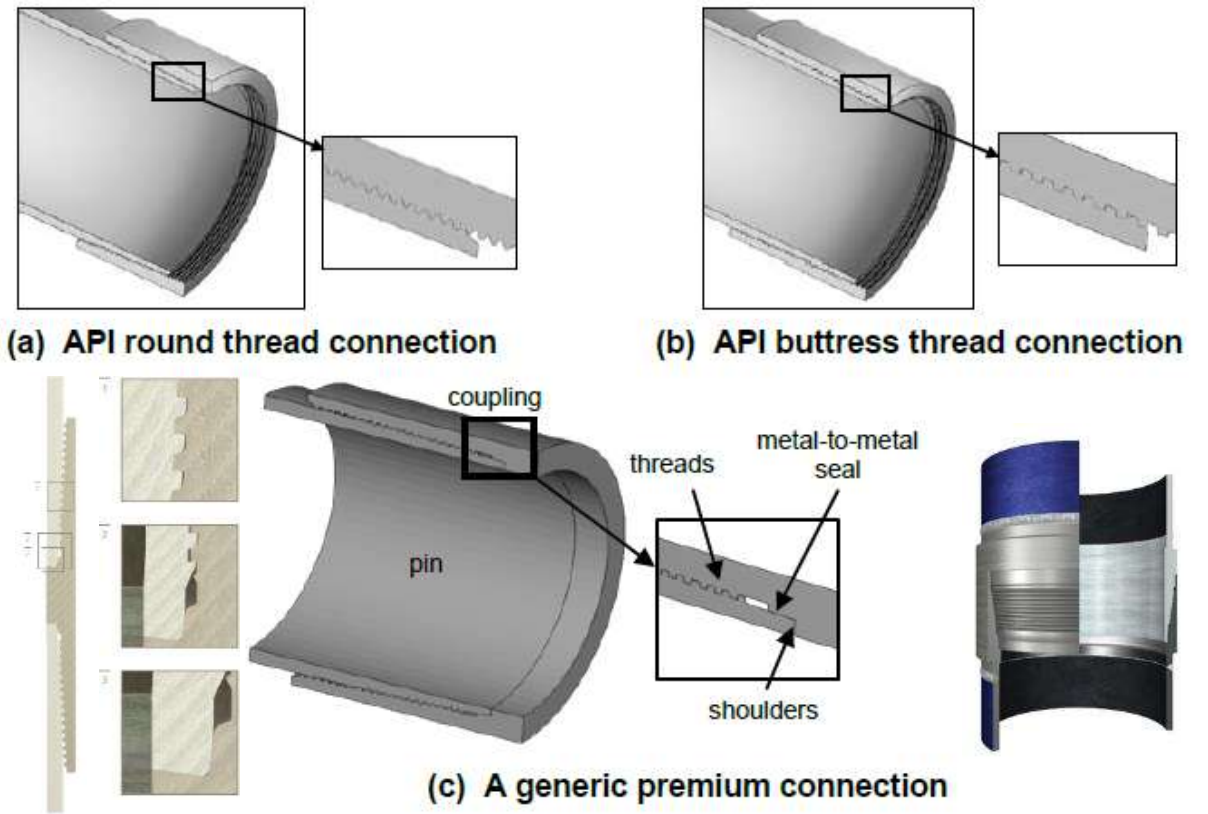
Tipikal spesifikasi casing yang umum dipakai sumur Big Bore

Section	Size	PPF	Grade	PSL	Range	Connection	Type
Conductor	30"	0.75" WT	X56	1	R3	Beveled End without Connection	Spiral SAW ERW
Surface Casing	20"	133	X56 or K55	1	R3	T&C Semi Premium with 3 TPI (Big Thread)	ERW SAW
	18-5/8"	87.5					
Production Casing	13-3/8"	68 or 72	L80	2 or higher	R3	T&C Premium qualified to TWCCCEP	Seamless
Perforated Liner	10-3/4"	40.5	K55 or L80	1 or 2	R3	BTC Special Clearance or Semi Premium Special Clearance	Seamless
	8-5/8"	24 or 32	K55	1 or 2	R3		
	7"	23 or 26 or 29	K55	1 or 2	R3		

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CASING CONNECTION



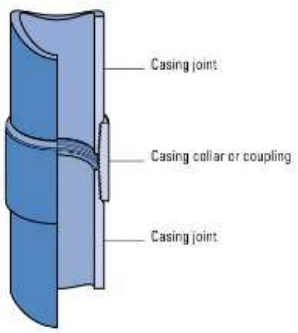
- 30” Casing, ERW or Spiral SAW, 0.75” WT, R-3, Plain End
- 20” Casing, ERW or SAW, 133 ppf, K-55 (X-56), R-3, 3TPI CONNECTOR
- 13-3/8” Casing, Seamless, 68-72 ppf, L-80, R-3, PREMIUM
- 10-3/4” Perforated Casing, 40 ppf, K-55/L-80, R-3, SCC BTC
- 8-5/8” Perforated Casing, 28 ppf, K-55/L-80, R-3, SCC BTC
- 7” Perforated Casing, 23 ppf, K-55/L-80, R-3, SCC BTC

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BUTRESS vs QUICK Connection Thread

Pros - Butress	Cons - Butress
• Cheap price casing connection • MU to tri-angle mark -> no torque turn monitoring • API	• Not for gas well • Low Temperature and pressure • Limit DLS • **Longer time to MU (i.e. 20" casing -> 20-30 minutes per joint**







- 3 TPI (Thread Per Inch) Connection
- < 5 minutes per joint
- 15 minutes rig time saving.
- Rig Cost = USD 3000 /hrs
- Cost saving per joint: USD 750.
  - Additional Cost for 3 TPI connection: USD 200
  - Net cost saving: USD 550 /jt
  - 45 joint per well: **USD 24750**

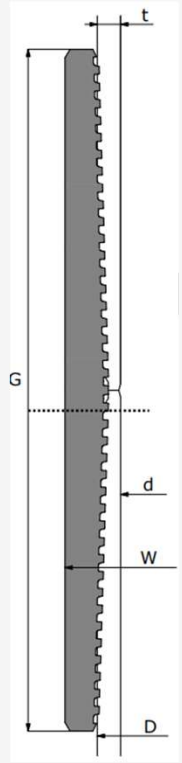
COST SAVINGS

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20” Casing Thread List and Comparison

Thread Name	Number of Threads	Remark
BTC, Geoconn, GPButt, TMK UP CWB	5 TPI (Thread Per Inch)	BTC Compatible BTC Compatible BTC Compatible
GP3P	3TPI	
Big Omega, Big Omega E	3TPI	
Tenaris ER	3TPI	
Hunting Seal Lock Boss	3TPI	
TMK UP Magna	3TPI	



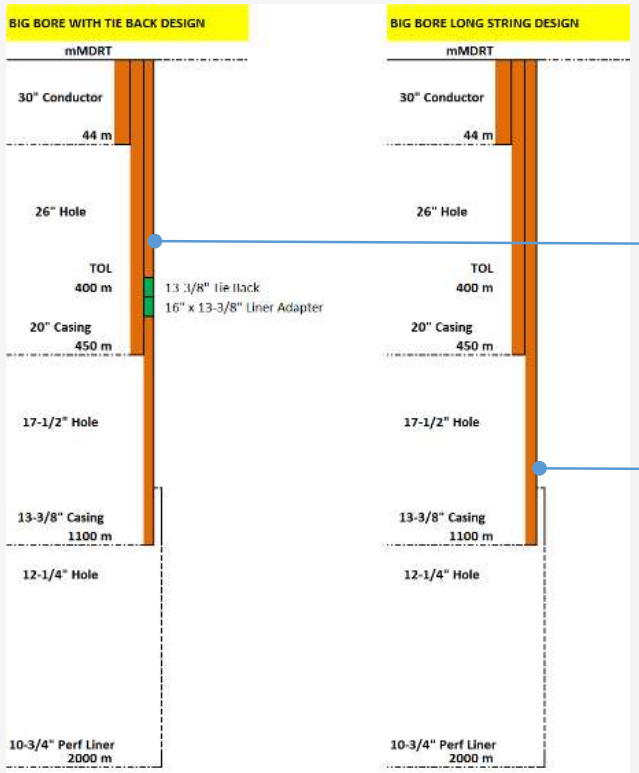
Geoconn

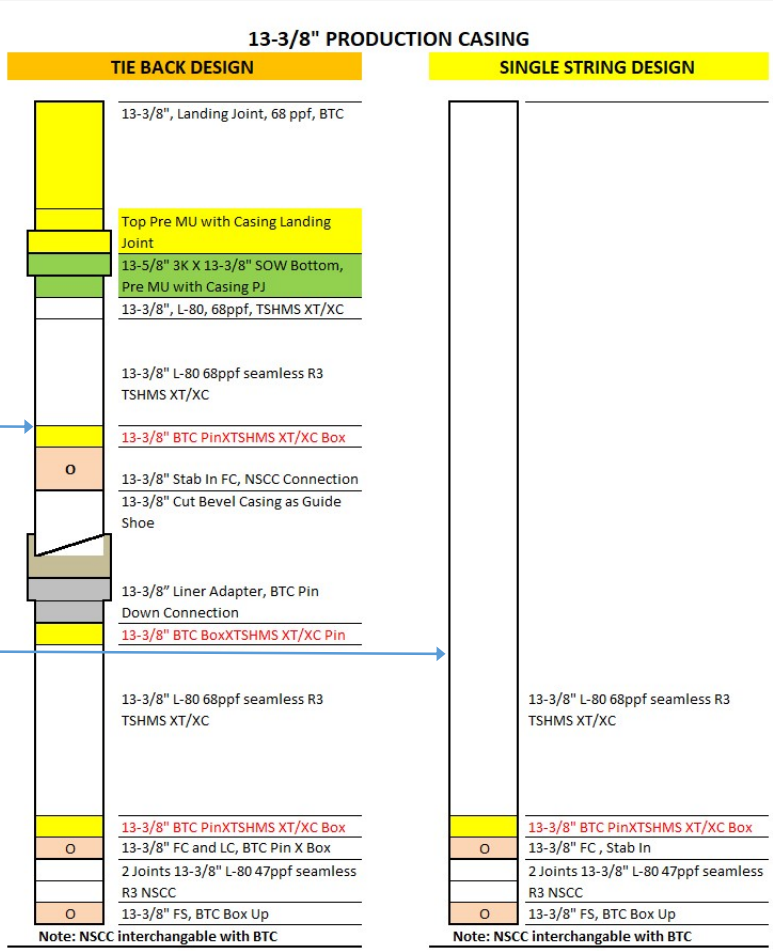
13-3/8” Casing Design: Tie Back instead of Long String

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Well Design Optimization

Casing Schematic





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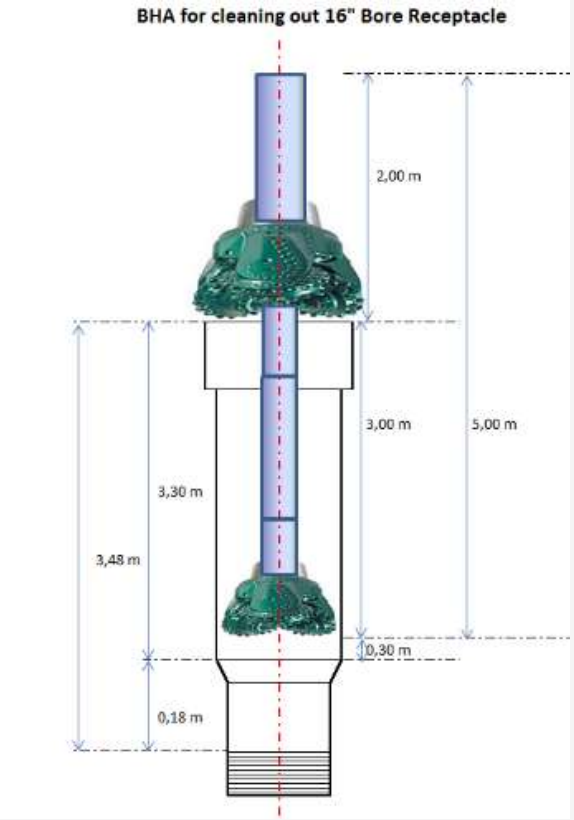
Running Operation

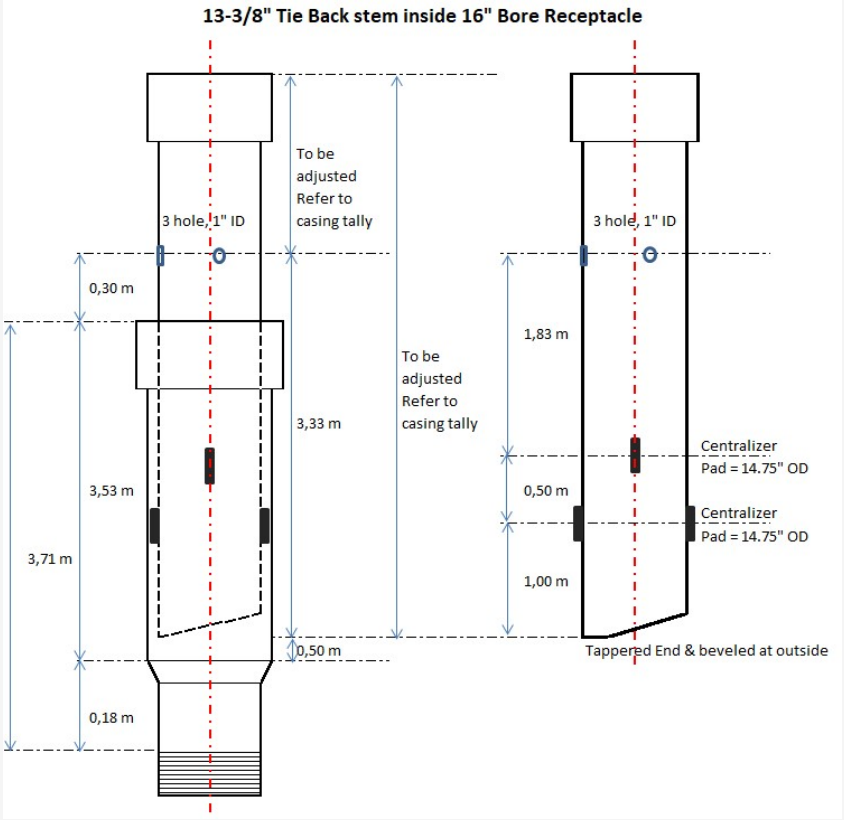
<table>
<tr><th>Tie Back Design</th><th>Long String Design</th></tr>
<tr><td>Liner Run:
MU and run 13-3/8” Liner (TOL – 50 m above 20” FS)
MU 20” X 13-3/8” Liner Adapter
Run with running string (HWDP + DP)
Cementing
POOH Running String
Wait On Cement
Run Clean Out BHA
Remedial Cement if required
Pressure Liner Lap</td><td rowspan="2">Single Run:
MU and run 13-3/8” Casing to surface
Run with Inner string (HWDP + DP)
Cementing
Remedial Cement if required
POOH Inner String
Wait On Cement
Remedial Cement if required
Lift BOP and Cut 13-3/8” Casing and Laid Down
ND BOP
Cut 20” casing and Final Cut 13-3/8” casing.
Risk of heat expose 13-3/8” casing due to cutting torch
Weld section A to 13-3/8” casing
More rig Time (No pre-welded)
NU Master Valve
NU BOP</td></tr>
<tr><td>Tie Back Run:
ND BOP
Cut 20” casing
MU and run 13-3/8” Tie Back Casing to surface
MU Pre-Welded Wellhead and Land on C Plate
Run inner string and stab into FC
Cementing. NO Wait On Cement (Casing has been landed on C Plate prior cementing job).
NU Master Valve
NU BOP</td></tr>
</table>

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Running Operation – 13-3/8” Liner & Tie Back

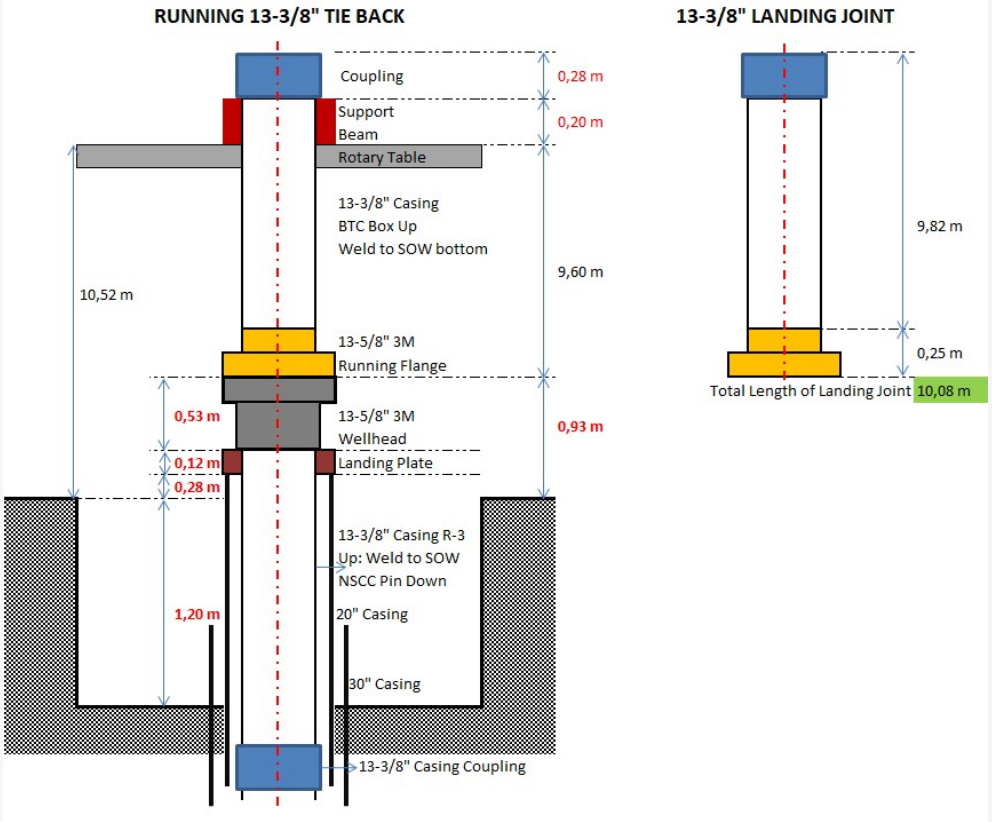




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Running Operation – 13-3/8” Tie Back Casing



- The 13-3/8” Tie Back Casing will be run with PRE-WELDED Wellhead for rig time saving.
- PRE-WELDED Wellhead will be landed on C Plate on 20” casing before cementing -> No Wait On Cement is required, we could NU BOP direct after cementing Job.
- The 13-3/8” Tie Back Casing will be cemented with **heavy cement slurry 16.2 ppg**, almost double than water gradient.
  - Heavy Cement slurry will easier to remove water in annulus comparing lighter cement used in long string cementing (13.5 ppg).
  - Easier to get Free Water in 16.2 ppg Cement slurry comparing lighter cement used in long string cementing (13.5 ppg).
  - Much less risk of water trap along 20” X 13-3/8” casing annulus -> No well integrity issue.

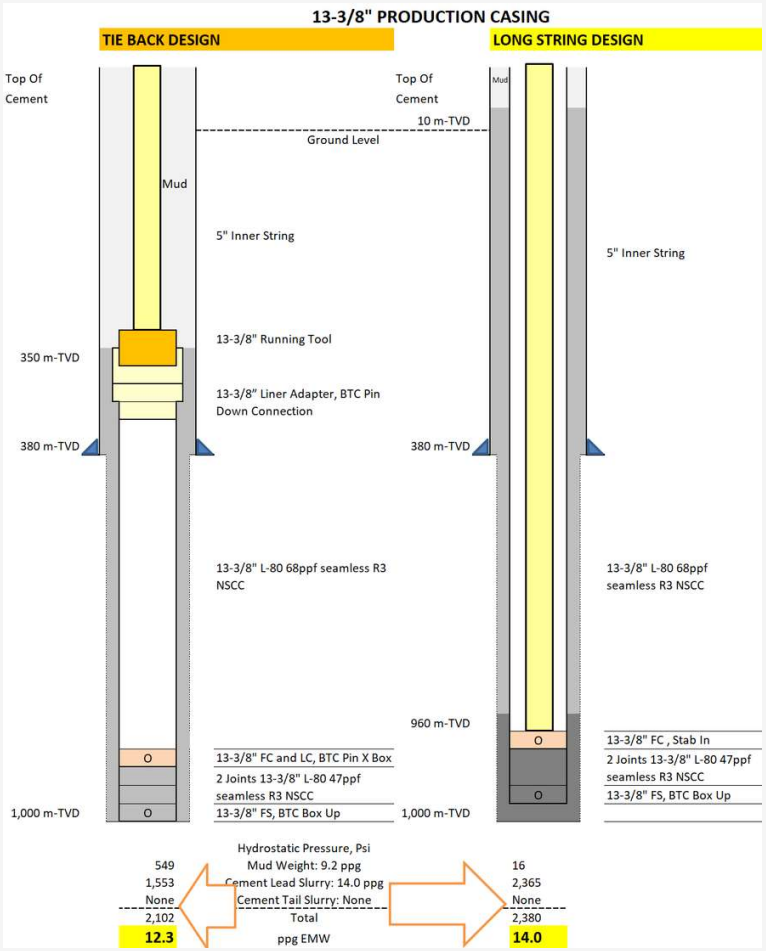
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Risk Of Losses While Cementing

- Tie Back System: Hydrostatic pressure during 13-3/8” Liner Casing cementing is **278 psi (1,7 ppg)** less -> Less risk of loss circulation while cementing -> lower probability for remedial cementing job.

- Tie Back System: Single 16.2 ppg Cement Slurry: easier to get Free Water Recipe -> less risk of water trap in cement when hard -> less risk casing collapse when expose to hot temperature when flowing well.

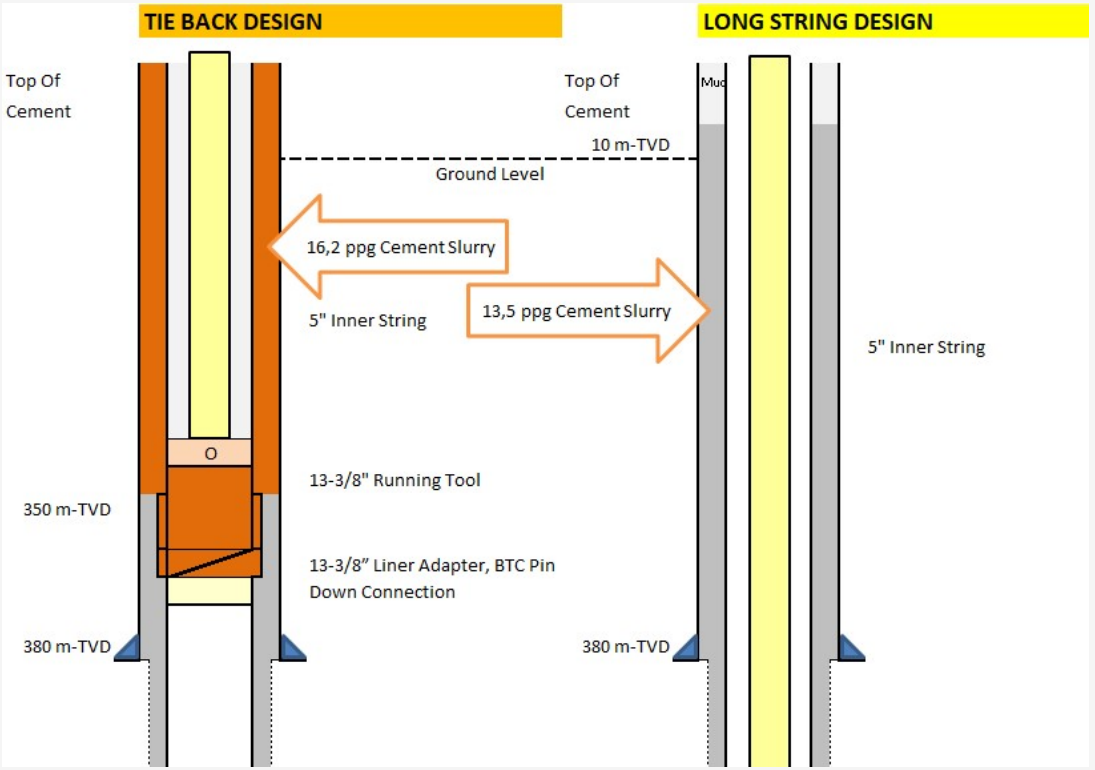


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Cement Quality in 20” X 13-3/8” Case Hole

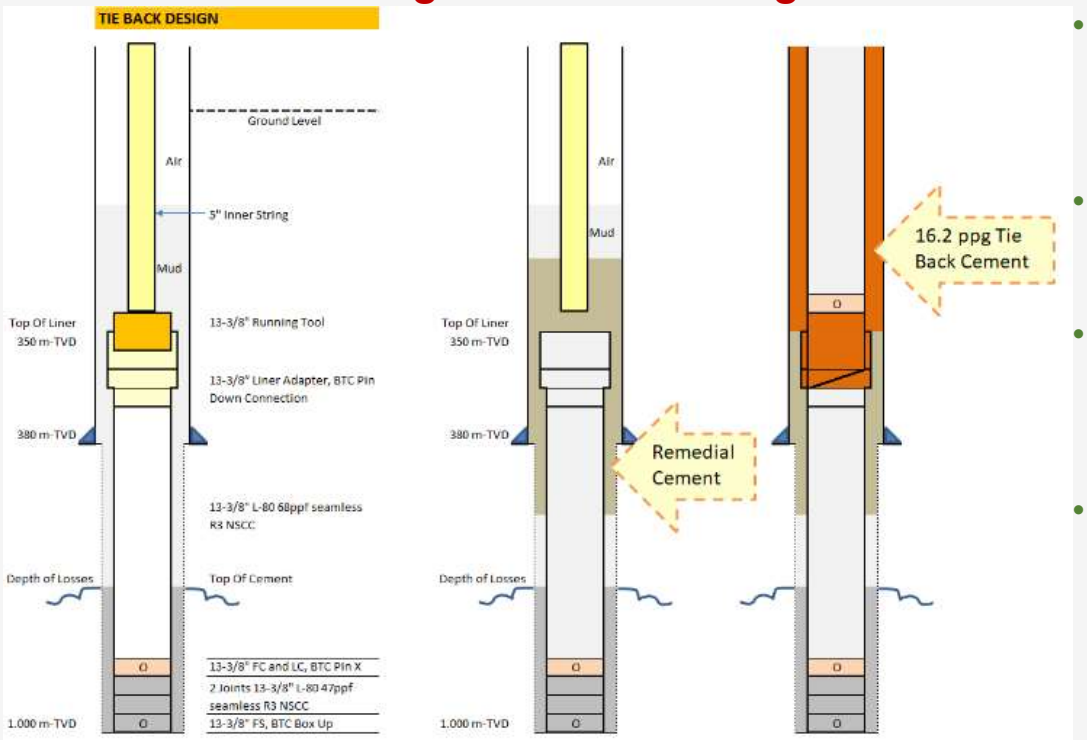
- The 13-3/8” Tie Back Casing will be cemented with **heavy cement slurry 16.2 ppg**, almost double than water gradient.
  - Heavy Cement slurry will easier to remove water in annulus comparing lighter cement used in long string cementing (13.5 ppg).
  - Easier to get Free Water in 16.2 ppg Cement slurry comparing lighter cement used in long string cementing (13.5 ppg).
  - Less risk of water trap along 20” X 13-3/8” casing annulus (short Liner Lap) -> No well integrity issue.
- The 13-3/8” Tie Back casing will cemented after liner lap being pressure tested to 16,5 ppg EMW -> to ensure no losses when pumping 16,2 ppg cement slurry. Until today, no fail on Tie back cementing job.
- Good cementing quality with 16.2 ppg slurry -> better anchoring for 13-3/8” production casing -> less casing and wellhead expansion growth.



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Well Design Optimization

Remedial Cementing – Tie Back Design



- If any losses during cementing, do remedial cementing by dropping or squeezing 50-150 bbls of 15.8 pp cement -> until we could pressure test liner lap to 16.5 ppg EMW.
- It is easier since shorter length for remedial cement -> 30-50 m liner lap and open hole annulus.
- If no injectivity rate:
  - Pressure test liner lap to 16.5 ppg EMW, if leaked, squeeze cement.
  - Run and cement 13-3/8" Tie Back.
- Good Casing and Cementing Integrity.

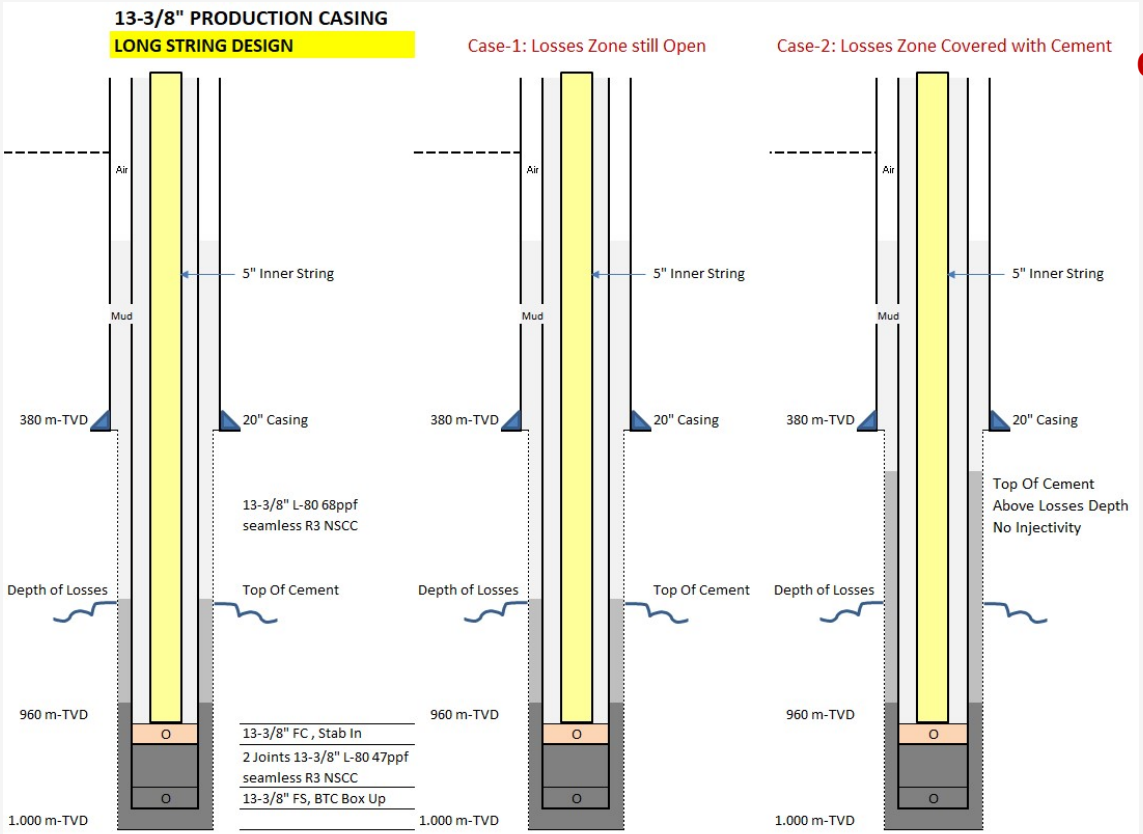
If any Total Losses during remedial cement and fail cured with cement plug:

- **Pump Fast set cement Recipe (Econolite + Spacer + 15.8 ppg Cement**

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Well Design Optimization

Remedial Cementing – Long String Design



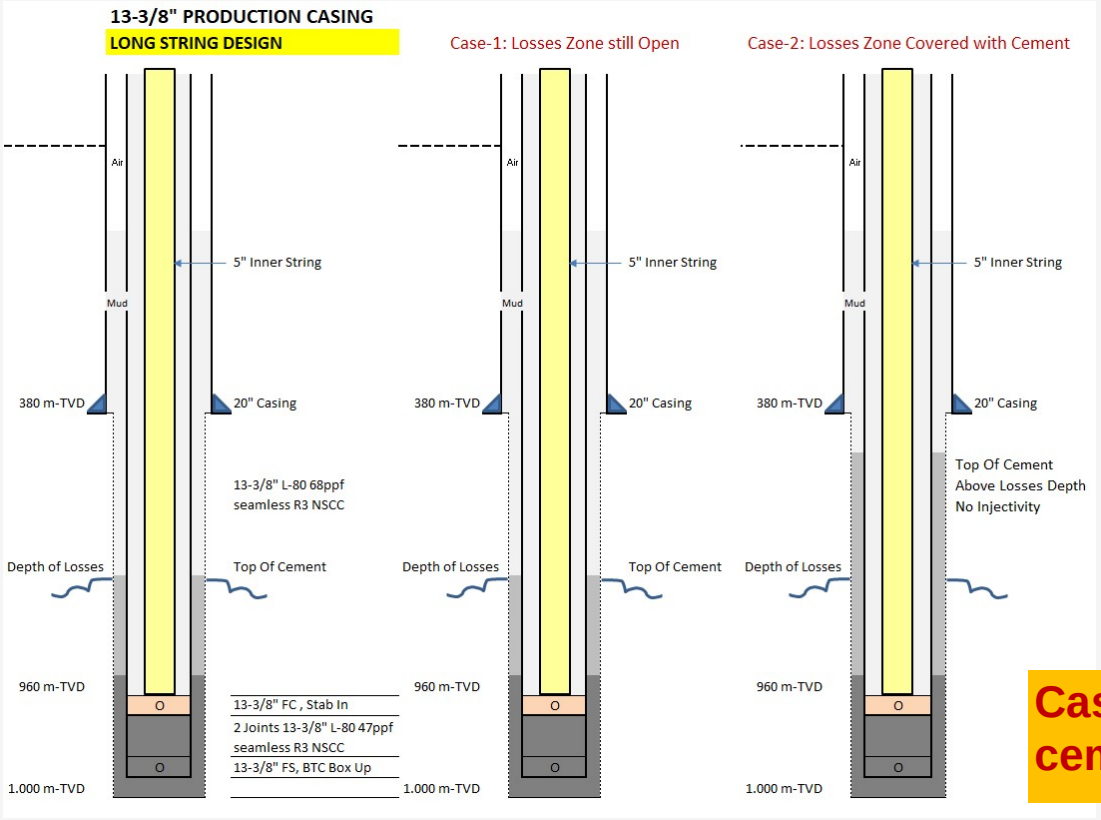
Case-1: Losses Zone still open

- Cement Top Job by pumping cement slurry thru annulus.
- The heavy cement slurry will drop into long annulus section, can not control cement speed -> risk of mixed with water inside annulus -> Risk of free water > nil -> water trap and cause well integrity issue.
- If still losses: Pump Thixotropic cement ahead and follow with remedial cement slurry -> risk of premature set if any stop pumping -> lack of cement quality along case hole section.
- All above points could initiate well integrity issue.

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Well Design Optimization

Remedial Cementing – Long String Design



Case-2: Top Of Cement Above Losses Depth causing No Injectivity:

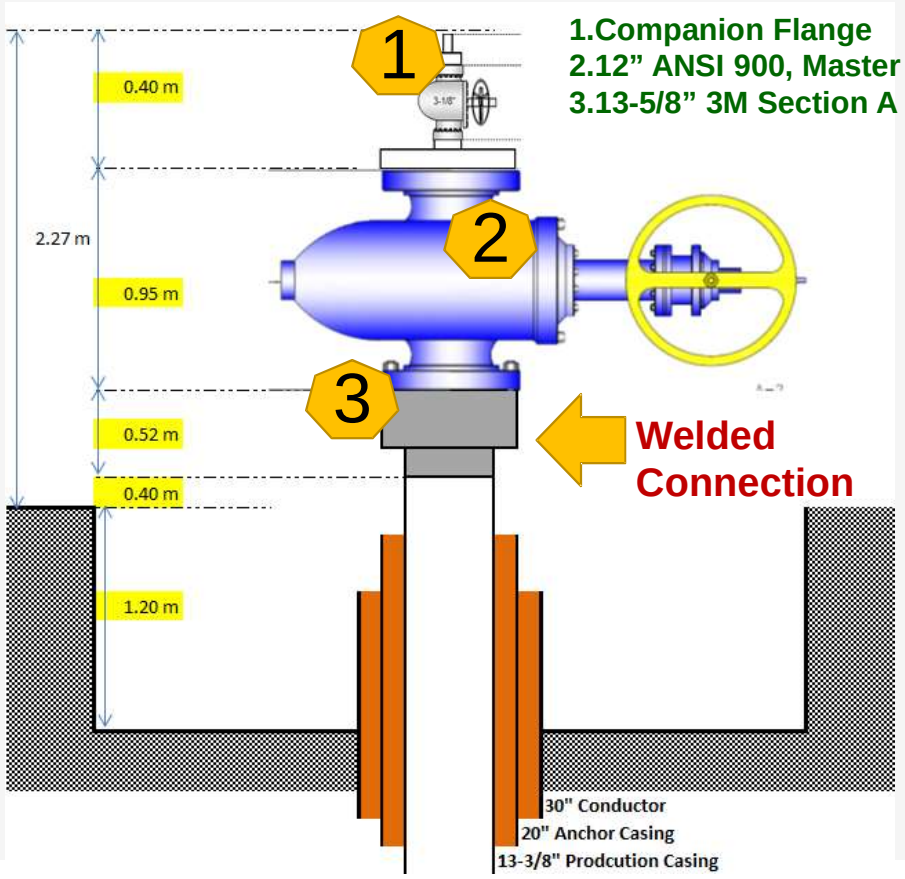
- Unable to pump remedial cement slurry.
- Long Un-cemented 13-3/8” Production Casing
- When flowing during well test or production phase, cause higher casing expansion and cause wellhead growth -> damage on surface pipe line connection.
- Again the above points could initiate well integrity issue.

Case-1 and 2 are more difficult Remedial cement job, may need longer rig time.

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Well Design Optimization

Wellhead & MV Installation Comparison



1.Companion Flange
2.12” ANSI 900, Master Valve
3.13-5/8” 3M Section A Wellhead

13-5/8” 3M Section A Wellhead Installation:

a.13-3/8” Tie Back Design:

- Pre-welded in horizontal position
- 32-40 hours
- Off line job -> no use Rig time

b.13-3/8” Long String Design

- To be welded in Vertical position, after cementing job and final cut 13-3/8” casing
- Rig Time: 32-40 hours
- Can not do Off Line -> Use Rig time
- Option to expedite welding job -> by pressure testing in hot condition -> risk of lack on welding quality

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Rig Time Comparison

TIE BACK SYSTEM		T-01	T-03	
No	Activity	Hours	Hours	
1	RU Casing Handling	1,5	2,5	
2	MU Shoe Track	2,5	3,0	
3	MU and Run Casing	9,5	9,0	
4	MU Liner Adapter	1,5	1,5	
5	Run on DP	3,5	3,5	
6	Cementing	11,5	9,0	
7	Release and POOH DP	3,0	4,0	
8	RIG Clean Out BHA	12,5	13,0	
9	Pressure test	0,5	0,5	
10	POOH	3,0	1,5	
11	ND BOP	5,0	6,0	
12	RU Casing Handling	3,0	3,5	
13	MU and Run Tie Back Casing	6,5	6,0	
14	Space out, Run Pre-weld wellhead	4,0	1,5	
15	Run Inner string	3,0	3,5	
16	Cementing	3,0	2,5	
17	POOH DP	1,5	2,5	
18	NU Master Valve and BOP	15,5	12,0	
19	LD and PU Pipe	11,5	0,0	
20	MU and RIH 12-1/4" BHA	9,0	9,5	
21	Pressure test BOP Rams	1,0	0,5	
22	Drill out FC	4,0	4,5	
23	MU and RIH BHA	5,0	5,5	
24	Drill out Shoe Track + New Formation	8,5	5,5	
25	FIT	1,0	1,0	
		130,5	111,5	Hours
		5,4	4,6	Days

ESTIMATED RIG TIME LONG STRING SYSTEM			
No	Activity	Hours	
1	RU Casing Handling	1,5	
2	MU Shoe Track	2,5	
3	MU and Run Casing	15,0	
4	RU spider slip and elevator	1,0	
5	Run DP Inner String	4,0	
6	Cementing	8,0	
7	POOH DP	2,0	
8	WOC	6,0	
9	Lift BOP	1,0	
10	Rough Cut 13-3/8" casing, LD	1,0	
11	ND BOP	5,5	
12	Final Cut 13-3/8" Casing	1,0	
13	Weld Wellhead	32,0	
14	NU Master Valve and BOP	14,0	
15	LD and PU Pipe	5,5	
16	MU and RIH BHA	14,0	
17	Pressure test BOP Rams	1,0	
18	Drill out Shoe Track + New Formation	8,5	
19	FIT	1,0	
		124,5	Hours
		5,2	Days
	RIG TIME SAVING	-6,0	13,0 Hours

Rig Time Comparison – No NPT and No Remedial Cement:

- **Comparing SMP T-01: Long String could create 6 hours rig time Saving.**
- **Comparing SMP T-02: Long String could be 13 hours Longer rig time.**

Risk of longer rig time IF run Long string method:

- **Problem when weld 13-5/8” SOW Wellhead to 13-3/8” L-80 Casing.**
- **Difficulties for Remedial Cementing Job.**

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Additional Cost Comparison

13-3/8” Tie Back System	13-3/8” Long String System
Run 13-3/8” Casing into two section -> Less weight of running casing. It is sufficient to use 150 T Single Casing Elevator. This Equipment is part in the contract, no additional cost.	Run 13-3/8” casing in single run. It is still within 150 T Single Casing Elevator without any Safety factor. It is recommend to run 13-3/8” Long String with 350 T Spider Elevator. This equipment was not in the casing handling contract. Additional cost: • Equipment Rental = USD 18000 per well • Mob/Demobilization = USD 6000 per call
Need 20” X 13-3/8” Liner Adapter, additional Cost: • 20” X 13-3/8” Liner Adapter: USD 15000 / Set • Running Tool: USD 5000 / Well • 17-1/2” Hole Opener: USD 21000 for 3 well • 14-3/4” Mill Tooth Bit: USD 12000 for 3 well	No 20” X 13-3/8” Liner Adapter
• Total Additional Cost: USD 45,500 / Well	Total Additional Cost: USD 18000 + 6000/3 = USD 20000



- Overall less risk
- Much better Cement quality along case hole



Cost Different between Tie Back vs Long String:
- USD 25,500
- NOT Significant

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Conclusion

Strongly NOT Recommend to run Long String Design for Geothermal Well:

- Cementing 13-3/8” Casing is the most critical job to be main isolation and anchoring system. Remedial cement in Long string is more complicated and difficult comparing Tie back design.
- No Rig Time saving expected from Long String Method, since we must weld 13-5/8” SOW wellhead on line (Rig Time). Risk longer rig time if any NPT during welding the wellhead.
- Higher risk of longer rig time due to difficulties to do Remedial Cementing on 13-3/8” Long Production Casing string.
- In 13-3/8” Tie Back System where the casing is divided into 2 section, It could make better items of casing stress distribution due to casing expansion during well flowing.
- It was experienced that Long string Design contributed to the Lack of well integrity.
- Most Geothermal Operators in Indonesia have abandoned to use Long string method due to well integrity concerns.

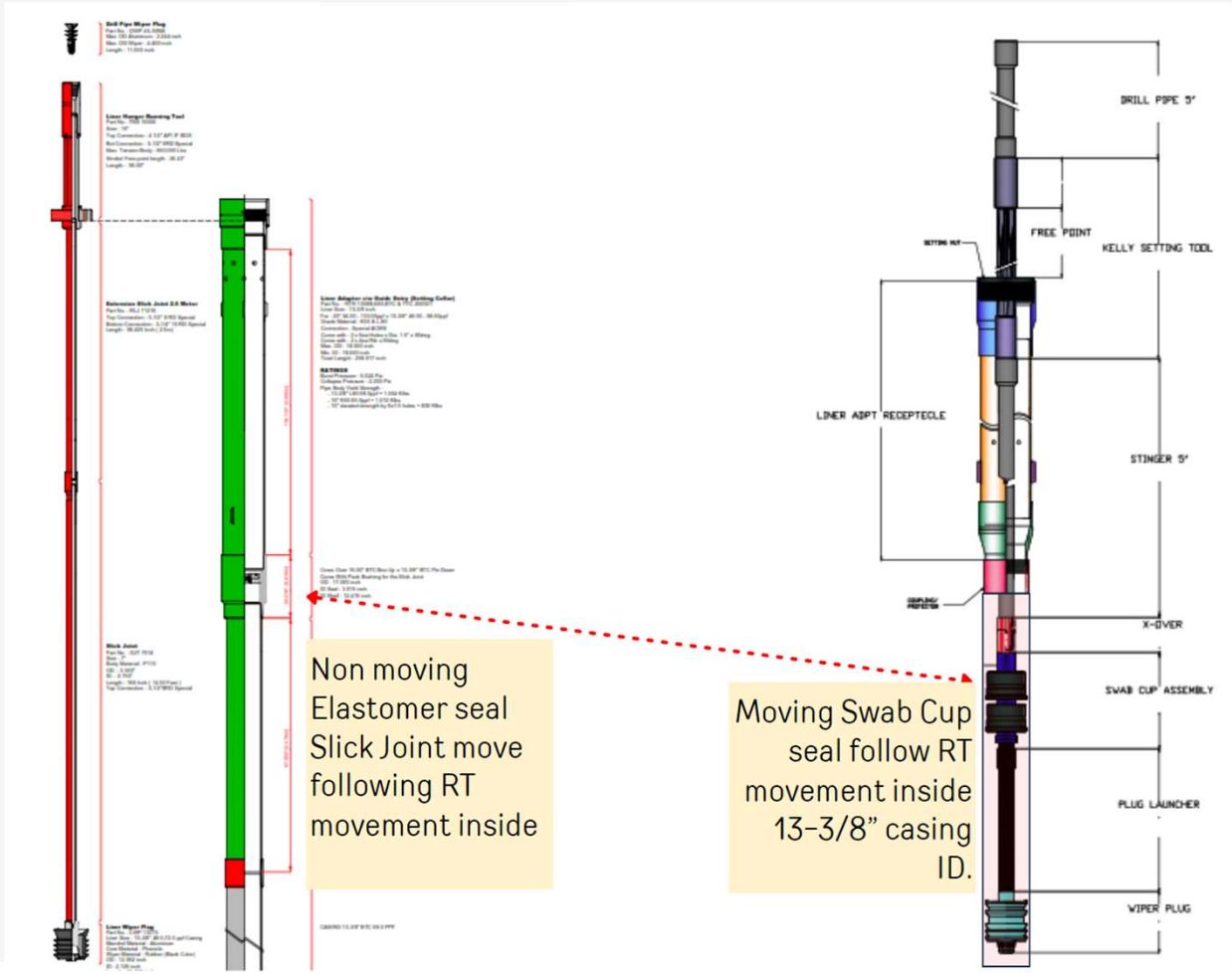
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20” X 13-3/8” Liner Adapter and Guide Shoe Design

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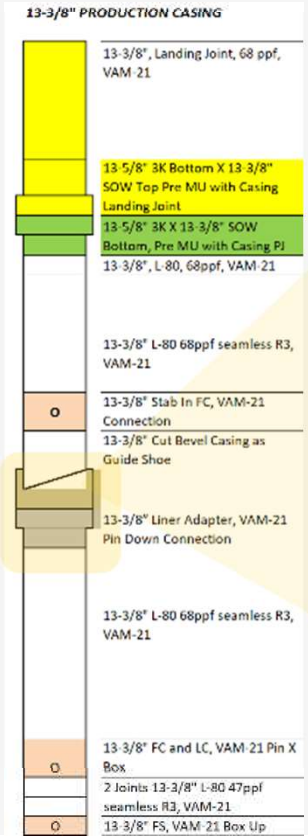
20” X 13-3/8” Liner Adapter Design

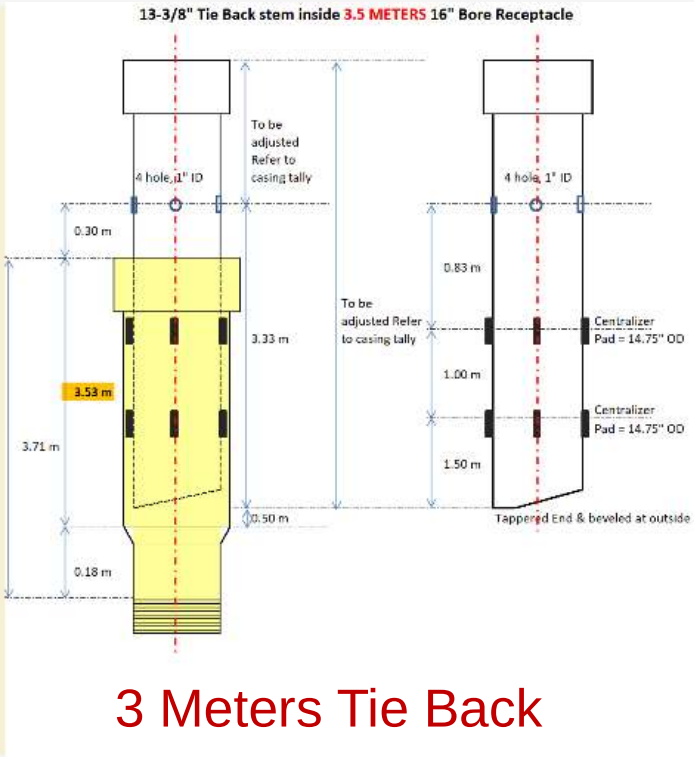


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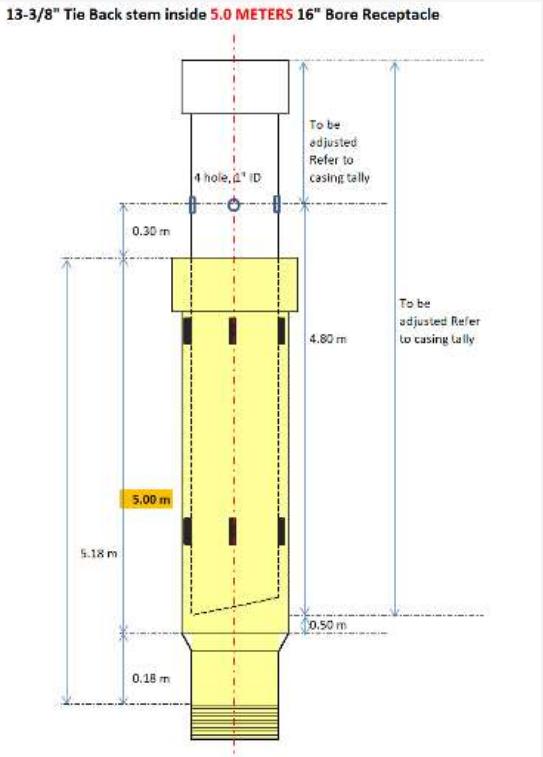
Well Design Optimization

20” X 13-3/8” Liner Adapter Design – Tie Back Receptacle Length





3 Meters Tie Back

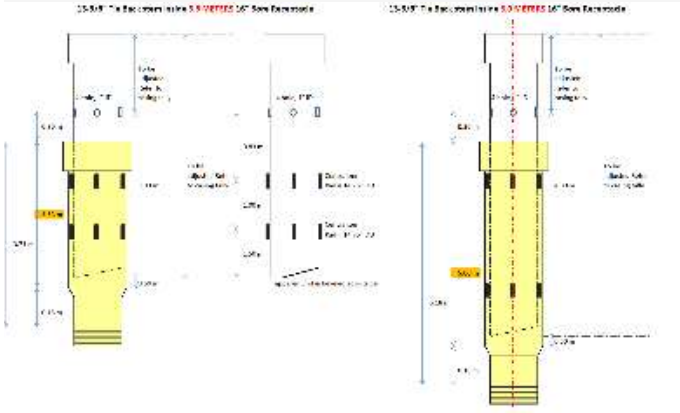


5 Meters Tie Back

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20” X 13-3/8” Liner Adapter Design – Bore Receptacle Lenght



Bore Receptacle Length – 3 Meters	Bore Receptacle Length – 5 Meters
Run 13-3/8” Tie Back Liner MU 85% make torque last 2 joints Tag to bottom Space out POOH and Laid down last 2 joints MU Pre-welded Wellhead Land on C plate LD Landing Joint RIH Inner string Cementing	Run 13-3/8” Tie Back Liner **Rig time = 4 hours** MU Pre-welded Wellhead Land on C plate LD Landing Joint RIH Inner string Cementing
	Rig Time Cost Saving = USD 11.7 K Cost for longer BR = USD 2 K **Net Cost Saving = USD 9.7 K**

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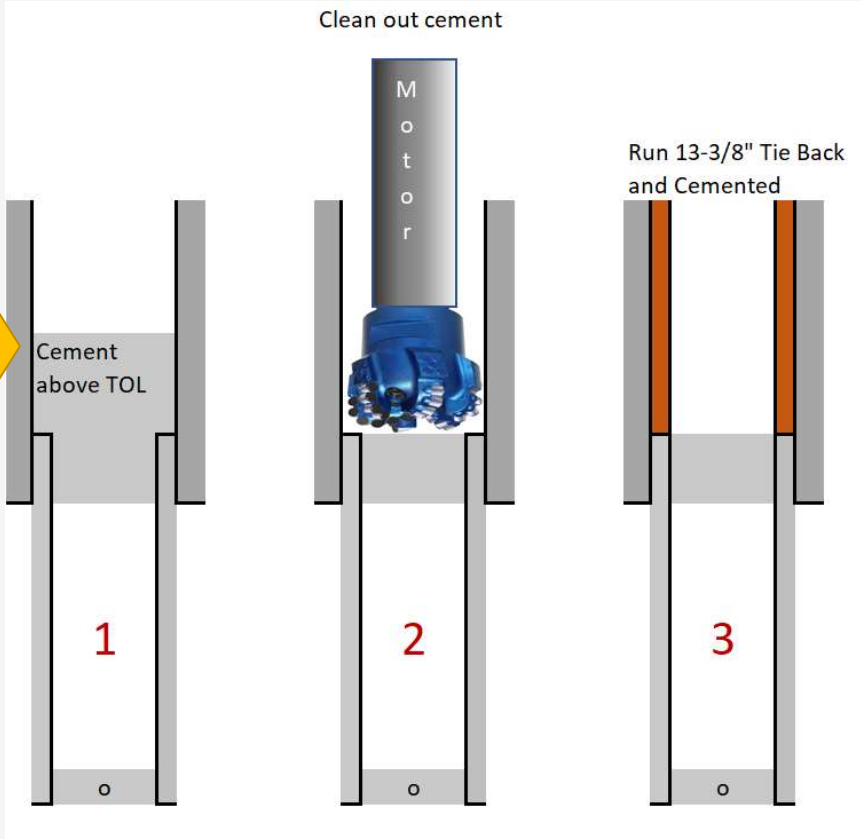
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Tie Back Cementing Bond Improvement



- During primary cementing of 13-3/8” Liner, any possibility cement exist above TOL (return during cementing or drop cement above TOL). It’s must be cleaned out before running 13-3/8” Tie Back casing.

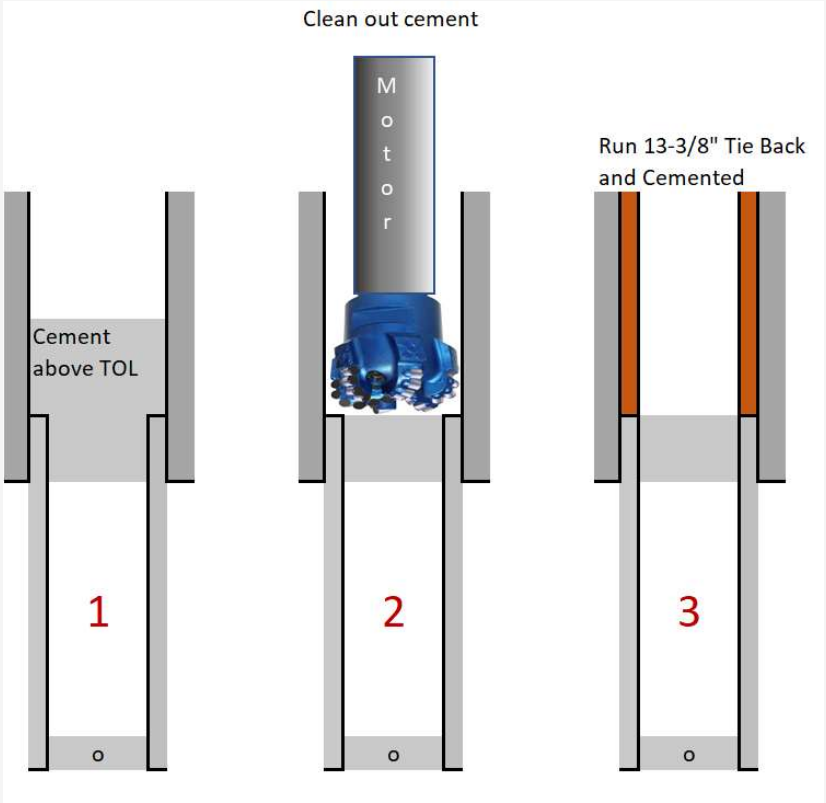
- **Concerns:** Clean out using 17-1/2” bit, it could be un-able to remove all cement and may remain cement layer along 20” casing ID (0.615” - 1.23” thickness).



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Tie Back Cementing Bond Improvement



As mitigation the concerns: Clean out using 17-1/2” bit, it could be un-able to remove all cement and may remain cement layer along 20” casing ID (0.615” - 1.23” thickness):

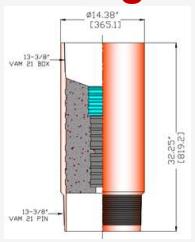
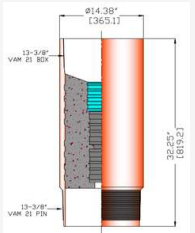
- 17-1/2” ROTARY Clean Out BHA (Straight BHA) may not able to clean out hard cement.
- Option-1: **Run 20” casing scraper** to remove the remaining cement seat, however the availability of 20” casing scraper is rare in this country. If available will be costly.
- Option-2: To run 17-1/2” Clean Out BHA with **9-1/2” Mud Motor with 1.15° bend** -> hopefully could remove remaining cement seat.



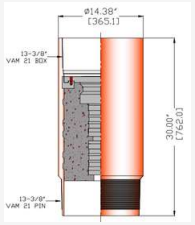
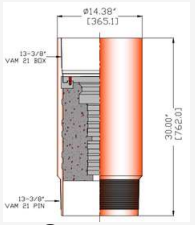
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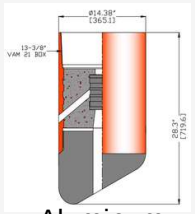
Casing Configuration, Float Equipment and Guide Shoe Design

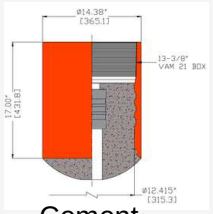
13-3/8” Stab In Float Collar, Cement

13-3/8” Landing Collar, Cement

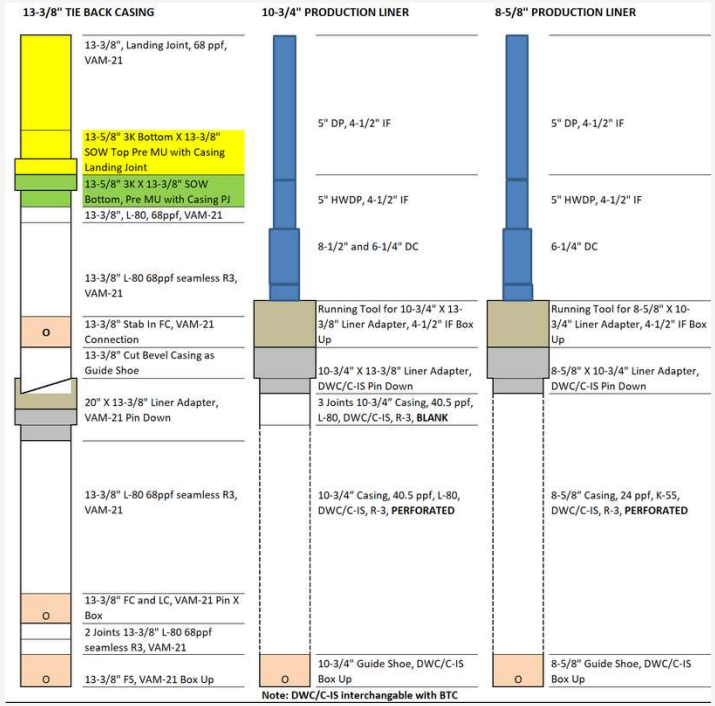


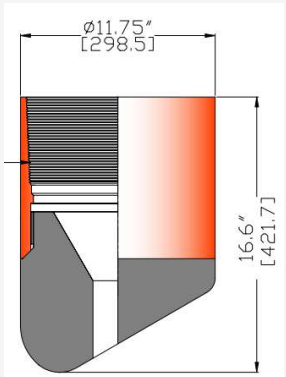
Aluminum



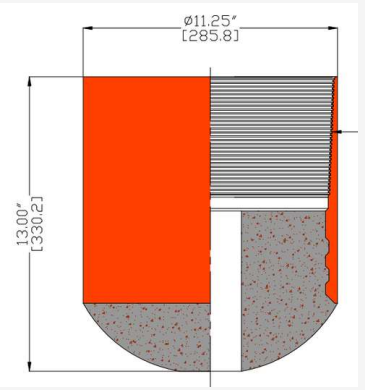
Cement

13-3/8” Float Shoe

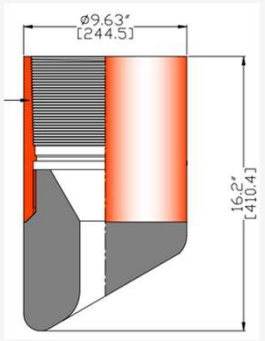




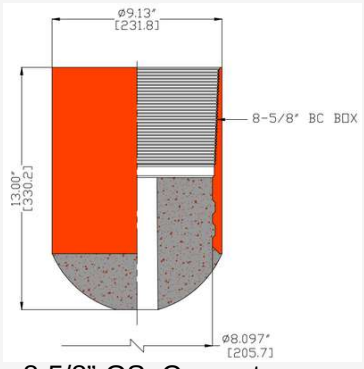
10-3/4” GS. Aluminum



10-3/4” GS. Cement



8-5/8” GS. Aluminum



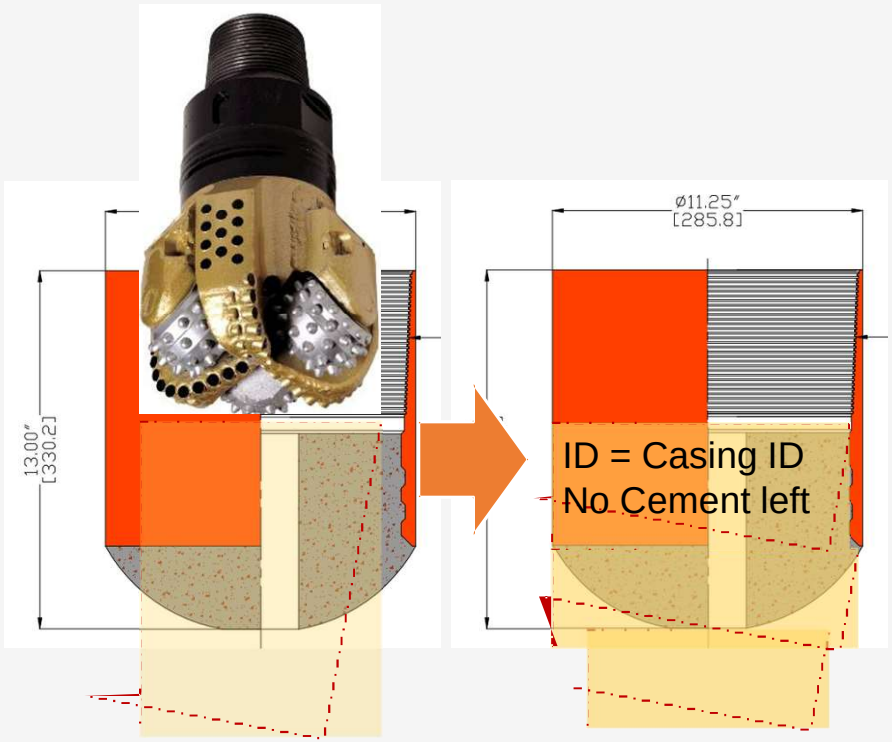
8-5/8” GS. Cement

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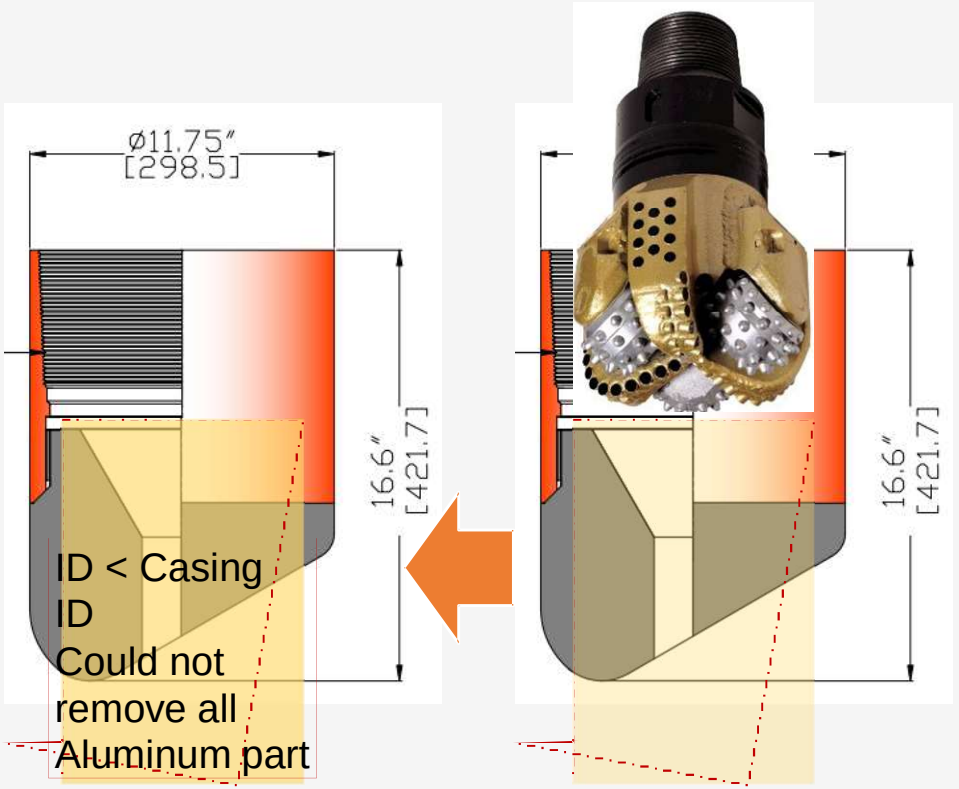
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Float Equipment and Guide Shoe Design

Drill Out Cement



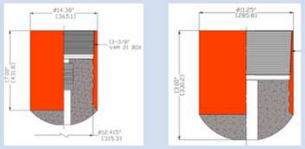
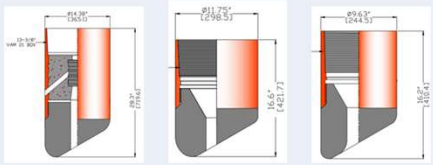
Drill Out Aluminum



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Float Equipment and Guide Shoe Design

Float Equipment and Guide Shoe Type	Advantage	Dis-Advantage
Made from Concrete Cement 	• Simple and Cheaper • Easy to be drilled out • ID after being clean out = ID casing • Less risk of stuck pipe	• Weaker that Aluminum • Run at normal speed in open hole section • Prefer not be run when bad hole condition
Made from Aluminum 	• Stronger • Run at faster speed in open hole section (2000 – 3000 mph) • Prefer to be run when bad hole condition	• More Expensive • Difficult to be drilled out • ID after being clean out = OD of the bit, not ID casing • Higher risk of stuck pipe
Reamer Shoe 	• Capable for rotating and reaming with casing/liner • Capable to be run in bad hole condition or ruble formation.	• Most Expensive • Difficult to be drilled out • ID after being clean out = OD of the bit, not ID casing • Risk of stuck pipe

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Float Equipment and Guide Shoe Design

Float Equipment and Guide Shoe are made from Cement and Aluminum. Refer the advantage and disadvantages, below are the recommendation:

- If hole condition is in good shape, run Float Equipment and Guide Shoe made from Cement. Easy to be drilled out and no ID reduction

- If hole condition is NOT in good shape, run Float Equipment and Guide Shoe made from Aluminum. We could run faster and the bottom profile of FS (GS) will self-position following the hole profile. However it has risk difficult to be drilled out and final ID will be bit size used for drilling out (ID reduction, less than casing ID). Than it will increasing risk of stuck pipe due to tight hole clearance to the BHA OD.

As mitigation, drill out with mud motor (slightly bend) and use PDC or Mill Tooth Bit. Do not ever use Insert Bit.

- In case we want to rotate casing or liner due to operation dictate, we could use Reamer shoe. However reamer shoe is similar disadvantage with FS / GS made from Aluminum.

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Casing Procurement Strategy

Casing Unit: Feet (Meter) vs Joint
Casing Size Grouping
Casing Sources
Casing Perforation

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Casing Pricing: Joint vs Feet

No	Description	Qty, Joints		Qty in Feet, R-3			Different, feet	
FOR DIENG 2				38 ft	41 ft	43 ft	Min-Max	Min - Avg
1	20", K-55, Semi Premium, 133 ppf, R-3, PSL-1	436	joint	16,568	17,658	18,748	2,180	1,090
2	13-3/8", L-80, Premium, 68 ppf, R-3, PSL-1	1124	joint	42,712	45,522	48,332	5,620	2,810
3	13-3/8", L-80, Premium, 72 ppf, R-3, PSL-1, special drift ID to 12.25"	413	joint	15,694	16,727	17,759	2,065	1,033
4	10-3/4", L-80, Semi Premium, 40.5 ppf, R-3, PSL-1, SCC Coupling OD 11.250"	1460	joint	55,480	59,130	62,780	7,300	3,650
5	8-5/8", K-55, Semi Premium, 24 ppf, R3, PSL-1, SCC Coupling OD 9.125"	285	joint	10,830	11,543	12,255	1,425	713
6	7", K-55, Semi Premium, 23 ppf, R-3, PSL-1, SCC Coupling OD 7.375"	91	joint	3,458	3,686	3,913	455	228
7	9-5/8", L-80, Semi Premium, 47 ppf, R-3, PSL-1	136	joint	5,168	5,508	5,848	680	340
						Feet	19,725	9,863
						Meter	6,012	3,006
FOR PATUHA 2								
1	20", K-55, Semi Premium, 133 ppf, R-3, PSL-1	447	joint	16,986	18,104	19,221	2,235	1,118
2	13-3/8", L-80, Premium, 68 ppf, R-3, PSL-1	1125	joint	42,750	45,563	48,375	5,625	2,813
3	10-3/4", L-80, Semi Premium, 40.5 ppf, R-3, PSL-1, SCC Coupling OD 11.250"	1159	joint	44,042	46,940	49,837	5,795	2,898
4	8-5/8", K-55, Semi Premium, 24 ppf, R3, PSL-1, SCC Coupling OD 9.125"	348	joint	13,224	14,094	14,964	1,740	870
5	7", K-55, Semi Premium, 23 ppf, R-3, PSL-1, SCC Coupling OD 7.375"	166	joint	6,308	6,723	7,138	830	415
6	9-5/8", L-80, Semi Premium, 47 ppf, R-3, PSL-1	159	joint	6,042	6,440	6,837	795	398
						Feet	14,785	7,393
						Meter	4,506	2,253

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CASING VENDOR SURVEY: Casing Size Grouping for Procurement

NO	SURVEY DESCRIPTION	CITRA TUBINDO	TENARIS SPIJ	APPIPA	PIPA MAS PUTIH	TELAGA MAS M	TPCO Pan ASIA	ELNUSA. F.K	HUNTING / SMB	CT ADVANCE	PATRAINDO	SERVOTECH	INDOKAYA
1	Location	Batam	Cilegon & Batam	Batam	Batam	Batam	Batam	Batam	Batam	Cilegon	Batam	Batam	Batam
2	Pipe Size	2 3/8" - 20"	2 3/8" - 20"	2 3/8" - 20"	2 3/8" - 20"	2 3/8" - 20"	2 3/8" - 20"	2 3/8" - 13 3/8"	2 3/8" - 20"	2 3/8" - 20"	2 3/8" - 13 3/8"	2 3/8" - 20"	Big Casing Size
3	Capacity					35.000 MT / Year							
4	Open Yard Storage	Yes	Yes	Yes	Yes	15.000 M2	?	Yes	Yes	Yes	?	?	Yes
5	Heat Treatment Facility	YES	YES	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
6	Threading Quick Connection	OMEGA	TENARIS ER	OMS (Welding)	NO	GB3P	NO	NO	SL BOSS	NO	NO	NO	LEOPARD (welding)
7	Threading BTC (API Conn)	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
8	Threading Premium Conn (T&C)	VAM Family	TenarisHydril	JFE Connection	NO	JFE Connection	TPCQ	NO	SL APEX	HENGYANG	NO	TMK PF	?
9	Threading Premium (Flush)	VAM FJL / SLIJ	TenarisHydril	NO	NO	TMK FJ	NO	NO	SLF / SLSF	NO	NO	TMK FJ	NO
10	Mill Sources	VAM, Anhui, Sumitomo, etc	Tenaris Group	JFE, Baotao, etc	Jiangyin City, Jiangsu Chendu, etc	JFE, Jing Jiang, Molong, Jesco, Baosteel, TPCO, etc	TPCO	?	Hengyang, Baosteel, Jiangsu Yulong, Jiangsu Changbao, etc	Jing Jiang, Hegyang, etc	Jing Jiang, Molong	Jing Jiang, Pangang Chende, etc	?
11	Perforated Facility	NO	NO	NO	YES	NO	NO	NO	NO	NO	NO	NO	NO
12	Contact Person	Joe Barlett	Dodi. S	Fuad M	Septi	Rahman		Danny	Danny	Kurniawan	Hamijah	Sigit	Kiki
13	Contact Number	0812 1062 774	0816 756 449	0812 1020 065	0857 8104 0505	0819 0808 1395		0812 6615 4909	08170501752	0812 1054 5401	081261310209	822 1062 774	08172329869

Partial Award refer to Casing Sizes

Group-1	Group-2	Group-3	Group-4
30” Conductor	20” Casing	13-3/8” Casing 9-5/8” Casing	10-3/4” Casing 8-5/8” Casing 7” Casing

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CASING SOURCES - Casing from China

1. Most opinion: NO due to lack of quality of green pipe
2. Second Opinion: YES due to cheap price. Accepted with mitigation.
3. Third Opinion: Have to, If I was working in KS ORKA.



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CASING PRICE COMPARISON (2010)



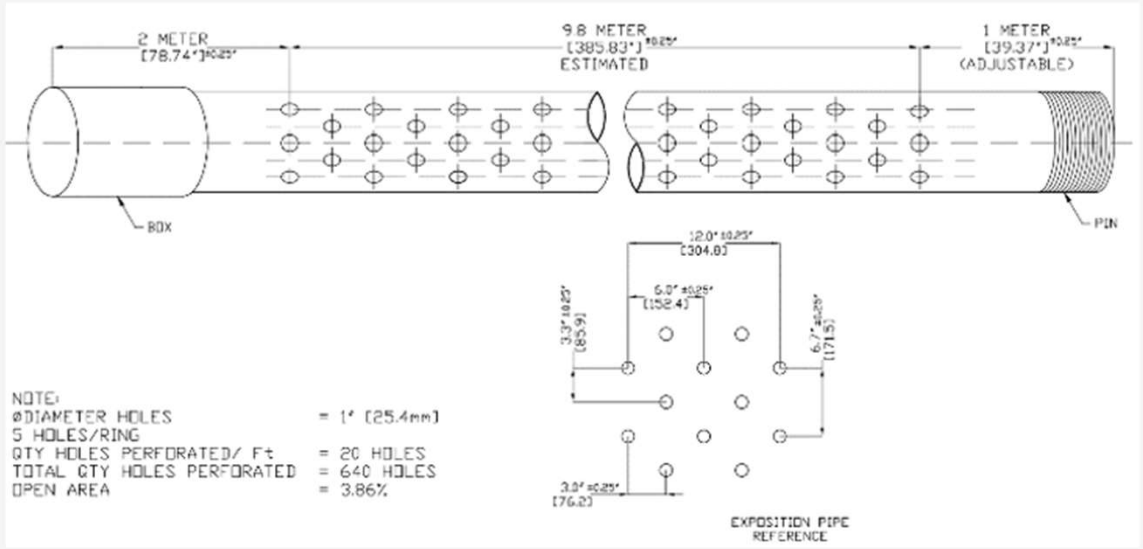
- How to ensure the casing quality from China



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Perforated Casing Method: At Workshop











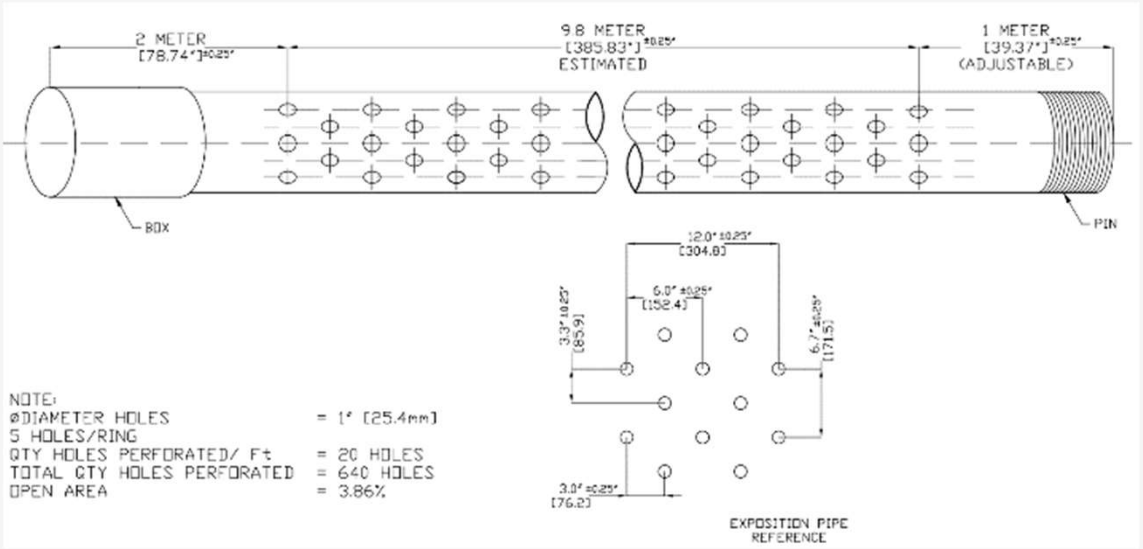




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Perforated Casing Method: At Rig Site



NOTE:
ØDIAMETER HOLES = 1" [25.4mm]
5 HOLES/RING
QTY HOLES PERFORATED/ Ft = 20 HOLES
TOTAL QTY HOLES PERFORATED = 640 HOLES
OPEN AREA = 3.86%











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Perforated Casing Method: At Workshop vs At Rig Site

Perforated Casing Size	Unit	Perforation Cost at Workshop	Perforation Cost at Rig Site
10-3/4”	USD/joint		128.5
8-5/8”	USD/joint		109.5
7”	USD/joint		109.5



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4 Efficiency in Wellhead Installation

Wellhead Configuration: Wellhead Housing and Master Valve:

Wellhead Housing:

- Double wellhead housing: 21-1/4” Wellhead dan Expansion Spool
- Single wellhead housing: 13-5/8” Wellhead Housing

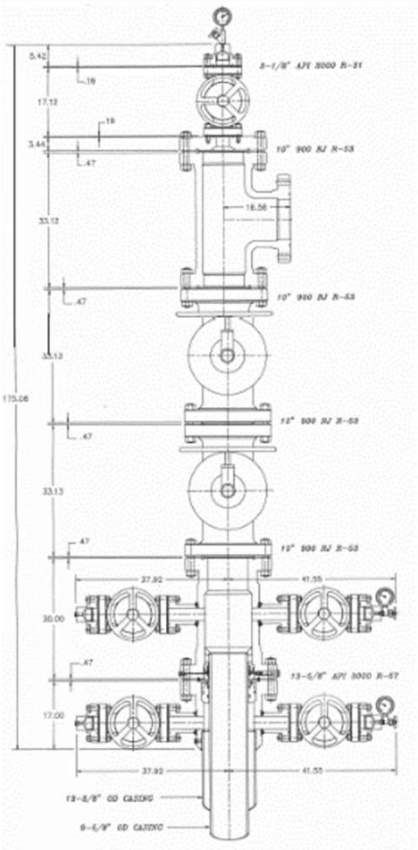
Master Valve:

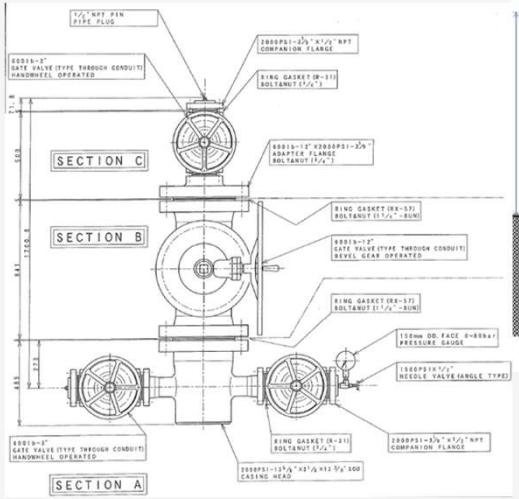
- Single Master Valve
- Double Master Valve (Drilling MV + Production MV)

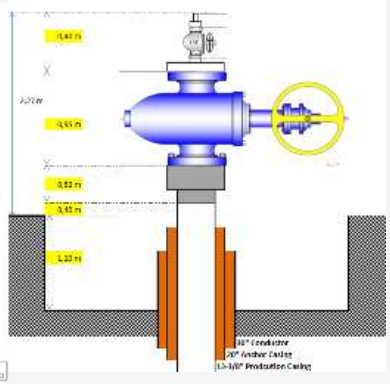
ARKA DATA
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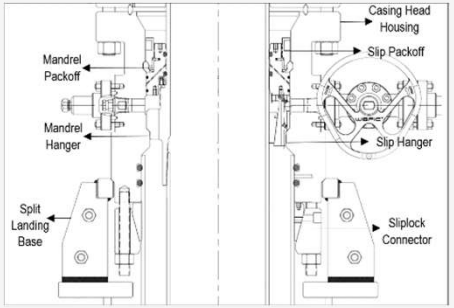
Well Design Optimization

WELLHEAD for GEOTHERMAL WELL











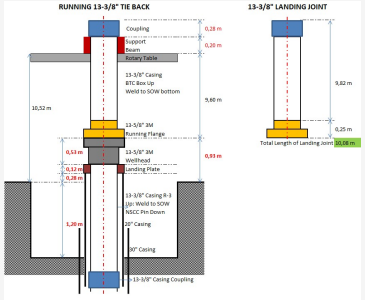
- Pre-weld wellhead
- Slip on Weld vs Slip Lock
- Casing To Casing Welded
- Wellhead Installation during WOC

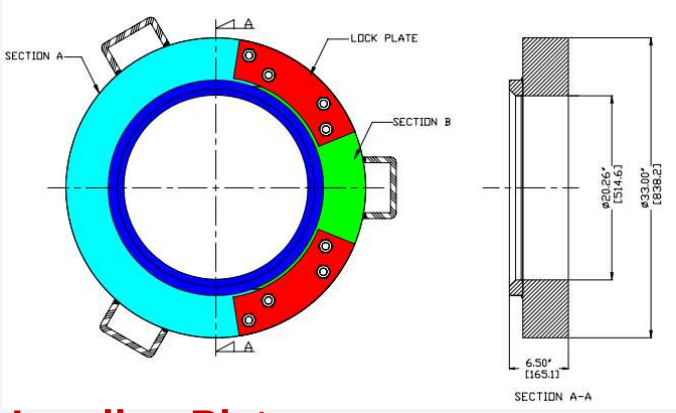
ARKA DATA
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Well Design Optimization

Landing Slip for 20” casing installation







Landing Slip:

- Propose for **hanging Casing**
- No WOC
- Seated on Cut Off Casing
- Hang casing on **Casing Body**
- Heavy, longer installation

Landing Plate:

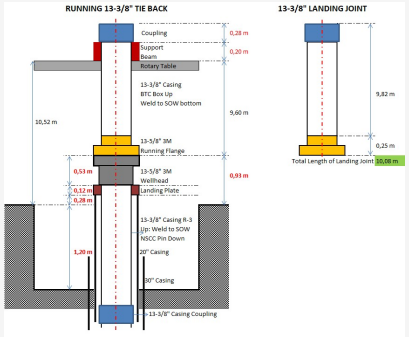
- **Hanging Pre-Welded Wellhead**
- No WOC
- Seated on Cut Off Casing
- Hang casing on **Casing Collar**
- Lighter, easy installation
- Need to ND BOP prior cementing

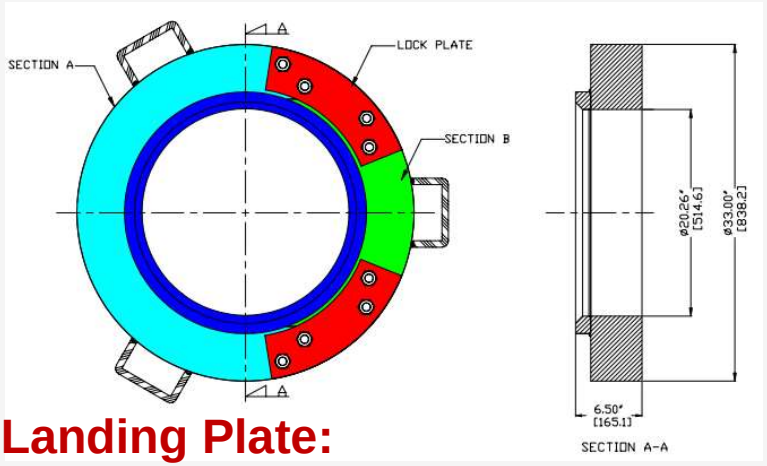
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Well Design Optimization

Landing Slip for 20” casing installation







Landing Slip:

- Drill 26” hole
- Run 20” casing
- Run Inner string
- Primary cementing
- Cut 30” casing, hang out 29” Diverter -> **Well control risk: Low**
- Install Landing Slip and land 20” casing
- Cut 20” casing and ND 29” Diverter
- Final cut 20” and NU Wellhead & BOP

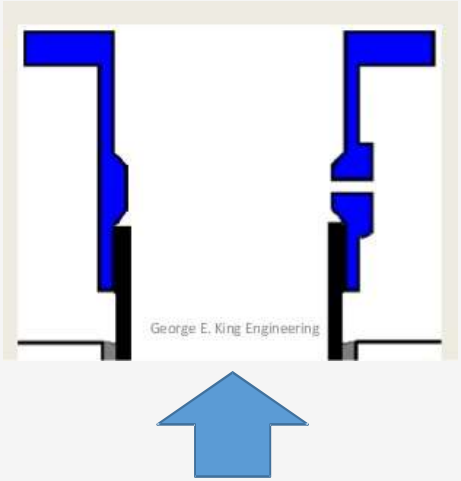
Landing Plate:

- Drill 26” hole
- Cut 30” casing, ND 29” Diverter -> **Well control risk: High**
- Run 20” casing and Pre-welded wellhead
- Install Landing Plate and land 20” casing
- Run Inner string
- Primary cementing
- NU BOP

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STARTING HEAD: 21-1/4” Housing
Slip on Weld vs Slip Lock System



Slip on Weld (SOW) Wellhead:
- Final Cut casing
- Slip wellhead on casing
- Do weld ID and OD

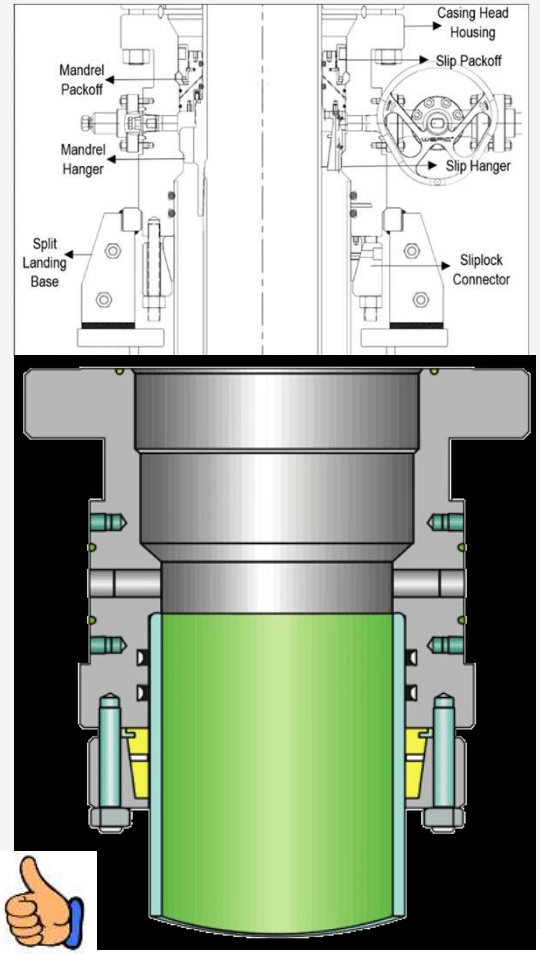
- **Cut Casing** to Nipple Down wellhead

Slip Lock Wellhead:
- Final Cut casing
- Slip wellhead on casing
- **Tighten the bolt**

- **Loose the bolt** to Nipple Down wellhead



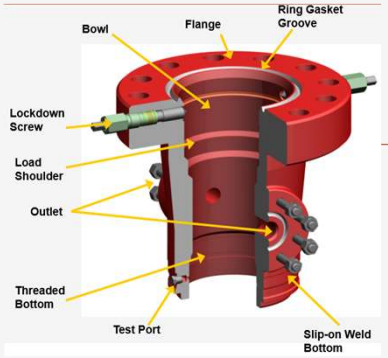
3-6 Rig Hours

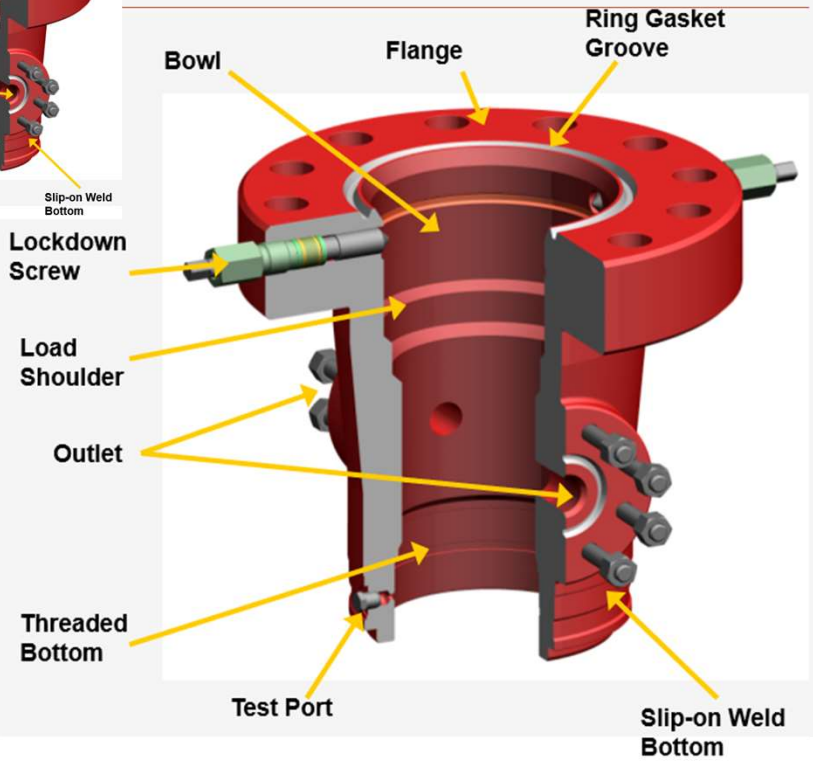


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Pre-Welded 13-5/8” Wellhead Housing





Pemasangan 13-5/8” Wellhead Housing ada 2 cara:

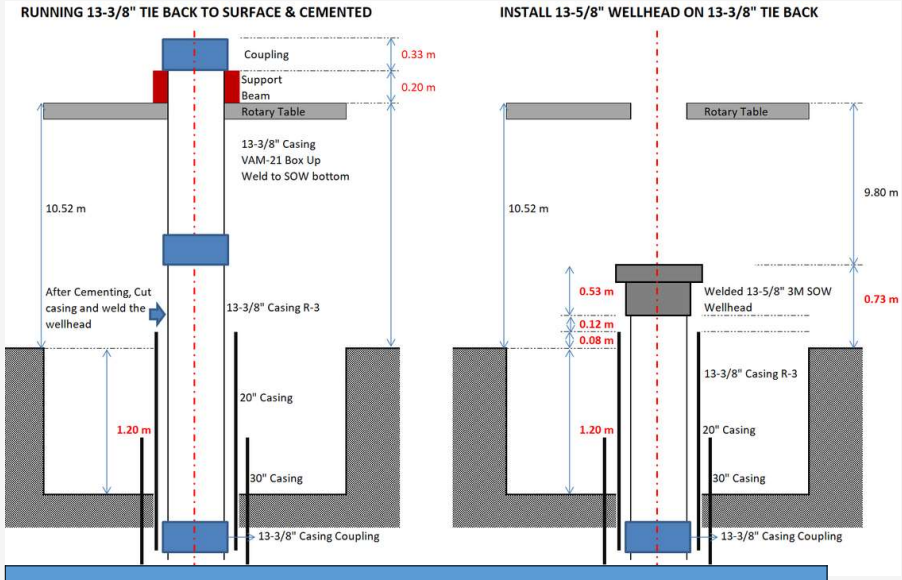
i. Online Wellhead Installation (Pemasangan langsung) -> Pengelasan Vertikal

i. Run Pre-Welded Wellhead Installation -> Pengelasan Horizontal

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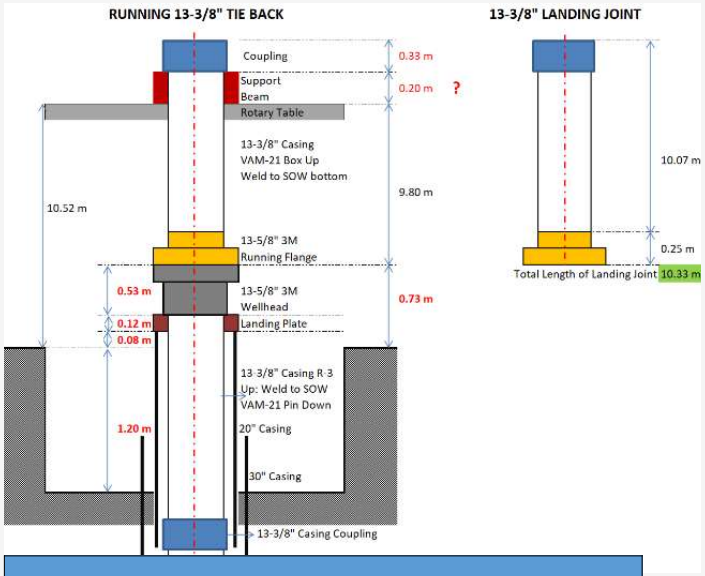
Well Design Optimization

Non Pre-Welded vs Pre-Welded 13-5/8” Wellhead Housing



Without Pre-Welded Operation

ND BOP
Run Casing Tie back
Run Inner string with casing stick up -> 3 hrs
Cementing
WOC -> 8 hrs
Cut casing ->
Install Wellhead -> 35 hrs



Pre-Welded Operation

ND BOP
Run casing Tie back
Run Pre-welded wellhead -> 1 hrs
Run Inner string on Rotary Table
Cementing
No WOC

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Data Driven Drilling Advisory

Well Design Optimization

Non Pre-Welded vs Pre-Welded 13-5/8” Wellhead Housing



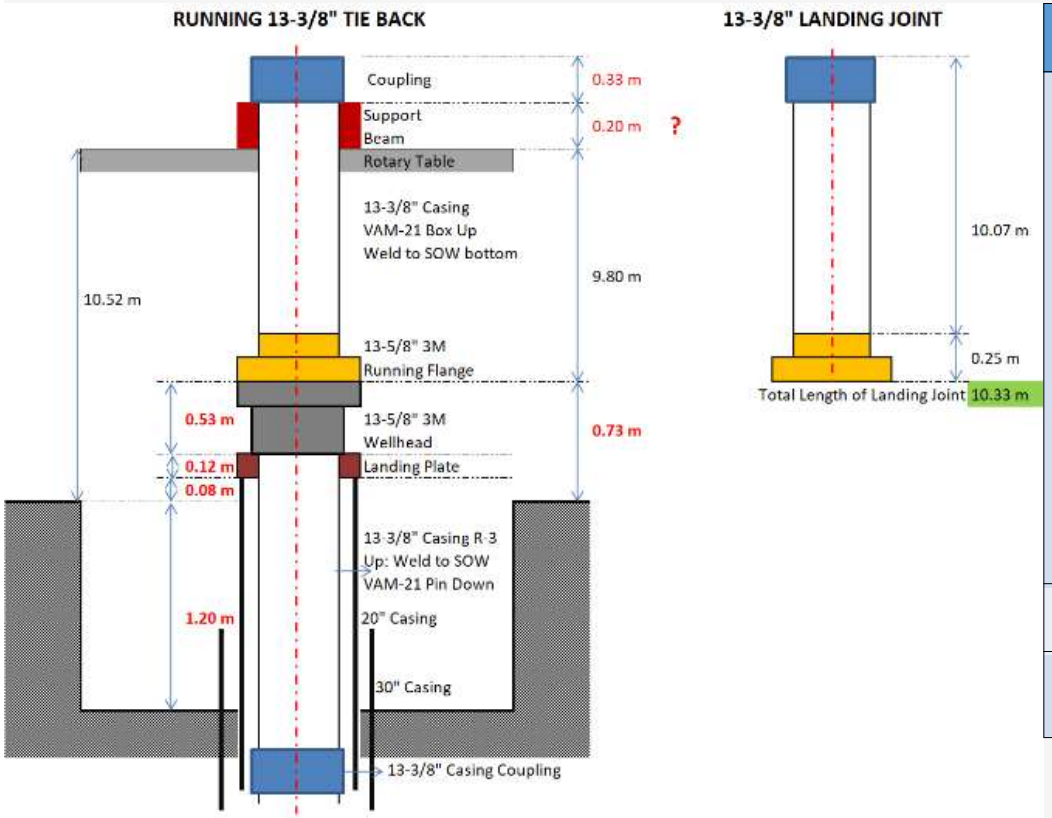




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Pre-Welded 13-5/8” Wellhead Housing

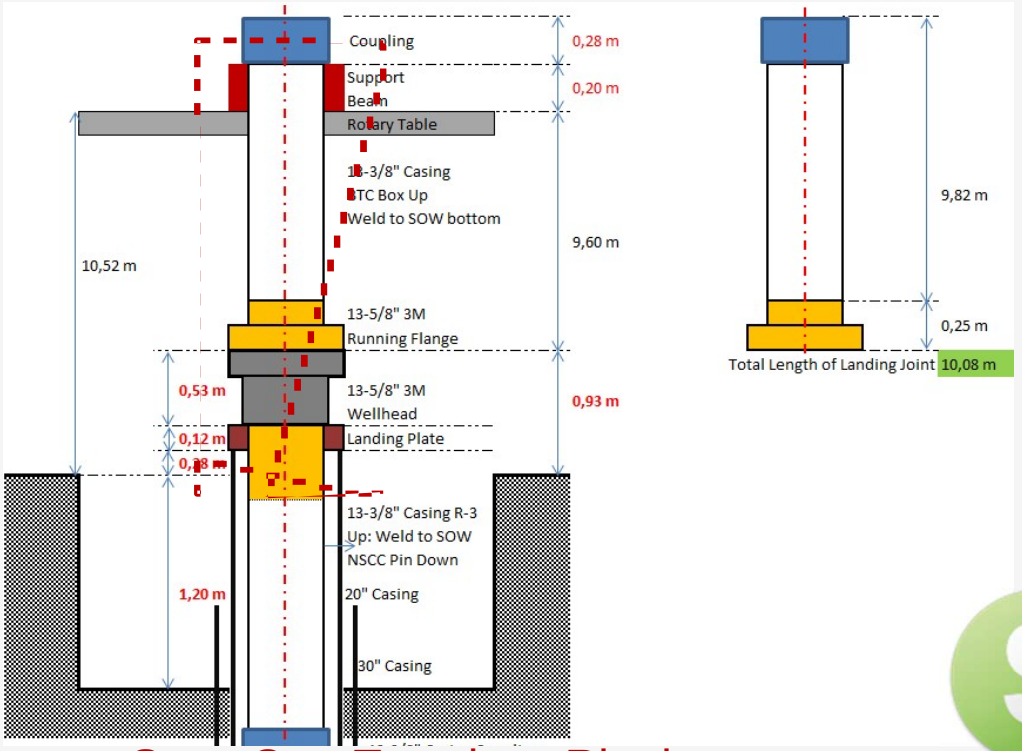


Without Pre-Welded	Pre-Welded
ND BOP Run Casing Tie back Run Inner string with casing stick up -> 3 hrs Cementing WOC -> 8 hrs ND BOP Cut casing -> Install Wellhead -> 35 hrs NU BOP	ND BOP Run casing Tie back Run Pre-welded wellhead No WOC NU BOP
3 + 8 + 35 hrs longer	Rig time saving: 46 hours
	Cost saving: **USD 134 K**

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PRE-WELDED WELLHEAD
13-3/8” Casing + 13-5/8” 3M Section A Wellhead



- RT opening 37.5”
- Split type RT bushing is preferable.
- Pre-welded wellhead in horizontal position.
- Pre-welded 13-5/8” Flange housing and 13-3/8” casing Landing Joint.



- Save One-Two days Rig time

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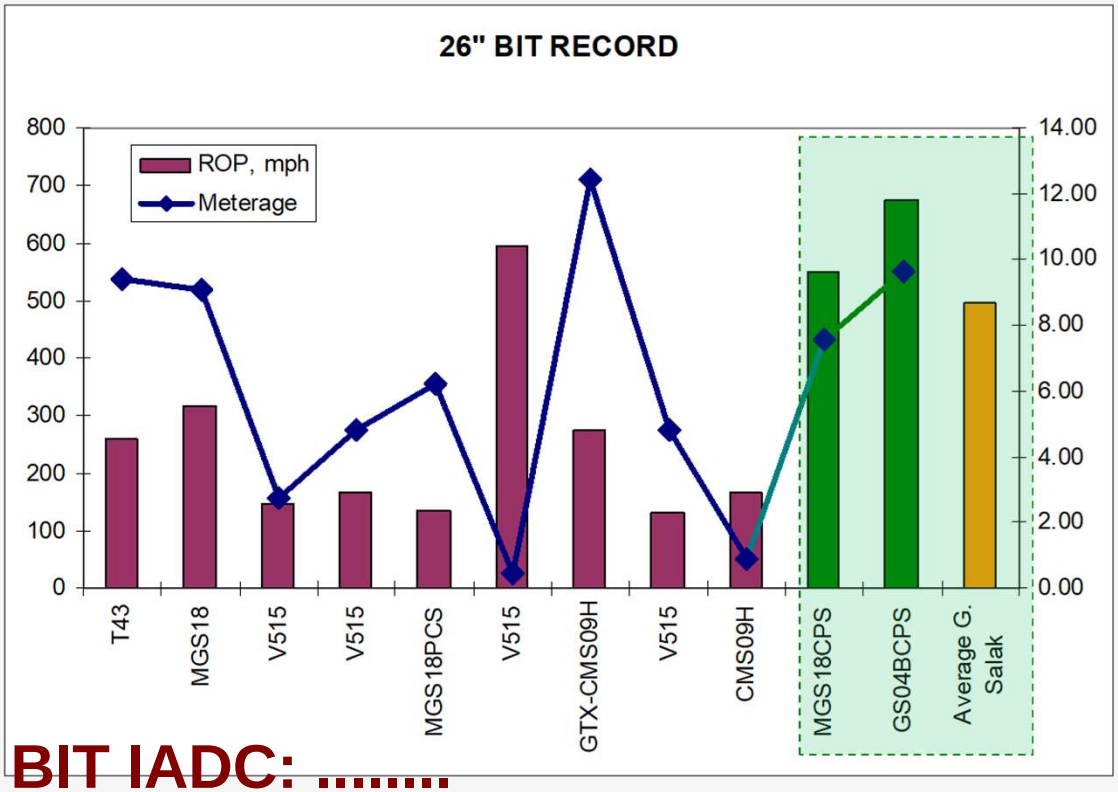
5 Selection of Latest Drilling Bit Technology

- Rate Of Penetration (ROP)
- Footage, one hole section one bit
- Cost per Meter

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Well Design Optimization

DRILLING BIT SELECTION – CASE STUDY





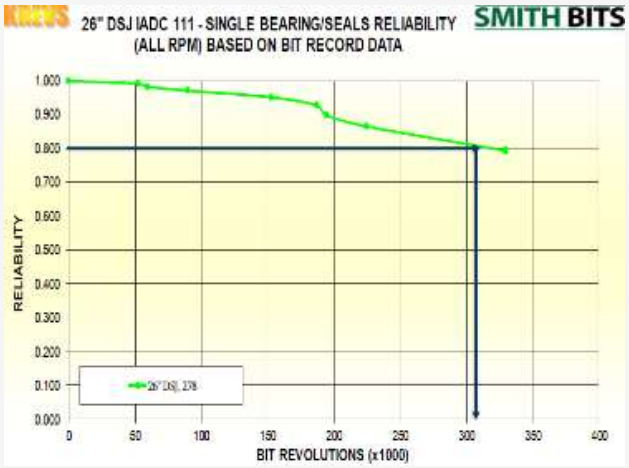


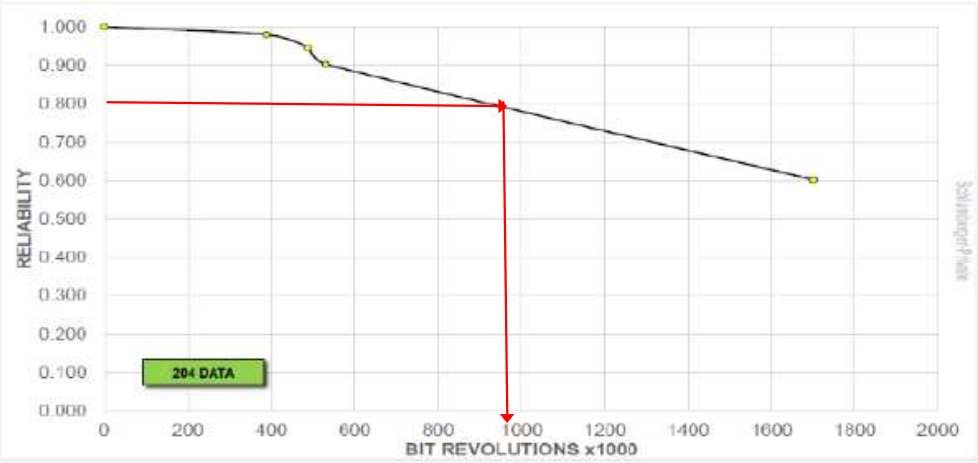
BIT IADC: .....

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ROCK BIT KREV:
Single vs Double Bearing (premium)





Drilling Bit Comparison: Same IADC but different Bearing Type:

- 80% Reliability: 305 KRev versus 950-1200 Krev (3.1-4 times)
- Two Bit vs Half Bit per section
- Two vs One BHA trip per section

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New Enhance Bit Type: Faster and Longer Run







Sting Blade
PDC Bit

Kymera
Hybrid Bit



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FuseTek Bit Series for Geothermal Drilling



Tier 1 cutter ~ ION 3D cutters

- Superior abrasive, thermal and impact resistance

HDI cutters as DiamondBack cutters with strategic tip offset setup

- Adding higher impact resistance and act as secondary cutting element (hybrid element)



Max TFA for each bit size ~ min 3.00 in²

- Support for high flow rate application ~ air drilling





Tier 1 gauge cutters

- Prevent UG condition when pulled out



Slick gauge

- Reduce drilling torque in poor hole quality environment



ReedHycalog | NOV Wellbore Technologies

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ReedHycalog – FuseTek Bit Series





Total Cost Saving after 2 runs of 12 ¼" FTKC76 :
US$ 349,842 ≈ 5 Rig Days

- Drill **1000 m** with **one run**
- Save 2 bit run if using Insert Bit
- Bit Dull: **1-1-BT**-A-X-3-PN-BHA
  -> **Re-run able**



Bit condition after being pulled to surface

Well Design Optimization

SMGP: Drilling time Saving – New technology Bit



12.25” NOV Fusetek bit

Saved **USD 142,109** by drilling 1,000 mMD with 11.12 mph ROP



17.5” Baker Hughes Kymera Hybrid bit

Saved **USD 129,834** by drilling 876 mMD with 10.68 mph ROP

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6

Optimum Design of Bottom Hole Assembly (BHA)

- Best BHA Configuration
- Hard Facing For Geothermal Drilling
- Drill Pipe Selection for Deep drilling
- Drill String Inspection Level
- Competitive Renumeration Table (Procurement)

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Mud Motor and MWD Size Selection

BHA Configuration	Parameters	26" Hole	17-1/2" Hole	12-1/4" Hole
9-½ MM + 9-½ EM-MWD	Max GPM	1200	1200	1200
	MU Time	Longer	Longer	Longer
	Directional Control	Good	Good	Fair (*)
9-½ MM + 8-½ EM-MWD	Max GPM	1200	1200	1200
	MU Time	Fair	Fair	Fair
	Directional Control	**Poor: too limber for nudging.**	Good but Risk of Twist off	Good, MM 6-5/8" Reg
8-½ MM + 8-½ MWD	Max GPM	900	900	900
	MU Time	Faster	Faster	Faster
	Directional Control	**Not Applicable, Bit: 7-5/8" Reg Connection, GPM too low**		Good

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Mud Motor and MWD Size Selection

BHA Configuration	Parameters	26" Hole	17-1/2" Hole	12-1/4" Hole
9-½ MM + 9-½ EM-MWD	Max GPM	1200	1200	1200
	MU Time	Longer	Longer	Longer
	Directional Control	Good	Good	Good
9-½ MM + 8-½ EM-MWD	Max GPM	1200	1200	1200
	MU Time	Fair	Fair	Fair
	Directional Control	Poor	Good	Good

26" BHA:

- **9-½ MM + 9-½ EM-MWD** → easy to control surface kick off and easy to fight if any surface formation Deeping
- **9-½ MM + 8-½ EM-MWD** → less rig time to make BHA and could be use for next 17-1/2" hole section, however it was experienced un-able to control directional AZ due to surface formation Deeping. Finally need longer for making directional correction → 1 – 2 BHA Trip in and out = 24 hours = 1 day additional rig time → USD 70 K

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Directional BHA for 12-1/4" Production Hole Section – Typical Slant Section

BHA Configuration	Parameters	12-1/4" Hole
9-½ MM + 8-½ EM-MWD	Max GPM	1200
	MU Time	Longer
	Weakness	9-½ MM usually had 7-5/8" Reg Box Down. Need X/O or special design with 6-5/8" REG Box down to meet with 12-¼ Bit Connection
8-½ MM + 8-½ EM-MWD • Motor Bend: 0° – 0.78° VS • Motor Bend: 1.15°	Max GPM	900
	MU Time	Faster
	Directional Control	Recommend to use Motor Bend: 1.15°: • Less sliding length if any correction • Enlarge the 12-¼ → easier to run 10-¾" casing with Special Clearance Coupling

Potential NPT:

- Run Push BHA, drillout shoe and push Liner to bottom, POOH
- Estimated NPT = 1 day = USD 70K

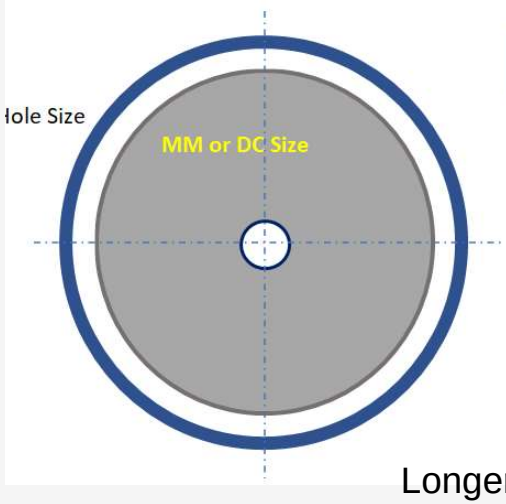


Hole: 12-¼
SCC Coupling: 11-¼
Hole vs SCC Coupling Clearance: ½"

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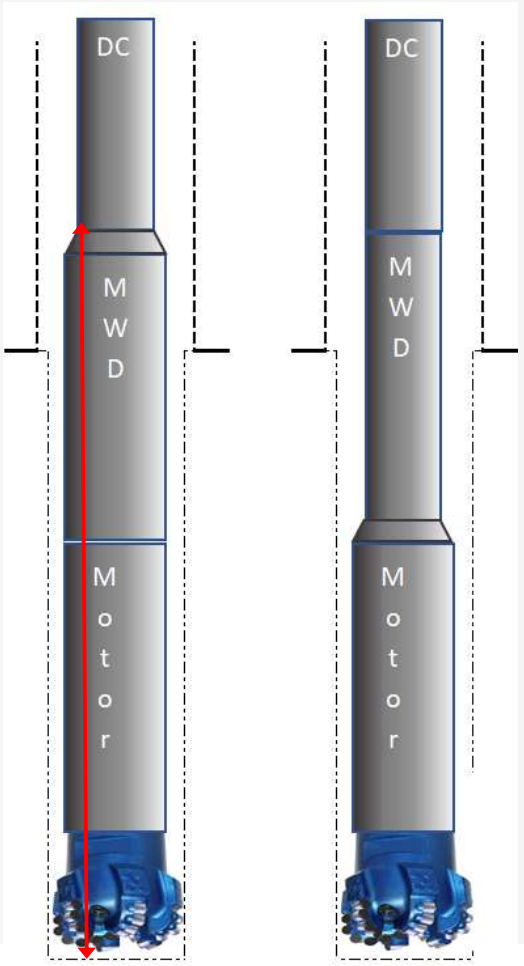
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Hole Size vs Mud Motor and MWD Size Selection



Hole Size in	MM Size in	MWD Size in	Clearance, in With DC	Clearance, in With MWD	Remark
26	9.5	9.5	8.2500	8.2500	
17.5	9.5	9.5	4.0000	4.0000	
12.25	8.5	8.5	1.8750	1.8750	
9.875	8.5	8.5	0.6875	0.6875	Not reccomended
9.875	8.5	6.5	0.6875	1.6875	Reccomended
7.875	6.5	6.5	0.6875	0.6875	Prefer not use MWD

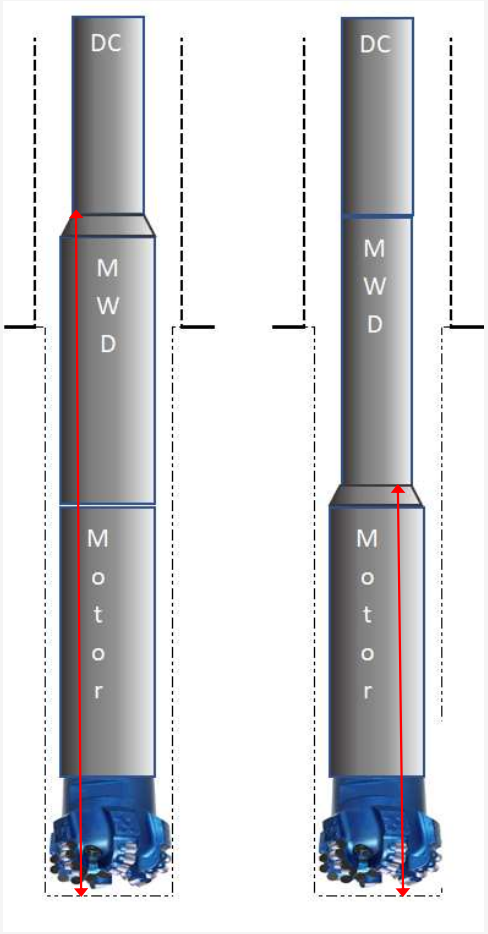
Longer MM + MWD -> Longer tight clearance interval -> risk of stuck pipe



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BHA Configuration	Parameters	9-7/8" Hole	7-7/8" Hole
8-½ MM + 8-½ EM-MWD	Max GPM	900	Not Applicable
	MU Time	Longer	
	Directional Control	Good	
8-½ MM + 6-½ EM-MWD	Max GPM	900	Not Applicable
	MU Time	Fair	
	Directional Control	Poor	
6-½ MM + 6-½ EM-MWD	Max GPM	Not Applicable	700
	MU Time		Fair
	Directional Control		Fair



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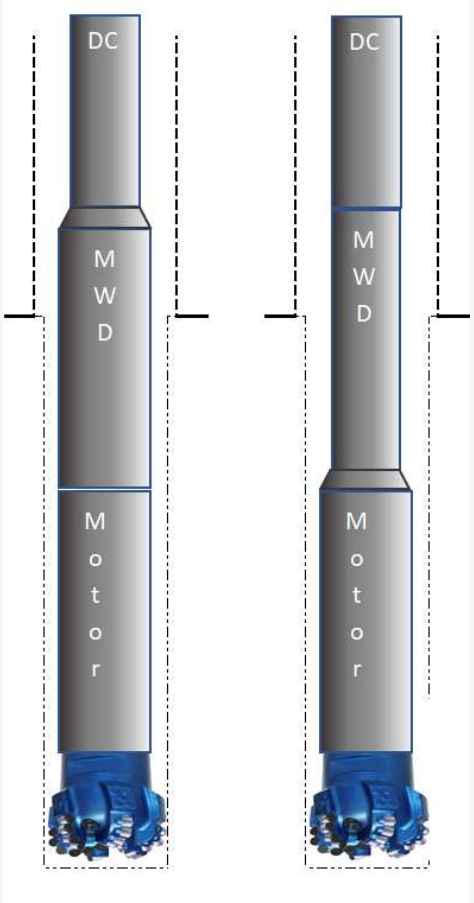
Well Design Optimization

Mud Motor and MWD Size Selection: 9-7/8” & 7-7/8” Hole

- **Lubang 9-7/8”:** 8-1/2” Mud Motor dan 8-1/2” MWD -> menambah panjang section annulus dengan tight clearance (0.6875”). **Hal ini akan menimbulkan resiko stuck pipe.**

- **Lubang 9-7/8”:** 8-1/2” Mud Motor dan 6-1/2” MWD -> memperpendek panjang section annulus dengan tight clearance (0.6875”). Sehingga hal ini akan **menurunkan resiko stuck pipe.**

- **Lubang 7-7/8”:** karena sifatnya untuk kotingency saja, kami menyarankan untuk run **rotary BHA saja** tanpa MWD.



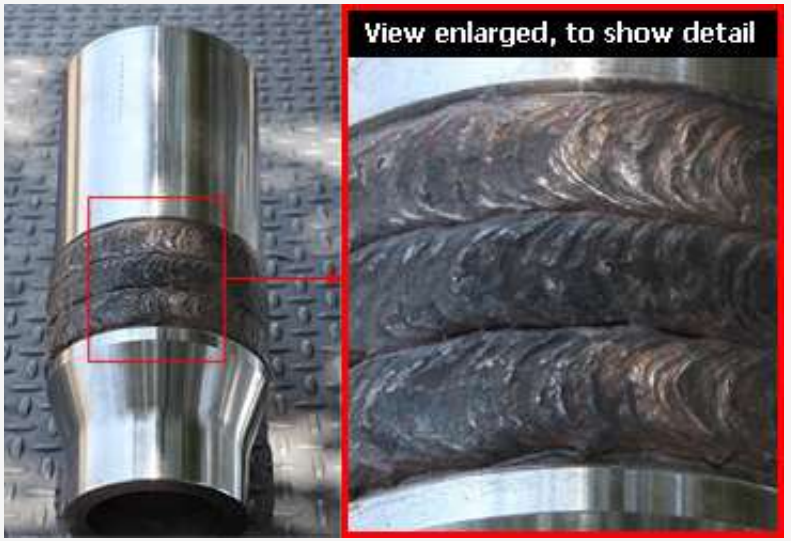
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Well Design Optimization

Hard Facing and Drill String Hard Banding



Stabilizer Blade with Hard Facing



Drill String with Hard Banding

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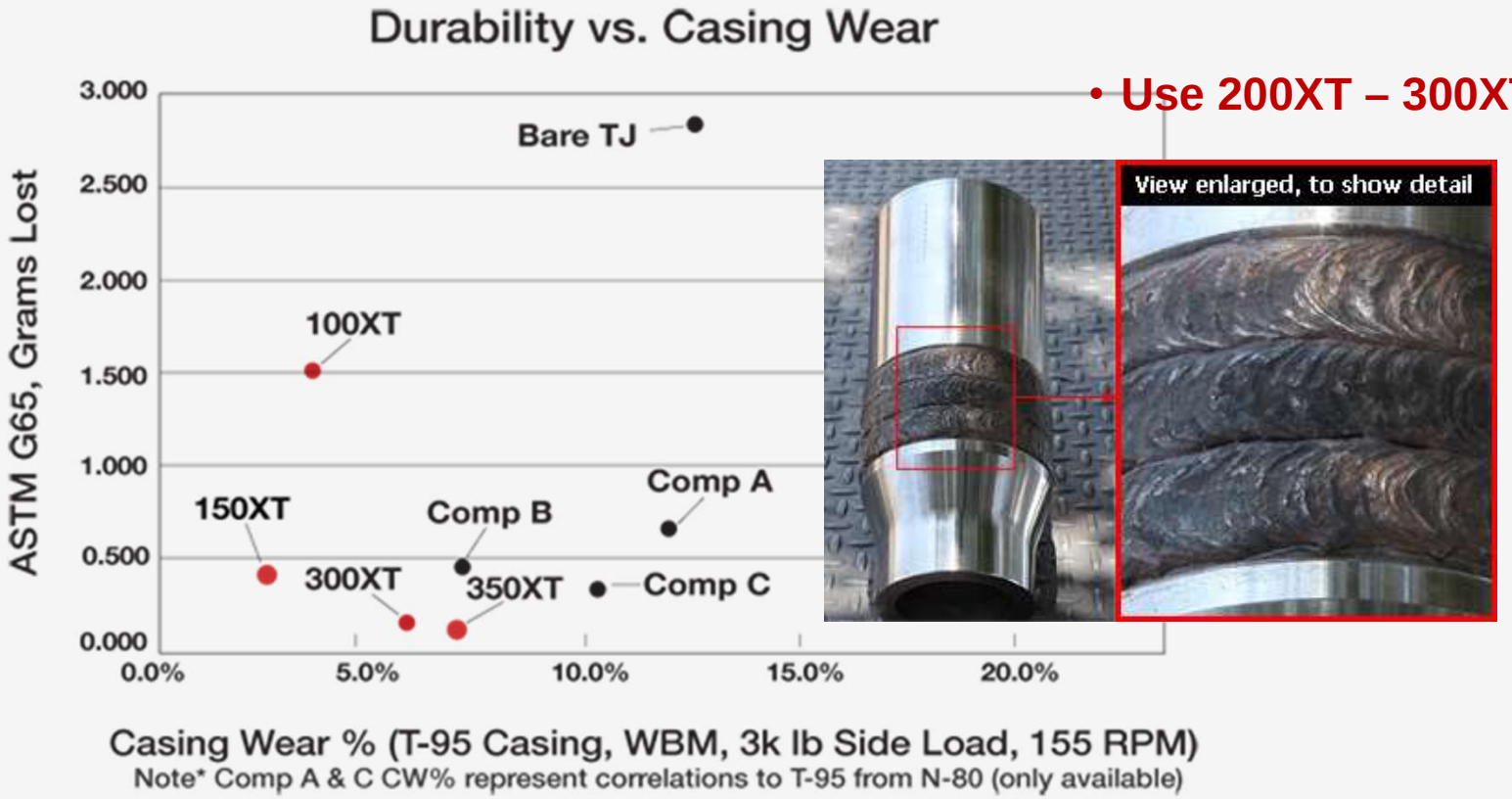
HARD FACING FOR BLADES STABILIZER

HardFacing Type	Picture
HF 4000 Tungsten carbide inserts (button type). The inserts have been developed to allow cold insertion and maintain close fit. A greater concentration of inserts on the bottom third of the blade and leading edge will increase surface contact to reduce wear in highly abrasive formations	
HF 5000 This oxy-acetylene process applies tough molten carbide particles of varying sizes held in a nickel chrome matrix which provides excellent bonding properties and greater surface wear characteristics. Surface hardness levels over 40HRC. **Ideal for GEO-THERMAL applications over 350°C.**	
HF 6000 HF6000 is a specially developed hardfacing deposit that is applied to Stabilizers and other Oilfield Industry products by the Plasma Transfer Arc (PTA) process. The process is fully automated. The movement of the hardfacing torch and simultaneous movement of the work-piece is controlled by a 6 axis Robot and an additional synchronised axis.	

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Hard Banding Durability vs. Casing Wear



- **Use 200XT – 300XT**

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Deep Geothermal Well

Drilling Engineering for Deep Geothermal Well:

- Simulate drilling hydraulic -> mud pump sizing, pump liner size.
- Torque and Drag Simulation -> DP torque and tension limit -> DP selection.
  - Surface Torque to be kept less than Drill Pipe Make Up Torque
  - Surface tension to capable hanging simulated Pick Up Load + Over Pulled to activate Drilling Jars (if any un-expected stuck pipe event)

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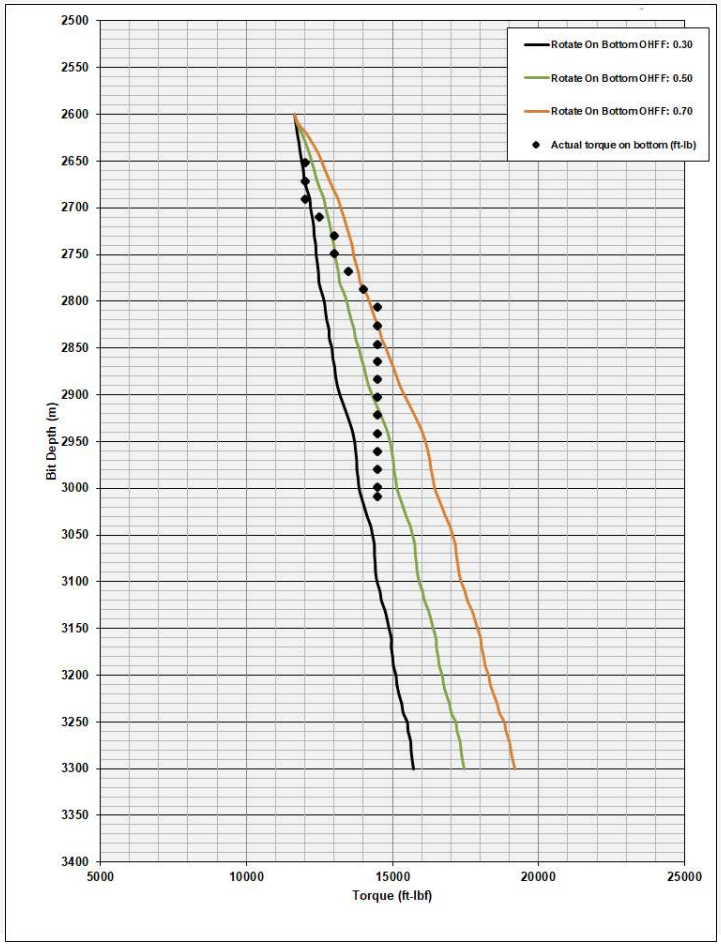
Drill Pipe Selection

Drill pipe function:

- To hang all drill string tension load (Pick up and slack off)
- To rotate the drill string and transfer drilling torque
- To be line for circulation drilling fluid, allow drilling fluid to be pumped down the hole through the bit and back up the annulus

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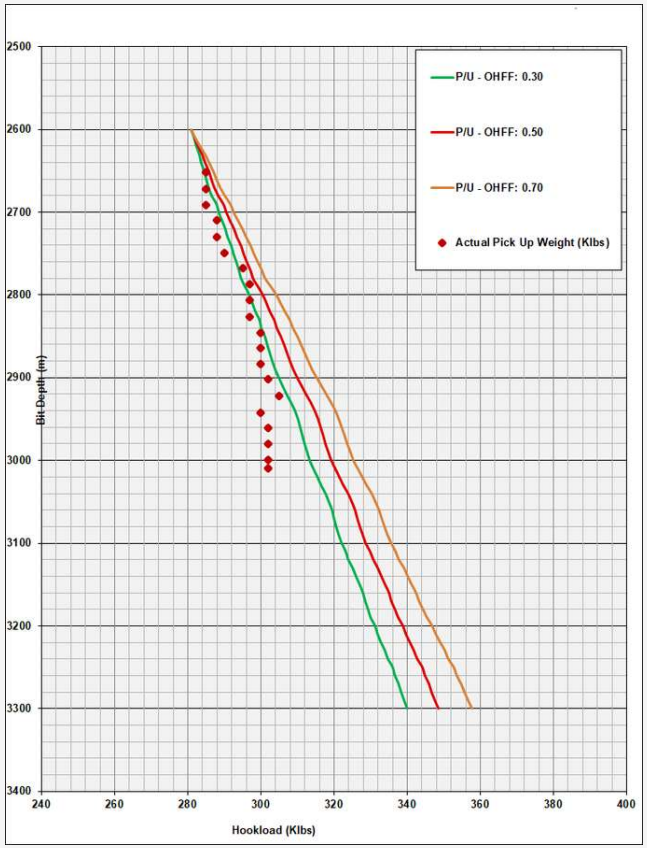
Drilling Torque simulation:

- Using Friction Factor 0.7, the surface torque was less than 20 Klbs ft
- Accepted range to use 5” 19 ppf, G-105, 4-1/2” IF

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Drill Pipe Tensile Load



Pick Up Load:

- 80% Drill Pipe Capacity, 5”, 19 ppf, G-105, Premium Class = **349 Klbs**
- Using Friction Factor 0.7, the pick up load 326 klbs (at 3000 m-MD) and 356 klbs (at 3300 m-MD).
- If any pipe stuck and need to activated the drilling Jars with **120 klbs** over pulled.
- Need to pick up the drill string **446 – 476 klbs -> exceeded DP tension capacity.**
- **Need to combine the DP with higher tensile capacity**

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Well Design Optimization

Drill Pipe Selection

Drill pipe commonly used (TD < ± 2500 m-MD):

- 5” 19 ppf, G-105, 4-1/2” IF

Drill pipe commonly combine used for Deep well:

- 5” 19 ppf, G-105, 4-1/2” IF
- 5” 19 ppf, S-135, 4-1/2” IF
- 5” 24 ppf, G-105, 4-1/2” IF
- 5.5” 19 ppf, G-105, 5-1/2” IF

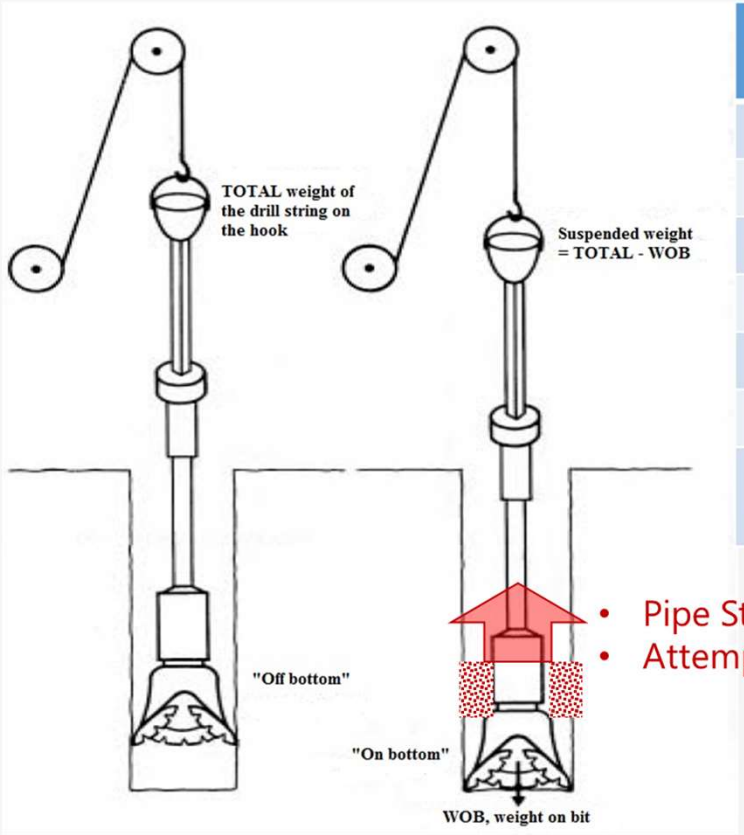
To be selected for the most economic



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OVER PULLED – at 3000 m-MD



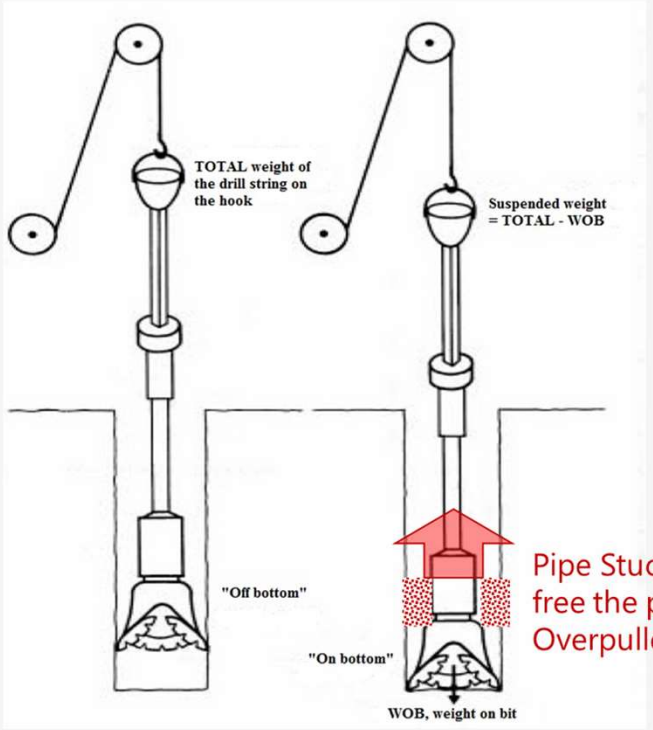
Description	Off Bottom	On Bottom
Travelling Block	30 Klbs	30 Klbs
TDS	25 Klbs	25 Klbs
BHA Weight	326 Klbs	326 Klbs
Over Pulled	None	120 Klbs
Weight Indicator	381 Klbs	501 Klbs
Axial Load at DP Wt below TDS	326 Klbs	446 Klbs

- Pipe Stuck
- Attempt to free the pipe with 120 Klbs Overpulled

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OVER PULLED – 3200 m-MD



Description	Off Bottom	On Bottom
Travelling Block	30 Klbs	30 Klbs
TDS	25 Klbs	25 Klbs
BHA Weight	356 Klbs	356 Klbs
Over Pulled	None	**120 Klbs**
Weight Indicator	411 Klbs	531 Klbs
Axial Load at DP Wt below TDS	356 Klbs	**436 Klbs**

Pipe Stuck, Attempt to free the pipe with 120 Klbs Overpulled

Need to higher grade Drill Pipe

Size, Weight, Grade	New	Premium	Class 2	ID
5 in, 19.5 lb/ft, Grade G	**553,833**	**436,150**	**378,605**	**4.276**
5.5 in, 21.9 lb/ft, Grade G	611,963	482,692	419,346	4.778
5.5 in, 24.7 lb/ft, Grade G	696,111	547,799	475,347	4.670
5 in, 19.5 lb/ft, Grade S	712,070	560,764	486,778	4.276
5 in, 25.6 lb/ft, Grade G	742,201	580,566	502,223	4.000
5 in, 25.6 lb/ft, Grade S	954,259	746,443	654,715	4.000

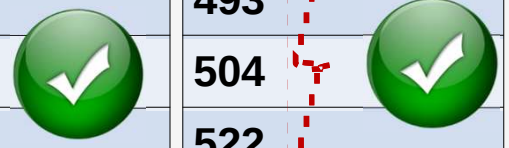
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Drill Pipe Tension Yield

Size, Weight, Grade	New	Premium
5 in, 19.5 lb/ft, Grade G	**553,833**	**436,150**
5.5 in, 21.9 lb/ft, Grade G	611,963	482,692
5.5 in, 24.7 lb/ft, Grade G	696,111	547,799
5 in, 19.5 lb/ft, Grade S	712,070	560,764
5 in, 25.6 lb/ft, Grade G	742,201	580,566
5 in, 25.6 lb/ft, Grade S	954,259	746,443

80% Pipe Capacity, Klbs	90% Pipe Capacity, Klbs
349	393
386	434
438	493
448	504
464	522
597	671



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Well Design Optimization

Drill Pipe Selection

- 5” Drill pipe S-135 -> Not H2S Resistance -> Cracking 

  a. Need to change Saver Sub at TDS

- Drill pipe G-105 -> H2S Resistance -> recommended for drilling Geothermal well
  a. Combine 5” DP 19.5 ppf and **5.5”**, 21.9 and 24.7 ppf:
     - Different size and Thread-> different handling tool and saver sub -> much longer rig time to make up pipe.



  b. Combine 5” DP 19.5 ppf and 5”, 25.6 ppf:
     - **Same size and same thread -> one handling tool and saver sub -> no additional rig time to make up.**



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Downhole Inspection Level

Features covered by inspection

Drill Pipe Inspection	DS-1 Category 1	2	3	4	5	API RP7G
Tube Body						
Visual Tube						Not Addressed
OD Gauging						Table only
UT Wall Thickness checks						
Electromagnetic Scanning						Not Addressed
MPI/Slip Area						
UT Slip Area						
Tool Joint						
Visual connection						Optional requirement
Fluorescent MPI of connection						
Thread profiling w/gauge						Not identified
OD/ID Gauging						OD only
Seal width/Condition						Condition only
Shoulder width						Table only
Bevel Diameter						Not Addressed
Seal Flatness						
Counter-bore depth						
Counter-bore Diameter						
Pin stretch						Option on request
Box Swell						
Tong Space						

Cat.	Remark
1	
2	• Well Depth < 5000 feet • Mud Weight < 9.0 ppg
3	• Well Depth 5000 – 10000 feet • Mud Weight 9.1 – 11.0 ppg
4	• Well Depth 10000 – 16000 feet • Mud Weight 11.1 – 14.5 ppg • Offshore, Inland water and remote location
5	• Well Depth > 16000 feet • Mud Weight > 14.5 ppg • Formation H2S concetration > 0.05 psia

API Inspection has no real universally understood meaning. DS-1 Inspection can vary from Category-1 to 5. DS-1 Inspection has a precise meaning, a category must be stated

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Directional – MWD Pricing System: Conventional Renumeration Table

COST ITEM DESCRIPTION	COST COMPONENT	UNIT PRICE	QTY	TOTAL COST
Mob	DD		1	
Demob	DD		1	
Moving Inter Well	DD		6.0	
26" Hole Section (Operating)				
9 5/8" Streeable Mud Motor, (2 ea)	DD		6	
25-3/4" Bottom Stabilizer of mud motor (2 ea)	DD		6	
9 1/2" Float Sub, (2 ea)	DD		6	
24 - 25-3/4" Top Stabilizer of Mud Motor, (2 ea)	DD		6	
9-1/2" EM-MWD, (2 ea)	DD		6	
9-1/2" Shock Sub, (2 ea)	DD		6	
9 1/2" , 30 feet NMDC, (2 ea)	DD		6	
Pony NMDC, (2 ea)	DD		6	
8" Drilling Jar, (1 ea)	DD		6	
26" Hole Section (Standby)				
9 5/8" Streeable Mud Motor, (2 ea)	DD		1	
25-3/4" Bottom Stabilizer of mud motor, (2 ea)	DD		1	
9 1/2" Float Sub, (2 ea)	DD		1	
24 - 25-3/4" Top Stabilizer of Mud Motor, (2 ea)	DD		1	
9-1/2" EM-MWD, (2 ea)	DD		1	
9-1/2" Shock Sub, (2 ea)	DD		1	
9 1/2" , 30 feet NMDC, (2 ea)	DD		1	
Pony NMDC, (2 ea)	DD		1	
8" Drilling Jar, (1 ea)	DD		1	
17-1/2" Hole Section (Operating)				
9 5/8" Streeable Mud Motor, (2 ea)	DD		29	
17-1/4" Bottom Stabilizer of mud motor, (2 ea)	DD		29	
9 1/2" Float Sub, (2 ea)	DD		29	
14-3/4" - 17" Top Stabilizer of Mud Motor, (2 ea)	DD		0	
14-3/4" - 171/4" String Stabilizer of Mud Motor, (2 ea)	DD		29	
9-1/2" EM-MWD, (2 ea)	DD		29	
9-1/2" Shock Sub, (2 ea)	DD		29	
9 1/2" , 30 feet NMDC, (2 ea)	DD		29	
Pony NMDC, (2 ea)	DD		29	
8" Drilling Jar, (1 ea)	DD		29	

Page 1

- Not Optimum Equipment Inventory
- Company Man, Material Man to manage own logistic -> less time for drilling operation control.
- Difficulty in checking invoices -> more man hours
- More risk for audit findings.

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Directional – MWD Pricing System: Lumpsum Per Day for Each Hole Section

NO	DIRECTIONAL CHARGE		Unit	Qty a	Unit Price b	Total Price c=a*b
A	**Mobilization and Demobilization Cost**					
1	Mobilization Cost for First well		LS	1		0
2	Demobilization Cost for Last well		LS	1		0
3	Mobilization and Demobilization Cost between well		LS/Well	9		0
					Total A	0
NO	DIRECTIONAL CHARGE	Qty Per Well a	Unit	Number Of Well b	Unit Price c	Total Price c=a*b
B	**Service Charge**					
1	Directional Drilling Equipment, MWD, Directional Driller and MWD personnel:					
1.1	Directional Drilling Hole Section 26"	9	Days	10		0
1.2	Directional Drilling Hole Section 17-1/2"	20	Days	10		0
1.3	Drill Out Cement with 12-1/4" Rotary BHA	1	Days	10		0
1.4	Directional Drilling Hole Section 12-1/4"	20	Days	10		0
1.5	Directional Drilling Hole Section 9-7/8"	5	Days	10		0
1.6	Rotary Drilling Hole Section 7-7/8" or 8-1/2"	2	Days	5		0
1.7	Directional Drilling Hole Section 7-7/8" or 8-1/2"	2	Days	5		0
1.8	Run Push BHA for 10-3/4" or 8-5/8" or 7" perforated Liner	1	LS/Run	5		0
2	Case Hole Sidetrack in 13-3/8" casing:					
2.a	13-3/8" Whipstock	1	Set	2		0
2.b	Sidetrack Drilling Hole Section 12-1/4" (include MWD Mud Pulse)	4	Days	2		0
3	17-1/2" Hole Opener for clean out 13-3/8" Liner Adapter					
3.a	Lumsump Charges per well	1	LS/Well	10		0
3.b	Cone Cutter for Hole Opener (Consumable)	1	Set	1		0
4	LumpSum Gyro Survey per Run, Including Mob/Demob	1	Lot	5		0
					Total A	0

- Vendor will do optimization and manage for their tool inventory to meet their best LS Unit price
- Simple timesheet and invoicing, less man hours.

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Well Design Optimization

Directional – MWD Pricing System: Lumpsum Per Day for Each Hole Section

NO	DIRECTIONAL CHARGE	Unit	Probability a	Unit Price b	Total Price c=a*b
C	**Lost in Hole Charges**				
1	Mud Motor with 9-1/4" to 9-3/4" OD	Each	0,25		0
2	Mud Motor with 8" to 8-1/2" OD	Each	0,25		0
3	Mud Motor with 6-1/2" to 7" OD	Each	0,25		0
4	Electromagnectic Measurement While Drilling 9-1/4" OD or equivalent	Each	0,25		0
5	Electromagnectic Measurement While Drilling 8" OD or equivalent	Each	0,25		0
6	Electromagnectic Measurement While Drilling 6-3/4" OD or equivalent	Each	0,25		0
7	Drilling Jars, Hydro Mechanical, 8" OD body or equivalent	Each	0,25		0
8	Drilling Jars, Hydro Mechanical, 6-3/4" OD body or equivalent	Each	0,25		0
9	NMDC, 9-1/4" OD body or equivalent	Each	0,25		0
10	NMDC, 8" OD body or equivalent	Each	0,25		0
11	NMDC, 6-3/4" OD body or equivalent	Each	0,25		0
12	Pony NMDC, 9-1/4" OD body or equivalent	Each	0,25		0
13	Pony NMDC, 8" OD body or equivalent	Each	0,25		0
14	Pony NMDC, 6-3/4" OD body or equivalent	Each	0,25		0
15	Near Bit Stabilizer, 9-1/4" OD body or equivalent	Each	0,25		0
16	Near Bit Stabilizer, 8" OD body or equivalent	Each	0,25		0
17	Near Bit Stabilizer, 6-3/4" OD body or equivalent	Each	0,25		0
18	String Bit Stabilizer, 9-1/4" OD body or equivalent	Each	0,25		0
19	String Bit Stabilizer, 8" OD body or equivalent	Each	0,25		0
20	String Bit Stabilizer, 6-3/4" OD body or equivalent	Each	0,25		0
21	Float Sub or Bit Sub, 9-1/4" OD body or equivalent	Each	0,25		0
22	Float Sub or Bit Sub, 8" OD body or equivalent	Each	0,25		0
23	Float Sub or Bit Sub, 6-3/4" OD body or equivalent	Each	0,25		0
24	Cross Over Sub, All type and sizes	Each	0,25		0
25	Shock Absorber Tool for 26" and 17-1/2" BHA	Each	0,25		0
26	17-1/2" Hole Opener for clean out 13-3/8" Liner Adapter	Each	0,1		0
27	Gyro Survey Tool	Each	0,1		0
28	String Stabilizer Mill for 13-3/8" Whipstock	Each	0,1		0
				Total C	0
				ESTIMATED CONTRACT VALUE FOR PATUHA (Total A+B+C)	0

- To include LIH price and estimate LIH probability in commercial evaluation
- Vendor to provide best LIH price to win this bid.
- Otherwise Vendor will give un-reasonable unit price -> and potential contract dispute if any LIH event.

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Efficiency in Drilling Operations

1. Handling Losses in the 26" Section
2. Stuck Pipe Prevention
3. Pipe Recovery Method
4. Post-Drilling Technical Evaluation

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Efficiency in Drilling Operations

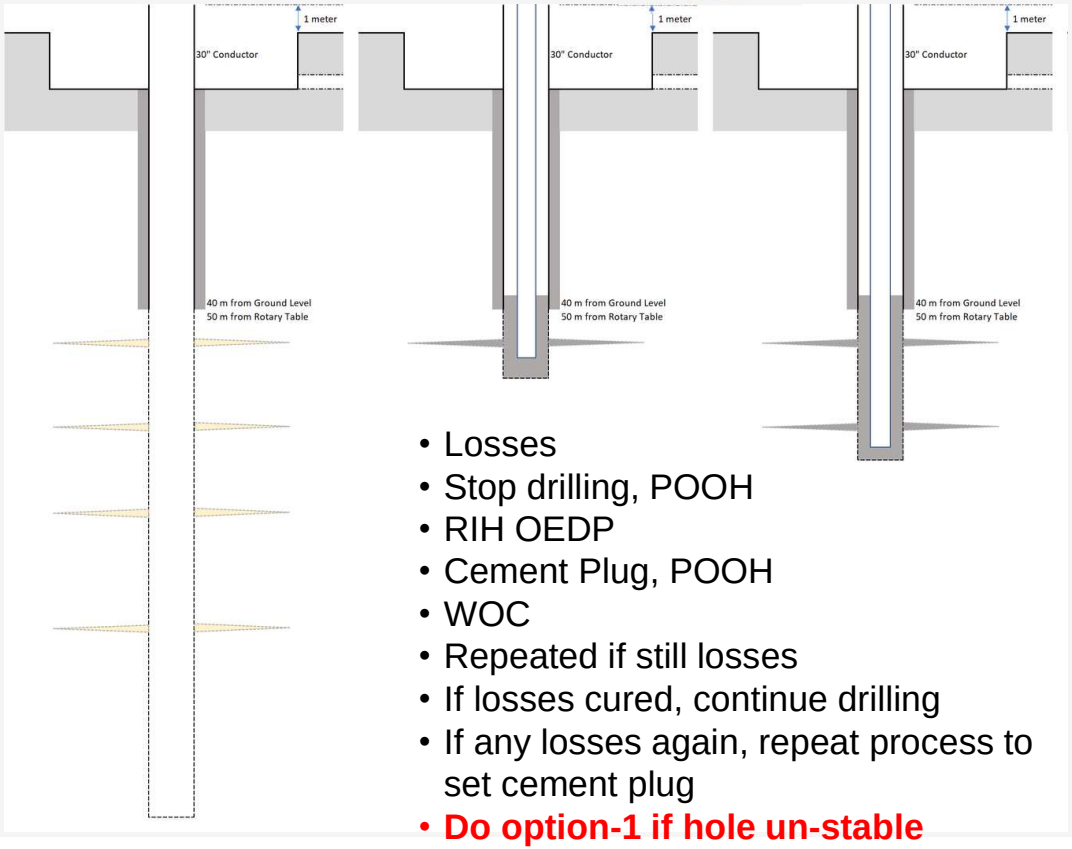
26” and 17-1/2” Curing Losses

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Efficiency in Drilling Operations

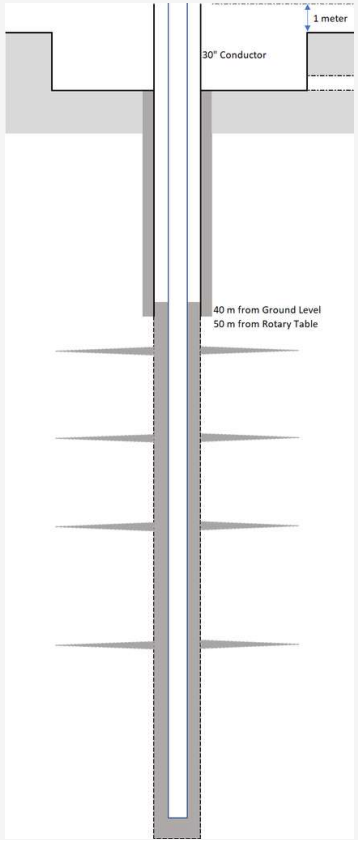
Surface Losses

1



- Losses
- Stop drilling, POOH
- RIH OEDP
- Cement Plug, POOH
- WOC
- Repeated if still losses
- If losses cured, continue drilling
- If any losses again, repeat process to set cement plug
- **Do option-1 if hole un-stable**

2



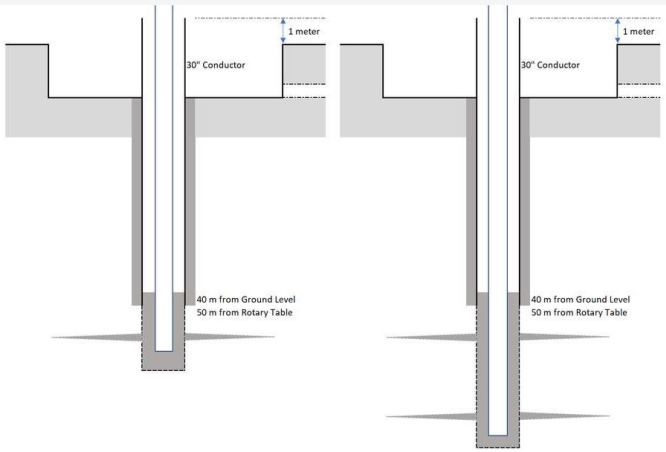
- Losses
- Spot LCM
- Keep drilling with losses to bottom losses depth
- Keep low concentration LCM in the system
- If losses cured by drilled cutting -> continue drilling
- If still severe losses, stop drilling -> do cement plug as per option-1.

Advantage:
- cement plug could be easier since drilled cutting has plugged the loss zone.

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Efficiency in Drilling Operations

Surface Losses – Cost Comparison

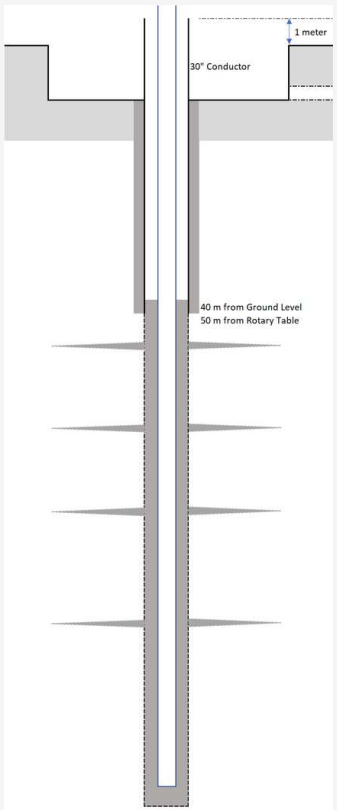


1

No	Activity Description	Hours
1	MU BHA, RIH	3
2	Drill 26" Hole to 50 m, losses	5
3	POOH	2
4	RIH OEDP, set cement plug	1
5	POOH, WOC	5
6	MU BHA, RIH	3
7	Drill 26" Hole to 100 m, losses	25
8	POOH	2
9	RIH OEDP, set cement plug	1
10	POOH, WOC	5
11	MU BHA, RIH	3
12	Drill 26" Hole to 150 m, losses	25
13	POOH	2
14	RIH OEDP, set cement plug	1
15	POOH, WOC	5
16	MU BHA, RIH	3
17	Drill 26" Hole to 200 m, losses	25
18	POOH	2
19	RIH OEDP, set cement plug	1
20	POOH, WOC	5
21	MU BHA, RIH	3
22	Drill 26" Hole to 250 m, losses	25
23	POOH	2
24	RIH OEDP, set cement plug	1
25	POOH, WOC	5
26	MU BHA, RIH	3
27	Drill 26" Hole to 300 m, losses	25
28	POOH	2
29	RIH OEDP, set cement plug	1
30	POOH, WOC	5
	TOTAL RIG TIME	196

2

No	Activity Description	Hours
1	MU BHA, RIH	3
2	Drill 26" Hole to 50 m, losses	5
3	Spot LCM	2
4	Drill 26" Hole to 300 m, losses	125
5	POOH	2
6	RIH OEDP, set cement plug	1
7	POOH, WOC	5
8	RIH OEDP, set cement plug	1
9	POOH, WOC	5
	TOTAL RIG TIME	149
	TOTAL RIG TIME SAVING	47



Cost saving: **USD 140 K**

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Efficiency in Drilling Operations

Stuck Pipe Prevention





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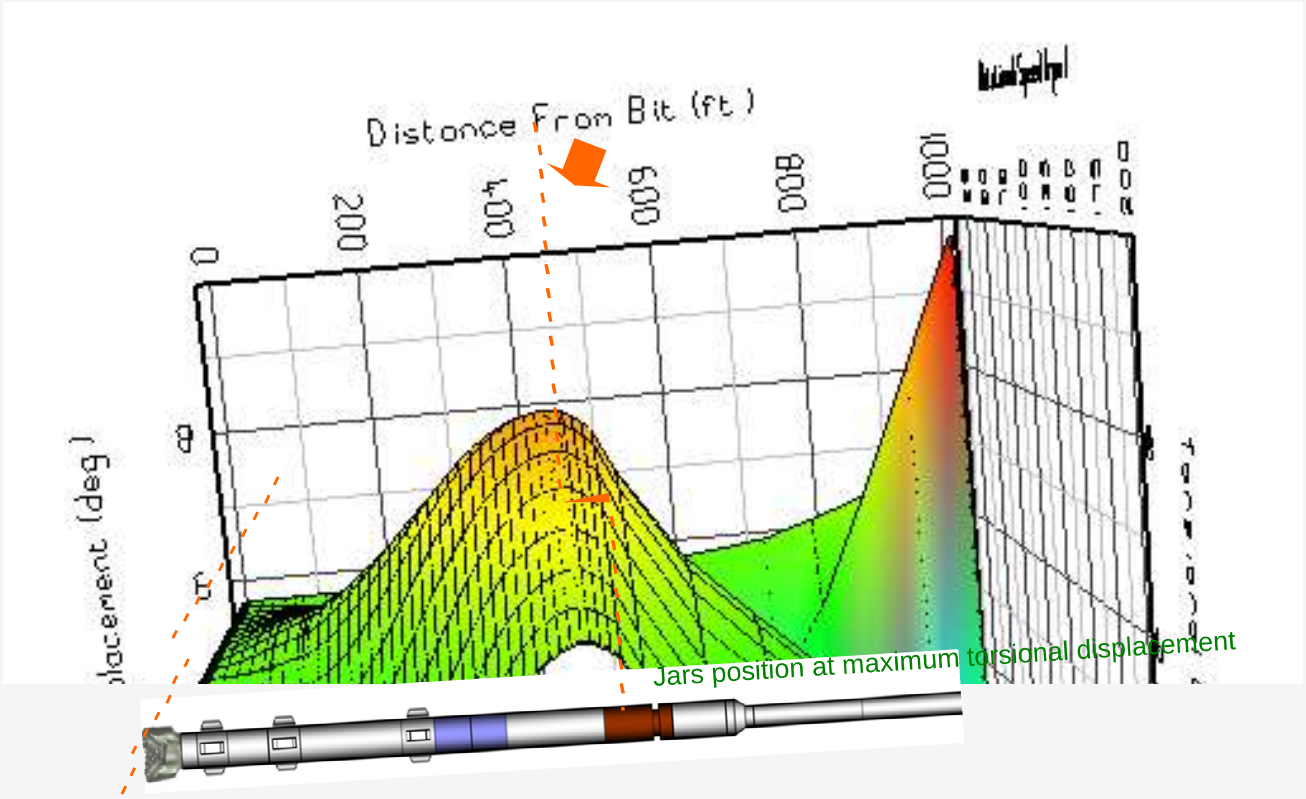
Stuck Pipe Prevention

Non Production Hole (26” & 17-1/2”)	Production Hole (12-1/4”)
• Enhance hole cleaning: ◦ High flow rate (900 – 1200 gpm) ◦ Sweep with fiber Hi-Vis ◦ Pump Tandem Hi-Vis if required • Use asphaltic mud chemical if any un-stable formation (ruble zone). • PLC-TLC: Pump 3 X LCM (50 -100 ppb) and drill blind to let drill cutting cure losses. • Closely monitor Off Bottom Torque • Only Use 1 or 2 string stabilizers • Use strong Hydro Mechanical Drilling Jars (**8” Jars Yield Torque > 100000 ft.lbs**	• Enhance hole cleaning: ◦ High flow rate (900 – 1000 gpm) ◦ Pump Tandem Hi-Vis • Drill with losses: ◦ PLC up to 200 bph: Keep drilling with mud ◦ PLC > 200 blh: Drill with aerated mud ◦ TLC: Drill blind with aerated mud or hi-rate technic. • Closely monitor Off Bottom Torque • Only Use 1 or 2 string stabilizers • Use strong Hydro Mechanical Drilling Jars • Monitor pick up weight to keep less pipe pulling capacity with 100-150 klbs over pulled to activated Drilling Jars up.

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Efficiency in Drilling Operations

BHA STUDY
Why Drilling Jars fail..?

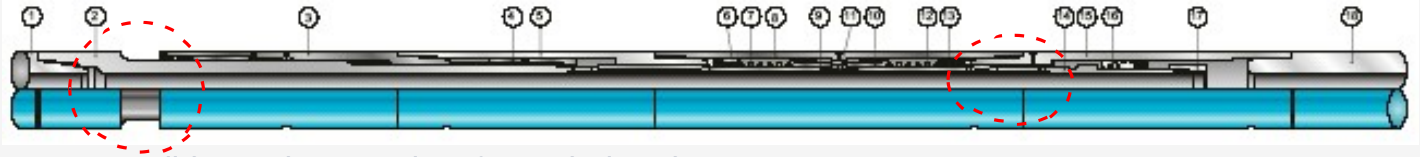


Wellplan 2000, Critical Speed Module

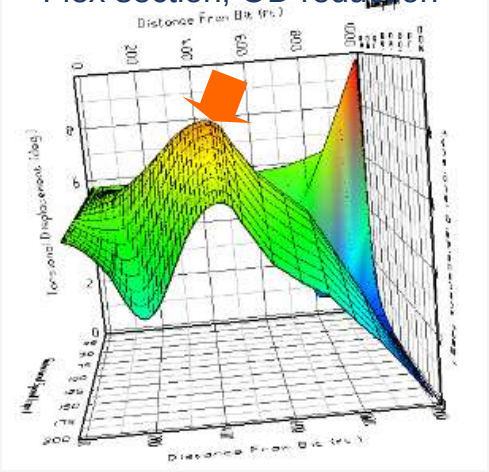
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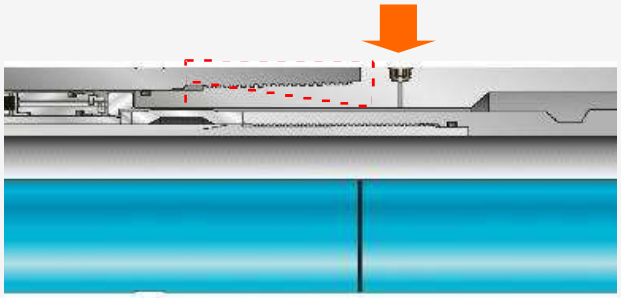
Efficiency in Drilling Operations

BHA STUDY: Jars Construction



- Not solid part, but consist of mandrel and housing
- Flex section, OD reduction





Weak Points (T Drilling Dynamics Study):

- ACME or non-API thread with sharper root radii than API RSC thread root
- Mid body connection have smaller cross-section areas
- Notch or hole in jars housing will accelerate fatigue
- Various internal component age traceability

Jars Placement

- Most of the jars were placed at distance from bit where torque displacement is maximum

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Efficiency in Drilling Operations

Drilling Jars Specification Comparison

Jars Type	Dailey L.I. Mechanical	Dailey Hydraulic Jars		Houston Engineering		Griffith-Vector Mechanical Jars		
Outside Diameter	7.75	7.75	8	7.75	8	7	8	
Inside diameter	2.75	3	3	3	3	2.5	2.81	
Tool Joint Size	6-5/8" Reg	6-5/8" Reg	6-5/8" Reg	6-5/8" Reg	6-5/8" Reg			
Tensile Yield" (x1000 lbs)	1148	1600	1750	1300	1600	828	965	Load after Jarring
Torsional Yield" (x1000 ft-lbs)	66.3	76.4	105	118	118	40	68	To yield body connection
Max. Overpull Up/Down (x1000 lbs)	125	260	300					
Approx. Length Extended (ft-in)	35'-6"	33'-0"	33'-0"	32'	32'	10.5	12.5	Latched
Approx. Weigth (lbs)	4000	3500	3800	3200	3550	1210	1730	
Free Travel Upstroke (in)	7-3/4"	7	7	8	8	4.75	5	
Free Travel Downstroke (in)	8-1/4"	7	7	7	7	6	6	
Total Stroke (in)								
Max. BHT (°F)	325	400	400					
Pump Open Area (sq. in.)	17.7	28.3	28.3			11	14.2	

Source: Jars Technical Specification from Manufacturer

Less Torsional Yield strength in its class

Jars yang kuat: Drilling Jars ukuran 8", sebaiknya mempunyai Torsional Yield lebih dari 100 ft-Klbs.

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Pipe Recovery Method

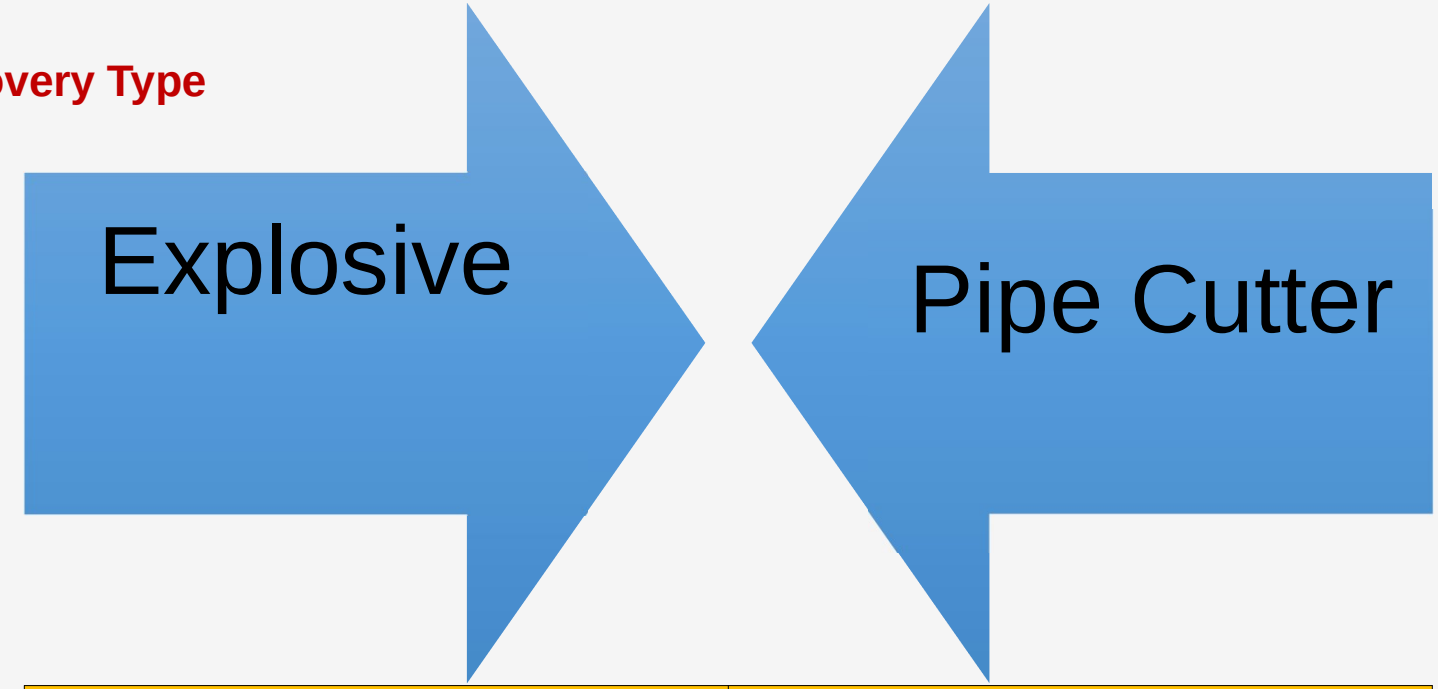




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Pipe Recovery Type

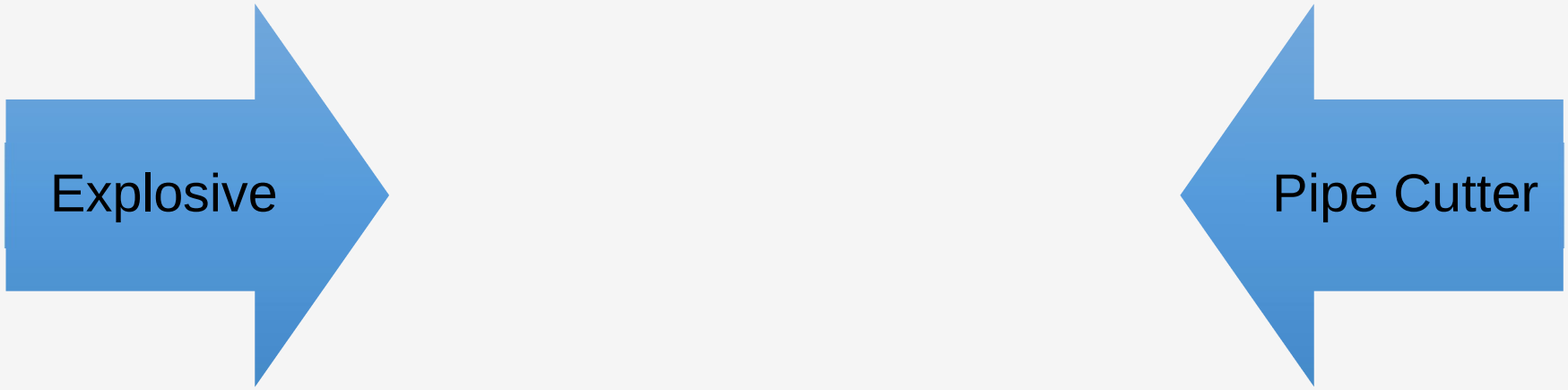


Could cut all type of BHA Component as long Explosive OD could pass drift ID of BHA component **Need various Permits**	**Only could cut 5” Drill Pipe or Smaller.** **Un-able to cut 5” HWDP and all DC type and sizes**

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Efficiency in Drilling Operations

Pipe Recovery Type: Cost Impact

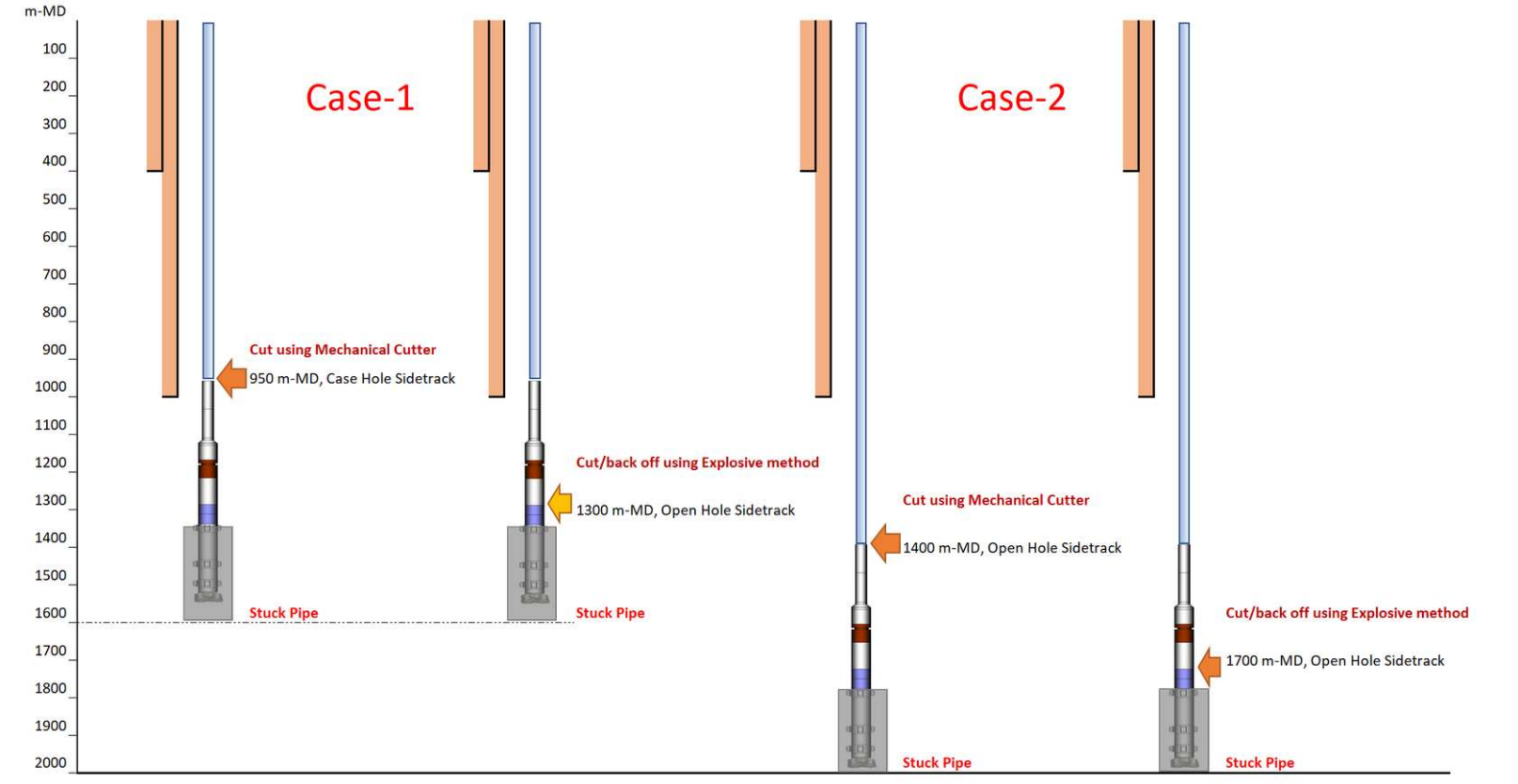


Explosive	Pipe Cutter
• Less BHA component left in hole ◦ More chance to recover BHA left in hole ◦ Less LIH Prices ◦ Deeper top of fish (TOF) -> deeper sidetrack point ◦ More chance TOF still in open hole -> easier to side track in open hole	• More (longer) BHA component left in hole ◦ Less chance to recover BHA left in hole ◦ Expensive LIH Prices ◦ Shallower top of fish (TOF) -> Shallow sidetrack point -> costly ◦ Risk of TOF in CASE hole -> Need Whipstock to side track -> costly

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Efficiency in Drilling Operations

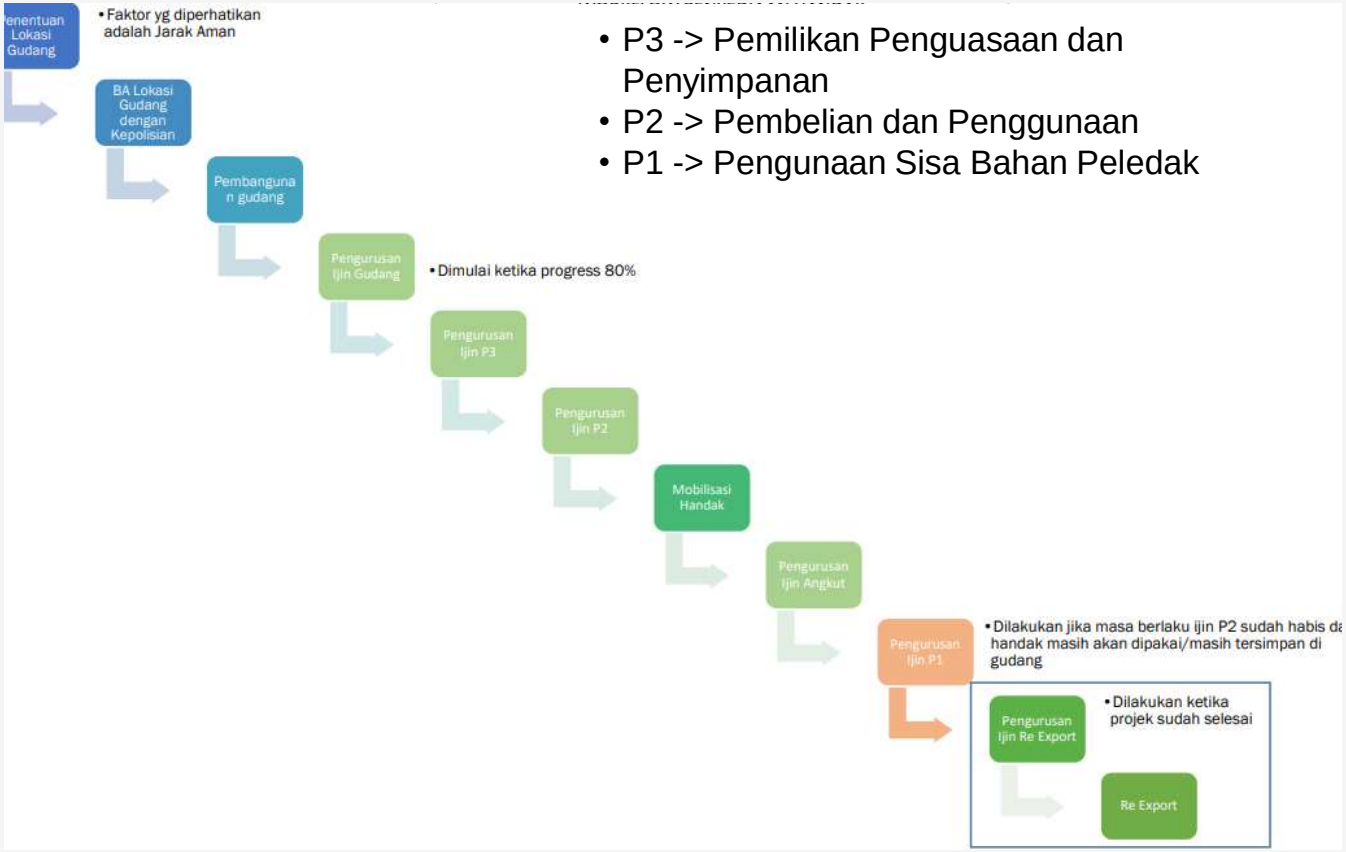
Pipe Recovery Type: Cost Impact



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Efficiency in Drilling Operations

Workflow For Explosive Handling



- P3 -> Pemilikan Penguasaan dan Penyimpanan
- P2 -> Pembelian dan Penggunaan
- P1 -> Pengunaan Sisa Bahan Peledak

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Procurement Strategies for Goods and Services

1. List of Equipment and Drilling Services Procurement
2. Review of Procurement Methods for Equipment and Drilling Services
3. Strategy for IPM/Turnkey Procurement Method
4. Strategy for Discrete or Bundle Procurement Method
5. Summary of Drilling Contracts and Drilling Cost Guidelines

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Procurement Strategies for Goods and Services

Procurement strategy for Drilling Material and Services



Purchase	Services	Support
1.Casing 2.Wellhead 3.Liner Adapter 4.Drilling Bits 5.Casing Accessories 6.DSA and X/O	1.Drilling Rig 2.Cement and Mud 3.Directional Drilling 4.Air Drilling 5.Mud Coller 6.Wellhead Installation 7.Drilling Personnel 8.Heavy Equipment 9.Drilling Waste Handling 10.VSAT 11.Drilling Report 12.Casing Handling 13.H2S Service 14.Mud Logging 15.Drill String Inspection 16.Hard Bending 17.Solid Control Services 18.Fishing Tool Services	1.Drilling Site Construction 2.Diesel Fuel 3.Water Pump Services 4.UKL / UPL 5.Security Services 6.Road Survey 7.Tubular Inspection 8.Explosive Handling 9.Permits 10.Perforated Liner 11.Logistic Services 12.General Services

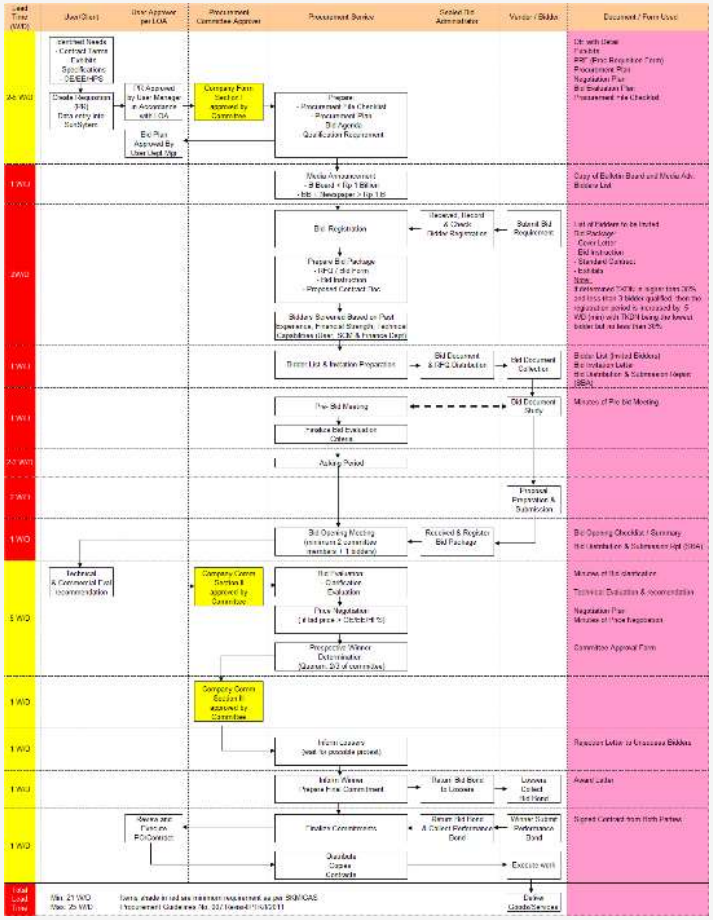
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Procurement Strategies for Goods and Services

COMPETITIVE BIDDING PROCESS

12 – 25 weeks:

- Pra Qualification
- CSMS Qualification
- Legal (Terms and Condition) Clarification
- Protest Clarification
- Technical Evaluation
- Commercial Evaluation
- Contract Award
- Delivery time
- Rig Up or Installation time
- Internal Approval
- Avoid any Conflict of interest



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Procurement Strategies for Goods and Services

Common Type of Drilling Procurement Method in Indonesia

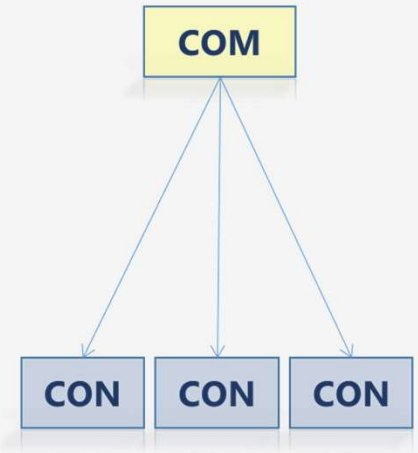
1. **Individual or Discrete Contract**
   - Company has drilling team to do tender process to select each drilling services.
2. **Bundled Contract -> Semi IPM**
   - Simplified Discrete Contract type by bundling 2 or more drilling services.
3. **Integrated Project Management (IPM)**
   - One Contractor do all drilling services with drilling risk at Company.
4. **Turnkey Project Contract**
   - Contractor do all drilling project (including drilling risk) with Lumpsum price / well.

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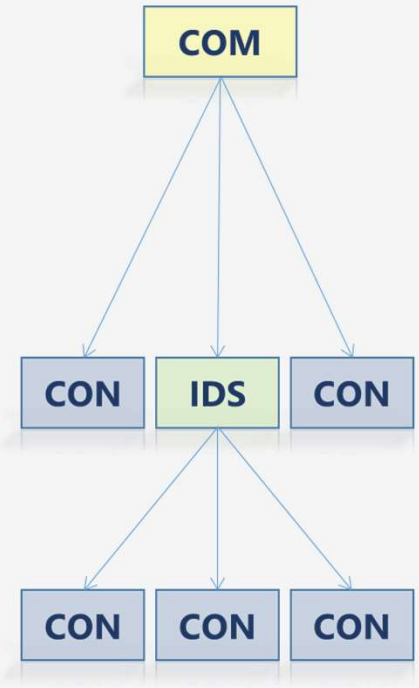
Procurement Strategies for Goods and Services

TYPE OF CONTRACT for Procuring Drilling Services

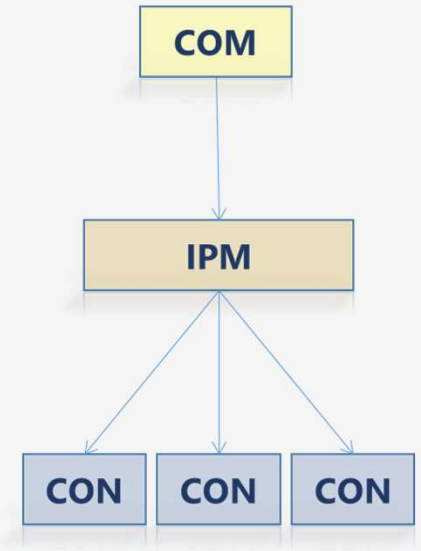
Discrete



Bundled



IPM / Turnkey





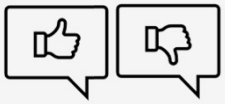


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Procurement Strategies for Goods and Services

Common in our mindset of Non DRILLER

- Discrete
- Bundle







- Semi IPM
- IPM, TurnKey

No risk, less risk



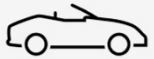
More experiences



Less man hours, less of paper work



Trend of procurement method



Faster in completion time





Take longer time



More man hours, resources



Lot of paper works



Old procurement method

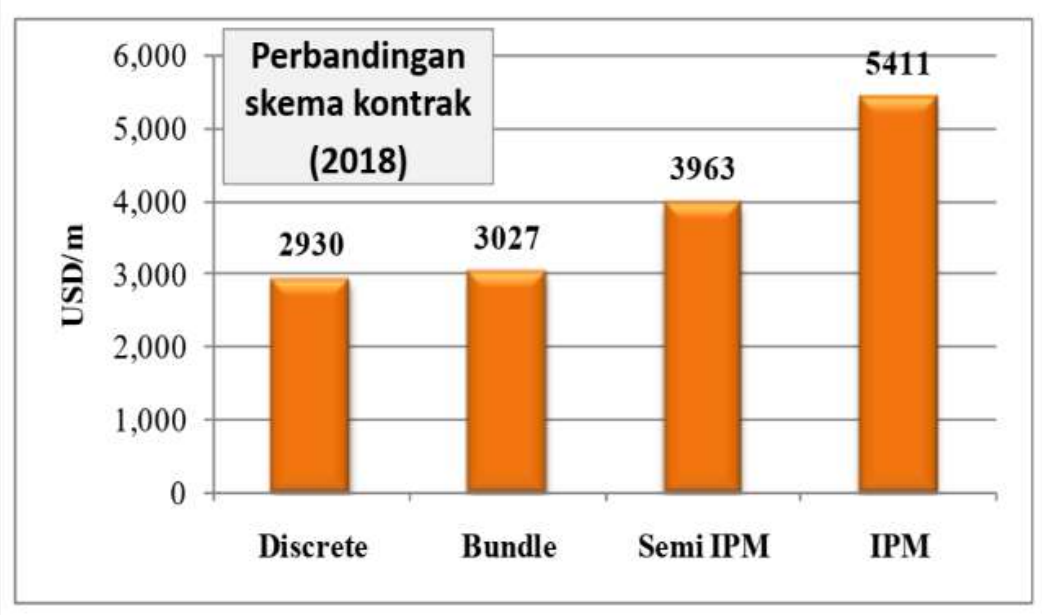


Manage Own Risk

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Procurement Strategies for Goods and Services

Why Drilling Procurement Method is critical?.



- Bundle vs IPM -> Cost/m Different: USD 2384 /m
- Bundle vs Semi IPM -> Cost/m Different: USD 936 /m

Drilling Contract Scheme Evaluation indicate the following trend:

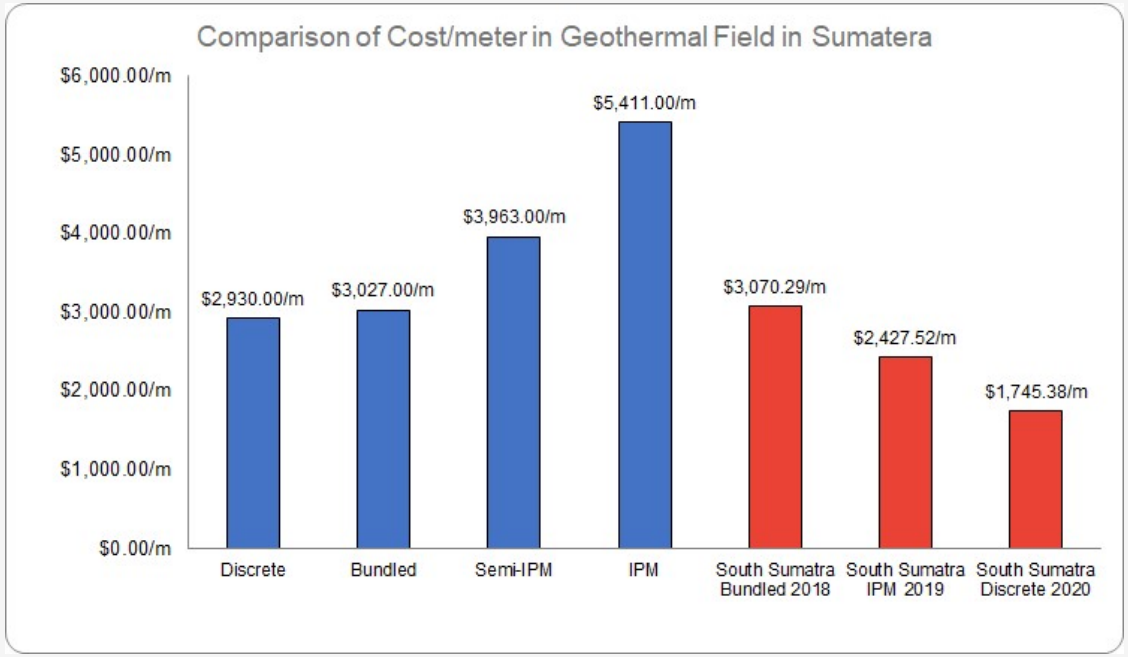
- 2012-2016: Majority use Semi-IPM and Full IPM (Avg. 2180 m Depth)
- 2017-2019: Majority use Discrete and Bundled (Avg. 2290 m Depth)
- Discrete and Bundled Contract scheme provide cheaper drilling cost.

Potential Cost Reduction: **USD 5.4 – 2.1 Million per Well**

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Procurement Strategies for Goods and Services

Why Drilling Procurement Method is critical?.



South Sumatra IPM in 2019:

- The IPM pricing was Lump Sum refer to TD range and capped with maximum drilling days.
- If exceeding maximum drilling days, Contractor will apply daily rate.

Cost Saving using Discrete for drilling in South Sumatra:

USD 682 per meter OR 1.5 Million per Well

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Procurement Strategies for Goods and Services

Drilling IPM (Integrated Project Management)



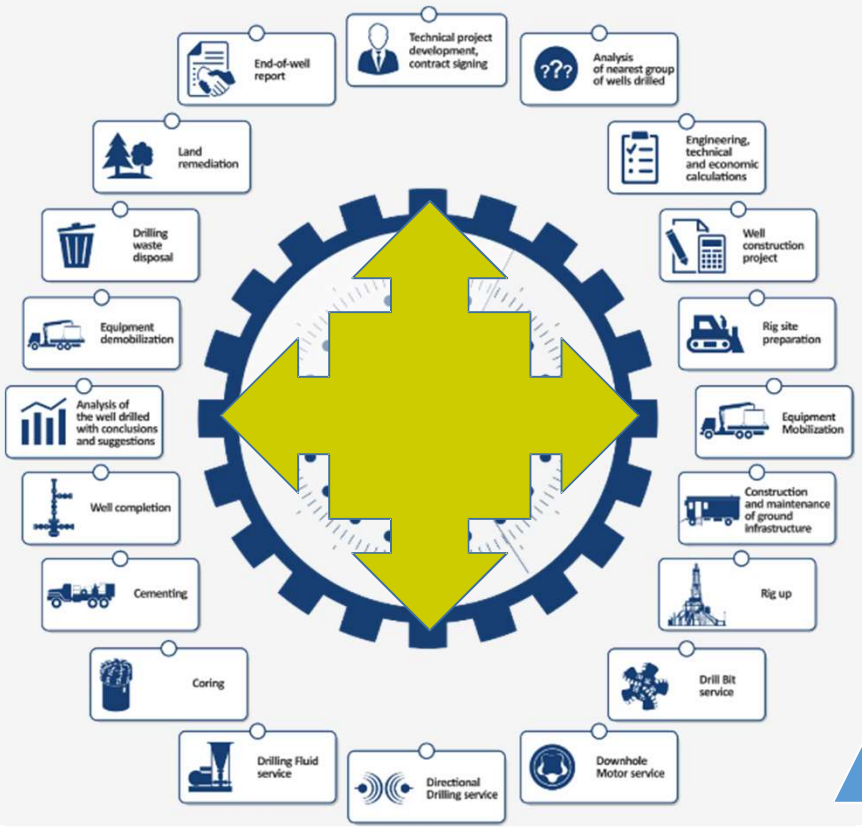


IPM
Moving your projects forward

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Procurement Strategies for Goods and Services

Drilling Integrated Project Management



- What are they offer
- Pros and Cons
- IPM Pricing & Risk Sharing
- IPM Selection & Control

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Procurement Strategies for Goods and Services

IDS / IPM ADVANTAGES

- COMPANY are fucusing on their core business and managing fewer CONTRACTORS
- CONTRACTORS are becoming more flexible in adapting to COMPANY requirement
- Experienced team members to provide best design and services

None of the above IPM promises mentioned the competitive drilling well cost ???

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IPM Contractors Market Survey

A major overspent for IPM providers comes from services that **are not their core business and outsourced**.

Those include, drilling rigs, casing, wellheads, logistics, vessels, civil works and a like. On some projects, the degree of outsourcing can reach as much as **40-90% of IPM scope**.

No	Services	Schlumberger	Halliburton	Baker	Greatwall	PDSI	PRA	Sari Pari	ADA	APS	Elnusa	Lekomars	Bormindo	Harman	Drilco	Teaka	Sucofindo	Surveyor Indonesia	Global Asia Petroleum	Tiga Musim Mas Jaya
1	IPM	•	•	•	•	•	•	•	•	•	•		•	•	•	•	•	•	•	•
2	Bundle	•	•	•	•	•	•			•	•	•								
3	Drilling Rig				•	•	•			•	•									
4	HWU / WO Rig				•	•	•			•	•	•	•							
5	Cement	•	•	•	•															
6	Drilling Fluid	•	•	•	•	•					•									
7	Directional Drilling	•	•	•	•	•														
8	Air Drilling					•			•											
9	Mud Coller				•	•														
10	Wellhead Installation																			
11	Drilling Personnel																			
12	Heavy Equipment																			
13	Drilling Waste Handling		•																	
14	VSAT		•			•														
15	Drilling Report	•	•																	
16	Casing Handling			•																
17	H2S Service					•														
18	Mud Logging	•		•		•														
19	Drill String Inspection	•										•								
20	Hard Bending	•																		
21	Solid Control Services	•	•																	
22	Fishing Tool Services	•		•																
23	Coring Services	•	•	•		•														
24	E-Line & Explosive	•	•	•																
25	Diesel Fuel																			
26	Water Pump Services																			
27	UKL / UPL																			
28	Security Services																			
29	Road Survey																			
30	Civil work																			
31	Tubular Inspection																			
32	Explosive Handling	•	•	•																
33	Permits																			
34	Perforated Liner																			
35	Logistic Services																			
36	General Services																			

Ownership of Drilling Services:

- IPM Vendors: 9%
- Sub Contractor: 91%

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Procurement Strategies for Goods and Services

Review of Drilling Contract Types

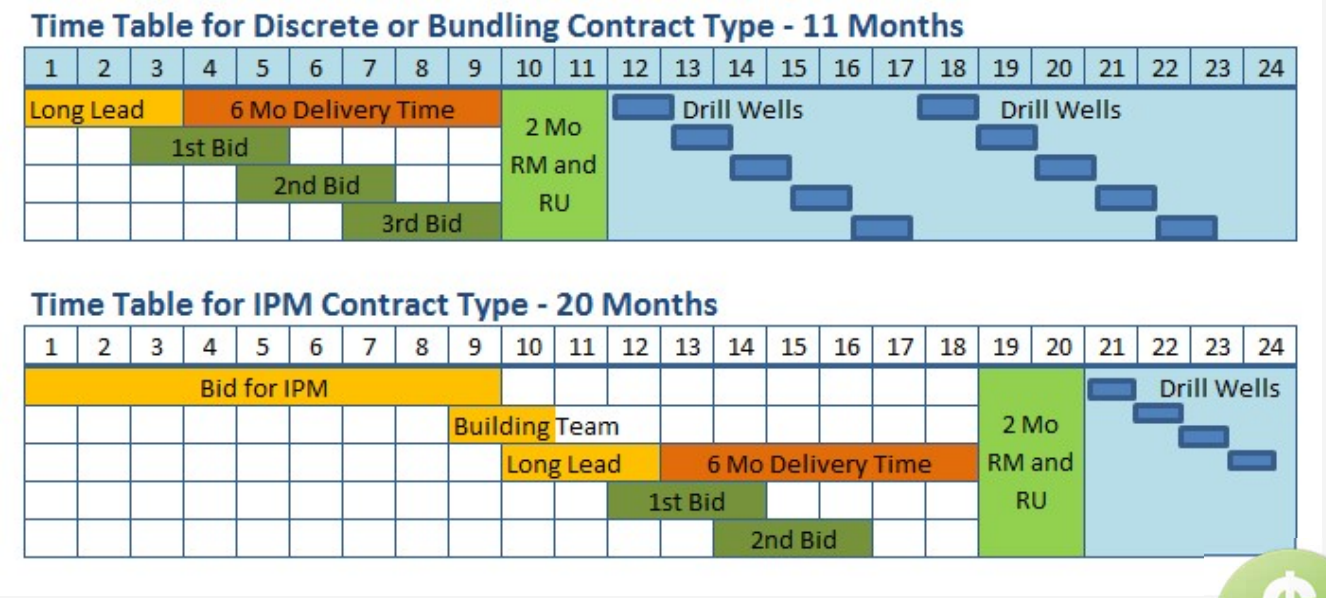
Description Turnkey Project	Discrete	Bundle	IPM	
Project Manager	Company Rep	Company Rep	Contractor Rep	Contractor Rep
Well Cost	5-6 M	4-5 M	7-8 M	7-10 M
Resources	> 10 person	3-8 person	3-5 person	2-3 person
No of Contract	30-50	< 10	2-5	1
Dispute Risk	Very Low	Low	Medium	High
Procurment Time				
Materials	6-9 months	6-9 months	6-12 months	6-12 months
Services	9-12 months	5-8 months	9-18 months	9-15 months
Interface	> 15 Reps	3-10 Reps	3-5 Reps	1 Reps
Pinalty	Individual	Individual	All Services	All Services
Excess Material	Yes	Yes	Yes	None
Master List Facility	Company To provide	Company To provide	Company To provide	None
Tax Over Tax	None	None - Minimum	Sub Contractors to SIPM	Sub Contractors to IPM
Double Profit	None	None - Minimum	Sub Contractors to SIPM	Sub Contractors to IPM
Mob/Demob	Individual	Individual-Share	Individual-Share	Share
Lesson Learnt	Minimum	Available	Available	Available
Vendor List	Lot	Lot	Adequate	Adequate

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Procurement Strategies for Goods and Services

Drilling Project Time Table

DISCRETE Versus IPM Services



Longer project delay -> more overhead cost



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Procurement Strategies for Goods and Services

Company Expectation from IPM contracts Scheme



Resources



Sharing Risk

Optimum Price



IPM

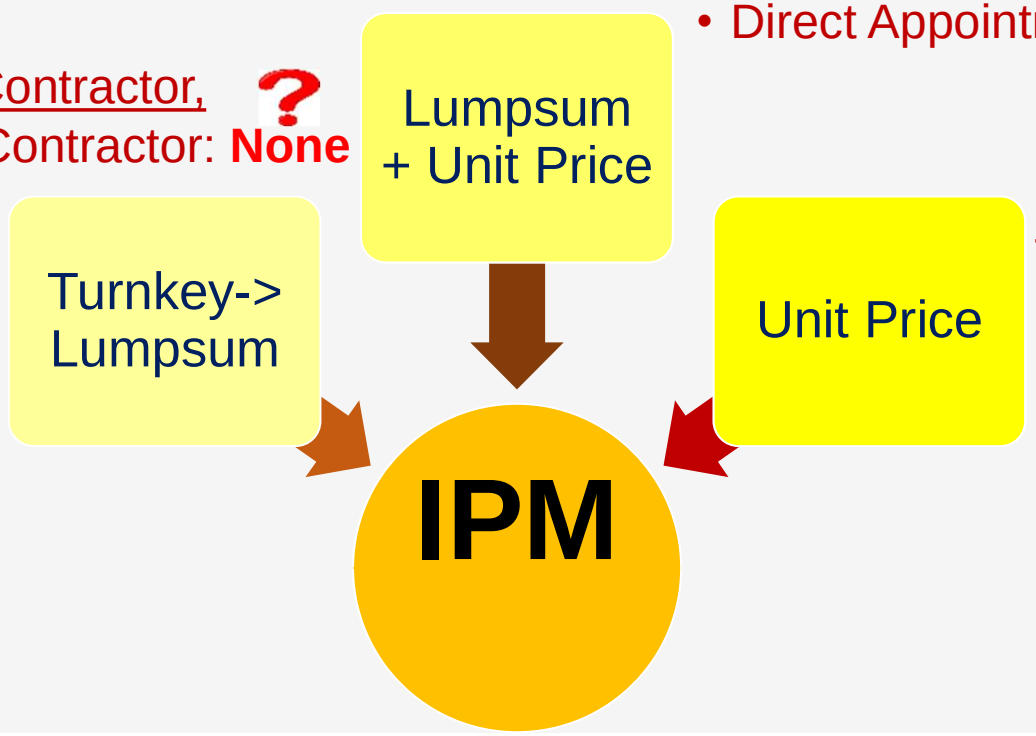
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Procurement Strategies for Goods and Services

Commercial Elements of IPM contracts

Pain-share & Gain-share,
- IPM Contractor: **GWDC** (the only one)
- Direct Appointment, less Competitive Price

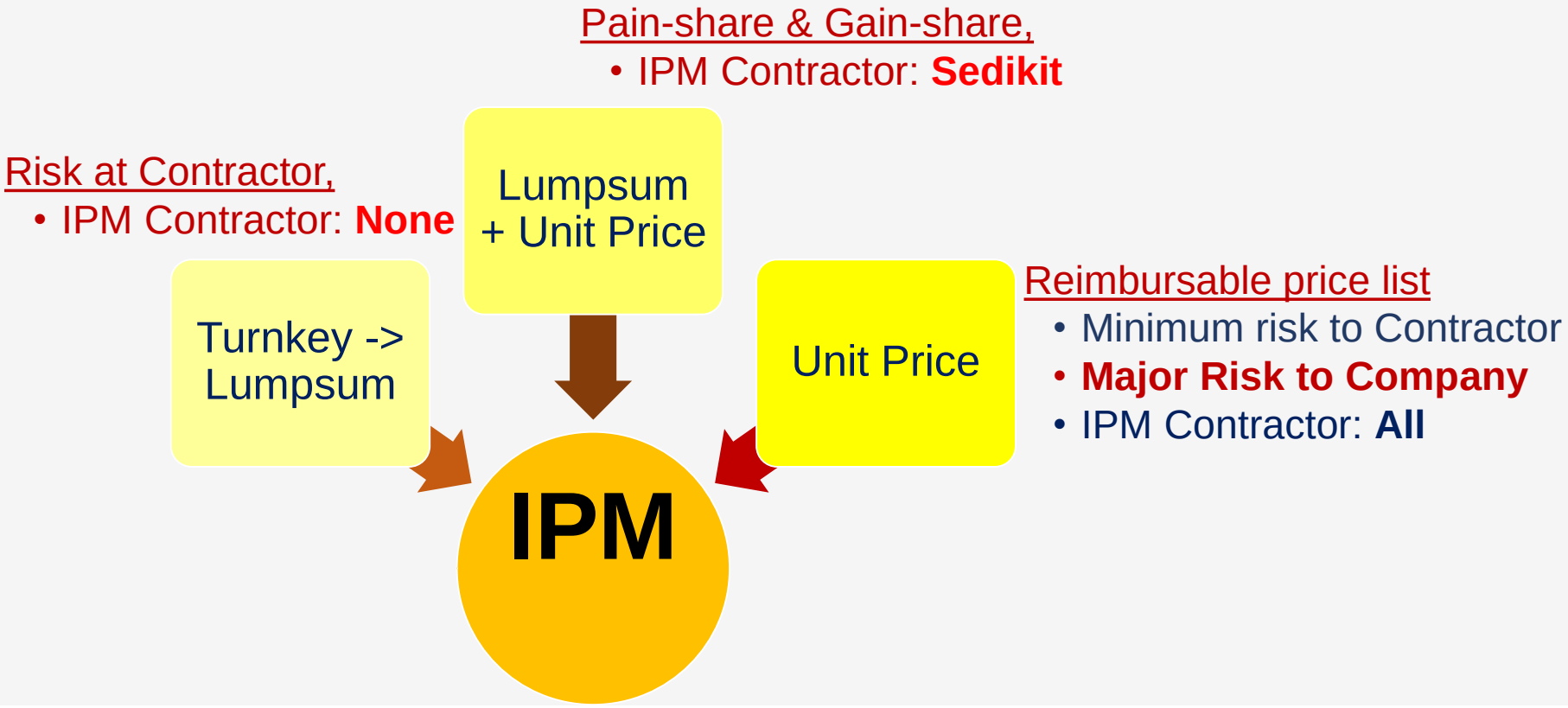
Risk at Contractor,
- IPM Contractor: **None**



Reimbursable price list
- Minimum risk to Contractor
- **Major Risk to Company**
- IPM Contractor: **All**

Procurement Strategies for Goods and Services

Commercial Elements of IPM contracts



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Procurement Strategies for Goods and Services

How IPM Contractor to build best Pricing ?



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Typical problem – IPM Drilling (Lumpsum)

1. **Dispute – additional cost -> un-predictable drilling events**
2. Less Flexibility to change -> Tight IPM contract
3. Low quality of drilling services -> Sister company of IPM, cheap sub-contract price
4. **Book Price utilization -> in-complete pricing system**
5. Delay on payment to Sub Contractor
6. Lack of HSE awareness
7. Exceed maximum working days -> less control from Company
8. High price on LIH -> not be evaluated in bidding
9. Community issue -> not part of IPM business
10. **Conflict of interest -> Big Contract Value**
11. Logistic issue -> Weak of financial capability
12. Timesheet manipulation -> less control from Company
13. In complete responsible matrix -> to big scope of work
14. Commencement date dispute
15. Excess Rig equipment problem -> limited drilling area
16. Use unnecessary services, material or chemicals -> less control from Company.
17. **Longer decision in handling drilling problem which cost related.**

Procurement Strategies for Goods and Services

Strong Legal Expert – IPM Contract

Drilling Manager



Legal



Drilling Engineer

Profit

Procurement & Logistic

Cost Control


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DO THE
THING
RIGHT
DO THE
RIGHT
THING

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Procurement Strategies for Goods and Services

Strategi Metode Pengadaan Discrete atau Bundle

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Procurement Strategies for Goods and Services

CONTRACT PRIORITY RANK - Delivery Time

Description	Moths
Drilling Personnel	2
Drilling Report	2
Water	2
General Services	2
Road Survey	2
Explosive Handling	2
Mud Coller	3
Wellhead Installation	3
Drilling Waste Handling	3
VSAT	3
Casing Handling	3
H2S Service	3
Mud Logging	3
Drill String Inspection	3
Hard Bending	3
Solid Control Services	3
Fishing Tool Services	3
Coring Services	3
Diesel Fuel	3
UKL / UPL	3
Civil work	3
Tubular Inspection	3
Perforated Liner	3
Cement and Mud	4
Directional Drilling	4
Heavy Equipment	4
Air Drilling	5
Drilling Bits	6
Liner Adapter	7
E-Line & Explosive	10
Drilling Rig	11
Casing	13
Wellhead	17





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Procurement Strategies for Goods and Services

CONTRACT PRIORITY RANK

FIRST	SECOND	NEXT
1.Casing 2.Wellhead 3.Liner Adapter 4.Drilling Rig 5.Drilling Personnel 6.E-Line & Explosive 7.Explosive Handling 8.UKL / UPL 9.Civil work 10.Permits 11.Road Survey	1.Drilling Bits 2.Cement and Mud 3.Directional Drilling 4.Heavy Equipment 5.Drill String Inspection 6.Perforated Liner 7.Diesel Fuel 8.Air Drilling 9.Mud Coller 10.Wellhead Installation	1.Water 2.General Services 3.Tubular Inspection 4.Drilling Waste Handling 5.VSAT 6.Drilling Report 7.Casing Handling 8.H2S Service 9.Mud Logging 10.Hard Bending 11.Solid Control Services 12.Fishing Tool Services 13.Coring Services

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Procurement Strategies for Goods and Services

Bundling and Discrete Contract Selection

A Non IPM Contract Type

Bundling

No	Description	Contractor Name			
		A	B	C	D
1	Directional Drilling				
	Coring Services				
	Cement and Mud				
	Mud Logging				
	Solid Control Services				
	E-Line & Explosive				
	Drilling Report				
	Explosive Handling				
	VSAT				
	Fishing Tool Services				

Discrete – Drilling Services

No	Description	Contractor Name			
		A	B	C	D
1	Drilling Rig + Mud Cooler				
2	Drilling Personnel				
3	Drilling Waste Handling				
4	H2S Service				
5	Drill String Inspection & Hard Banding				
6	Air Drilling				
7	Casing Handling				
8	Rig Inspection				
9	Wellhead Welding Services				
10	Heavy Equipment				
11	UKL / UPL				
12	Well Pad Construction				

Discrete – Purchase Material

1	OCTG
2	Liner Adapter
3	Drilling Bits
4	Wellhead and MV
5	Casing Accessories
6	Shaker Screen
7	DSA, Crossover, etc
8	Diesel Fuel

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Procurement Strategies for Goods and Services

MANAGING DRILLING CONTRACT



Procurement Strategies for Goods and Services

CONTRACT OWNER Responsibilities:

- Prepare SOW
- Prequalification Evaluation
- Pre-bid meeting and clarification
- Provide Justification (DA/DS)
- Tender technical evaluation
- Mobilization note
- Contract Summary
- Contract Monitoring
- Contract Amendment
- Check invoice

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Procurement Strategies for Goods and Services

CONTRACT OWNER

PIC	Drilling Contract
DE 3-4	1.Casing 2.Wellhead 3.Liner Adapter 4.Drilling Rig 5.Drilling Bits 6.Cement and Mud 7.Directional Drilling 8.Perforated Liner 9.Air Drilling 10.Mud Coller 11.Wellhead Installation 12.VSAT 13.Drilling Report 14.Casing Handling 15.Hard Bending 16.Solid Control Services 17.Fishing Tool Services 18.Coring Services 19.Drill String Inspection

PIC	Drilling Contract
Logistic & Procurement	1.Permits 2.Heavy Equipment 3.Diesel Fuel 4.General Services
Facility/ Civil	1.Permits 2.Road Survey 3.Water Pump Services
HSE	1.UKL / UPL 2.Drilling Waste Handling 3.H2S Service
G & G	1.E-Line & Explosive 2.Mud Logging
HRD	1.Drilling Personnel

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Procurement Strategies for Goods and Services

COMPETITIVE RENUMERATION TABLE

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Procurement Strategies for Goods and Services

Competitive Renumeration Table - Drilling Services

No	Drilling Services	Cost Saving Initiative
1	Directional Drilling	• LS Cost per day for each hole section • LIH include in contract price
2	Cementing Drilling Fluid	• Itemize unit price for each mud chemical • Consignment basis
4	Wire Line	• Stand rate for Unit and personnel • Tool: Call Out, no stand by rental cost • LS cost for only successful run job • Consignment Explosive material • LIH include in contract price
5	Aerated Drilling	• Stand by and Operating Monthly Rate
6	Fishing Services	• Monthly rate for dedicated fishing equipment • Call out daily rate for secondary fishing equipment
7	Water Supply	• Use Vitolic or steel pipe (1000 psi working pressure) instead of HDPE Pipe (300 psi working pressure) -> reduce no of water pumps -> less price.
8	Drilling Rig	• LS Rig Mob/Demob, Skidding, Move interpad • Include Fuel consumption • LIH include in contract price

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Procurement Strategies for Goods and Services

Competitive Renumeration Table - Drilling Services

No	Drilling Services	Cost Saving Initiative
9	Drilling Bit	• Consignment basis
10	Explosive	• Cost per run for back off and severing • Re-export remaining explosive
11	Cutting Disposal	• Lump Sum price per well
12		
13	Mud Logging, H2S, Casing Handling,	• Include in rig daily rate
14	Rig Inspection	• Lum sump per rig inspection
15	Tubular Inspection	• Daily Rate • USD per joint
16	Aerated Drilling	• LS Mob/Demob, Skidding, Move interpad • Use Synpol and Injection pump

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Procurement Strategies for Goods and Services

Ringkasan Kontrak Drilling dan Pedoman Biaya Pemboran

a. **Mengkompile** semua kontrak drilling dalam betuk soft copy.
b. **Membuat resume masing kontrak drilling** yang meliputi: point on/off hire, klausal pinalti, table harga, scope pekerjaan, matrik tanggung jawab, definisi, nomenklatur, ketentuan hilang dalam lubang, barang konsumable dan lain-lain
c. **Buatkan pedoman biaya pemboran (Drilling Cost Guide)** yang isinya semua table biaya jasa pemboran dan harga satuan material pemboran.
d. **Sosialisasikan** poin-poin diatas kepada seluruh tim drilling, kususnya kepada tim drilling operation yang ada di lapangan. Dengan sosialisasi ini akan meningkatkan awareness dan kepedulian semua tim pemboran atas besaran biaya yang akan timbul dan tentunya hal ini akan mempengaruhi cara memanage operasi pemboran secara lebih effisien.
e. Membuat rangkuman pedoman **tentang cara dan syarat-syarat pengajuan penagihan (invoicing)** dan proses pembayaran.
f. Membuat template untuk **pengontrolan spending** setiap kontrak pemboran.
g. Melakukan **rapat bulanan untuk monitor progress** setiap kontrak pemboran termasuk penyelesaian semua tagihan bulanan.
h. **Melakukan SQM (Service Quality Meeting)** untuk meningkatkan kinerja setiap kontrak pemboran. Termasuk mendiskusikan langkah-langkah efisiensi biaya pemboran kedepan.

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Conclusion

1. Membangun Departemen Drilling adalah suatu keharusan
2. Pengembangan Program penghematan biaya pemboran secara berkelanjutan
3. Efisiensi Biaya Pemboran merupakan dokumen hidup yang akan selalu terupdate oleh ide-ide baru
4. Biaya sumur murah, proyek lebih ekonomis, menarik investor, lebih banyak sumur yang dibor
5. Ucapan terima kasih kepada EBTKE yang sangat mensupport program Efisiensi Biaya Pemboran

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Conclusion



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Thank You!

+62-813 988 66556
yhartono.67@gmail.com



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Procurement Strategies for Goods and Services

1. Establishing a Drilling Department is Essential
2. Developing a Sustainable Drilling Cost Reduction Program
3. Drilling Cost Efficiency is a Living Document That Will Continuously Be Updated with New Ideas
4. Lower Well Costs Lead to More Economical Projects, Attract Investors, and Result in More Wells Being Drilled
5. Thank You to EBTKE for Supporting the Drilling Cost Efficiency Program

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Conclusion

Wellhead and BOP Installation



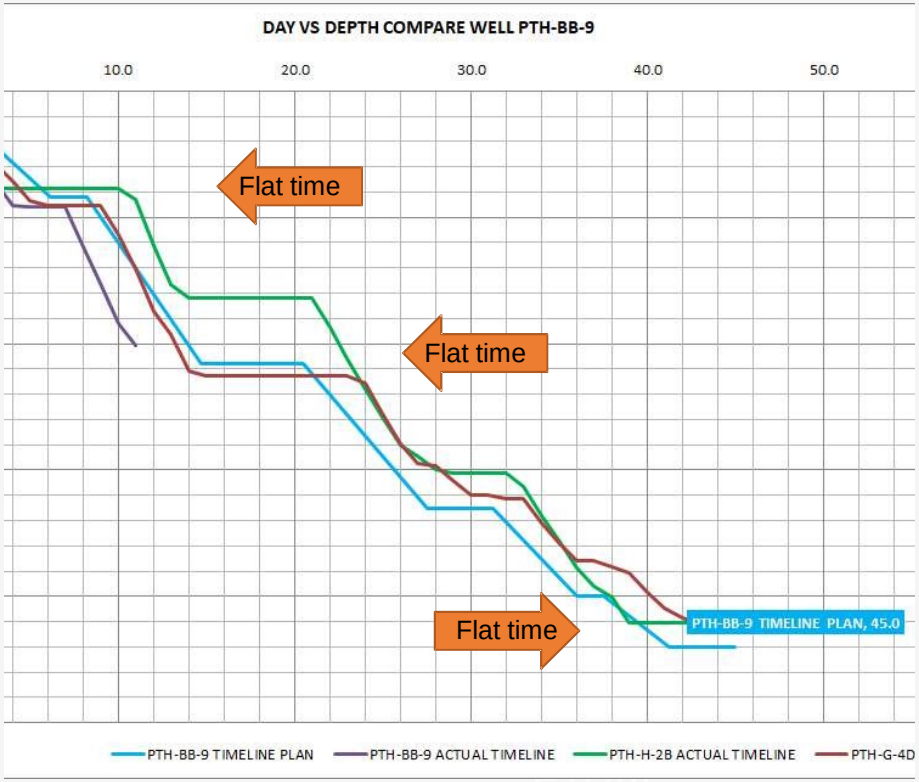


Profit

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Conclusion

Flat Time Improvement



Rig Flat Time Operation:

1. Wiper Trip and BHA trip out
2. Run Casing and Cementing Job (include WOC)
   - Include Clean out cement above TOL
   - Run 13-3/8” Tie Back Casing and Cementing
3. Nipple Down BOP and Install Wellhead
4. Nipple Up BOP (include BOP Test)
5. Make Up BHA and RIH BHA
6. LOT, FIT

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Faster Running Casing and Cementing


Breakout Machine


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Rig Power System – Use Own Electricity





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Rig Power System – Use Own Electricity

Fuel to Electric		
Base Camp Electricity		
Mini Camp		
Rig Power Unit		



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Effective in Managing Drilling Operation





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Other Potential Cost Saving





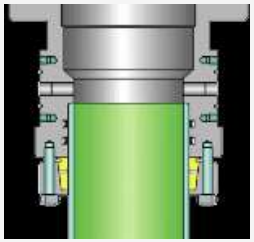
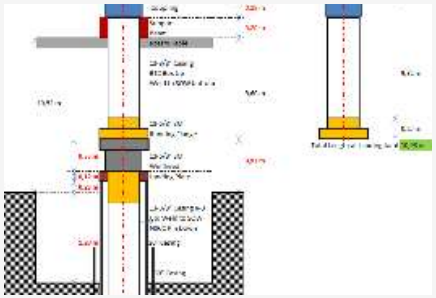
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Geothermal Drilling Operation Practice


1. Pre-installed 30” Conductor
2. 20-3/4” Wellhead Installation:
   - Pre-weld 20-3/4” Wellhead
   - Casing to casing welding
   - Use Slip Lock system bottom connection
3. Pre-welded 13-5/8” 3K Wellhead
4. Use 21-1/4” Single bop for all hole section
5. Pre-MU Float Equipment (FS, FC, LC)
6. PGE: Use Direct XCD Drillable Alloy Casing Bit -> consider not to perform wiper BHA trip.
7. SGI: Use Econolite or fast set cement to seal off 20” X 13-3/8” Liner Lap
8. Use CMC Polymer for viscosifier at 26” and 17-1/2” hole: cheaper, coarse -> additional advantage to cure losses
9. BHA Design:
   - BHA design: prefer Jars in Compression, Utilize HWDP to provide BHA weight for WOB
   - Always check critical speed RPM
   - Always check off bottom torque, PU and Slack off weight
   - Drill production section w/ directional BHA (Motor bend 1.15°) to enlarge hole -> smoothing with running perforated casing liner, SCC BTC connection

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Efficient Mud Design





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Most Effective Geothermal Mud Design

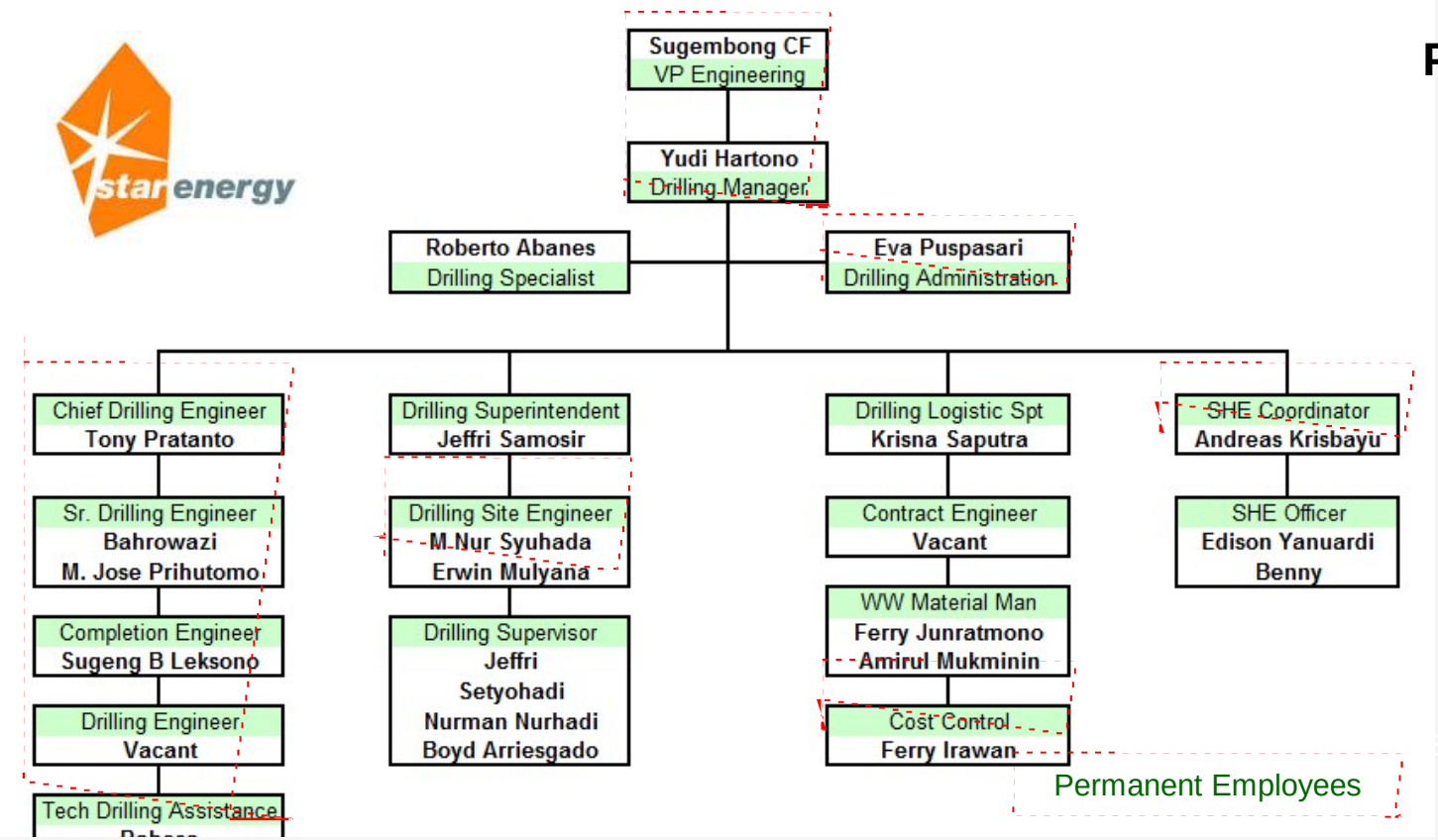
- Mud Type: KCL, NaCL or Lime?
- Hi Vis Chemicals: Bentonite vs XCD Polymer or mixed
- Effective Hi Vis Design: Single Hi-Vis, Tandem Hi Vis, Tandem with Light Mud. Option using Lifting agent (Fiber material)
- Corrosion Inhibitor: use from Mud or Aerated Drilling Company
- KCL (high price) vs Polyamin base
- Production section: when TLC, isolate remaining mud to be covert for Hi-Vis.
- Centrifuge: need or no need?
- Shaker Screen Type: Pyramid vs Flat

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Drilling Resources – Ideal



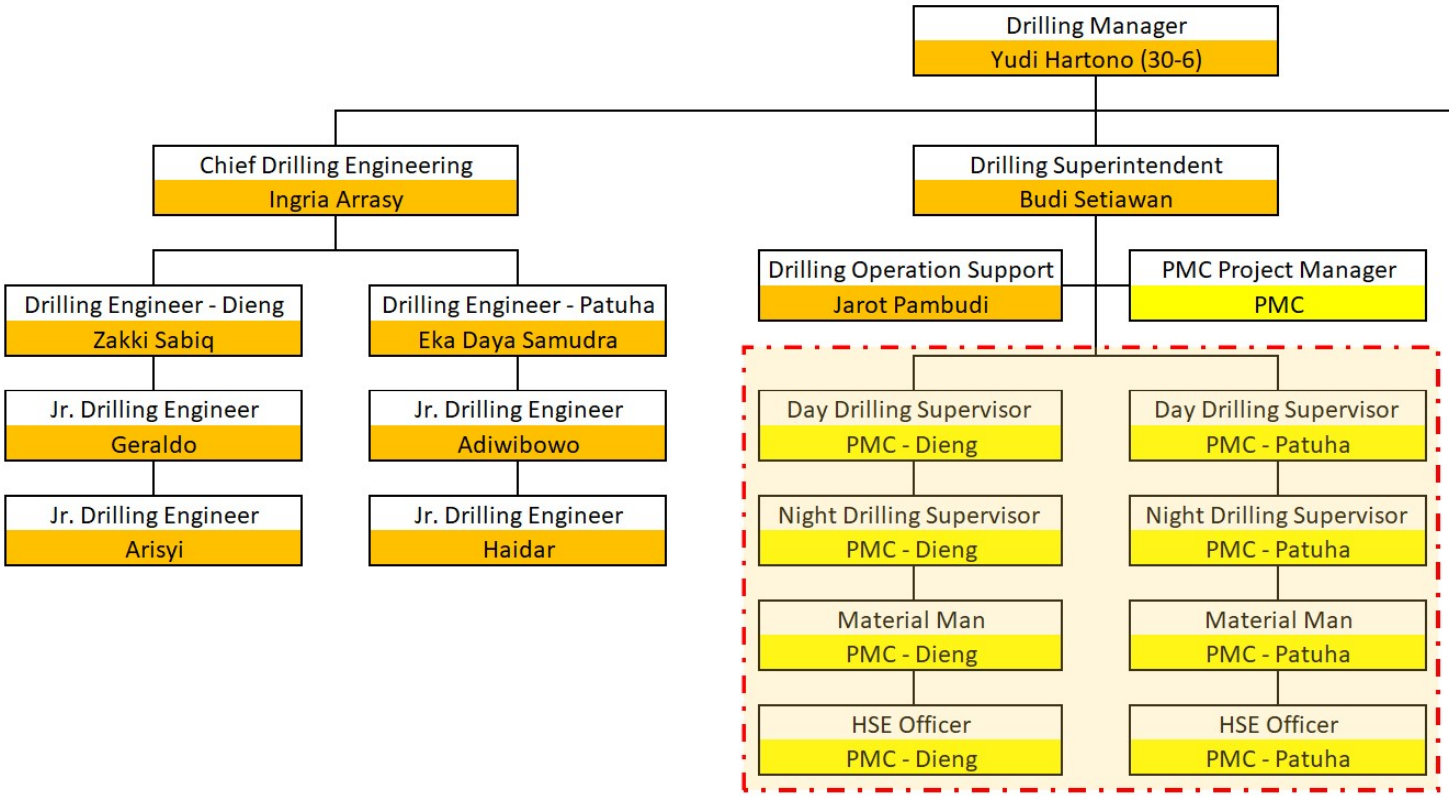
Project type:

- Drilling Oil/Gas (Onshore + Offshore)
- Drilling Geothermal
- Capable to perform Discrete drilling contract type

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Drilling Organization Structure - Optimum



Project List:

- Drilling Geothermal
  - Dieng-2
  - Patuha-2
- Capable to perform Bundled drilling contract type

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Use Left Hand Thread DP to Recover Pipe







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Effective Casing Running





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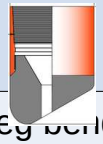
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Effective Run Casing

Casing	Topic	Concern
20”	Connection	• Do not use BTC or modified BTC (5 TPI = 5 thread per inch). Difficult to make up. Small thread is very sensitive to the casing alignment while being make up. • Use bigger thread, 3 TPI.
	Float Shoe	• No side port type. If any side port, closed it all. If any cutting fill, it will easier to wash down to bottom. • Preferable to have double float.
10-3/4” Perforated casing	Guide Shoe	• With eccentric nose 
	Drill 12-1/4” Hole	• Use Mud Motor with 1.15 deg bend to enlarge the hole size. It will minimize the casing stop off bottom.
	Running Speed	• Once the casing entering open hole section, increase running speed … mph. It will

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Drilling With Casing Option





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Drilling Contingency Plan





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List of Contingency Plan Equipment and Services

Equipment	Services	Objectives	Remark
24” XCD Bit	• Engineer • Pre-Make Up with Pup Joint Casing or Full Joint		
Casing XO: • 10-3/4” x 8-5/8” • 8-5/8” x 7”			
Wellhead Component: • Ring Gasket • Bolt and Nuts • Pressure Gauge • 3-1/8” Gate Valve			